Senior Thesis

Theophilus Bose

2026-09-07

```
rhfs.original <- read.csv("RHFS_2024_Public_Variables.csv")

# Turning Categorical Variables into Factors
rhfs.original <- rhfs.original |>
  mutate(
    across(
      c(NUMCAT_R, OUTCOME, PROPCON, PROPTOWN,
        COMPLEX, NUMBLD_R, YRNEWBLG_R, YRPROPACQ_R, APPRWAIVER,
        CASHDWN, HOWACQ, HOWMVAL, OFFSTPARKAHS, UNITSTAT,
        ADAGENCY, ADAGENT, ADNEWSPAPER, ADONLINE, ADOTHONLINE,
        ADVERTISE, ADWEBSITE, MLS, MNGMNT, NUMOTHPROP,
        NUMOTHUNITS, OWNENT, OWNOTHPROP, COMMSPACE, CONCSS,
        ICABLE, IELEC, IGAS, IPARKING, IPOOL,
        ISEWER, ITRASH, IWATER, IWIFI, LEASLNG,
        LNGTRMPRV, NORNTAPPFEE, PAYAPP, PAYCASH, PAYCHK,
        PAYCREDIT, PAYONLINE, PAYOTH, PAYTHIRD, RCONTROLAHS,
        RECCOMMREN, RNTAPPFEEWAIV, SECDEPAMT, SENIOR, WEEKLNG,
        OPTAX_R, PROPANS, ACCEPT, BENEOTH, ECONDEVLP,
        ELDERLY, GOVBELOW, GOVGRANT, HISTORIC, LOWINAREA,
        LOWINCTEN, OTHRCV, RSUBSIDY, STLOWINC, YRMORT_R,
        DEBT, DEBT1, GREENMORT, LOCTYP, MORTGET,
        PMTFREQ, REGPAY, RPIPMI, RPIPRIN, RPIPRPINS,
        RPIPRPTAX, WHYNOMRT, SAMPSTRATA, MORTAMT_R, OUTBAL_R, RNTAPPFEE_R,
        YRPROPREHAB_R),
      as.factor
    )
   )

# Renaming some factor levels for later convenience
rhfs.original <- rhfs.original |>
  mutate(
    NUMCAT_R = fct_recode(
      NUMCAT_R,
      "1 Unit" = "0",
      "2-4 Units" = "1",
      "5-24 Units" = "2",
      "25-49 Units" = "3",
      "50+ Units" = "4"
    ),
    OUTCOME = fct_recode(
      OUTCOME,
      "Interview Compl; self response" = "1",
```

```
      "Assisted living limited interview; self response" = "3",
      "Conversion from partial to full interview" = "5",
      "Interview Compl; field interview" = "201",
      "Assisted living limited interview; field interview" = "203",
      "Sufficient partial; field interview" = "204",
      "Military limited field interview" = "206"
    )
  )

# Rearranging variables greatly improves View(rhfs)
questionnaire_order <- c(
  # 1c-1e and property definition
  "PROPTOWN", "PROPCON", "COMPLEX",
  "NUMBLD_R", "NUMUNITS_R", "NUMCAT_R",

  # 2-4e
  "UNITSTAT", "YRNEWBLG_R", "HOWACQ", "YRPROPACQ_R", "PURPRICE_R",
  "CASHDWN", "CASHDWNAMT", "MRKTVAL_R", "HOWMVAL", "APPRWAIVER",

  # 5-6: occupancy, rents, and rent payment methods
  "BROOMS", "TOTRENTVAC", "TRENOC_R", "TRENVA_R", "TOWNOC_R",
  "TEMPOC_R", "TOTHVA_R", "LOWRENT", "HIRENT", "RENT",
  "RENTNO_R", "RENTLT249_R", "RENT274_R", "RENT299_R", "RENT349_R",
  "RENT399_R", "RENT449_R", "RENT499_R", "RENT549_R", "RENT599_R",
  "RENT649_R", "RENT699_R", "RENT749_R", "RENT799_R", "RENT899_R",
  "RENT999_R", "RENT1099_R", "RENT1249_R", "RENT1499_R", "RENT1999_R",
  "RENT2499_R", "RENT2999_R", "RENT3499_R", "RENT3999_R", "RENT4499_R",
  "RENT4999_R", "RENT5000_R", "BED0_R", "BED1_R", "BED2_R", "BED3_R",
  "PAYONLINE", "PAYTHIRD", "PAYAPP", "PAYCREDIT", "PAYCHK", "PAYCASH", "PAYOTH",

  # 6a-11c
  "LEASLNG", "LNGTRMPRV", "WEEKLNG", "WEEKLNGNUM", "WEEKLNGPCT",
  "RNTAPPFEE_R", "NORNTAPPFEE", "RNTAPPFEEWAIV", "PCTFEEWAIV_R", "SECDEPAMT",
  "CONCSS", "CONCSSNUM", "SENIOR", "RCONTROL", "RCONTROLAHS",
  "OFFSTPARK", "OFFSTPARKAHS", "IELEC", "IGAS", "IWATER", "ISEWER", "ITRASH",
  "ICABLE", "IWIFI", "IPOOL", "IPARKING", "PROPANS",

  # 12a-15e
  "TOTCOLL", "COMMSPACE", "RECCOMMREN", "COMMRENT", "MNGMNT", "HRSMNGMNT",
  "ADVERTISE", "ADWEBSITE", "ADONLINE", "ADOTHONLINE", "ADNEWSPAPER",
  "ADAGENCY", "ADAGENT", "MLS",

  # 16: operating expenses
  "OPPAY", "OPTAX_R", "OPINSUR", "OPELGAS", "OPWTRSWR", "OPREP", "OPMNG",
  "OPPROTH", "OPSECUR", "OPLAND", "OPOTHEX", "OPEX_R",

  # 17-18c: capital expenses and ownership
  "CAPEX_R", "CPHEATAMT", "CPRFAMT", "CPWINAMT", "CPDOORAMT", "CPFLRAMT",
  "CPEXTAMT", "CPPLBAMT", "CPELEAMT", "CPKITAMT", "CPBATAMT", "CPACCAMT",
  "CPOTHAMT", "CPRESAMT_R", "OWNENT", "OWNOTHPROP", "NUMOTHPROP", "NUMOTHUNITS",

  # 19-23: subsidies, tax credits, and tax reductions
  "HUDUNITS", "GOVBELOW", "RSUBSIDY", "ELDERLY", "GOVGRANT", "BENEOTH",
```

```
  "LIHTCUNITS", "STLOWINC", "HISTORIC", "LOWINCTEN", "LOWINAREA", "ECONDEVLP", "OTHRCV",

  # 24-37: debt/mortgage section and HUD voucher history
  "DEBT", "WHYNOMRT", "DEBTNUM", "DEBT1", "YRMORT_R", "MORTAMT_R", "OUTBAL_R",
  "APPVAL_R", "GREENMORT", "MORTGET", "LOCTYP", "REGPAY", "PMTAMT_R", "PMTFREQ",
  "RPIPRIN", "RPIPRPINS", "RPIPMI", "RPIPRPTAX", "ACCEPT"
)

rhfs.original <- rhfs.original |>
  relocate(
    CONTROLPUF, OUTCOME, SAMPSTRATA, SURVYR,
    all_of(questionnaire_order)
  ) |>
  # Keep the full-sample and replicate weights together at the far right.
  relocate(
    WEIGHT, starts_with("REPWGT"), .after = last_col()
  )
```

```
# Makes View(rhfs) much more productive.
rhfs <- rhfs.original |>
  select(-starts_with("REPWGT"), -WEIGHT,
         -SURVYR, # is always 2024
         # J flag variables mark values that were manually reviewed; note that
         # they may become relevant if outliers are under scrutiny.
         -starts_with("J"),
         # A continuous variable coded as a factor without documentation; may
         # be useful for pure prediction
         # -PCTFEEWAIV_R,
         # Poorly documented variable with 99.5% of its values missing
         -WEEKLNGPCT
         )

# NAs in the dataset
missingvalues <- function(df) {
  NAs <- sum(sapply(df, \(x) sum(x == -8))) # Not applicable
  NRs <- sum(sapply(df, \(x) sum(x == -9))) # Not reported

  tibble(
    NAs = NAs, # Not applicable
    NRs = NRs, # Not reported
    perc.missing = (NAs + NRs) / prod(dim(df)) # % of missing values in the df
  )
}
missingvalues(rhfs)
```

```
## # A tibble: 1 x 3
##      NAs    NRs perc.missing
##    <int>  <int>        <dbl>
## 1 170275 144234        0.399
```

```
# Missing values by interview type
rhfs |>
  summarize(
```

```
    # pick(everything()) extracts a data frame of all non-grouping columns
    missingvalues(pick(everything())),
    .by = OUTCOME
  )
```

```
##                                              OUTCOME   NAs   NRs perc.missing
## 1                  Interview Compl; field interview 97135 73188    0.3911694
## 2               Sufficient partial; field interview 29948 36570    0.5148054
## 3                    Interview Compl; self response 37769 21028    0.3212635
## 4         Conversion from partial to full interview  3442  7740    0.4512510
## 5                  Military limited field interview   297   839    0.7131199
## 6 Assisted living limited interview; field interview  1286  3761    0.7128531
## 7   Assisted living limited interview; self response   398  1108    0.7090395
```

```
# Handling known Trivially Missing values
# Certain values in the columns below can only exist if REGPAY is 1, so -9 is
# wrongly assigned to the corresponding REGPAY values.
trivially.missing.regpay <- rhfs |>
  mutate(id = row_number()) |>
  filter(
    REGPAY == -9 &
      # In a given observation, if any of these columns contain a value other than
      # -8 or -9 return True.
      if_any(c(PMTAMT_R, PMTFREQ, RPIPRIN, RPIPRPINS, RPIPMI, RPIPRPTAX), \(x) !x %in% c(-8, -9))
    ) |>
  select(id, REGPAY, PMTAMT_R, PMTFREQ, RPIPRIN, RPIPRPINS, RPIPMI, RPIPRPTAX) |>
  pull(id)

rhfs$REGPAY[trivially.missing.regpay] <- 1


trivially.missing.ownothprop <- rhfs |>
  mutate(id = row_number()) |>
  filter(
    OWNOTHPROP == -9 &
      if_any(c(NUMOTHPROP, NUMOTHUNITS), \(x) !x %in% c(-8, -9))
    ) |>
  select(id, NUMOTHPROP, NUMOTHUNITS) |>
  pull(id)

rhfs$OWNOTHPROP[trivially.missing.ownothprop] <- 1


trivially.missing.rntappfeewaiv <- rhfs |>
  mutate(id = row_number()) |>
  filter(
    RNTAPPFEEWAIV == -9 &
      if_any(c(PCTFEEWAIV_R), \(x) !x %in% c(-8, -9))
    ) |>
  select(id, PCTFEEWAIV_R) |>
  pull(id)

rhfs$RNTAPPFEEWAIV[trivially.missing.rntappfeewaiv] <- 1
```

```r
# Total rental receipts as a percent of property value (TRENTRCTPV_R) is
# calculated with (TOTCOLL + COMMRENT) / MRKTVAL. If COMMSPACE = 2, then
# COMMRENT = 0, so TRENTRCTPV_R = TOTCOLL / MRKTVAL as derived below.
trivially.missing.trentrctpv_r <- rhfs |>
  mutate(id = row_number()) |>
  select(id, TRENTRCTPV_R, COMMSPACE, COMMRENT, RECCOMMREN, TOTCOLL, MRKTVAL_R) |>
  filter(TRENTRCTPV_R == -8 & COMMSPACE == 2 &
      if_all(c(TOTCOLL, MRKTVAL_R), \(x) !x %in% c(-8, -9))
    ) |>
  pull(id)

rhfs$TRENTRCTPV_R[trivially.missing.trentrctpv_r] <-
  rhfs$TOTCOLL[trivially.missing.trentrctpv_r] /
  rhfs$MRKTVAL_R[trivially.missing.trentrctpv_r]


# When CASHDWN is 2, CASHDWNAMT is indeed always -9
rhfs |>
  mutate(id = row_number()) |>
  select(id, CASHDWN, CASHDWNAMT) |>
  filter(CASHDWN == 2 & CASHDWNAMT != -9)
```

```
## [1] id         CASHDWN    CASHDWNAMT
## <0 rows> (or 0-length row.names)
```

```r
# If a CASHDWN payment wasn't made, then the CASHDWNAMT should be 0
rhfs$CASHDWNAMT <- ifelse(rhfs$CASHDWN == 2, 0, rhfs$CASHDWNAMT)


# Same idea as with CASHDWN/CASHDWNAMT
rhfs |>
  mutate(id = row_number()) |>
  select(id, WEEKLNG, WEEKLNGNUM) |>
  filter(WEEKLNG == 2 & WEEKLNGNUM != -9)
```

```
##     id WEEKLNG WEEKLNGNUM
## 1 2173       2          0
## 2 3507       2          0
## 3 4197       2          0
```

```r
# If no units are offered for daily/weekly rental, then WEEKLNGNUM (the count
# of those that are offered) should be 0
rhfs$WEEKLNGNUM <- ifelse(rhfs$WEEKLNG == 2, 0, rhfs$WEEKLNGNUM)


# WEEKLNG and WEEKLNGNUM (daily/weekly rental availability variables) are -8
# when NUMUNITS != 1. For 7 of these properties, LEASLNG = 3 (lease length for
# single-units is weekly) implies that WEEKLNG and WEEKLNGNUM are 1.
weekly_targets <- rhfs |>
  mutate(row = row_number()) |>
  filter(NUMUNITS_R == 1, LEASLNG == "3") |>
```

```
  select(row, LEASLNG, WEEKLNG, WEEKLNGNUM)

rhfs <- rhfs |>
  mutate(
    WEEKLNG = replace(WEEKLNG, weekly_targets$row, "1"),
    WEEKLNGNUM = replace(WEEKLNGNUM, weekly_targets$row, 1)
  )


# Viewing targets for imputation
rhfs |>
  mutate(id = row_number()) |>
  select(COMMSPACE, COMMRENT, RECCOMMREN) |>
  filter(COMMSPACE == 2)
```

```
##      COMMSPACE COMMRENT RECCOMMREN
## 1            2       -9         -9
## 2            2       -9         -9
## 3            2       -9         -9
## 4            2       -9         -9
## 5            2       -9         -9
## 6            2       -9         -9
## 7            2       -9         -9
## 8            2       -9         -9
## 9            2       -9         -9
## 10           2       -9         -9
## 11           2       -9         -9
## 12           2       -9         -9
## 13           2       -9         -9
## 14           2       -9         -9
## 15           2       -9         -9
## 16           2       -9         -9
## 17           2       -9         -9
## 18           2       -9         -9
## 19           2       -9         -9
## 20           2       -9         -9
## 21           2       -9         -9
## 22           2       -9         -9
## 23           2       -9         -9
## 24           2       -9         -9
## 25           2       -9         -9
## 26           2       -9         -9
## 27           2       -9         -9
## 28           2       -9         -9
## 29           2       -9         -9
## 30           2       -9         -9
## 31           2       -9         -9
## 32           2       -9         -8
## 33           2       -9         -9
## 34           2       -9         -9
## 35           2       -9         -9
## 36           2       -9         -9
## 37           2       -9         -9
```

```
## 38          2       -9        -9
## 39          2       -9        -9
## 40          2       -9        -9
## 41          2       -9        -9
## 42          2       -9        -8
## 43          2       -9        -9
## 44          2       -9        -9
## 45          2       -9        -9
## 46          2       -9        -9
## 47          2       -9        -9
## 48          2       -9        -9
## 49          2       -9        -9
## 50          2       -9        -9
## 51          2       -9        -8
## 52          2       -9        -9
## 53          2       -9        -9
## 54          2       -9        -9
## 55          2       -9        -8
## 56          2       -9        -9
## 57          2       -9        -9
## 58          2       -9        -9
## 59          2       -9        -9
## 60          2       -9        -9
## 61          2       -9        -9
## 62          2       -9        -9
## 63          2       -9        -9
## 64          2       -9        -9
## 65          2       -9        -9
## 66          2       -9        -9
## 67          2       -9        -9
## 68          2       -9        -8
## 69          2       -9        -9
## 70          2       -9        -9
## 71          2       -9        -9
## 72          2       -9        -9
## 73          2       -9        -9
## 74          2       -9        -9
## 75          2       -9        -9
## 76          2       -9        -9
## 77          2       -9        -9
## 78          2       -9        -9
## 79          2       -9        -9
## 80          2       -9        -8
## 81          2       -9        -9
## 82          2       -9        -9
## 83          2       -9        -9
## 84          2       -9        -8
## 85          2       -9        -9
## 86          2       -9        -9
## 87          2       -9        -9
## 88          2       -9        -9
## 89          2       -9        -9
## 90          2       -9        -9
## 91          2       -9        -9
```

```
## 92          2       -9        -9
## 93          2       -9        -9
## 94          2       -9        -8
## 95          2       -9        -9
## 96          2       -9        -9
## 97          2       -9        -9
## 98          2       -9        -9
## 99          2       -9        -9
## 100         2       -9        -9
## 101         2       -9        -9
## 102         2       -9        -9
## 103         2       -9        -9
## 104         2       -9        -9
## 105         2       -9        -9
## 106         2       -9        -9
## 107         2       -9        -9
## 108         2       -9        -9
## 109         2       -9        -9
## 110         2       -9        -9
## 111         2       -9        -9
## 112         2       -9        -9
## 113         2       -9        -9
## 114         2       -9        -9
## 115         2       -9        -9
## 116         2       -9        -9
## 117         2       -9        -9
## 118         2       -9        -9
## 119         2       -9        -9
## 120         2       -9        -9
## 121         2       -9        -9
## 122         2       -9        -9
## 123         2       -9        -9
## 124         2       -9        -9
## 125         2       -9        -9
## 126         2       -9        -9
## 127         2       -9        -9
## 128         2       -9        -9
## 129         2       -9        -9
## 130         2       -9        -9
## 131         2       -9        -9
## 132         2       -9        -9
## 133         2       -9        -9
## 134         2       -9        -9
## 135         2       -9        -9
## 136         2       -9        -9
## 137         2       -9        -9
## 138         2       -9        -9
## 139         2       -9        -9
## 140         2       -9        -9
## 141         2       -9        -9
## 142         2       -9        -9
## 143         2       -9        -9
## 144         2       -9        -9
## 145         2       -9        -9
```

```
## 146          2       -9         -9
## 147          2       -9         -9
## 148          2       -9         -9
## 149          2       -9         -9
## 150          2       -9         -9
## 151          2       -9         -9
## 152          2       -9         -9
## 153          2       -9         -9
## 154          2       -9         -9
## 155          2       -9         -9
## 156          2       -9         -9
## 157          2       -9         -9
## 158          2       -9         -9
## 159          2       -9         -9
## 160          2       -9         -9
## 161          2       -9         -9
## 162          2       -9         -9
## 163          2       -9         -9
## 164          2       -9         -9
## 165          2       -9         -9
## 166          2       -9         -9
## 167          2       -9         -9
## 168          2       -9         -9
## 169          2       -9         -9
## 170          2       -9         -9
## 171          2       -9         -9
## 172          2       -9         -9
## 173          2       -9         -9
## 174          2       -9         -9
## 175          2       -9         -9
## 176          2       -9         -9
## 177          2       -9         -9
## 178          2       -9         -9
## 179          2       -9         -9
## 180          2       -9         -9
## 181          2       -9         -9
## 182          2       -9         -9
## 183          2       -9         -9
## 184          2       -9         -9
## 185          2       -9         -9
## 186          2       -9         -9
## 187          2       -9         -9
## 188          2       -9         -9
## 189          2       -9         -9
## 190          2       -9         -9
## 191          2       -9         -9
## 192          2       -9         -9
## 193          2       -9         -9
## 194          2       -9         -9
## 195          2       -9         -9
## 196          2       -9         -9
## 197          2       -9         -9
## 198          2       -9         -9
## 199          2       -9         -9
```

```
## 200          2       -9         -9
## 201          2       -9         -9
## 202          2       -9         -9
## 203          2       -9         -9
## 204          2       -9         -9
## 205          2       -9         -9
## 206          2       -9         -9
## 207          2       -9         -9
## 208          2       -9         -9
## 209          2       -9         -9
## 210          2       -9         -9
## 211          2       -9         -9
## 212          2       -9         -9
## 213          2       -9         -9
## 214          2       -9         -9
## 215          2       -9         -9
## 216          2       -9         -9
## 217          2       -9         -9
## 218          2       -9         -9
## 219          2       -9         -9
## 220          2       -9         -9
## 221          2       -9         -9
## 222          2       -9         -9
## 223          2       -9         -9
## 224          2       -9         -9
## 225          2       -9         -9
## 226          2       -9         -9
## 227          2       -9         -9
## 228          2       -9         -9
## 229          2       -9         -9
## 230          2       -9         -9
## 231          2       -9         -9
## 232          2       -9         -9
## 233          2       -9         -9
## 234          2       -9         -9
## 235          2       -9         -9
## 236          2       -9         -8
## 237          2       -9         -9
## 238          2       -9         -9
## 239          2       -9         -9
## 240          2       -9         -9
## 241          2       -9         -9
## 242          2       -9         -9
## 243          2       -9         -9
## 244          2       -9         -9
## 245          2       -9         -9
## 246          2       -9         -9
## 247          2       -9         -9
## 248          2       -9         -9
## 249          2       -9         -9
## 250          2       -9         -9
## 251          2       -9         -9
## 252          2       -9         -9
## 253          2       -9         -9
```

```
## 254          2       -9         -9
## 255          2       -9         -9
## 256          2       -9         -9
## 257          2       -9         -9
## 258          2       -9         -9
## 259          2       -9         -9
## 260          2       -9         -9
## 261          2       -9         -9
## 262          2       -9         -9
## 263          2       -9         -8
## 264          2       -9         -9
## 265          2       -9         -9
## 266          2       -9         -8
## 267          2       -9         -9
## 268          2       -9         -9
## 269          2       -9         -9
## 270          2       -9         -9
## 271          2       -9         -9
## 272          2       -9         -9
## 273          2       -9         -9
## 274          2       -9         -9
## 275          2       -9         -9
## 276          2       -9         -9
## 277          2       -9         -9
## 278          2       -9         -9
## 279          2       -9         -9
## 280          2       -9         -9
## 281          2       -9         -9
## 282          2       -9         -9
## 283          2       -9         -9
## 284          2       -9         -9
## 285          2       -9         -9
## 286          2       -9         -9
## 287          2       -9         -9
## 288          2       -9         -9
## 289          2       -9         -9
## 290          2       -9         -9
## 291          2       -9         -9
## 292          2       -9         -9
## 293          2       -9         -9
## 294          2       -9         -9
## 295          2       -9         -9
## 296          2       -9         -9
## 297          2       -9         -9
## 298          2       -9         -9
## 299          2       -9         -9
## 300          2       -9         -9
## 301          2       -9         -9
## 302          2       -9         -9
## 303          2       -9         -9
## 304          2       -9         -9
## 305          2       -9         -9
## 306          2       -9         -9
## 307          2       -9         -9
```

```
## 308          2       -9         -9
## 309          2       -9         -9
## 310          2       -9         -9
## 311          2       -9         -9
## 312          2       -9         -9
## 313          2       -9         -9
## 314          2       -9         -9
## 315          2       -9         -9
## 316          2       -9         -9
## 317          2       -9         -9
## 318          2       -9         -9
## 319          2       -9         -9
## 320          2       -9         -9
## 321          2       -9         -9
## 322          2       -9         -9
## 323          2       -9         -9
## 324          2       -9         -9
## 325          2       -9         -9
## 326          2       -9         -9
## 327          2       -9         -9
## 328          2       -9         -9
## 329          2       -9         -9
## 330          2       -9         -9
## 331          2       -9         -9
## 332          2       -9         -9
## 333          2       -9         -9
## 334          2       -9         -9
## 335          2       -9         -9
## 336          2       -9         -8
## 337          2       -9         -9
## 338          2       -9         -9
## 339          2       -9         -9
## 340          2       -9         -9
## 341          2       -9         -9
## 342          2       -9         -9
## 343          2       -9         -9
## 344          2       -9         -9
## 345          2       -9         -9
## 346          2       -9         -9
## 347          2       -9         -9
## 348          2       -9         -9
## 349          2       -9         -9
## 350          2       -9         -9
## 351          2       -9         -9
## 352          2       -9         -9
## 353          2       -9         -9
## 354          2       -9         -9
## 355          2       -9         -9
## 356          2       -9         -9
## 357          2       -9         -9
## 358          2       -9         -9
## 359          2       -9         -9
## 360          2       -9         -9
## 361          2       -9         -9
```

```
## 362          2       -9         -9
## 363          2       -9         -9
## 364          2       -9         -9
## 365          2       -9         -9
## 366          2       -9         -9
## 367          2       -9         -9
## 368          2       -9         -9
## 369          2       -9         -9
## 370          2       -9         -9
## 371          2       -9         -9
## 372          2       -9         -9
## 373          2       -9         -9
## 374          2       -9         -9
## 375          2       -9         -9
## 376          2       -9         -9
## 377          2       -9         -8
## 378          2       -9         -9
## 379          2       -9         -9
## 380          2       -9         -9
## 381          2       -9         -9
## 382          2       -9         -9
## 383          2       -9         -9
## 384          2       -9         -9
## 385          2       -9         -9
## 386          2       -9         -9
## 387          2       -9         -9
## 388          2       -9         -9
## 389          2       -9         -9
## 390          2       -9         -9
## 391          2       -9         -9
## 392          2       -9         -9
## 393          2       -9         -8
## 394          2       -9         -8
## 395          2       -9         -9
## 396          2       -9         -8
## 397          2       -9         -9
## 398          2       -9         -9
## 399          2       -9         -9
## 400          2       -9         -9
## 401          2       -9         -9
## 402          2       -9         -9
## 403          2       -9         -9
## 404          2       -9         -8
## 405          2       -9         -9
## 406          2       -9         -9
## 407          2       -9         -9
## 408          2       -9         -8
## 409          2       -9         -9
## 410          2       -9         -9
## 411          2       -9         -9
## 412          2       -9         -9
## 413          2       -9         -9
## 414          2       -9         -9
## 415          2       -9         -9
```

```
## 416          2       -9         -9
## 417          2       -9         -8
## 418          2       -9         -9
## 419          2       -9         -8
## 420          2       -9         -8
## 421          2       -9         -9
## 422          2       -9         -9
## 423          2       -9         -9
## 424          2       -9         -8
## 425          2       -9         -9
## 426          2       -9         -8
## 427          2       -9         -9
## 428          2       -9         -9
## 429          2       -9         -9
## 430          2       -9         -9
## 431          2       -9         -9
## 432          2       -9         -9
## 433          2       -9         -9
## 434          2       -9         -8
## 435          2       -9         -9
## 436          2       -9         -8
## 437          2       -9         -9
## 438          2       -9         -9
## 439          2       -9         -9
## 440          2       -9         -8
## 441          2       -9         -9
## 442          2       -9         -9
## 443          2       -9         -8
## 444          2       -9         -9
## 445          2       -9         -9
## 446          2       -9         -9
## 447          2       -9         -8
## 448          2       -9         -9
## 449          2       -9         -8
## 450          2       -9         -8
## 451          2       -9         -9
## 452          2       -9         -9
## 453          2       -9         -9
## 454          2       -9         -9
## 455          2       -9         -9
## 456          2       -9         -9
## 457          2       -9         -8
## 458          2       -9         -8
## 459          2       -9         -9
## 460          2       -9         -9
## 461          2       -9         -9
## 462          2       -9         -8
## 463          2       -9         -9
## 464          2       -9         -9
## 465          2       -9         -8
## 466          2       -9         -9
## 467          2       -9         -9
## 468          2       -9         -9
## 469          2       -9         -8
```

```
## 470          2       -9         -9
## 471          2       -9         -9
## 472          2       -9         -8
## 473          2       -9         -9
## 474          2       -9         -9
## 475          2       -9         -9
## 476          2       -9         -9
## 477          2       -9         -9
## 478          2       -9         -9
## 479          2       -9         -9
## 480          2       -9         -8
## 481          2       -9         -9
## 482          2       -9         -9
## 483          2       -9         -9
## 484          2       -9         -9
## 485          2       -9         -9
## 486          2       -9         -8
## 487          2       -9         -8
## 488          2       -9         -9
## 489          2       -9         -9
## 490          2       -9         -9
## 491          2       -9         -9
## 492          2       -9         -9
## 493          2       -9         -9
## 494          2       -9         -9
## 495          2       -9         -9
## 496          2       -9         -9
## 497          2       -9         -9
## 498          2       -9         -9
## 499          2       -9         -9
## 500          2       -9         -9
## 501          2       -9         -9
## 502          2       -9         -8
## 503          2       -9         -8
## 504          2       -9         -9
## 505          2       -9         -9
## 506          2       -9         -9
## 507          2       -9         -9
## 508          2       -9         -9
## 509          2       -9         -9
## 510          2       -9         -9
## 511          2       -9         -9
## 512          2       -9         -9
## 513          2       -9         -9
## 514          2       -9         -9
## 515          2       -9         -9
## 516          2       -9         -9
## 517          2       -9         -9
## 518          2       -9         -9
## 519          2       -9         -9
## 520          2       -9         -9
## 521          2       -9         -9
## 522          2       -9         -9
## 523          2       -9         -9
```

```
## 524          2       -9         -9
## 525          2       -9         -9
## 526          2       -9         -9
## 527          2       -9         -9
## 528          2       -9         -9
## 529          2       -9         -9
## 530          2       -9         -9
## 531          2       -9         -9
## 532          2       -9         -9
## 533          2       -9         -9
## 534          2       -9         -9
## 535          2       -9         -9
## 536          2       -9         -9
## 537          2       -9         -9
## 538          2       -9         -9
## 539          2       -9         -9
## 540          2       -9         -9
## 541          2       -9         -9
## 542          2       -9         -9
## 543          2       -9         -9
## 544          2       -9         -9
## 545          2       -9         -9
## 546          2       -9         -9
## 547          2       -9         -9
## 548          2       -9         -9
## 549          2       -9         -9
## 550          2       -9         -9
## 551          2       -9         -9
## 552          2       -9         -9
## 553          2       -9         -9
## 554          2       -9         -9
## 555          2       -9         -9
## 556          2       -9         -9
## 557          2       -9         -9
## 558          2       -9         -9
## 559          2       -9         -9
## 560          2       -9         -9
## 561          2       -9         -9
## 562          2       -9         -9
## 563          2       -9         -9
## 564          2       -9         -9
## 565          2       -9         -9
## 566          2       -9         -9
## 567          2       -9         -9
## 568          2       -9         -9
## 569          2       -9         -9
## 570          2       -9         -9
## 571          2       -9         -9
## 572          2       -9         -9
## 573          2       -9         -9
## 574          2       -9         -9
## 575          2       -9         -9
## 576          2       -9         -9
## 577          2       -9         -9
```

```
## 578          2       -9         -9
## 579          2       -9         -9
## 580          2       -9         -8
## 581          2       -9         -9
## 582          2       -9         -9
## 583          2       -9         -9
## 584          2       -9         -9
## 585          2       -9         -9
## 586          2       -9         -9
## 587          2       -9         -9
## 588          2       -9         -9
## 589          2       -9         -9
## 590          2       -9         -9
## 591          2       -9         -9
## 592          2       -9         -9
## 593          2       -9         -9
## 594          2       -9         -9
## 595          2       -9         -9
## 596          2       -9         -9
## 597          2       -9         -9
## 598          2       -9         -9
## 599          2       -9         -9
## 600          2       -9         -9
## 601          2       -9         -9
## 602          2       -9         -9
## 603          2       -9         -9
## 604          2       -9         -9
## 605          2       -9         -9
## 606          2       -9         -9
## 607          2       -9         -9
## 608          2       -9         -9
## 609          2       -9         -9
## 610          2       -9         -9
## 611          2       -9         -9
## 612          2       -9         -9
## 613          2       -9         -9
## 614          2       -9         -9
## 615          2       -9         -9
## 616          2       -9         -9
## 617          2       -9         -9
## 618          2       -9         -9
## 619          2       -9         -9
## 620          2       -9         -9
## 621          2       -9         -9
## 622          2       -9         -9
## 623          2       -9         -9
## 624          2       -9         -9
## 625          2       -9         -9
## 626          2       -9         -9
## 627          2       -9         -9
## 628          2       -9         -9
## 629          2       -9         -9
## 630          2       -9         -9
## 631          2       -9         -9
```

```
## 632          2       -9         -9
## 633          2       -9         -9
## 634          2       -9         -9
## 635          2       -9         -9
## 636          2       -9         -9
## 637          2       -9         -9
## 638          2       -9         -9
## 639          2       -9         -9
## 640          2       -9         -9
## 641          2       -9         -9
## 642          2       -9         -9
## 643          2       -9         -9
## 644          2       -9         -9
## 645          2       -9         -9
## 646          2       -9         -9
## 647          2       -9         -9
## 648          2       -9         -9
## 649          2       -9         -9
## 650          2       -9         -9
## 651          2       -9         -9
## 652          2       -9         -9
## 653          2       -9         -9
## 654          2       -9         -9
## 655          2       -9         -9
## 656          2       -9         -9
## 657          2       -9         -9
## 658          2       -9         -9
## 659          2       -9         -9
## 660          2       -9         -9
## 661          2       -9         -9
## 662          2       -9         -8
## 663          2       -9         -9
## 664          2       -9         -9
## 665          2       -9         -9
## 666          2       -9         -9
## 667          2       -9         -9
## 668          2       -9         -9
## 669          2       -9         -9
## 670          2       -9         -9
## 671          2       -9         -9
## 672          2       -9         -9
## 673          2       -9         -9
## 674          2       -9         -9
## 675          2       -9         -9
## 676          2       -9         -9
## 677          2       -9         -9
## 678          2       -9         -9
## 679          2       -9         -9
## 680          2       -9         -9
## 681          2       -9         -9
## 682          2       -9         -9
## 683          2       -9         -9
## 684          2       -9         -9
## 685          2       -9         -9
```

```
## 686          2       -9         -9
## 687          2       -9         -9
## 688          2       -9         -9
## 689          2       -9         -9
## 690          2       -9         -9
## 691          2       -9         -9
## 692          2       -9         -9
## 693          2       -9         -9
## 694          2       -9         -9
## 695          2       -9         -9
## 696          2       -9         -9
## 697          2       -9         -9
## 698          2       -9         -9
## 699          2       -9         -9
## 700          2       -9         -9
## 701          2       -9         -9
## 702          2       -9         -9
## 703          2       -9         -9
## 704          2       -9         -9
## 705          2       -9         -9
## 706          2       -9         -9
## 707          2       -9         -9
## 708          2       -9         -9
## 709          2       -9         -9
## 710          2       -9         -9
## 711          2       -9         -9
## 712          2       -9         -9
## 713          2       -9         -9
## 714          2       -9         -9
## 715          2       -9         -9
## 716          2       -9         -9
## 717          2       -9         -9
## 718          2       -9         -9
## 719          2       -9         -9
## 720          2       -9         -9
## 721          2       -9         -9
## 722          2       -9         -9
## 723          2       -9         -9
## 724          2       -9         -9
## 725          2       -9         -9
## 726          2       -9         -9
## 727          2       -9         -9
## 728          2       -9         -9
## 729          2       -9         -9
## 730          2       -9         -9
## 731          2       -9         -9
## 732          2       -9         -9
## 733          2       -9         -9
## 734          2       -9         -9
## 735          2       -9         -9
## 736          2       -9         -9
## 737          2       -9         -9
## 738          2       -9         -9
## 739          2       -9         -9
```

```
## 740          2       -9         -9
## 741          2       -9         -9
## 742          2       -9         -9
## 743          2       -9         -9
## 744          2       -9         -9
## 745          2       -9         -9
## 746          2       -9         -9
## 747          2       -9         -8
## 748          2       -9         -9
## 749          2       -9         -9
## 750          2       -9         -9
## 751          2       -9         -9
## 752          2       -9         -9
## 753          2       -9         -9
## 754          2       -9         -9
## 755          2       -9         -9
## 756          2       -9         -9
## 757          2       -9         -9
## 758          2       -9         -9
## 759          2       -9         -9
## 760          2       -9         -9
## 761          2       -9         -9
## 762          2       -9         -9
## 763          2       -9         -9
## 764          2       -9         -9
## 765          2       -9         -9
## 766          2       -9         -9
## 767          2       -9         -9
## 768          2       -9         -9
## 769          2       -9         -9
## 770          2       -9         -9
## 771          2       -9         -9
## 772          2       -9         -9
## 773          2       -8         -9
## 774          2       -8          2
## 775          2       -8         -9
## 776          2       -8         -9
## 777          2       -8         -9
## 778          2       -8         -9
## 779          2       -8         -9
## 780          2       -8         -9
## 781          2       -8         -9
## 782          2       -8         -9
## 783          2       -8         -9
## 784          2       -8         -9
## 785          2       -8         -9
## 786          2       -8         -9
## 787          2       -8         -9
## 788          2       -8         -9
## 789          2       -8         -9
## 790          2       -8         -9
## 791          2       -8         -9
## 792          2       -8         -9
## 793          2       -8         -9
```

```
## 794          2       -8         -9
## 795          2       -8         -9
## 796          2       -8         -9
## 797          2       -8         -9
## 798          2       -8         -9
## 799          2       -8         -9
## 800          2       -8         -9
## 801          2       -8         -9
## 802          2       -8         -9
## 803          2       -8         -9
## 804          2       -8         -9
## 805          2       -8         -9
## 806          2       -8         -9
## 807          2       -8         -9
## 808          2       -8         -9
## 809          2       -8         -9
## 810          2       -8         -9
## 811          2       -8         -9
## 812          2       -8         -9
## 813          2       -8         -9
## 814          2       -8         -9
## 815          2       -8         -9
## 816          2       -8         -9
## 817          2       -8         -9
## 818          2       -8         -9
## 819          2       -8         -9
## 820          2       -8         -9
## 821          2       -8         -9
## 822          2       -8         -9
## 823          2       -8         -9
## 824          2       -8         -9
## 825          2       -8         -9
## 826          2       -8         -9
## 827          2       -8         -9
## 828          2       -8         -9
## 829          2       -8         -9
## 830          2       -8         -9
## 831          2       -8         -9
## 832          2       -8         -9
## 833          2       -8         -9
## 834          2       -8         -9
## 835          2       -8         -9
## 836          2       -8         -9
## 837          2       -8         -9
## 838          2       -8         -9
## 839          2       -8         -9
## 840          2       -8         -9
## 841          2       -8         -9
## 842          2       -8         -9
## 843          2       -8         -9
## 844          2       -8         -9
## 845          2       -8         -9
## 846          2       -8         -9
## 847          2       -8         -9
```

```
## 848          2       -8         -9
## 849          2       -8         -9
## 850          2       -8         -9
## 851          2       -8         -9
## 852          2       -8         -9
## 853          2       -8         -9
## 854          2       -8         -9
## 855          2       -8         -9
## 856          2       -8         -8
## 857          2       -8         -9
## 858          2       -8         -9
## 859          2       -8         -9
## 860          2       -8         -9
## 861          2       -8         -9
## 862          2       -8         -9
## 863          2       -8         -9
## 864          2       -8         -9
## 865          2       -8         -9
## 866          2       -8         -9
## 867          2       -8         -9
## 868          2       -8         -9
## 869          2       -8         -9
## 870          2       -8         -9
## 871          2       -8         -9
## 872          2       -8         -9
## 873          2       -8         -9
## 874          2       -8         -9
## 875          2       -8         -8
## 876          2       -8         -8
## 877          2       -8         -9
## 878          2       -8         -8
## 879          2       -8         -9
## 880          2       -8         -8
## 881          2       -8         -8
## 882          2       -8         -8
## 883          2       -8         -9
## 884          2       -8         -9
## 885          2       -8         -8
## 886          2       -8         -9
## 887          2       -8         -9
## 888          2       -8         -8
## 889          2       -8         -8
## 890          2       -8         -9
## 891          2       -8         -9
## 892          2       -8         -8
## 893          2       -8         -9
## 894          2       -8         -9
## 895          2       -8         -8
## 896          2       -8         -9
## 897          2       -8         -9
## 898          2       -8         -8
## 899          2       -8         -8
## 900          2       -8         -8
## 901          2       -8         -8
```

```
## 902          2       -8         -9
## 903          2       -8         -8
## 904          2       -8         -8
## 905          2       -8         -8
## 906          2       -8         -9
## 907          2       -8         -9
## 908          2       -8         -9
## 909          2       -8         -8
## 910          2       -8         -9
## 911          2       -8         -9
## 912          2       -8         -9
## 913          2       -8         -9
## 914          2       -8         -9
## 915          2       -8         -9
## 916          2       -8         -9
## 917          2       -8         -9
## 918          2       -8         -9
## 919          2       -8         -9
## 920          2       -8         -9
## 921          2       -8         -8
## 922          2       -8         -9
## 923          2       -8         -9
## 924          2       -8         -8
## 925          2       -8         -9
## 926          2       -8         -8
## 927          2       -8         -9
## 928          2       -8         -8
## 929          2       -8         -9
## 930          2       -8         -9
## 931          2       -8         -9
## 932          2       -8         -9
## 933          2       -8         -9
## 934          2       -8         -8
## 935          2       -8         -8
## 936          2       -8         -8
## 937          2       -8         -9
## 938          2       -8         -9
## 939          2       -8         -8
## 940          2       -8         -9
## 941          2       -8         -8
## 942          2       -8         -8
## 943          2       -8         -8
## 944          2       -8         -9
## 945          2       -8         -9
## 946          2       -8         -8
## 947          2       -8         -9
## 948          2       -8         -9
## 949          2       -8         -9
## 950          2       -8         -9
## 951          2       -8         -8
## 952          2       -8         -8
## 953          2       -8         -9
## 954          2       -8         -9
## 955          2       -8         -9
```

```
## 956          2       -8         -9
## 957          2       -8         -9
## 958          2       -8         -9
## 959          2       -8         -9
## 960          2       -8         -9
## 961          2       -8         -8
## 962          2       -8         -9
## 963          2       -8         -9
## 964          2       -8         -9
## 965          2       -8         -8
## 966          2       -8         -9
## 967          2       -8         -9
## 968          2       -8         -8
## 969          2       -8         -9
## 970          2       -8         -9
## 971          2       -8         -9
## 972          2       -8         -9
## 973          2       -8         -8
## 974          2       -8         -9
## 975          2       -8         -9
## 976          2       -8         -9
## 977          2       -8         -9
## 978          2       -8         -8
## 979          2       -8         -8
## 980          2       -8         -9
## 981          2       -8         -9
## 982          2       -8         -9
## 983          2       -8         -9
## 984          2       -8         -9
## 985          2       -8         -9
## 986          2       -8         -9
## 987          2       -8         -9
## 988          2       -8         -9
## 989          2       -8         -8
## 990          2       -8         -8
## 991          2       -8         -9
## 992          2       -8         -9
## 993          2       -8         -9
## 994          2       -8         -9
## 995          2       -8         -9
## 996          2       -8         -8
## 997          2       -8         -9
## 998          2       -8         -9
## 999          2       -8         -8
## 1000         2       -8         -9
## 1001         2       -8         -8
## 1002         2       -8         -9
## 1003         2       -8         -8
## 1004         2       -8         -9
## 1005         2       -8         -9
## 1006         2       -8         -8
## 1007         2       -8         -9
## 1008         2       -8         -8
## 1009         2       -8         -9
```

```
## 1010         2       -8         -9
## 1011         2       -8         -9
## 1012         2       -8         -9
## 1013         2       -8         -9
## 1014         2       -8         -8
## 1015         2       -8         -9
## 1016         2       -8         -9
## 1017         2       -8         -9
## 1018         2       -8         -9
## 1019         2       -8         -9
## 1020         2       -8         -9
## 1021         2       -8         -9
## 1022         2       -8         -9
## 1023         2       -8         -8
## 1024         2       -8         -9
## 1025         2       -8         -9
## 1026         2       -8         -9
## 1027         2       -8         -9
## 1028         2       -8         -8
## 1029         2       -8         -9
## 1030         2       -8         -8
## 1031         2       -8         -9
## 1032         2       -8         -9
## 1033         2       -8         -9
## 1034         2       -8         -8
## 1035         2       -8         -8
## 1036         2       -8         -9
## 1037         2       -8         -9
## 1038         2       -8         -9
## 1039         2       -8         -9
## 1040         2       -8         -8
## 1041         2       -8         -9
## 1042         2       -8         -9
## 1043         2       -8         -9
## 1044         2       -8         -9
## 1045         2       -8         -9
## 1046         2       -8         -8
## 1047         2       -8         -8
## 1048         2       -8         -8
## 1049         2       -8         -9
## 1050         2       -8         -8
## 1051         2       -8         -8
## 1052         2       -8         -9
## 1053         2       -8         -9
## 1054         2       -8         -9
## 1055         2       -8         -9
## 1056         2       -8         -9
## 1057         2       -8         -9
## 1058         2       -8         -9
## 1059         2       -8         -9
## 1060         2       -8         -8
## 1061         2       -8         -9
## 1062         2       -8         -9
## 1063         2       -8         -9
```

```
## 1064         2       -8         -9
## 1065         2       -8         -9
## 1066         2       -8         -9
## 1067         2       -8         -9
## 1068         2       -8         -9
## 1069         2       -8         -9
## 1070         2       -8         -9
## 1071         2       -8         -9
## 1072         2       -8         -9
## 1073         2       -8         -9
## 1074         2       -8         -9
## 1075         2       -8         -9
## 1076         2       -8         -8
## 1077         2       -8         -9
## 1078         2       -8         -9
## 1079         2       -8         -9
## 1080         2       -8         -9
## 1081         2       -8         -9
## 1082         2       -8         -9
## 1083         2       -8         -9
## 1084         2       -8         -9
## 1085         2       -8         -9
## 1086         2       -8         -9
## 1087         2       -8         -9
## 1088         2       -8         -9
## 1089         2       -8         -9
## 1090         2       -8         -9
## 1091         2       -8         -9
## 1092         2       -8         -9
## 1093         2       -8         -9
## 1094         2       -8         -9
## 1095         2       -8         -9
## 1096         2       -8         -9
## 1097         2       -8         -9
## 1098         2       -8         -9
## 1099         2       -8         -9
## 1100         2       -8         -9
## 1101         2       -8         -8
## 1102         2       -8         -9
## 1103         2       -8         -9
## 1104         2       -8         -9
## 1105         2       -8         -9
## 1106         2       -8         -9
## 1107         2       -8         -9
## 1108         2       -8         -9
## 1109         2       -8         -9
## 1110         2       -8         -9
## 1111         2       -8         -9
## 1112         2       -8         -9
## 1113         2       -8         -9
## 1114         2       -8         -9
## 1115         2       -8         -9
## 1116         2       -8         -9
## 1117         2       -8         -9
```

```
## 1118         2       -8         -8
## 1119         2       -8         -9
## 1120         2       -8         -9
## 1121         2       -8         -9
## 1122         2       -8         -9
## 1123         2       -8         -9
## 1124         2       -8         -9
## 1125         2       -8         -9
## 1126         2       -8         -9
## 1127         2       -8         -9
## 1128         2       -8         -9
## 1129         2       -8         -9
## 1130         2       -8         -9
## 1131         2       -8         -9
## 1132         2       -8         -9
## 1133         2       -8         -9
## 1134         2       -8         -9
## 1135         2       -8         -9
## 1136         2       -8         -9
## 1137         2       -8         -9
## 1138         2       -8         -9
## 1139         2       -8         -9
## 1140         2       -8         -9
## 1141         2       -8         -9
## 1142         2       -8         -9
## 1143         2       -8         -9
## 1144         2       -8         -9
## 1145         2       -8         -9
## 1146         2       -8         -9
## 1147         2       -8         -9
## 1148         2       -8         -9
## 1149         2       -8         -9
## 1150         2       -8         -9
## 1151         2       -8         -9
## 1152         2       -8         -9
## 1153         2       -8         -9
## 1154         2       -8         -9
## 1155         2       -8         -9
## 1156         2       -8         -9
## 1157         2       -8         -9
## 1158         2       -8         -9
## 1159         2       -8         -9
## 1160         2       -8         -9
## 1161         2       -8         -9
## 1162         2       -8         -9
## 1163         2       -8         -9
## 1164         2       -8         -9
## 1165         2       -8         -9
## 1166         2       -8         -9
## 1167         2       -8         -9
## 1168         2       -8         -9
## 1169         2       -8         -9
## 1170         2       -8         -9
## 1171         2       -8         -9
```

```
## 1172         2       -8         -8
## 1173         2       -8         -9
## 1174         2       -8         -9
## 1175         2       -8         -9
## 1176         2       -8         -9
## 1177         2       -8         -8
## 1178         2       -8         -9
## 1179         2       -8         -9
## 1180         2       -8         -9
## 1181         2       -8         -9
## 1182         2       -8         -9
## 1183         2       -8         -9
## 1184         2       -8         -9
## 1185         2       -8         -9
## 1186         2       -8         -9
## 1187         2       -8         -9
## 1188         2       -8         -9
## 1189         2       -8         -9
## 1190         2       -8         -9
## 1191         2       -8         -9
## 1192         2       -8         -9
## 1193         2       -8         -9
## 1194         2       -8         -9
## 1195         2       -8         -9
## 1196         2       -8         -9
## 1197         2       -8         -9
## 1198         2       -8         -9
## 1199         2       -8         -9
## 1200         2       -8         -9
## 1201         2       -8         -9
## 1202         2       -8         -9
## 1203         2       -8         -9
## 1204         2       -8         -9
## 1205         2       -8         -9
## 1206         2       -8         -9
## 1207         2       -8         -9
## 1208         2       -8         -9
## 1209         2       -8         -9
## 1210         2       -8         -9
## 1211         2       -8         -9
## 1212         2       -8         -9
## 1213         2       -8         -9
## 1214         2       -8         -9
## 1215         2       -8         -9
## 1216         2       -8         -9
## 1217         2       -8         -9
## 1218         2       -8         -9
## 1219         2       -8         -9
## 1220         2       -8         -9
## 1221         2       -8         -9
## 1222         2       -8         -9
## 1223         2       -8         -9
## 1224         2       -8         -9
## 1225         2       -8         -9
```

```
## 1226         2       -8         -9
## 1227         2       -8         -9
## 1228         2       -8         -9
## 1229         2       -8         -9
## 1230         2       -8         -9
## 1231         2       -8         -9
## 1232         2       -8         -9
## 1233         2       -8         -9
## 1234         2       -8         -9
## 1235         2       -8         -8
## 1236         2       -8         -9
## 1237         2       -8         -9
## 1238         2       -8         -9
## 1239         2       -8         -9
## 1240         2       -8         -9
## 1241         2       -8         -9
## 1242         2       -8         -9
## 1243         2       -8         -9
## 1244         2       -8         -9
## 1245         2       -8         -9
## 1246         2       -8         -9
## 1247         2       -8         -9
## 1248         2       -8         -9
## 1249         2       -8         -9
## 1250         2       -8         -9
## 1251         2       -8         -9
## 1252         2       -8         -9
## 1253         2       -8         -9
## 1254         2       -8         -9
## 1255         2       -8         -9
## 1256         2       -8         -9
## 1257         2       -8         -9
## 1258         2       -8         -9
## 1259         2       -8         -9
## 1260         2       -8         -9
## 1261         2       -8         -9
## 1262         2       -8         -9
## 1263         2       -8         -9
## 1264         2       -8         -9
## 1265         2       -8         -9
## 1266         2       -8         -9
## 1267         2       -8         -9
## 1268         2       -8         -9
## 1269         2       -8         -9
## 1270         2       -8         -9
## 1271         2       -8         -9
## 1272         2       -8         -9
## 1273         2       -8         -9
## 1274         2       -8         -9
## 1275         2       -8         -9
## 1276         2       -8         -9
## 1277         2       -8         -9
## 1278         2       -8         -9
## 1279         2       -8         -9
```

```
## 1280         2       -8         -9
## 1281         2       -8         -9
## 1282         2       -8         -9
## 1283         2       -8         -9
## 1284         2       -8         -9
## 1285         2       -8         -9
## 1286         2       -8         -9
## 1287         2       -8         -9
## 1288         2       -8         -9
## 1289         2       -8         -9
## 1290         2       -8         -9
## 1291         2       -8         -9
## 1292         2       -8         -9
## 1293         2       -8         -9
## 1294         2       -8         -9
## 1295         2       -8         -9
## 1296         2       -8         -9
## 1297         2       -8         -9
## 1298         2       -8         -9
## 1299         2       -8         -9
## 1300         2       -8         -9
## 1301         2       -8         -9
## 1302         2       -8         -9
## 1303         2       -8         -9
## 1304         2       -8         -9
## 1305         2       -8         -9
## 1306         2       -8         -9
## 1307         2       -8         -9
## 1308         2       -8         -9
## 1309         2       -8         -9
## 1310         2       -8         -9
## 1311         2       -8         -9
## 1312         2       -8         -9
## 1313         2       -8         -9
## 1314         2       -8         -9
## 1315         2       -8         -9
## 1316         2       -8         -9
## 1317         2       -8         -9
## 1318         2       -8         -9
## 1319         2       -8         -9
## 1320         2       -8         -9
## 1321         2       -8         -9
## 1322         2       -8         -9
## 1323         2       -8         -9
## 1324         2       -8         -9
## 1325         2       -8         -9
## 1326         2       -8         -9
## 1327         2       -8         -9
## 1328         2       -8         -9
## 1329         2       -8         -9
## 1330         2       -8         -9
## 1331         2       -8         -9
## 1332         2       -8         -9
## 1333         2       -8         -9
```

```
## 1334         2       -8         -9
## 1335         2       -8         -9
## 1336         2       -8         -9
## 1337         2       -8         -9
## 1338         2       -8         -8
## 1339         2       -8         -9
## 1340         2       -8         -9
## 1341         2       -8         -9
## 1342         2       -8         -9
## 1343         2       -8         -9
## 1344         2       -8         -9
## 1345         2       -8         -9
## 1346         2       -8         -9
## 1347         2       -8         -9
## 1348         2       -8         -9
## 1349         2       -8         -9
## 1350         2       -8         -9
## 1351         2       -8         -9
## 1352         2       -8         -9
## 1353         2       -8         -9
## 1354         2       -8         -9
## 1355         2       -8         -9
## 1356         2       -8         -9
## 1357         2       -8         -9
## 1358         2       -8         -9
## 1359         2       -8         -9
## 1360         2       -8         -9
## 1361         2       -8         -9
## 1362         2       -8         -9
## 1363         2       -8         -9
## 1364         2       -8         -9
## 1365         2       -8         -9
## 1366         2       -8         -9
## 1367         2       -8         -9
## 1368         2       -8         -9
## 1369         2       -8         -9
## 1370         2       -8         -9
## 1371         2       -8         -9
## 1372         2       -8         -9
## 1373         2       -8         -9
## 1374         2       -8         -9
## 1375         2       -8         -8
## 1376         2       -8         -9
## 1377         2       -8         -9
## 1378         2       -8         -9
## 1379         2       -8         -9
## 1380         2       -8         -9
## 1381         2       -8         -9
## 1382         2       -8         -9
## 1383         2       -8         -9
## 1384         2       -8         -9
## 1385         2       -8         -9
## 1386         2       -8         -9
## 1387         2       -8         -9
```

```
## 1388         2       -8         -9
## 1389         2       -8         -9
## 1390         2       -8         -9
## 1391         2       -8         -9
## 1392         2       -8         -9
## 1393         2       -8         -9
## 1394         2       -8         -9
## 1395         2       -8         -9
## 1396         2       -8         -9
## 1397         2       -8         -9
## 1398         2       -8         -9
## 1399         2       -8         -9
## 1400         2       -8         -9
## 1401         2       -8         -9
## 1402         2       -8         -9
## 1403         2       -8         -9
## 1404         2       -8         -9
## 1405         2       -8         -9
## 1406         2       -8         -9
## 1407         2       -8         -9
## 1408         2       -8         -9
## 1409         2       -8         -9
## 1410         2       -8         -9
## 1411         2       -8         -9
## 1412         2       -8         -9
## 1413         2       -8         -9
## 1414         2       -8         -9
## 1415         2       -8         -9
## 1416         2       -8         -9
## 1417         2       -8         -9
## 1418         2       -8         -9
## 1419         2       -8         -9
## 1420         2       -8         -8
## 1421         2       -8         -8
## 1422         2       -8         -8
## 1423         2       -8         -8
## 1424         2       -8         -8
## 1425         2       -8         -9
## 1426         2       -8         -8
## 1427         2       -8         -8
## 1428         2       -8         -9
## 1429         2       -8         -9
## 1430         2       -8         -9
## 1431         2       -8         -9
## 1432         2       -8         -8
## 1433         2       -8         -9
## 1434         2       -8         -8
## 1435         2       -8         -9
## 1436         2       -8         -8
## 1437         2       -8         -8
## 1438         2       -8         -9
## 1439         2       -8         -9
## 1440         2       -8         -8
## 1441         2       -8         -8
```

```
## 1442        2       -8        -8
## 1443        2       -8        -8
## 1444        2       -8        -9
## 1445        2       -8        -9
## 1446        2       -8        -9
## 1447        2       -8        -9
## 1448        2       -8        -9
## 1449        2       -8        -8
## 1450        2       -8        -9
## 1451        2       -8        -8
## 1452        2       -8        -8
## 1453        2       -8        -9
## 1454        2       -8        -9
## 1455        2       -8        -8
## 1456        2       -8        -9
## 1457        2       -8        -9
## 1458        2       -8        -8
## 1459        2       -8        -8
## 1460        2       -8        -8
## 1461        2       -8        -8
## 1462        2       -8        -8
## 1463        2       -8        -9
## 1464        2       -8        -9
## 1465        2       -8        -8
## 1466        2       -8        -9
## 1467        2       -8        -8
## 1468        2       -8        -9
## 1469        2       -8        -8
## 1470        2       -8        -9
## 1471        2       -8        -8
## 1472        2       -8        -8
## 1473        2       -8        -9
## 1474        2       -8        -8
## 1475        2       -8        -9
## 1476        2       -8        -9
## 1477        2       -8        -8
## 1478        2       -8        -9
## 1479        2       -8        -8
## 1480        2       -8        -9
## 1481        2       -8        -8
## 1482        2       -8        -9
## 1483        2       -8        -8
## 1484        2       -8        -9
## 1485        2       -8        -8
## 1486        2       -8        -9
## 1487        2       -8        -8
## 1488        2       -8        -8
## 1489        2       -8        -9
## 1490        2       -8        -8
## 1491        2       -8        -8
## 1492        2       -8        -9
## 1493        2       -8        -9
## 1494        2       -8        -9
## 1495        2       -8        -9
```

```
## 1496         2       -8         -8
## 1497         2       -8         -9
## 1498         2       -8         -9
## 1499         2       -8         -9
## 1500         2       -8         -8
## 1501         2       -8         -9
## 1502         2       -8         -8
## 1503         2       -8         -8
## 1504         2       -8         -9
## 1505         2       -8         -8
## 1506         2       -8         -9
## 1507         2       -8         -8
## 1508         2       -8         -9
## 1509         2       -8         -8
## 1510         2       -8         -9
## 1511         2       -8         -8
## 1512         2       -8         -8
## 1513         2       -8         -9
## 1514         2       -8         -8
## 1515         2       -8         -9
## 1516         2       -8         -8
## 1517         2       -8         -9
## 1518         2       -8         -9
## 1519         2       -8         -9
## 1520         2       -8         -9
## 1521         2       -8         -9
## 1522         2       -8         -9
## 1523         2       -8         -9
## 1524         2       -8         -9
## 1525         2       -8         -9
## 1526         2       -8         -8
## 1527         2       -8         -9
## 1528         2       -8         -9
## 1529         2       -8         -9
## 1530         2       -8         -9
## 1531         2       -8         -9
## 1532         2       -8         -9
## 1533         2       -8         -8
## 1534         2       -8         -9
## 1535         2       -8         -9
## 1536         2       -8         -9
## 1537         2       -8         -8
## 1538         2       -8         -9
## 1539         2       -8         -9
## 1540         2       -8         -8
## 1541         2       -8         -9
## 1542         2       -8         -9
## 1543         2       -8         -9
## 1544         2       -8         -9
## 1545         2       -8         -8
## 1546         2       -8         -9
## 1547         2       -8         -9
## 1548         2       -8         -9
## 1549         2       -8         -9
```

```
## 1550         2       -8         -9
## 1551         2       -8         -9
## 1552         2       -8         -9
## 1553         2       -8         -8
## 1554         2       -8         -9
## 1555         2       -8         -9
## 1556         2       -8         -9
## 1557         2       -8         -8
## 1558         2       -8         -9
## 1559         2       -8         -9
## 1560         2       -8         -9
## 1561         2       -8         -9
## 1562         2       -8         -9
## 1563         2       -8         -9
## 1564         2       -8         -8
## 1565         2       -8         -9
## 1566         2       -8         -9
## 1567         2       -8         -9
## 1568         2       -8         -8
## 1569         2       -8         -8
## 1570         2       -8         -9
## 1571         2       -8         -9
## 1572         2       -8         -8
## 1573         2       -8         -9
## 1574         2       -8         -9
## 1575         2       -8         -9
## 1576         2       -8         -8
## 1577         2       -8         -9
## 1578         2       -8         -8
## 1579         2       -8         -9
## 1580         2       -8         -8
## 1581         2       -8         -9
## 1582         2       -8         -9
## 1583         2       -8         -9
## 1584         2       -8         -8
## 1585         2       -8         -8
## 1586         2       -8         -9
## 1587         2       -8         -9
## 1588         2       -8         -8
## 1589         2       -8         -9
## 1590         2       -8         -8
## 1591         2       -8         -9
## 1592         2       -8         -8
## 1593         2       -8         -9
## 1594         2       -8         -8
## 1595         2       -8         -9
## 1596         2       -8         -8
## 1597         2       -8         -9
## 1598         2       -8         -9
## 1599         2       -8         -9
## 1600         2       -8         -9
## 1601         2       -8         -9
## 1602         2       -8         -8
## 1603         2       -8         -9
```

```
## 1604         2       -8         -9
## 1605         2       -8         -8
## 1606         2       -8         -9
## 1607         2       -8         -9
## 1608         2       -8         -9
## 1609         2       -8         -9
## 1610         2       -8         -8
## 1611         2       -8         -9
## 1612         2       -8         -9
## 1613         2       -8         -9
## 1614         2       -8         -9
## 1615         2       -8         -9
## 1616         2       -8         -9
## 1617         2       -8         -9
## 1618         2       -8         -9
## 1619         2       -8         -9
## 1620         2       -8         -9
## 1621         2       -8         -9
## 1622         2       -8         -9
## 1623         2       -8         -9
## 1624         2       -8         -9
## 1625         2       -8         -9
## 1626         2       -8         -9
## 1627         2       -8         -9
## 1628         2       -8         -9
## 1629         2       -8         -9
## 1630         2       -8         -9
## 1631         2       -8         -9
## 1632         2       -8         -9
## 1633         2       -8         -9
## 1634         2       -8         -9
## 1635         2       -8         -9
## 1636         2       -8         -9
## 1637         2       -8         -9
## 1638         2       -8         -9
## 1639         2       -8         -9
## 1640         2       -8         -9
## 1641         2       -8         -9
## 1642         2       -8         -9
## 1643         2       -8         -9
## 1644         2       -8         -9
## 1645         2       -8         -9
## 1646         2       -8         -9
## 1647         2       -8         -9
## 1648         2       -8         -9
## 1649         2       -8         -9
## 1650         2       -8         -9
## 1651         2       -8         -9
## 1652         2       -8         -9
## 1653         2       -8         -9
## 1654         2       -8         -9
## 1655         2       -8         -9
## 1656         2       -8         -9
## 1657         2       -8         -9
```

```
## 1658         2       -8         -9
## 1659         2       -8         -9
## 1660         2       -8         -9
## 1661         2       -8         -9
## 1662         2       -8         -9
## 1663         2       -8         -8
## 1664         2       -8         -9
## 1665         2       -8         -9
## 1666         2       -8         -9
## 1667         2       -8         -9
## 1668         2       -8         -9
## 1669         2       -8         -9
## 1670         2       -8         -9
## 1671         2       -8         -9
## 1672         2       -8         -9
## 1673         2       -8         -9
## 1674         2       -8         -9
## 1675         2       -8         -9
## 1676         2       -8         -9
## 1677         2       -8         -9
## 1678         2       -8         -9
## 1679         2       -8         -9
## 1680         2       -8         -9
## 1681         2       -8         -9
## 1682         2       -8         -9
## 1683         2       -8         -9
## 1684         2       -8         -9
## 1685         2       -8         -9
## 1686         2       -8         -9
## 1687         2       -8         -9
## 1688         2       -8         -9
## 1689         2       -8         -9
## 1690         2       -8         -9
## 1691         2       -8         -9
## 1692         2       -8         -9
## 1693         2       -8         -9
## 1694         2       -8         -9
## 1695         2       -8         -9
## 1696         2       -8         -9
## 1697         2       -8         -9
## 1698         2       -8         -9
## 1699         2       -8         -9
## 1700         2       -8         -9
## 1701         2       -8         -9
## 1702         2       -8         -9
## 1703         2       -8         -9
## 1704         2       -8         -9
## 1705         2       -8         -9
## 1706         2       -8         -9
## 1707         2       -8         -9
## 1708         2       -8         -9
## 1709         2       -8         -9
## 1710         2       -8         -9
## 1711         2       -8         -9
```

```
## 1712        2       -8        -9
## 1713        2       -8        -9
## 1714        2       -8        -9
## 1715        2       -8        -9
## 1716        2       -8        -9
## 1717        2       -8        -9
## 1718        2       -8        -9
## 1719        2       -8        -9
## 1720        2       -8         2
## 1721        2       -8        -9
## 1722        2       -8        -9
## 1723        2       -8        -9
## 1724        2       -8        -9
## 1725        2       -8        -9
## 1726        2       -8        -9
## 1727        2       -8        -9
## 1728        2       -8        -9
## 1729        2       -8        -9
## 1730        2       -8        -9
## 1731        2       -8        -9
## 1732        2       -8        -9
## 1733        2       -8        -9
## 1734        2       -8        -9
## 1735        2       -8        -9
## 1736        2       -8         2
## 1737        2       -8        -9
## 1738        2       -8        -9
## 1739        2       -8        -8
## 1740        2       -8        -9
## 1741        2       -8        -9
## 1742        2       -8        -9
## 1743        2       -8        -9
## 1744        2       -8        -9
## 1745        2       -8        -9
## 1746        2       -8        -9
## 1747        2       -8        -9
## 1748        2       -8        -9
## 1749        2       -8        -9
## 1750        2       -8        -9
## 1751        2       -8        -9
## 1752        2       -8        -9
## 1753        2       -8        -9
## 1754        2       -8        -9
## 1755        2       -8        -9
## 1756        2       -8        -9
## 1757        2       -8        -9
## 1758        2       -8        -9
## 1759        2       -8        -9
## 1760        2       -8        -9
## 1761        2       -8        -9
## 1762        2       -8        -9
## 1763        2       -8        -9
## 1764        2       -8        -9
## 1765        2       -8        -9
```

```
## 1766         2       -8         -9
## 1767         2       -8         -9
## 1768         2       -8         -9
## 1769         2       -8         -9
## 1770         2       -8         -9
## 1771         2       -8         -9
## 1772         2       -8         -9
## 1773         2       -8         -9
## 1774         2       -8         -9
## 1775         2       -8         -9
## 1776         2       -8         -9
## 1777         2       -8         -9
## 1778         2       -8         -8
## 1779         2       -8         -9
## 1780         2       -8          2
## 1781         2       -8         -9
## 1782         2       -8         -9
## 1783         2       -8         -9
## 1784         2       -8         -9
## 1785         2       -8         -9
## 1786         2       -8         -9
## 1787         2       -8         -9
## 1788         2       -8         -9
## 1789         2       -8         -9
## 1790         2       -8         -9
## 1791         2       -8         -9
## 1792         2       -8         -9
## 1793         2       -8         -9
## 1794         2       -8         -9
## 1795         2       -8         -9
## 1796         2       -8         -9
## 1797         2       -8         -9
## 1798         2       -8         -9
## 1799         2       -8         -9
## 1800         2       -8         -9
## 1801         2       -8         -9
## 1802         2       -8         -9
## 1803         2       -8         -9
## 1804         2       -8         -9
## 1805         2       -8         -9
## 1806         2       -8         -8
## 1807         2       -8         -9
## 1808         2       -8         -9
## 1809         2       -8         -9
## 1810         2       -8         -9
## 1811         2       -8         -9
## 1812         2       -8         -9
## 1813         2       -8         -9
## 1814         2       -8         -9
## 1815         2       -8         -9
## 1816         2       -8         -9
## 1817         2       -8         -9
## 1818         2       -8         -9
## 1819         2       -8         -9
```

```
## 1820         2       -8         -9
## 1821         2       -8         -9
## 1822         2       -8         -9
## 1823         2       -8         -9
## 1824         2       -8         -9
## 1825         2       -8         -9
## 1826         2       -8         -9
## 1827         2       -8         -9
## 1828         2       -8         -9
## 1829         2       -8         -9
## 1830         2       -8         -9
## 1831         2       -8          2
## 1832         2       -8         -9
## 1833         2       -8         -9
## 1834         2       -8         -9
## 1835         2       -8         -9
## 1836         2       -8         -9
## 1837         2       -8         -9
## 1838         2       -8         -9
## 1839         2       -8         -9
## 1840         2       -8         -9
## 1841         2       -8         -9
## 1842         2       -8         -9
## 1843         2       -8         -9
## 1844         2       -8         -9
## 1845         2       -8         -9
## 1846         2       -8         -9
## 1847         2       -8         -9
## 1848         2       -8         -9
## 1849         2       -8         -9
## 1850         2       -8         -9
## 1851         2       -8         -9
## 1852         2       -8         -9
## 1853         2       -8         -9
## 1854         2       -8         -9
## 1855         2       -8         -9
## 1856         2       -8         -9
## 1857         2       -8         -9
## 1858         2       -8         -9
## 1859         2       -8         -9
## 1860         2       -8         -9
## 1861         2       -8         -9
## 1862         2       -8         -9
## 1863         2       -8         -9
## 1864         2       -8         -8
## 1865         2       -8         -9
## 1866         2       -8         -9
## 1867         2       -8         -9
## 1868         2       -8         -9
## 1869         2       -8         -9
## 1870         2       -8         -9
## 1871         2       -8         -9
## 1872         2       -8         -9
## 1873         2       -8         -9
```

```
## 1874         2       -8         -9
## 1875         2       -8         -9
## 1876         2       -8         -9
## 1877         2       -8         -9
## 1878         2       -8         -9
## 1879         2       -8         -9
## 1880         2       -8         -9
## 1881         2       -8         -9
## 1882         2       -8          2
## 1883         2       -8         -9
## 1884         2       -8         -9
## 1885         2       -8         -9
## 1886         2       -8         -9
## 1887         2       -8         -9
## 1888         2       -8         -9
## 1889         2       -8         -9
## 1890         2       -8         -9
## 1891         2       -8         -9
## 1892         2       -8         -9
## 1893         2       -8         -9
## 1894         2       -8         -9
## 1895         2       -8         -9
## 1896         2       -8         -9
## 1897         2       -8         -9
## 1898         2       -8         -9
## 1899         2       -8         -9
## 1900         2       -8         -9
## 1901         2       -8         -9
## 1902         2       -8         -9
## 1903         2       -8         -9
## 1904         2       -8         -9
## 1905         2       -8         -9
## 1906         2       -8         -9
## 1907         2       -8         -9
## 1908         2       -8         -9
## 1909         2       -8         -9
## 1910         2       -8         -9
## 1911         2       -8         -9
## 1912         2       -8         -9
## 1913         2       -8         -9
## 1914         2       -8         -9
## 1915         2       -8         -9
## 1916         2       -8         -9
## 1917         2       -8         -9
## 1918         2       -8         -9
## 1919         2       -8         -9
## 1920         2       -8         -9
## 1921         2       -8         -9
## 1922         2       -8         -9
## 1923         2       -8         -9
## 1924         2       -8         -9
## 1925         2       -8         -9
## 1926         2       -8         -9
## 1927         2       -8         -9
```

```
## 1928         2       -8         -9
## 1929         2       -8         -9
## 1930         2       -8         -9
## 1931         2       -8         -9
## 1932         2       -8         -9
## 1933         2       -8         -9
## 1934         2       -8         -9
## 1935         2       -8         -9
## 1936         2       -8         -9
## 1937         2       -8         -9
## 1938         2       -8         -9
## 1939         2       -8         -9
## 1940         2       -8         -9
## 1941         2       -8         -8
## 1942         2       -8         -9
## 1943         2       -8         -9
## 1944         2       -8         -9
## 1945         2       -8         -9
## 1946         2       -8         -9
## 1947         2       -8         -9
## 1948         2       -8         -9
## 1949         2       -8         -9
## 1950         2       -8         -9
## 1951         2       -8         -9
## 1952         2       -8         -9
## 1953         2       -8         -9
## 1954         2       -8         -9
## 1955         2       -8         -9
## 1956         2       -8         -9
## 1957         2       -8         -9
## 1958         2       -8         -9
## 1959         2       -8         -9
## 1960         2       -8         -9
## 1961         2       -8         -9
## 1962         2       -8         -9
## 1963         2       -8         -9
## 1964         2       -8         -9
## 1965         2       -8         -9
## 1966         2       -8         -9
## 1967         2       -8         -9
## 1968         2       -8         -9
## 1969         2       -8         -9
## 1970         2       -8         -9
## 1971         2       -8         -9
## 1972         2       -8         -9
## 1973         2       -8         -9
## 1974         2       -8         -9
## 1975         2       -8         -9
## 1976         2       -8         -9
## 1977         2       -8         -9
## 1978         2       -8         -9
## 1979         2       -8         -9
## 1980         2       -8         -9
## 1981         2       -8         -9
```

```
## 1982         2       -8         -9
## 1983         2       -8         -9
## 1984         2       -8         -9
## 1985         2       -8         -9
## 1986         2       -8         -9
## 1987         2       -8         -9
## 1988         2       -8         -9
## 1989         2       -8         -9
## 1990         2       -8         -9
## 1991         2       -8         -9
## 1992         2       -8         -9
## 1993         2       -8         -9
## 1994         2       -8         -9
## 1995         2       -8         -9
## 1996         2       -8         -9
## 1997         2       -8         -9
## 1998         2       -8         -9
## 1999         2       -8         -9
## 2000         2       -8         -9
## 2001         2       -8         -9
## 2002         2       -8         -9
## 2003         2       -8         -9
## 2004         2       -8         -9
## 2005         2       -8         -9
## 2006         2       -8         -9
## 2007         2       -8         -9
## 2008         2       -8         -9
## 2009         2       -8         -9
## 2010         2       -8         -9
## 2011         2       -8         -9
## 2012         2       -8         -8
## 2013         2       -8         -9
## 2014         2       -8         -9
## 2015         2       -8         -9
## 2016         2       -8         -9
## 2017         2       -8         -9
## 2018         2       -8         -9
## 2019         2       -8         -9
## 2020         2       -8         -9
## 2021         2       -8         -9
## 2022         2       -8         -9
## 2023         2       -8         -9
## 2024         2       -8         -9
## 2025         2       -8         -9
## 2026         2       -8         -9
## 2027         2       -8         -9
## 2028         2       -8         -9
## 2029         2       -8         -9
## 2030         2       -8         -9
## 2031         2       -8         -9
## 2032         2       -8         -8
## 2033         2       -8         -9
## 2034         2       -8         -9
## 2035         2       -8         -9
```

```
## 2036         2       -8         -9
## 2037         2       -8         -9
## 2038         2       -8         -9
## 2039         2       -8         -9
## 2040         2       -8         -9
## 2041         2       -8         -9
## 2042         2       -8         -9
## 2043         2       -8         -9
## 2044         2       -8         -9
## 2045         2       -8         -9
## 2046         2       -8         -9
## 2047         2       -8         -9
## 2048         2       -8         -9
## 2049         2       -8         -9
## 2050         2       -8         -9
## 2051         2       -8         -9
## 2052         2       -8         -9
## 2053         2       -8         -9
## 2054         2       -8         -9
## 2055         2       -8         -9
## 2056         2       -8         -9
## 2057         2       -8         -9
## 2058         2       -8         -9
## 2059         2       -8         -9
## 2060         2       -8         -9
## 2061         2       -8         -9
## 2062         2       -8         -9
## 2063         2       -8         -9
## 2064         2       -8         -9
## 2065         2       -8         -9
## 2066         2       -8         -9
## 2067         2       -8         -9
## 2068         2       -8         -9
## 2069         2       -8         -9
## 2070         2       -8         -9
## 2071         2       -8         -9
## 2072         2       -8         -9
## 2073         2       -8         -9
## 2074         2       -8         -9
## 2075         2       -8         -9
## 2076         2       -8         -9
## 2077         2       -8         -9
## 2078         2       -8         -9
## 2079         2       -8         -9
## 2080         2       -8         -9
## 2081         2       -8         -8
## 2082         2       -8         -9
## 2083         2       -8         -9
## 2084         2       -8         -9
## 2085         2       -8         -9
## 2086         2       -8         -9
## 2087         2       -8         -9
## 2088         2       -8         -9
## 2089         2       -8         -9
```

```
## 2090        2       -8        -9
## 2091        2       -8        -9
## 2092        2       -8        -9
## 2093        2       -8        -9
## 2094        2       -8        -9
## 2095        2       -8        -9
## 2096        2       -8        -9
## 2097        2       -8        -9
## 2098        2       -8        -9
## 2099        2       -8        -9
## 2100        2       -8        -9
## 2101        2       -8        -9
## 2102        2       -8        -9
## 2103        2       -8        -9
## 2104        2       -8        -9
## 2105        2       -8        -9
## 2106        2       -8        -9
## 2107        2       -8        -9
## 2108        2       -8        -9
## 2109        2       -8        -9
## 2110        2       -8        -9
## 2111        2       -8        -9
## 2112        2       -8        -9
## 2113        2       -8        -9
## 2114        2       -8        -9
## 2115        2       -8        -8
## 2116        2       -8        -8
## 2117        2       -8        -9
## 2118        2       -8        -9
## 2119        2       -8        -9
## 2120        2       -8        -9
## 2121        2       -8        -9
## 2122        2       -8        -9
## 2123        2       -8        -9
## 2124        2       -8        -9
## 2125        2       -8        -9
## 2126        2       -8        -9
## 2127        2       -8        -9
## 2128        2       -8        -9
## 2129        2       -8        -9
## 2130        2       -8        -9
## 2131        2       -8        -9
## 2132        2       -8        -9
## 2133        2       -8        -9
## 2134        2       -8        -9
## 2135        2       -8        -8
## 2136        2       -8        -9
## 2137        2       -8        -9
## 2138        2       -8        -9
## 2139        2       -8        -9
## 2140        2       -8        -9
## 2141        2       -8        -9
## 2142        2       -8        -9
## 2143        2       -8        -9
```

```
## 2144        2       -8        -9
## 2145        2       -8        -9
## 2146        2       -8        -8
## 2147        2       -8        -9
## 2148        2       -8        -9
## 2149        2       -8        -9
## 2150        2       -8        -9
## 2151        2       -8        -9
## 2152        2       -8        -9
## 2153        2       -8        -9
## 2154        2       -8        -9
## 2155        2       -8        -8
## 2156        2       -8        -8
## 2157        2       -8        -9
## 2158        2       -8        -9
## 2159        2       -8        -9
## 2160        2       -8        -9
## 2161        2       -8        -9
## 2162        2       -8        -9
## 2163        2       -8        -9
## 2164        2       -8        -9
## 2165        2       -8        -9
## 2166        2       -8        -9
## 2167        2       -8        -9
## 2168        2       -8        -9
## 2169        2       -8        -9
## 2170        2       -8        -9
## 2171        2       -8        -9
## 2172        2       -8        -9
## 2173        2       -8        -9
## 2174        2       -8        -9
## 2175        2       -8        -9
## 2176        2       -8        -9
## 2177        2       -8        -9
## 2178        2       -8        -9
## 2179        2       -8        -9
## 2180        2       -8        -9
## 2181        2       -8        -9
## 2182        2       -8        -9
## 2183        2       -8        -9
## 2184        2       -8        -9
## 2185        2       -8        -9
## 2186        2       -8        -9
## 2187        2       -8        -9
## 2188        2       -8        -9
## 2189        2       -8        -9
## 2190        2       -8        -9
## 2191        2       -8        -9
## 2192        2       -8        -9
## 2193        2       -8        -9
## 2194        2       -8        -9
## 2195        2       -8        -9
## 2196        2       -8        -9
## 2197        2       -8        -9
```

```
## 2198         2       -8         -9
## 2199         2       -8         -9
## 2200         2       -8         -9
## 2201         2       -8         -9
## 2202         2       -8         -9
## 2203         2       -8         -9
## 2204         2       -8         -9
## 2205         2       -8         -9
## 2206         2       -8         -9
## 2207         2       -8         -9
## 2208         2       -8         -9
## 2209         2       -8         -9
## 2210         2       -8         -9
## 2211         2       -8         -9
## 2212         2       -8         -9
## 2213         2       -8         -9
## 2214         2       -8         -9
## 2215         2       -8         -8
## 2216         2       -8         -9
## 2217         2       -8         -9
## 2218         2       -8         -9
## 2219         2       -8         -9
## 2220         2       -8         -9
## 2221         2       -8         -9
## 2222         2       -8         -9
## 2223         2       -8         -9
## 2224         2       -8         -9
## 2225         2       -8         -9
## 2226         2       -8         -9
## 2227         2       -8         -9
## 2228         2       -8         -9
## 2229         2       -8         -9
## 2230         2       -8         -9
## 2231         2       -8         -9
## 2232         2       -8         -9
## 2233         2       -8         -9
## 2234         2       -8         -9
## 2235         2       -8         -9
## 2236         2       -8         -9
## 2237         2       -8         -9
## 2238         2       -8         -9
## 2239         2       -8         -9
## 2240         2       -8         -9
## 2241         2       -8         -9
## 2242         2       -8         -9
## 2243         2       -8         -9
## 2244         2       -8         -9
## 2245         2       -8         -9
## 2246         2       -8         -9
## 2247         2       -8         -9
## 2248         2       -8         -9
## 2249         2       -8         -9
## 2250         2       -8         -9
## 2251         2       -8         -9
```

```
## 2252        2       -8        -9
## 2253        2       -8        -9
## 2254        2       -8        -9
## 2255        2       -8        -9
## 2256        2       -8        -9
## 2257        2       -8        -9
## 2258        2       -8        -9
## 2259        2       -8        -9
## 2260        2       -8        -9
## 2261        2       -8        -9
## 2262        2       -8        -9
## 2263        2       -8        -9
## 2264        2       -8        -9
## 2265        2       -8        -9
## 2266        2       -8        -9
## 2267        2       -8        -9
## 2268        2       -8        -9
## 2269        2       -8        -9
## 2270        2       -8        -9
## 2271        2       -8        -9
## 2272        2       -8        -9
## 2273        2       -8        -9
## 2274        2       -8        -9
## 2275        2       -8        -9
## 2276        2       -8        -9
## 2277        2       -8        -9
## 2278        2       -8        -9
## 2279        2       -8        -9
## 2280        2       -8        -9
## 2281        2       -8        -9
## 2282        2       -8        -9
## 2283        2       -8        -9
## 2284        2       -8        -9
## 2285        2       -8        -9
## 2286        2       -8        -9
## 2287        2       -8        -9
## 2288        2       -8        -9
## 2289        2       -8        -9
## 2290        2       -8        -9
## 2291        2       -8        -9
## 2292        2       -8        -8
## 2293        2       -8        -9
## 2294        2       -8        -9
## 2295        2       -8        -9
## 2296        2       -8        -9
## 2297        2       -8        -9
## 2298        2       -8        -9
## 2299        2       -8        -9
## 2300        2       -8        -8
## 2301        2       -8        -9
## 2302        2       -8        -9
## 2303        2       -8        -9
## 2304        2       -8        -9
## 2305        2       -8        -9
```

```
## 2306         2       -8         -9
## 2307         2       -8         -9
## 2308         2       -8         -9
## 2309         2       -8         -9
## 2310         2       -8         -9
## 2311         2       -8         -9
## 2312         2       -8         -9
## 2313         2       -8         -9
## 2314         2       -8         -8
## 2315         2       -8         -9
## 2316         2       -8         -9
## 2317         2       -8         -9
## 2318         2       -8         -9
## 2319         2       -8         -8
## 2320         2       -8         -9
## 2321         2       -8         -9
## 2322         2       -8         -9
## 2323         2       -8         -9
## 2324         2       -8         -9
## 2325         2       -8         -9
## 2326         2       -8         -9
## 2327         2       -8         -9
## 2328         2       -8         -9
## 2329         2       -8         -9
## 2330         2       -8         -9
## 2331         2       -8         -9
## 2332         2       -8         -9
## 2333         2       -8         -9
## 2334         2       -8         -9
## 2335         2       -8         -9
## 2336         2       -8         -9
## 2337         2       -8         -9
## 2338         2       -8         -9
## 2339         2       -8         -9
## 2340         2       -8         -9
## 2341         2       -8         -9
## 2342         2       -8         -9
## 2343         2       -8         -9
## 2344         2       -8         -9
## 2345         2       -8         -9
## 2346         2       -8         -9
## 2347         2       -8         -9
## 2348         2       -8         -9
## 2349         2       -8         -9
## 2350         2       -8         -9
## 2351         2       -8         -9
## 2352         2       -8         -9
## 2353         2       -8         -9
## 2354         2       -8         -9
## 2355         2       -8         -9
## 2356         2       -8         -9
## 2357         2       -8         -9
## 2358         2       -8         -9
## 2359         2       -8         -9
```

```
## 2360        2       -8        -9
## 2361        2       -8        -9
## 2362        2       -8        -9
## 2363        2       -8        -9
## 2364        2       -8        -9
## 2365        2       -8        -9
## 2366        2       -8        -9
## 2367        2       -8        -9
## 2368        2       -8        -9
## 2369        2       -8        -9
## 2370        2       -8        -9
## 2371        2       -8        -9
## 2372        2       -8        -8
## 2373        2       -8        -8
## 2374        2       -8        -8
## 2375        2       -8        -9
## 2376        2       -8        -9
## 2377        2       -8        -8
## 2378        2       -8        -8
## 2379        2       -8        -8
## 2380        2       -8        -8
## 2381        2       -8        -9
## 2382        2       -8        -9
## 2383        2       -8        -8
## 2384        2       -8        -9
## 2385        2       -8        -9
## 2386        2       -8        -9
## 2387        2       -8        -8
## 2388        2       -8        -9
## 2389        2       -8        -8
## 2390        2       -8        -8
## 2391        2       -8        -9
## 2392        2       -8        -9
## 2393        2       -8        -9
## 2394        2       -8        -9
## 2395        2       -8        -9
## 2396        2       -8        -8
## 2397        2       -8        -8
## 2398        2       -8        -9
## 2399        2       -8        -8
## 2400        2       -8        -8
## 2401        2       -8        -8
## 2402        2       -8        -9
## 2403        2       -8        -9
## 2404        2       -8        -8
## 2405        2       -8        -8
## 2406        2       -8        -9
## 2407        2       -8        -9
## 2408        2       -8        -8
## 2409        2       -8        -9
## 2410        2       -8        -9
## 2411        2       -8        -9
## 2412        2       -8        -8
## 2413        2       -8        -8
```

```
## 2414        2       -8        -9
## 2415        2       -8        -9
## 2416        2       -8        -8
## 2417        2       -8        -9
## 2418        2       -8        -9
## 2419        2       -8        -9
## 2420        2       -8        -8
## 2421        2       -8        -9
## 2422        2       -8        -9
## 2423        2       -8        -8
## 2424        2       -8        -9
## 2425        2       -8        -9
## 2426        2       -8        -9
## 2427        2       -8        -8
## 2428        2       -8        -8
## 2429        2       -8        -8
## 2430        2       -8        -9
## 2431        2       -8        -9
## 2432        2       -8        -8
## 2433        2       -8        -8
## 2434        2       -8        -8
## 2435        2       -8        -9
## 2436        2       -8        -9
## 2437        2       -8        -8
## 2438        2       -8        -8
## 2439        2       -8        -8
## 2440        2       -8        -8
## 2441        2       -8        -9
## 2442        2       -8        -9
## 2443        2       -8        -8
## 2444        2       -8        -9
## 2445        2       -8        -8
## 2446        2       -8        -9
## 2447        2       -8        -8
## 2448        2       -8        -8
## 2449        2       -8        -9
## 2450        2       -8        -9
## 2451        2       -8        -9
## 2452        2       -8        -8
## 2453        2       -8        -8
## 2454        2       -8        -9
## 2455        2       -8        -9
## 2456        2       -8        -8
## 2457        2       -8        -9
## 2458        2       -8        -8
## 2459        2       -8        -8
## 2460        2       -8        -8
## 2461        2       -8        -9
## 2462        2       -8        -9
## 2463        2       -8        -8
## 2464        2       -8        -9
## 2465        2       -8        -9
## 2466        2       -8        -8
## 2467        2       -8        -9
```

```
## 2468        2       -8        -8
## 2469        2       -8        -8
## 2470        2       -8        -9
## 2471        2       -8        -8
## 2472        2       -8        -8
## 2473        2       -8        -8
## 2474        2       -8        -8
## 2475        2       -8        -8
## 2476        2       -8        -9
## 2477        2       -8        -9
## 2478        2       -8        -9
## 2479        2       -8        -8
## 2480        2       -8        -8
## 2481        2       -8        -9
## 2482        2       -8        -9
## 2483        2       -8        -8
## 2484        2       -8        -8
## 2485        2       -8        -8
## 2486        2       -8        -8
## 2487        2       -8        -9
## 2488        2       -8        -8
## 2489        2       -8        -9
## 2490        2       -8        -8
## 2491        2       -8        -8
## 2492        2       -8        -8
## 2493        2       -8        -8
## 2494        2       -8        -9
## 2495        2       -8        -8
## 2496        2       -8        -8
## 2497        2       -8        -9
## 2498        2       -8        -8
## 2499        2       -8        -9
## 2500        2       -8        -9
## 2501        2       -8        -8
## 2502        2       -8        -9
## 2503        2       -8        -9
## 2504        2       -8        -8
## 2505        2       -8        -8
## 2506        2       -8        -8
## 2507        2       -8        -9
## 2508        2       -8        -8
## 2509        2       -8        -8
## 2510        2       -8        -8
## 2511        2       -8        -8
## 2512        2       -8        -9
## 2513        2       -8        -8
## 2514        2       -8        -8
## 2515        2       -8        -8
## 2516        2       -8        -8
## 2517        2       -8        -9
## 2518        2       -8        -8
## 2519        2       -8        -8
## 2520        2       -8        -8
## 2521        2       -8        -8
```

```
## 2522         2       -8         -8
## 2523         2       -8         -9
## 2524         2       -8         -8
## 2525         2       -8         -8
## 2526         2       -8         -9
## 2527         2       -8         -9
## 2528         2       -8         -8
## 2529         2       -8         -9
## 2530         2       -8         -8
## 2531         2       -8         -9
## 2532         2       -8         -9
## 2533         2       -8         -8
## 2534         2       -8         -9
## 2535         2       -8         -9
## 2536         2       -8         -9
## 2537         2       -8         -8
## 2538         2       -8         -9
## 2539         2       -8         -9
## 2540         2       -8         -9
## 2541         2       -8         -8
## 2542         2       -8         -8
## 2543         2       -8         -8
## 2544         2       -8         -8
## 2545         2       -8         -9
## 2546         2       -8         -8
## 2547         2       -8         -9
## 2548         2       -8         -8
## 2549         2       -8         -8
## 2550         2       -8         -8
## 2551         2       -8         -9
## 2552         2       -8         -8
## 2553         2       -8         -9
## 2554         2       -8         -9
## 2555         2       -8         -8
## 2556         2       -8         -8
## 2557         2       -8         -8
## 2558         2       -8         -8
## 2559         2       -8         -8
## 2560         2       -8         -8
## 2561         2       -8         -8
## 2562         2       -8         -9
## 2563         2       -8         -9
## 2564         2       -8         -9
## 2565         2       -8         -8
## 2566         2       -8         -9
## 2567         2       -8         -9
## 2568         2       -8         -9
## 2569         2       -8         -8
## 2570         2       -8         -8
## 2571         2       -8         -9
## 2572         2       -8         -9
## 2573         2       -8         -8
## 2574         2       -8         -9
## 2575         2       -8         -9
```

```
## 2576         2       -8         -8
## 2577         2       -8         -8
## 2578         2       -8         -8
## 2579         2       -8         -8
## 2580         2       -8         -8
## 2581         2       -8         -8
## 2582         2       -8         -9
## 2583         2       -8         -9
## 2584         2       -8         -8
## 2585         2       -8         -9
## 2586         2       -8         -8
## 2587         2       -8         -8
## 2588         2       -8         -8
## 2589         2       -8         -9
## 2590         2       -8         -8
## 2591         2       -8         -8
## 2592         2       -8         -8
## 2593         2       -8         -8
## 2594         2       -8         -8
## 2595         2       -8         -8
## 2596         2       -8         -9
## 2597         2       -8         -9
## 2598         2       -8         -9
## 2599         2       -8         -9
## 2600         2       -8         -9
## 2601         2       -8         -8
## 2602         2       -8         -8
## 2603         2       -8         -9
## 2604         2       -8         -8
## 2605         2       -8         -8
## 2606         2       -8         -9
## 2607         2       -8         -9
## 2608         2       -8         -8
## 2609         2       -8         -8
## 2610         2       -8         -9
## 2611         2       -8         -9
## 2612         2       -8         -8
## 2613         2       -8         -9
## 2614         2       -8         -8
## 2615         2       -8         -9
## 2616         2       -8         -9
## 2617         2       -8         -9
## 2618         2       -8         -9
## 2619         2       -8         -9
## 2620         2       -8         -8
## 2621         2       -8         -8
## 2622         2       -8         -8
## 2623         2       -8         -9
## 2624         2       -8         -9
## 2625         2       -8         -8
## 2626         2       -8         -9
## 2627         2       -8         -8
## 2628         2       -8         -9
## 2629         2       -8         -8
```

```
## 2630        2       -8        -8
## 2631        2       -8        -8
## 2632        2       -8        -9
## 2633        2       -8        -9
## 2634        2       -8        -9
## 2635        2       -8        -8
## 2636        2       -8        -9
## 2637        2       -8        -8
## 2638        2       -8        -8
## 2639        2       -8        -9
## 2640        2       -8        -8
## 2641        2       -8        -9
## 2642        2       -8        -8
## 2643        2       -8        -9
## 2644        2       -8        -9
## 2645        2       -8        -8
## 2646        2       -8        -9
## 2647        2       -8        -8
## 2648        2       -8        -8
## 2649        2       -8        -8
## 2650        2       -8        -9
## 2651        2       -8        -8
## 2652        2       -8        -8
## 2653        2       -8        -9
## 2654        2       -8        -9
## 2655        2       -8        -8
## 2656        2       -8        -8
## 2657        2       -8        -8
## 2658        2       -8        -8
## 2659        2       -8        -9
## 2660        2       -8        -8
## 2661        2       -8        -9
## 2662        2       -8        -9
## 2663        2       -8        -8
## 2664        2       -8        -9
## 2665        2       -8        -8
## 2666        2       -8        -9
## 2667        2       -8        -9
## 2668        2       -8        -9
## 2669        2       -8        -8
## 2670        2       -8        -8
## 2671        2       -8        -8
## 2672        2       -8        -9
## 2673        2       -8        -9
## 2674        2       -8        -8
## 2675        2       -8        -8
## 2676        2       -8        -9
## 2677        2       -8        -9
## 2678        2       -8        -9
## 2679        2       -8        -9
## 2680        2       -8        -8
## 2681        2       -8        -8
## 2682        2       -8        -9
## 2683        2       -8        -8
```

```
## 2684         2       -8         -8
## 2685         2       -8         -8
## 2686         2       -8         -8
## 2687         2       -8         -9
## 2688         2       -8         -9
## 2689         2       -8         -9
## 2690         2       -8         -8
## 2691         2       -8         -9
## 2692         2       -8         -8
## 2693         2       -8         -9
## 2694         2       -8         -8
## 2695         2       -8         -8
## 2696         2       -8         -8
## 2697         2       -8         -8
## 2698         2       -8         -8
## 2699         2       -8         -9
## 2700         2       -8         -8
## 2701         2       -8         -8
## 2702         2       -8         -8
## 2703         2       -8         -8
## 2704         2       -8         -9
## 2705         2       -8         -9
## 2706         2       -8         -8
## 2707         2       -8         -9
## 2708         2       -8         -9
## 2709         2       -8         -8
## 2710         2       -8         -8
## 2711         2       -8         -9
## 2712         2       -8         -9
## 2713         2       -8         -8
## 2714         2       -8         -9
## 2715         2       -8         -9
## 2716         2       -8         -8
## 2717         2       -8         -9
## 2718         2       -8         -9
## 2719         2       -8         -8
## 2720         2       -8         -8
## 2721         2       -8         -8
## 2722         2       -8         -9
## 2723         2       -8         -8
## 2724         2       -8         -9
## 2725         2       -8         -9
## 2726         2       -8         -8
## 2727         2       -8         -8
## 2728         2       -8         -8
## 2729         2       -8         -8
## 2730         2       -8         -8
## 2731         2       -8         -9
## 2732         2       -8         -8
## 2733         2       -8         -9
## 2734         2       -8         -9
## 2735         2       -8         -9
## 2736         2       -8         -8
## 2737         2       -8         -9
```

```
## 2738        2       -8        -9
## 2739        2       -8        -9
## 2740        2       -8        -9
## 2741        2       -8        -9
## 2742        2       -8        -9
## 2743        2       -8        -8
## 2744        2       -8        -9
## 2745        2       -8        -8
## 2746        2       -8        -8
## 2747        2       -8        -9
## 2748        2       -8        -8
## 2749        2       -8        -9
## 2750        2       -8        -9
## 2751        2       -8        -8
## 2752        2       -8        -9
## 2753        2       -8        -8
## 2754        2       -8        -9
## 2755        2       -8        -9
## 2756        2       -8        -9
## 2757        2       -8        -9
## 2758        2       -8        -9
## 2759        2       -8        -9
## 2760        2       -8        -8
## 2761        2       -8        -8
## 2762        2       -8        -9
## 2763        2       -8        -8
## 2764        2       -8        -9
## 2765        2       -8        -9
## 2766        2       -8        -9
## 2767        2       -8        -9
## 2768        2       -8        -9
## 2769        2       -8        -9
## 2770        2       -8        -9
## 2771        2       -8        -9
## 2772        2       -8        -9
## 2773        2       -8        -9
## 2774        2       -8        -9
## 2775        2       -8        -8
## 2776        2       -8        -9
## 2777        2       -8        -9
## 2778        2       -8        -9
## 2779        2       -8        -9
## 2780        2       -8        -9
## 2781        2       -8        -8
## 2782        2       -8        -9
## 2783        2       -8        -8
## 2784        2       -8        -9
## 2785        2       -8        -9
## 2786        2       -8        -9
## 2787        2       -8        -8
## 2788        2       -8        -9
## 2789        2       -8        -8
## 2790        2       -8        -9
## 2791        2       -8        -8
```

```
## 2792        2       -8        -9
## 2793        2       -8        -8
## 2794        2       -8        -9
## 2795        2       -8        -8
## 2796        2       -8        -9
## 2797        2       -8        -9
## 2798        2       -8        -9
## 2799        2       -8        -9
## 2800        2       -8        -9
## 2801        2       -8        -8
## 2802        2       -8        -8
## 2803        2       -8        -8
## 2804        2       -8        -8
## 2805        2       -8        -9
## 2806        2       -8        -8
## 2807        2       -8        -8
## 2808        2       -8        -9
## 2809        2       -8        -9
## 2810        2       -8        -9
## 2811        2       -8        -9
## 2812        2       -8        -8
## 2813        2       -8        -9
## 2814        2       -8        -8
## 2815        2       -8        -9
## 2816        2       -8        -9
## 2817        2       -8        -9
## 2818        2       -8        -8
## 2819        2       -8        -9
## 2820        2       -8        -9
## 2821        2       -8        -9
## 2822        2       -8        -9
## 2823        2       -8        -9
## 2824        2       -8        -9
## 2825        2       -8        -9
## 2826        2       -8        -9
## 2827        2       -8        -9
## 2828        2       -8        -9
## 2829        2       -8        -8
## 2830        2       -8        -8
## 2831        2       -8        -9
## 2832        2       -8        -9
## 2833        2       -8        -9
## 2834        2       -8        -8
## 2835        2       -8        -9
## 2836        2       -8        -9
## 2837        2       -8        -8
## 2838        2       -8        -9
## 2839        2       -8        -9
## 2840        2       -8        -9
## 2841        2       -8        -8
## 2842        2       -8        -9
## 2843        2       -8        -8
## 2844        2       -8        -9
## 2845        2       -8        -8
```

```
## 2846         2       -8         -9
## 2847         2       -8         -9
## 2848         2       -8         -9
## 2849         2       -8         -9
## 2850         2       -8         -8
## 2851         2       -8         -9
## 2852         2       -8         -8
## 2853         2       -8         -8
## 2854         2       -8         -9
## 2855         2       -8         -8
## 2856         2       -8         -8
## 2857         2       -8         -8
## 2858         2       -8         -8
## 2859         2       -8         -9
## 2860         2       -8         -9
## 2861         2       -8         -9
## 2862         2       -8         -9
## 2863         2       -8         -9
## 2864         2       -8         -9
## 2865         2       -8         -8
## 2866         2       -8         -8
## 2867         2       -8         -9
## 2868         2       -8         -8
## 2869         2       -8         -9
## 2870         2       -8         -9
## 2871         2       -8         -9
## 2872         2       -8         -9
## 2873         2       -8         -9
## 2874         2       -8         -9
## 2875         2       -8         -9
## 2876         2       -8         -9
## 2877         2       -8         -9
## 2878         2       -8         -9
## 2879         2       -8         -8
## 2880         2       -8         -9
## 2881         2       -8         -9
## 2882         2       -8         -8
## 2883         2       -8         -9
## 2884         2       -8         -8
## 2885         2       -8         -8
## 2886         2       -8         -8
## 2887         2       -8         -9
## 2888         2       -8         -9
## 2889         2       -8         -9
## 2890         2       -8         -8
## 2891         2       -8         -9
## 2892         2       -8         -9
## 2893         2       -8         -8
## 2894         2       -8         -9
## 2895         2       -8         -9
## 2896         2       -8         -9
## 2897         2       -8         -8
## 2898         2       -8         -8
## 2899         2       -8         -8
```

```
## 2900        2       -8        -8
## 2901        2       -8        -9
## 2902        2       -8        -9
## 2903        2       -8        -9
## 2904        2       -8        -9
## 2905        2       -8        -9
## 2906        2       -8        -8
## 2907        2       -8        -9
## 2908        2       -8        -9
## 2909        2       -8        -9
## 2910        2       -8        -9
## 2911        2       -8        -9
## 2912        2       -8        -9
## 2913        2       -8        -8
## 2914        2       -8        -9
## 2915        2       -8        -8
## 2916        2       -8        -9
## 2917        2       -8        -9
## 2918        2       -8        -8
## 2919        2       -8        -9
## 2920        2       -8        -8
## 2921        2       -8        -8
## 2922        2       -8        -8
## 2923        2       -8        -9
## 2924        2       -8        -9
## 2925        2       -8        -9
## 2926        2       -8        -9
## 2927        2       -8        -8
## 2928        2       -8        -9
## 2929        2       -8        -9
## 2930        2       -8        -9
## 2931        2       -8        -8
## 2932        2       -8        -9
## 2933        2       -8        -9
## 2934        2       -8        -9
## 2935        2       -8        -9
## 2936        2       -8        -9
## 2937        2       -8        -8
## 2938        2       -8        -8
## 2939        2       -8        -8
## 2940        2       -8        -8
## 2941        2       -8        -9
## 2942        2       -8        -8
## 2943        2       -8        -9
## 2944        2       -8        -8
## 2945        2       -8        -9
## 2946        2       -8        -9
## 2947        2       -8        -9
## 2948        2       -8        -9
## 2949        2       -8        -9
## 2950        2       -8        -9
## 2951        2       -8        -9
## 2952        2       -8        -9
## 2953        2       -8        -9
```

```
## 2954         2       -8         -9
## 2955         2       -8         -8
## 2956         2       -8         -8
## 2957         2       -8         -8
## 2958         2       -8         -8
## 2959         2       -8         -9
## 2960         2       -8         -8
## 2961         2       -8         -9
## 2962         2       -8         -9
## 2963         2       -8         -9
## 2964         2       -8         -9
## 2965         2       -8         -8
## 2966         2       -8         -8
## 2967         2       -8         -9
## 2968         2       -8         -8
## 2969         2       -8         -9
## 2970         2       -8         -9
## 2971         2       -8         -9
## 2972         2       -8         -9
## 2973         2       -8         -9
## 2974         2       -8         -9
## 2975         2       -8         -9
## 2976         2       -8         -9
## 2977         2       -8         -9
## 2978         2       -8         -9
## 2979         2       -8         -9
## 2980         2       -8         -8
## 2981         2       -8         -9
## 2982         2       -8         -9
## 2983         2       -8         -8
## 2984         2       -8         -9
## 2985         2       -8         -9
## 2986         2       -8         -8
## 2987         2       -8         -8
## 2988         2       -8         -9
## 2989         2       -8         -9
## 2990         2       -8         -8
## 2991         2       -8         -8
## 2992         2       -8         -8
## 2993         2       -8         -8
## 2994         2       -8         -9
## 2995         2       -8         -9
## 2996         2       -8         -9
## 2997         2       -8         -8
## 2998         2       -8         -9
## 2999         2       -8         -9
## 3000         2       -8         -9
## 3001         2       -8         -9
## 3002         2       -8         -8
## 3003         2       -8         -8
## 3004         2       -8         -9
## 3005         2       -8         -8
## 3006         2       -8         -9
## 3007         2       -8         -9
```

```
## 3008        2       -8        -9
## 3009        2       -8        -9
## 3010        2       -8        -9
## 3011        2       -8        -9
## 3012        2       -8        -9
## 3013        2       -8        -9
## 3014        2       -8        -9
## 3015        2       -8        -9
## 3016        2       -8        -9
## 3017        2       -8        -9
## 3018        2       -8        -8
## 3019        2       -8        -9
## 3020        2       -8        -9
## 3021        2       -8        -9
## 3022        2       -8        -9
## 3023        2       -8        -9
## 3024        2       -8        -9
## 3025        2       -8        -8
## 3026        2       -8        -8
## 3027        2       -8        -9
## 3028        2       -8        -8
## 3029        2       -8        -9
## 3030        2       -8        -9
## 3031        2       -8        -9
## 3032        2       -8        -8
## 3033        2       -8        -9
## 3034        2       -8        -9
## 3035        2       -8        -9
## 3036        2       -8        -8
## 3037        2       -8        -9
## 3038        2       -8        -8
## 3039        2       -8        -9
## 3040        2       -8        -9
## 3041        2       -8        -9
## 3042        2       -8        -8
## 3043        2       -8        -8
## 3044        2       -8        -9
## 3045        2       -8        -9
## 3046        2       -8        -9
## 3047        2       -8        -9
## 3048        2       -8        -8
## 3049        2       -8        -8
## 3050        2       -8        -9
## 3051        2       -8        -8
## 3052        2       -8        -9
## 3053        2       -8        -8
## 3054        2       -8        -8
## 3055        2       -8        -9
## 3056        2       -8        -8
## 3057        2       -8        -8
## 3058        2       -8        -9
## 3059        2       -8        -9
## 3060        2       -8        -8
## 3061        2       -8        -9
```

```
## 3062        2       -8        -9
## 3063        2       -8        -9
## 3064        2       -8        -9
## 3065        2       -8        -9
## 3066        2       -8        -9
## 3067        2       -8        -9
## 3068        2       -8        -9
## 3069        2       -8        -9
## 3070        2       -8        -8
## 3071        2       -8        -9
## 3072        2       -8        -9
## 3073        2       -8        -9
## 3074        2       -8        -9
## 3075        2       -8        -9
## 3076        2       -8        -8
## 3077        2       -8        -9
## 3078        2       -8        -9
## 3079        2       -8        -8
## 3080        2       -8        -9
## 3081        2       -8        -9
## 3082        2       -8        -8
## 3083        2       -8        -9
## 3084        2       -8        -9
## 3085        2       -8        -9
## 3086        2       -8        -9
## 3087        2       -8        -9
## 3088        2       -8        -9
## 3089        2       -8        -9
## 3090        2       -8        -8
## 3091        2       -8        -8
## 3092        2       -8        -8
## 3093        2       -8        -9
## 3094        2       -8        -9
## 3095        2       -8        -9
## 3096        2       -8        -9
## 3097        2       -8        -9
## 3098        2       -8        -8
## 3099        2       -8        -9
## 3100        2       -8        -9
## 3101        2       -8        -9
## 3102        2       -8        -9
## 3103        2       -8        -9
## 3104        2       -8        -8
## 3105        2       -8        -9
## 3106        2       -8        -8
## 3107        2       -8        -8
## 3108        2       -8        -9
## 3109        2       -8        -8
## 3110        2       -8        -9
## 3111        2       -8        -9
## 3112        2       -8        -9
## 3113        2       -8        -9
## 3114        2       -8        -9
## 3115        2       -8        -8
```

```
## 3116        2       -8        -8
## 3117        2       -8        -9
## 3118        2       -8        -9
## 3119        2       -8        -9
## 3120        2       -8        -9
## 3121        2       -8        -9
## 3122        2       -8        -9
## 3123        2       -8        -8
## 3124        2       -8        -9
## 3125        2       -8        -9
## 3126        2       -8        -9
## 3127        2       -8        -9
## 3128        2       -8        -9
## 3129        2       -8        -8
## 3130        2       -8        -9
## 3131        2       -8        -8
## 3132        2       -8        -9
## 3133        2       -8        -8
## 3134        2       -8        -9
## 3135        2       -8        -9
## 3136        2       -8        -8
## 3137        2       -8        -9
## 3138        2       -8        -9
## 3139        2       -8        -9
## 3140        2       -8        -9
## 3141        2       -8        -9
## 3142        2       -8        -9
## 3143        2       -8        -9
## 3144        2       -8        -9
## 3145        2       -8        -8
## 3146        2       -8        -8
## 3147        2       -8        -9
## 3148        2       -8        -9
## 3149        2       -8        -9
## 3150        2       -8        -9
## 3151        2       -8        -9
## 3152        2       -8        -8
## 3153        2       -8        -9
## 3154        2       -8        -9
## 3155        2       -8        -9
## 3156        2       -8        -9
## 3157        2       -8        -9
## 3158        2       -8        -8
## 3159        2       -8        -9
## 3160        2       -8        -9
## 3161        2       -8        -8
## 3162        2       -8        -9
## 3163        2       -8        -8
## 3164        2       -8        -8
## 3165        2       -8        -9
## 3166        2       -8        -8
## 3167        2       -8        -8
## 3168        2       -8        -9
## 3169        2       -8        -9
```

```
## 3170        2       -8        -9
## 3171        2       -8        -9
## 3172        2       -8        -9
## 3173        2       -8        -8
## 3174        2       -8        -9
## 3175        2       -8        -9
## 3176        2       -8        -9
## 3177        2       -8        -9
## 3178        2       -8        -9
## 3179        2       -8        -9
## 3180        2       -8        -9
## 3181        2       -8        -9
## 3182        2       -8        -9
## 3183        2       -8        -9
## 3184        2       -8        -9
## 3185        2       -8        -9
## 3186        2       -8        -9
## 3187        2       -8        -9
## 3188        2       -8        -9
## 3189        2       -8        -9
## 3190        2       -8        -8
## 3191        2       -8        -8
## 3192        2       -8        -9
## 3193        2       -8        -8
## 3194        2       -8        -9
## 3195        2       -8        -8
## 3196        2       -8        -9
## 3197        2       -8        -9
## 3198        2       -8        -9
## 3199        2       -8        -9
## 3200        2       -8        -8
## 3201        2       -8        -8
## 3202        2       -8        -8
## 3203        2       -8        -9
## 3204        2       -8        -9
## 3205        2       -8        -8
## 3206        2       -8        -9
## 3207        2       -8        -9
## 3208        2       -8        -9
## 3209        2       -8        -9
## 3210        2       -8        -9
## 3211        2       -8        -8
## 3212        2       -8        -9
## 3213        2       -8        -8
## 3214        2       -8        -9
## 3215        2       -8        -8
## 3216        2       -8        -8
## 3217        2       -8        -9
## 3218        2       -8        -9
## 3219        2       -8        -9
## 3220        2       -8        -9
## 3221        2       -8        -8
## 3222        2       -8        -8
## 3223        2       -8        -9
```

```
## 3224         2       -8         -9
## 3225         2       -8         -9
## 3226         2       -8         -9
## 3227         2       -8         -9
## 3228         2       -8         -9
## 3229         2       -8         -8
## 3230         2       -8         -8
## 3231         2       -8         -8
## 3232         2       -8         -9
## 3233         2       -8         -8
## 3234         2       -8         -8
## 3235         2       -8         -9
## 3236         2       -8         -8
## 3237         2       -8         -9
## 3238         2       -8         -9
## 3239         2       -8         -9
## 3240         2       -8         -9
## 3241         2       -8         -8
## 3242         2       -8         -8
## 3243         2       -8         -8
## 3244         2       -8         -8
## 3245         2       -8         -8
## 3246         2       -8         -9
## 3247         2       -8         -8
## 3248         2       -8         -9
## 3249         2       -8         -9
## 3250         2       -8         -8
## 3251         2       -8         -9
## 3252         2       -8         -9
## 3253         2       -8         -9
## 3254         2       -8         -9
## 3255         2       -8         -8
## 3256         2       -8         -8
## 3257         2       -8         -9
## 3258         2       -8         -9
## 3259         2       -8         -9
## 3260         2       -8         -9
## 3261         2       -8         -9
## 3262         2       -8         -9
## 3263         2       -8         -8
## 3264         2       -8         -9
## 3265         2       -8         -9
## 3266         2       -8         -8
## 3267         2       -8         -8
## 3268         2       -8         -9
## 3269         2       -8         -8
## 3270         2       -8         -9
## 3271         2       -8         -9
## 3272         2       -8         -8
## 3273         2       -8         -9
## 3274         2       -8         -9
## 3275         2       -8         -8
## 3276         2       -8         -8
## 3277         2       -8         -9
```

```
## 3278        2       -8        -8
## 3279        2       -8        -9
## 3280        2       -8        -9
## 3281        2       -8        -8
## 3282        2       -8        -9
## 3283        2       -8        -8
## 3284        2       -8        -9
## 3285        2       -8        -9
## 3286        2       -8        -8
## 3287        2       -8        -9
## 3288        2       -8        -9
## 3289        2       -8        -9
## 3290        2       -8        -8
## 3291        2       -8        -9
## 3292        2       -8        -9
## 3293        2       -8        -8
## 3294        2       -8        -9
## 3295        2       -8        -8
## 3296        2       -8        -9
## 3297        2       -8        -8
## 3298        2       -8        -8
## 3299        2       -8        -8
## 3300        2       -8        -8
## 3301        2       -8        -8
## 3302        2       -8        -9
## 3303        2       -8        -9
## 3304        2       -8        -9
## 3305        2       -8        -9
## 3306        2       -8        -9
## 3307        2       -8        -9
## 3308        2       -8        -9
## 3309        2       -8        -9
## 3310        2       -8        -9
## 3311        2       -8        -9
## 3312        2       -8        -8
## 3313        2       -8        -9
## 3314        2       -8        -9
## 3315        2       -8        -9
## 3316        2       -8        -9
## 3317        2       -8        -9
## 3318        2       -8        -9
## 3319        2       -8        -9
## 3320        2       -8        -9
## 3321        2       -8        -9
## 3322        2       -8        -9
## 3323        2       -8        -9
## 3324        2       -8        -9
## 3325        2       -8        -9
## 3326        2       -8        -9
## 3327        2       -8        -9
## 3328        2       -8        -9
## 3329        2       -8        -9
## 3330        2       -8        -9
## 3331        2       -8        -9
```

```
## 3332        2       -8        -9
## 3333        2       -8        -9
## 3334        2       -8        -9
## 3335        2       -8        -9
## 3336        2       -8        -9
## 3337        2       -8        -9
## 3338        2       -8        -9
## 3339        2       -8        -9
## 3340        2       -8        -9
## 3341        2       -8        -9
## 3342        2       -8        -9
## 3343        2       -8        -9
## 3344        2       -8        -9
## 3345        2       -8        -9
## 3346        2       -8        -9
## 3347        2       -8        -9
## 3348        2       -8        -9
## 3349        2       -8        -9
## 3350        2       -8        -9
## 3351        2       -8        -9
## 3352        2       -8        -9
## 3353        2       -8        -9
## 3354        2       -8        -9
## 3355        2       -8        -9
## 3356        2       -8        -9
## 3357        2       -8        -9
## 3358        2       -8        -9
## 3359        2       -8        -9
## 3360        2       -8        -9
## 3361        2       -8        -9
## 3362        2       -8        -9
## 3363        2       -8        -9
## 3364        2       -8        -9
## 3365        2       -8        -9
## 3366        2       -8        -9
## 3367        2       -8        -9
## 3368        2       -8        -9
## 3369        2       -8        -9
## 3370        2       -8        -9
## 3371        2       -8        -9
## 3372        2       -8        -9
## 3373        2       -8        -9
## 3374        2       -8        -9
## 3375        2       -8        -9
## 3376        2       -8        -9
## 3377        2       -8        -9
## 3378        2       -8        -9
## 3379        2       -8        -9
## 3380        2       -8        -9
## 3381        2       -8        -9
## 3382        2       -8        -9
## 3383        2       -8        -9
## 3384        2       -8        -9
## 3385        2       -8        -8
```

```
## 3386        2       -8        -9
## 3387        2       -8        -9
## 3388        2       -8        -9
## 3389        2       -8        -9
## 3390        2       -8        -9
## 3391        2       -8        -9
## 3392        2       -8        -9
## 3393        2       -8        -9
## 3394        2       -8        -9
## 3395        2       -8        -9
## 3396        2       -8        -9
## 3397        2       -8        -9
## 3398        2       -8        -9
## 3399        2       -8        -9
## 3400        2       -8        -9
## 3401        2       -8        -9
## 3402        2       -8        -9
## 3403        2       -8        -9
## 3404        2       -8        -9
## 3405        2       -8        -9
## 3406        2       -8        -9
## 3407        2       -8        -9
## 3408        2       -8        -9
## 3409        2       -8        -9
## 3410        2       -8        -9
## 3411        2       -8        -9
## 3412        2       -8        -9
## 3413        2       -8        -9
## 3414        2       -8        -9
## 3415        2       -8        -9
## 3416        2       -8        -8
## 3417        2       -8        -9
## 3418        2       -8        -9
## 3419        2       -8        -9
## 3420        2       -8        -8
## 3421        2       -8        -9
## 3422        2       -8        -9
## 3423        2       -8        -9
## 3424        2       -8        -9
## 3425        2       -8        -9
## 3426        2       -8        -9
## 3427        2       -8        -9
## 3428        2       -8        -9
## 3429        2       -8        -9
## 3430        2       -8        -9
## 3431        2       -8        -9
## 3432        2       -8        -9
## 3433        2       -8        -9
## 3434        2       -8        -9
## 3435        2       -8        -9
## 3436        2       -8        -9
## 3437        2       -8        -9
## 3438        2       -8        -9
## 3439        2       -8        -9
```

```
## 3440        2       -8        -9
## 3441        2       -8        -9
## 3442        2       -8        -9
## 3443        2       -8        -9
## 3444        2       -8        -9
## 3445        2       -8        -9
## 3446        2       -8        -9
## 3447        2       -8        -9
## 3448        2       -8        -9
## 3449        2       -8        -9
## 3450        2       -8        -9
## 3451        2       -8        -9
## 3452        2       -8        -9
## 3453        2       -8        -9
## 3454        2       -8        -9
## 3455        2       -8        -9
## 3456        2       -8        -9
## 3457        2       -8        -9
## 3458        2       -8        -9
## 3459        2       -8        -9
## 3460        2       -8        -9
## 3461        2       -8        -9
## 3462        2       -8        -9
## 3463        2       -8        -9
## 3464        2       -8        -9
## 3465        2       -8        -9
## 3466        2       -8        -9
## 3467        2       -8        -9
## 3468        2       -8        -9
## 3469        2       -8        -9
## 3470        2       -8        -9
## 3471        2       -8        -9
## 3472        2       -8        -9
## 3473        2       -8        -9
## 3474        2       -8        -9
## 3475        2       -8        -9
## 3476        2       -8        -9
## 3477        2       -8        -9
## 3478        2       -8        -9
## 3479        2       -8        -9
## 3480        2       -8        -9
## 3481        2       -8        -9
## 3482        2       -8        -9
## 3483        2       -8        -9
## 3484        2       -8        -9
## 3485        2       -8        -9
## 3486        2       -8        -9
## 3487        2       -8        -9
## 3488        2       -8        -9
## 3489        2       -8        -9
## 3490        2       -8        -9
## 3491        2       -8        -9
## 3492        2       -8        -9
## 3493        2       -8        -9
```

```
## 3494         2       -8         -9
## 3495         2       -8         -9
## 3496         2       -8         -9
## 3497         2       -8         -9
## 3498         2       -8         -9
## 3499         2       -8         -9
## 3500         2       -8         -9
## 3501         2       -8         -9
## 3502         2       -8         -9
## 3503         2       -8         -9
## 3504         2       -8         -9
## 3505         2       -8         -9
## 3506         2       -8         -9
## 3507         2       -8         -9
## 3508         2       -8         -9
## 3509         2       -8         -9
## 3510         2       -8         -9
## 3511         2       -8         -9
## 3512         2       -8         -9
## 3513         2       -8         -9
## 3514         2       -8         -9
## 3515         2       -8         -9
## 3516         2       -8         -9
## 3517         2       -8         -9
## 3518         2       -8         -9
## 3519         2       -8         -9
## 3520         2       -8         -9
## 3521         2       -8         -9
## 3522         2       -8         -9
## 3523         2       -8         -9
## 3524         2       -8         -9
## 3525         2       -8         -9
## 3526         2       -8         -9
## 3527         2       -8         -9
## 3528         2       -8         -9
## 3529         2       -8         -9
## 3530         2       -8         -9
## 3531         2       -8         -9
## 3532         2       -8         -9
## 3533         2       -8         -9
## 3534         2       -8         -9
## 3535         2       -8         -9
## 3536         2       -8         -9
## 3537         2       -8         -9
## 3538         2       -8         -9
## 3539         2       -8         -9
## 3540         2       -8         -9
## 3541         2       -8         -8
## 3542         2       -8         -9
## 3543         2       -8         -9
## 3544         2       -8         -9
## 3545         2       -8         -9
## 3546         2       -8         -9
## 3547         2       -8         -9
```

```
## 3548        2       -8        -9
## 3549        2       -8        -9
## 3550        2       -8        -9
## 3551        2       -8        -9
## 3552        2       -8        -8
## 3553        2       -8        -9
## 3554        2       -8        -9
## 3555        2       -8        -9
## 3556        2       -8        -9
## 3557        2       -8        -9
## 3558        2       -8        -9
## 3559        2       -8        -9
## 3560        2       -8        -9
## 3561        2       -8        -9
## 3562        2       -8        -9
## 3563        2       -8        -9
## 3564        2       -8        -9
## 3565        2       -8        -9
## 3566        2       -8        -9
## 3567        2       -8        -9
## 3568        2       -8        -9
## 3569        2       -8        -9
## 3570        2       -8        -9
## 3571        2       -8        -9
## 3572        2       -8        -9
## 3573        2       -8        -9
## 3574        2       -8        -9
## 3575        2       -8        -9
## 3576        2       -8        -9
## 3577        2       -8        -9
## 3578        2       -8        -9
## 3579        2       -8        -9
## 3580        2       -8        -9
## 3581        2       -8        -9
## 3582        2       -8        -9
## 3583        2       -8        -9
## 3584        2       -8        -9
## 3585        2       -8        -9
## 3586        2       -8        -9
## 3587        2       -8        -9
## 3588        2       -8        -9
## 3589        2       -8        -9
## 3590        2       -8        -9
## 3591        2       -8        -9
## 3592        2       -8        -9
## 3593        2       -8        -9
## 3594        2       -8        -9
## 3595        2       -8        -9
## 3596        2       -8        -9
## 3597        2       -8        -9
## 3598        2       -8        -9
## 3599        2       -8        -9
## 3600        2       -8        -9
## 3601        2       -8        -9
```

```
## 3602         2       -8         -9
## 3603         2       -8         -9
## 3604         2       -8         -9
## 3605         2       -8         -9
## 3606         2       -8         -9
## 3607         2       -8         -9
## 3608         2       -8         -9
## 3609         2       -8         -9
## 3610         2       -8         -9
## 3611         2       -8         -9
## 3612         2       -8         -9
## 3613         2       -8         -9
## 3614         2       -8         -9
## 3615         2       -8         -9
## 3616         2       -8         -9
## 3617         2       -8         -9
## 3618         2       -8         -9
## 3619         2       -8         -9
## 3620         2       -8         -9
## 3621         2       -8         -9
## 3622         2       -8         -9
## 3623         2       -8         -9
## 3624         2       -8         -9
## 3625         2       -8         -9
## 3626         2       -8         -9
## 3627         2       -8         -9
## 3628         2       -8         -9
## 3629         2       -8         -9
## 3630         2       -8         -9
## 3631         2       -8         -9
## 3632         2       -8         -9
## 3633         2       -8         -8
## 3634         2       -8         -8
## 3635         2       -8         -9
## 3636         2       -8         -9
## 3637         2       -8         -9
## 3638         2       -8         -9
## 3639         2       -8         -9
## 3640         2       -8         -9
## 3641         2       -8         -9
## 3642         2       -8         -9
## 3643         2       -8         -9
## 3644         2       -8         -9
## 3645         2       -8         -9
## 3646         2       -8         -9
## 3647         2       -8         -9
## 3648         2       -8         -9
## 3649         2       -8         -9
## 3650         2       -8         -9
## 3651         2       -8         -9
## 3652         2       -8         -9
## 3653         2       -8         -9
## 3654         2       -8         -9
## 3655         2       -8         -9
```

```
## 3656         2       -8         -9
## 3657         2       -8         -9
## 3658         2       -8         -9
## 3659         2       -8         -9
## 3660         2       -8         -9
## 3661         2       -8         -9
## 3662         2       -8         -9
## 3663         2       -8         -9
## 3664         2       -8         -9
## 3665         2       -8         -9
## 3666         2       -8         -9
## 3667         2       -8         -9
## 3668         2       -8         -9
## 3669         2       -8         -9
## 3670         2       -8         -9
## 3671         2       -8         -9
## 3672         2       -8         -9
## 3673         2       -8         -9
## 3674         2       -8         -9
## 3675         2       -8         -9
## 3676         2       -8         -9
## 3677         2       -8         -9
## 3678         2       -8         -9
## 3679         2       -8         -9
## 3680         2       -8         -9
## 3681         2       -8         -9
## 3682         2       -8         -9
## 3683         2       -8         -9
## 3684         2       -8         -9
## 3685         2       -8         -9
## 3686         2       -8         -9
## 3687         2       -8         -9
## 3688         2       -8         -9
## 3689         2       -8         -9
## 3690         2       -8         -9
## 3691         2       -8         -9
## 3692         2       -8         -9
## 3693         2       -8         -9
## 3694         2       -8         -9
## 3695         2       -8         -9
## 3696         2       -8         -9
## 3697         2       -8         -9
## 3698         2       -8         -9
## 3699         2       -8         -9
## 3700         2       -8         -9
## 3701         2       -8         -9
## 3702         2       -8         -9
## 3703         2       -8         -9
## 3704         2       -8         -9
## 3705         2       -8         -9
## 3706         2       -8         -9
## 3707         2       -8         -9
## 3708         2       -8         -9
## 3709         2       -8         -9
```

```
## 3710        2       -8        -9
## 3711        2       -8        -9
## 3712        2       -8        -9
## 3713        2       -8        -9
## 3714        2       -8        -9
## 3715        2       -8        -9
## 3716        2       -8        -9
## 3717        2       -8        -9
## 3718        2       -8        -9
## 3719        2       -8        -9
## 3720        2       -8        -9
## 3721        2       -8        -8
## 3722        2       -8        -9
## 3723        2       -8        -9
## 3724        2       -8        -9
## 3725        2       -8        -9
## 3726        2       -8        -9
## 3727        2       -8        -9
## 3728        2       -8        -9
## 3729        2       -8        -9
## 3730        2       -8        -9
## 3731        2       -8        -9
## 3732        2       -8        -9
## 3733        2       -8        -9
## 3734        2       -8        -9
## 3735        2       -8        -9
## 3736        2       -8        -9
## 3737        2       -8        -9
## 3738        2       -8        -9
## 3739        2       -8        -9
## 3740        2       -8        -9
## 3741        2       -8        -9
## 3742        2       -8        -9
## 3743        2       -8        -9
## 3744        2       -8        -9
## 3745        2       -8        -9
## 3746        2       -8        -9
## 3747        2       -8        -9
## 3748        2       -8        -9
## 3749        2       -8        -9
## 3750        2       -8        -9
## 3751        2       -8        -9
## 3752        2       -8        -9
## 3753        2       -8        -9
## 3754        2       -8        -9
## 3755        2       -8        -9
## 3756        2       -8        -9
## 3757        2       -8        -9
## 3758        2       -8        -9
## 3759        2       -8        -9
## 3760        2       -8        -9
## 3761        2       -8        -9
## 3762        2       -8        -9
## 3763        2       -8        -9
```

```
## 3764         2       -8         -9
## 3765         2       -8         -9
## 3766         2       -8         -9
## 3767         2       -8         -9
## 3768         2       -8         -9
## 3769         2       -8         -9
## 3770         2       -8         -9
## 3771         2       -8         -9
## 3772         2       -8         -9
## 3773         2       -8         -9
## 3774         2       -8         -9
## 3775         2       -8         -9
## 3776         2       -8         -9
## 3777         2       -8         -9
## 3778         2       -8         -9
## 3779         2       -8         -9
## 3780         2       -8         -9
## 3781         2       -8         -9
## 3782         2       -8         -9
## 3783         2       -8         -9
## 3784         2       -8         -9
## 3785         2       -8         -9
## 3786         2       -8         -8
## 3787         2       -8         -9
## 3788         2       -8         -9
## 3789         2       -8         -9
## 3790         2       -8         -9
## 3791         2       -8         -9
## 3792         2       -8         -9
## 3793         2       -8         -9
## 3794         2       -8         -9
## 3795         2       -8         -9
## 3796         2       -8         -9
## 3797         2       -8         -9
## 3798         2       -8         -8
## 3799         2       -8         -9
## 3800         2       -8         -9
## 3801         2       -8         -9
## 3802         2       -8         -9
## 3803         2       -8         -9
## 3804         2       -8         -9
## 3805         2       -8         -9
## 3806         2       -8         -9
## 3807         2       -8         -9
## 3808         2       -8         -9
## 3809         2       -8         -9
## 3810         2       -8         -9
## 3811         2       -8         -9
## 3812         2       -8         -9
## 3813         2       -8         -9
## 3814         2       -8         -9
## 3815         2       -8         -9
## 3816         2       -8         -9
## 3817         2       -8         -9
```

```r
# If there is no commercial space on this property, then the property did not
# receive any rental income from a commercial use of the property.
rhfs$RECCOMMREN[rhfs$COMMSPACE == 2] <- "2"
rhfs$COMMRENT <- ifelse(rhfs$COMMSPACE == 2, 0, rhfs$COMMRENT)


rhfs |>
  mutate(id = row_number()) |>
  select(RECCOMMREN, COMMRENT) |>
  filter(RECCOMMREN == 2) |>
  summarize(
    n(),
    .by = COMMRENT
  )
```

```
##   COMMRENT  n()
## 1        0 3874
## 2       -9    9
```

```r
# If commercial rent was not received, then commercial rent should be 0
rhfs$COMMRENT <- ifelse(rhfs$RECCOMMREN == 2, 0, rhfs$COMMRENT)


# More informative than "-8"
rhfs <- rhfs |>
  mutate(
    UNITSTAT = fct_recode(
      UNITSTAT,
      "Not a single unit" = "-8"
    )
  )


# TOTRENTVAC's (total units for rent or vacant-for rent) 377 missing values are
# due to them being outside its universe of NUMUNITS >= 2. UNITSTAT (status of
# single unit) indicates that all of these values are rented or vacant for rent,
# pointing towards the idea that the 377 missing values should have a TOTRENTVAC
# of 1.
table(rhfs$TOTRENTVAC)
```

```
##
##   -9    1    2    3    4    5    6    7    8    9   10   11   12   13   14   15
##  377   56  170   88  142   39   84   22   75   19   47   16   54   14   22   26
##   16   17   18   19   20   21   22   23   24   25   26   27   28   29   30   31
##   46    5   24   17   41   20   14   23   56   15   10   15   29   15   28   19
##   32   33   34   35   36   37   38   39   40   41   42   43   44   45   46   47
##   30    8   11   25   16   13    8   17   45   12   11    9   17   13   11   11
##   48   49   50   51   52   53   54   55   56   57   58   59   60   61   62   63
##   45   14   33    3    9    5    5    8   20    6    3   11   31    6   10    7
##   64   65   66   67   68   69   70   71   72   73   74   75   76   77   78   79
##   15    6    7    5   14    8   11    6   24    4    5   10    9    3    8    8
##   80   81   82   83   84   85   86   87   88   89   90   91   92   93   94   95
##   25    3    3   12   12    7    7    5   17    7   13    3    4    3   11    6
```

```
##   96   97   98   99  100  101  102  103  104  105  106  107  108  109  110  111
##   14    2   12   14   39    7    3    8   10    6    6    4   11    3   10    5
##  112  113  114  115  116  117  118  119  120  121  122  123  124  125  126  127
##   12    7    5    5    6    8    7   11   33    3    6    7    9    3    4    5
##  128  129  130  131  132  133  134  135  136  137  138  139  140  141  142  143
##   12    4   11    5    6    4    5    4   13    5    4    9   13    7    6    5
##  144  145  146  147  148  149  150  151  152  153  154  155  156  157  159  160
##   19    4    5    2   14    7   19    7   10    1    8    5    4    2    4   22
##  161  162  163  164  165  166  167  168  169  170  171  172  173  174  175  176
##    4    4    4    7    1    6    3   12    4    7    9    3    3    4    3   12
##  177  178  179  180  181  182  183  184  185  186  187  188  189  190  191  192
##    4    4    7   18    2    3    6    8    4    6    4   11    3    4    2   11
##  193  194  195  196  197  198  199  200  201  202  203  204  205  206  207  208
##    8    4    2   11    3   12    5   26   10    7    7    6    6    5    5   16
##  209  210  211  212  213  214  215  216  217  218  219  220  221  222  223  224
##    4   11    2    7    3    4    2   18    2    3    6    6    4    5    3    9
##  225  226  227  228  229  230  231  232  233  234  235  236  237  238  239  240
##    2    6    5    7    1    4    2    6    2    4    4    4    3    4    2   27
##  241  242  244  245  246  247  248  249  250  251  252  253  254  255  256  257
##    1    3    6    2    2    4    8    6    8    5   12    1    6    3    6    3
##  258  259  260  261  262  263  264  265  266  267  268  269  270  271  272  273
##    3    5    9    2    6    4    9    2    4    4    9    2    8    1    9    5
##  274  275  276  277  278  279  280  281  282  283  284  285  286  287  288  289
##    2    5    6    3    9    4   10    2    3    3    3    5    3    3   13    1
##  290  291  292  293  294  295  296  297  298  299  300  301  303  304  305  306
##    2    5    7    3    1    2    9    3    2    3   22    6    5   11    1    2
##  307  308  309  310  311  312  313  314  315  316  317  318  319  320  321  322
##    2    5    1    5    7   10    1    2    3    5    4    5    4   12    1    3
##  324  325  326  327  328  329  330  331  332  333  334  335  336  338  339  340
##    8    5    2    1    5    3    1    4    5    2    3    2   11    2    3    5
##  341  344  345  346  347  348  349  350  351  352  353  354  355  356  357  358
##    2    2    4    5    2    7    4    3    6    5    1    2    1    4    4    2
##  359  360  362  363  364  365  366  367  368  369  370  371  372  373  374  375
##    1   12    4    1    4    2    6    2    8    6   11    8   90    1    2    1
##  376  377  378  379  380  382  383  384  385  386  387  388  390  391  392  393
##    4    2    2    2    1    1    1    3    4    2    1    6    1    1    1    1
##  394  395  396  397  399  400  401  402  404  405  406  408  409  411  412  413
##    2    3    3    1    3    5    1    1    1    2    2    6    1    1    1    1
##  415  416  417  418  419  420  421  422  424  425  426  428  429  430  431  433
##    1    3    1    2    1    1    1    2    1    1    1    2    2    3    1    3
##  434  436  437  440  441  442  444  445  446  447  448  450  452  453  454  456
##    1    2    1    3    1    3    3    1    2    1    3    2    1    1    1    4
##  457  460  461  462  463  464  466  467  469  470  472  476  477  478  480  482
##    1    2    1    1    3    1    1    2    2    1    3    2    2    2    3    1
##  483  484  486  487  488  489  491  492  494  495  496  497  498  500  502  504
##    2    2    2    2    1    1    2    3    1    2    5    1    6    1    3    2
##  506  508  509  510  512  513  514  515  516  517  519  520  521  522  523  524
##    2    2    3    2    1    1    2    1    1    1    1    2    2    1    1    2
##  526  529  531  533  535  540  541  544  545  546  547  550  552  553  557  558
##    2    3    1    1    1    1    2    1    3    2   11    1    1    1    2    2
##  563  565  568  569  574  575  576  579  581  584  587  588  596  597  598  600
##    1    3    1    1    1    1    1    1    1    1    1    3    1    1    1    4
##  604  608  611  614  618  621  622  624  627  630  636  640  642  645  648  653
##    1    1    3    1    1    1    2    1    1    1    1    4    1    1    1    1
```

```
##  657  660  664  667  668  669  675  680  684  694  696  698  700  705  706  708
##    1    1    1    1    1    1    1    1    1    1    1    1    1    1    2    1
##  709  715  718  720  734  739  741  743  744  752  771  775  779  781  788  789
##    1    1    1    1    1    1    1    1    1    1    1    1    1    1    2    1
##  801  802  810  833  839  845  858  862  880  891  900  922 1000 1043 1097 1136
##    2    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1
## 1137 1215 1236 1241 1274 1293 1296 1322 1324 1350 1354 1416 1444 1474 1530 1730
##    1    1    1    1    2    2    1    1    1    1    1    1    1    1    1    1
## 1817
##    1
```

```
table(rhfs$UNITSTAT)
```

```
##
## Not a single unit                 1                 2
##              4048               360                17
```

```
rhfs$TOTRENTVAC <- ifelse(rhfs$TOTRENTVAC == -9, 1, rhfs$TOTRENTVAC)


# Reverse engineering TOTRENTVAC indicates that it is constructed by NUMUNITS -
# TOWNOC_R - TEMPOC_R - TOTHVA_R (accurate 99.66% of the time). To achieve this
# accuracy, values in TOWNOC_R, TEMPOC_R, and TOTHVA_R are assigned to 0.
rhfs$TOWNOC_R[rhfs$TOWNOC_R %in% c(-9, -8)] <- 0
rhfs$TEMPOC_R[rhfs$TEMPOC_R %in% c(-9, -8)] <- 0
rhfs$TOTHVA_R[rhfs$TOTHVA_R %in% c(-9, -8)] <- 0


# When all components of OPEX_R are explicitly 0, then OPEX_R is 0. The
# corresponding OPEXPU_R (OPEX_R / NUMUNITS_R) are therefore also 0.
trivially.missing.opex_r <- rhfs |>
  mutate(id = row_number()) |>
  filter(
    OPEX_R == -9 &
      if_all(c(OPPAY, OPINSUR, OPELGAS, OPWTRSWR, OPREP, OPMNG, OPPROTH,
               OPSECUR, OPLAND, OPOTHEX), \(x) x == 0)
    ) |>
  select(id, OPEX_R, OPEXPU_R, OPPAY, OPINSUR, OPELGAS, OPWTRSWR, OPREP, OPMNG,
         OPPROTH, OPSECUR, OPLAND, OPOTHEX) |>
  pull(id)

rhfs$OPEX_R[trivially.missing.opex_r] <- 0
rhfs$OPEXPU_R[trivially.missing.opex_r] <- 0


# When TRENOC_R is -8, then UNITSTAT is 1 (rented) or 2 (vacant, for rent).
rhfs |>
  select(TRENOC_R, UNITSTAT) |>
  filter(TRENOC_R == -8) |>
  summarize(
    n(),
    .by = UNITSTAT
  )
```

```
##   UNITSTAT n()
## 1        1 360
## 2        2  17
```

```
# TRENOC_R's (number of renter-occupied units) 377 -8s are due to them being
# outside of its universe of NUMUNITS >= 2. They can be derived from UNITSTAT.
rhfs$TRENOC_R <- ifelse(
  # If UNITSTAT is 1, then TRENOC_R is now 1.
  rhfs$UNITSTAT == "1", 1,
  # If UNITSTAT is 2, then TRENOC_R is now 0.
  ifelse(rhfs$UNITSTAT == "2", 0,
         # Otherwise, leave unchanged.
         rhfs$TRENOC_R))


# When TRENVA_R is -8, then UNITSTAT is 1 (rented) or 2 (vacant, for rent).
rhfs |>
  select(TRENVA_R, UNITSTAT) |>
  filter(TRENVA_R == -8) |>
  summarize(
    n(),
    .by = UNITSTAT
  )
```

```
##   UNITSTAT n()
## 1        1 360
## 2        2  17
```

```
# TRENVA_R's (number of vacant for rent units) 377 -8s are due to them being
# outside of its universe of NUMUNITS >= 2. They can be derived from UNITSTAT.
rhfs$TRENVA_R <- ifelse(
  # If UNITSTAT is 1, then TRENVA_R is now 0.
  rhfs$UNITSTAT == "1", 0,
  # If UNITSTAT is 2, then TRENVA_R is now 1.
  ifelse(rhfs$UNITSTAT == "2", 1,
         # Otherwise, leave unchanged.
         rhfs$TRENVA_R))


# BROOMS is observed for all but one single-unit property
rhfs |>
  select(BROOMS, BED0_R, BED1_R, BED2_R, BED3_R) |>
    filter(BED0_R == -8)
```

```
##     BROOMS BED0_R BED1_R BED2_R BED3_R
## 1        2     -8     -8     -8     -8
## 2        1     -8     -8     -8     -8
## 3        2     -8     -8     -8     -8
## 4        3     -8     -8     -8     -8
## 5        1     -8     -8     -8     -8
## 6       -9     -8     -8     -8     -8
## 7        1     -8     -8     -8     -8
## 8        2     -8     -8     -8     -8
```

```
## 9        1     -8     -8     -8     -8
## 10       3     -8     -8     -8     -8
## 11       3     -8     -8     -8     -8
## 12       1     -8     -8     -8     -8
## 13       3     -8     -8     -8     -8
## 14       2     -8     -8     -8     -8
## 15       1     -8     -8     -8     -8
## 16       2     -8     -8     -8     -8
## 17       3     -8     -8     -8     -8
## 18       2     -8     -8     -8     -8
## 19       1     -8     -8     -8     -8
## 20       4     -8     -8     -8     -8
## 21       3     -8     -8     -8     -8
## 22       4     -8     -8     -8     -8
## 23       2     -8     -8     -8     -8
## 24       4     -8     -8     -8     -8
## 25       2     -8     -8     -8     -8
## 26       3     -8     -8     -8     -8
## 27       2     -8     -8     -8     -8
## 28       3     -8     -8     -8     -8
## 29       2     -8     -8     -8     -8
## 30       2     -8     -8     -8     -8
## 31       3     -8     -8     -8     -8
## 32       3     -8     -8     -8     -8
## 33       0     -8     -8     -8     -8
## 34       2     -8     -8     -8     -8
## 35       3     -8     -8     -8     -8
## 36       1     -8     -8     -8     -8
## 37       2     -8     -8     -8     -8
## 38       3     -8     -8     -8     -8
## 39       2     -8     -8     -8     -8
## 40       2     -8     -8     -8     -8
## 41       1     -8     -8     -8     -8
## 42       2     -8     -8     -8     -8
## 43       3     -8     -8     -8     -8
## 44       3     -8     -8     -8     -8
## 45       3     -8     -8     -8     -8
## 46       3     -8     -8     -8     -8
## 47       2     -8     -8     -8     -8
## 48       2     -8     -8     -8     -8
## 49       2     -8     -8     -8     -8
## 50       1     -8     -8     -8     -8
## 51       3     -8     -8     -8     -8
## 52       2     -8     -8     -8     -8
## 53       3     -8     -8     -8     -8
## 54       1     -8     -8     -8     -8
## 55       3     -8     -8     -8     -8
## 56       2     -8     -8     -8     -8
## 57       3     -8     -8     -8     -8
## 58       3     -8     -8     -8     -8
## 59       1     -8     -8     -8     -8
## 60       3     -8     -8     -8     -8
## 61       3     -8     -8     -8     -8
## 62       3     -8     -8     -8     -8
```

```
## 63       2     -8     -8     -8     -8
## 64       3     -8     -8     -8     -8
## 65       3     -8     -8     -8     -8
## 66       2     -8     -8     -8     -8
## 67       3     -8     -8     -8     -8
## 68       1     -8     -8     -8     -8
## 69       2     -8     -8     -8     -8
## 70       2     -8     -8     -8     -8
## 71       2     -8     -8     -8     -8
## 72       3     -8     -8     -8     -8
## 73       1     -8     -8     -8     -8
## 74       2     -8     -8     -8     -8
## 75       2     -8     -8     -8     -8
## 76       3     -8     -8     -8     -8
## 77       3     -8     -8     -8     -8
## 78       1     -8     -8     -8     -8
## 79       2     -8     -8     -8     -8
## 80       4     -8     -8     -8     -8
## 81       2     -8     -8     -8     -8
## 82       3     -8     -8     -8     -8
## 83       2     -8     -8     -8     -8
## 84       1     -8     -8     -8     -8
## 85       1     -8     -8     -8     -8
## 86       3     -8     -8     -8     -8
## 87       3     -8     -8     -8     -8
## 88       3     -8     -8     -8     -8
## 89       2     -8     -8     -8     -8
## 90       4     -8     -8     -8     -8
## 91       5     -8     -8     -8     -8
## 92       3     -8     -8     -8     -8
## 93       1     -8     -8     -8     -8
## 94       1     -8     -8     -8     -8
## 95       2     -8     -8     -8     -8
## 96       3     -8     -8     -8     -8
## 97       3     -8     -8     -8     -8
## 98       2     -8     -8     -8     -8
## 99       5     -8     -8     -8     -8
## 100      2     -8     -8     -8     -8
## 101      4     -8     -8     -8     -8
## 102      5     -8     -8     -8     -8
## 103      2     -8     -8     -8     -8
## 104      4     -8     -8     -8     -8
## 105      1     -8     -8     -8     -8
## 106      1     -8     -8     -8     -8
## 107      3     -8     -8     -8     -8
## 108      3     -8     -8     -8     -8
## 109      3     -8     -8     -8     -8
## 110      1     -8     -8     -8     -8
## 111      3     -8     -8     -8     -8
## 112      1     -8     -8     -8     -8
## 113      3     -8     -8     -8     -8
## 114      3     -8     -8     -8     -8
## 115      2     -8     -8     -8     -8
## 116      3     -8     -8     -8     -8
```

```
## 117      3     -8     -8     -8     -8
## 118      2     -8     -8     -8     -8
## 119      3     -8     -8     -8     -8
## 120      3     -8     -8     -8     -8
## 121      1     -8     -8     -8     -8
## 122      3     -8     -8     -8     -8
## 123      1     -8     -8     -8     -8
## 124      1     -8     -8     -8     -8
## 125      1     -8     -8     -8     -8
## 126      2     -8     -8     -8     -8
## 127      2     -8     -8     -8     -8
## 128      2     -8     -8     -8     -8
## 129      2     -8     -8     -8     -8
## 130      0     -8     -8     -8     -8
## 131      1     -8     -8     -8     -8
## 132      2     -8     -8     -8     -8
## 133      2     -8     -8     -8     -8
## 134      2     -8     -8     -8     -8
## 135      3     -8     -8     -8     -8
## 136      1     -8     -8     -8     -8
## 137      2     -8     -8     -8     -8
## 138      3     -8     -8     -8     -8
## 139      3     -8     -8     -8     -8
## 140      1     -8     -8     -8     -8
## 141      2     -8     -8     -8     -8
## 142      2     -8     -8     -8     -8
## 143      2     -8     -8     -8     -8
## 144      2     -8     -8     -8     -8
## 145      1     -8     -8     -8     -8
## 146      3     -8     -8     -8     -8
## 147      4     -8     -8     -8     -8
## 148      4     -8     -8     -8     -8
## 149      2     -8     -8     -8     -8
## 150      2     -8     -8     -8     -8
## 151      1     -8     -8     -8     -8
## 152      1     -8     -8     -8     -8
## 153      2     -8     -8     -8     -8
## 154      2     -8     -8     -8     -8
## 155      2     -8     -8     -8     -8
## 156      1     -8     -8     -8     -8
## 157      3     -8     -8     -8     -8
## 158      3     -8     -8     -8     -8
## 159      2     -8     -8     -8     -8
## 160      2     -8     -8     -8     -8
## 161      2     -8     -8     -8     -8
## 162      3     -8     -8     -8     -8
## 163      3     -8     -8     -8     -8
## 164      0     -8     -8     -8     -8
## 165      2     -8     -8     -8     -8
## 166      2     -8     -8     -8     -8
## 167      2     -8     -8     -8     -8
## 168      1     -8     -8     -8     -8
## 169      2     -8     -8     -8     -8
## 170      3     -8     -8     -8     -8
```

```
## 171      2     -8     -8     -8     -8
## 172      2     -8     -8     -8     -8
## 173      2     -8     -8     -8     -8
## 174      2     -8     -8     -8     -8
## 175      3     -8     -8     -8     -8
## 176      3     -8     -8     -8     -8
## 177      3     -8     -8     -8     -8
## 178      2     -8     -8     -8     -8
## 179      1     -8     -8     -8     -8
## 180      3     -8     -8     -8     -8
## 181      2     -8     -8     -8     -8
## 182      3     -8     -8     -8     -8
## 183      2     -8     -8     -8     -8
## 184      3     -8     -8     -8     -8
## 185      1     -8     -8     -8     -8
## 186      2     -8     -8     -8     -8
## 187      2     -8     -8     -8     -8
## 188      2     -8     -8     -8     -8
## 189      1     -8     -8     -8     -8
## 190      2     -8     -8     -8     -8
## 191      2     -8     -8     -8     -8
## 192      2     -8     -8     -8     -8
## 193      4     -8     -8     -8     -8
## 194      3     -8     -8     -8     -8
## 195      2     -8     -8     -8     -8
## 196      2     -8     -8     -8     -8
## 197      2     -8     -8     -8     -8
## 198      1     -8     -8     -8     -8
## 199      2     -8     -8     -8     -8
## 200      2     -8     -8     -8     -8
## 201      2     -8     -8     -8     -8
## 202      3     -8     -8     -8     -8
## 203      3     -8     -8     -8     -8
## 204      2     -8     -8     -8     -8
## 205      4     -8     -8     -8     -8
## 206      2     -8     -8     -8     -8
## 207      2     -8     -8     -8     -8
## 208      1     -8     -8     -8     -8
## 209      3     -8     -8     -8     -8
## 210      2     -8     -8     -8     -8
## 211      2     -8     -8     -8     -8
## 212      2     -8     -8     -8     -8
## 213      3     -8     -8     -8     -8
## 214      1     -8     -8     -8     -8
## 215      3     -8     -8     -8     -8
## 216      3     -8     -8     -8     -8
## 217      3     -8     -8     -8     -8
## 218      1     -8     -8     -8     -8
## 219      2     -8     -8     -8     -8
## 220      3     -8     -8     -8     -8
## 221      1     -8     -8     -8     -8
## 222      2     -8     -8     -8     -8
## 223      0     -8     -8     -8     -8
## 224      2     -8     -8     -8     -8
```

```
## 225      2     -8     -8     -8     -8
## 226      3     -8     -8     -8     -8
## 227      3     -8     -8     -8     -8
## 228      2     -8     -8     -8     -8
## 229      3     -8     -8     -8     -8
## 230      2     -8     -8     -8     -8
## 231      2     -8     -8     -8     -8
## 232      2     -8     -8     -8     -8
## 233      2     -8     -8     -8     -8
## 234      1     -8     -8     -8     -8
## 235      2     -8     -8     -8     -8
## 236      3     -8     -8     -8     -8
## 237      2     -8     -8     -8     -8
## 238      2     -8     -8     -8     -8
## 239      2     -8     -8     -8     -8
## 240      4     -8     -8     -8     -8
## 241      2     -8     -8     -8     -8
## 242      2     -8     -8     -8     -8
## 243      2     -8     -8     -8     -8
## 244      3     -8     -8     -8     -8
## 245      3     -8     -8     -8     -8
## 246      2     -8     -8     -8     -8
## 247      1     -8     -8     -8     -8
## 248      3     -8     -8     -8     -8
## 249      1     -8     -8     -8     -8
## 250      3     -8     -8     -8     -8
## 251      2     -8     -8     -8     -8
## 252      2     -8     -8     -8     -8
## 253      2     -8     -8     -8     -8
## 254      1     -8     -8     -8     -8
## 255      3     -8     -8     -8     -8
## 256      4     -8     -8     -8     -8
## 257      3     -8     -8     -8     -8
## 258      2     -8     -8     -8     -8
## 259      3     -8     -8     -8     -8
## 260      4     -8     -8     -8     -8
## 261      1     -8     -8     -8     -8
## 262      1     -8     -8     -8     -8
## 263      2     -8     -8     -8     -8
## 264      2     -8     -8     -8     -8
## 265      3     -8     -8     -8     -8
## 266      3     -8     -8     -8     -8
## 267      3     -8     -8     -8     -8
## 268      1     -8     -8     -8     -8
## 269      4     -8     -8     -8     -8
## 270      3     -8     -8     -8     -8
## 271      1     -8     -8     -8     -8
## 272      2     -8     -8     -8     -8
## 273      2     -8     -8     -8     -8
## 274      3     -8     -8     -8     -8
## 275      2     -8     -8     -8     -8
## 276      2     -8     -8     -8     -8
## 277      2     -8     -8     -8     -8
## 278      2     -8     -8     -8     -8
```

```
## 279      2     -8     -8     -8     -8
## 280      3     -8     -8     -8     -8
## 281      2     -8     -8     -8     -8
## 282      2     -8     -8     -8     -8
## 283      3     -8     -8     -8     -8
## 284      2     -8     -8     -8     -8
## 285      2     -8     -8     -8     -8
## 286      3     -8     -8     -8     -8
## 287      3     -8     -8     -8     -8
## 288      3     -8     -8     -8     -8
## 289      4     -8     -8     -8     -8
## 290      2     -8     -8     -8     -8
## 291      1     -8     -8     -8     -8
## 292      3     -8     -8     -8     -8
## 293      1     -8     -8     -8     -8
## 294      2     -8     -8     -8     -8
## 295      3     -8     -8     -8     -8
## 296      2     -8     -8     -8     -8
## 297      1     -8     -8     -8     -8
## 298      3     -8     -8     -8     -8
## 299      1     -8     -8     -8     -8
## 300      2     -8     -8     -8     -8
## 301      3     -8     -8     -8     -8
## 302      2     -8     -8     -8     -8
## 303      2     -8     -8     -8     -8
## 304      3     -8     -8     -8     -8
## 305      2     -8     -8     -8     -8
## 306      1     -8     -8     -8     -8
## 307      3     -8     -8     -8     -8
## 308      2     -8     -8     -8     -8
## 309      1     -8     -8     -8     -8
## 310      2     -8     -8     -8     -8
## 311      1     -8     -8     -8     -8
## 312      3     -8     -8     -8     -8
## 313      2     -8     -8     -8     -8
## 314      2     -8     -8     -8     -8
## 315      1     -8     -8     -8     -8
## 316      2     -8     -8     -8     -8
## 317      3     -8     -8     -8     -8
## 318      3     -8     -8     -8     -8
## 319      3     -8     -8     -8     -8
## 320      3     -8     -8     -8     -8
## 321      2     -8     -8     -8     -8
## 322      3     -8     -8     -8     -8
## 323      2     -8     -8     -8     -8
## 324      3     -8     -8     -8     -8
## 325      1     -8     -8     -8     -8
## 326      2     -8     -8     -8     -8
## 327      3     -8     -8     -8     -8
## 328      2     -8     -8     -8     -8
## 329      5     -8     -8     -8     -8
## 330      3     -8     -8     -8     -8
## 331      2     -8     -8     -8     -8
## 332      2     -8     -8     -8     -8
```

```
## 333      2     -8     -8     -8     -8
## 334      3     -8     -8     -8     -8
## 335      2     -8     -8     -8     -8
## 336      2     -8     -8     -8     -8
## 337      4     -8     -8     -8     -8
## 338      1     -8     -8     -8     -8
## 339      2     -8     -8     -8     -8
## 340      2     -8     -8     -8     -8
## 341      1     -8     -8     -8     -8
## 342      4     -8     -8     -8     -8
## 343      4     -8     -8     -8     -8
## 344      2     -8     -8     -8     -8
## 345      2     -8     -8     -8     -8
## 346      2     -8     -8     -8     -8
## 347      2     -8     -8     -8     -8
## 348      3     -8     -8     -8     -8
## 349      3     -8     -8     -8     -8
## 350      4     -8     -8     -8     -8
## 351      2     -8     -8     -8     -8
## 352      2     -8     -8     -8     -8
## 353      3     -8     -8     -8     -8
## 354      3     -8     -8     -8     -8
## 355      4     -8     -8     -8     -8
## 356      4     -8     -8     -8     -8
## 357      2     -8     -8     -8     -8
## 358      2     -8     -8     -8     -8
## 359      4     -8     -8     -8     -8
## 360      2     -8     -8     -8     -8
## 361      2     -8     -8     -8     -8
## 362      3     -8     -8     -8     -8
## 363      2     -8     -8     -8     -8
## 364      3     -8     -8     -8     -8
## 365      2     -8     -8     -8     -8
## 366      2     -8     -8     -8     -8
## 367      2     -8     -8     -8     -8
## 368      4     -8     -8     -8     -8
## 369      3     -8     -8     -8     -8
## 370      2     -8     -8     -8     -8
## 371      2     -8     -8     -8     -8
## 372      2     -8     -8     -8     -8
## 373      3     -8     -8     -8     -8
## 374      2     -8     -8     -8     -8
## 375      3     -8     -8     -8     -8
## 376      4     -8     -8     -8     -8
## 377      4     -8     -8     -8     -8
```

```
# BED0_R, BED1_R, BED2_R, and BED3_R's 377 -8s are due to them being
# outside of their universe of NUMUNITS >= 2. They can be derived from BROOMS.
rhfs <- rhfs |>
  mutate(
    single = NUMUNITS_R == 1 & BROOMS >= 0,
    BED0_R = if_else(single, as.integer(BROOMS == 0), BED0_R),
    BED1_R = if_else(single, as.integer(BROOMS == 1), BED1_R),
    BED2_R = if_else(single, as.integer(BROOMS == 2), BED2_R),
```

```
    BED3_R = if_else(single, as.integer(BROOMS >= 3), BED3_R)
  ) |>
  select(-single)

rhfs |>
  select(starts_with("RENT")) |>
  filter(RENT5000_R > 0)
```

```
##    RENT RENTNO_R RENTLT249_R RENT274_R RENT299_R RENT349_R RENT399_R RENT449_R
## 1    -8        0           0         0         0         0         0         0
## 2    -8        0           0         0         0         0         0         0
## 3    -8        0           0         0         0         0         0         0
## 4    -8        0           0         0         0         0         0         0
## 5    -8        0           0         0         0         0         0         0
## 6    -8        0           0         0         0         0         0         0
## 7    -8        0           0         0         0         0         0         0
## 8    -8        0           0         0         0         0         0         0
## 9    -8        0           0         0         0         0         0         0
## 10   -8        0           0         0         0         0         0         0
## 11   -8        0           0         0         0         0         0         0
## 12   -8        0           0         0         0         0         0         0
## 13   -8        0           0         0         0         0         0         0
## 14   -8        0           0         0         0         0         0         0
## 15   -8        0           0         0         0         0         0         0
## 16   -8        0           0         0         0         0         0         0
## 17   -8        0           0         0         0         0         0         0
## 18   -8        0           0         0         0         0         0         0
## 19   -8        0           0         0         0         0         0         0
## 20   -8        0           0         0         0         0         0         0
## 21   -8        0           0         0         0         0         0         0
## 22   -8        0           0         0         0         0         0         0
## 23   -8        0           0         0         0         0         0         0
## 24   -8        0           0         0         0         0         0         0
## 25   -8        0           0         0         0         0         0         0
## 26   -8        0           0         0         0         0         0         0
## 27   -8        0           0         0         0         0         0         0
## 28   -8        0           0         0         0         0         0         0
## 29   -8        0           0         0         0         0         0         0
## 30   -8        0           0         0         0         0         0         0
## 31   -8        0           0         0         0         0         0         0
## 32   -8        0           0         0         0         0         0         0
## 33   -8        0           0         0         0         0         0         0
## 34   -8        0           0         0         0         0         0         0
## 35   -8        0           0         0         0         0         0         0
## 36   -8        0           0         0         0         0         0         0
## 37   -8        0           0         0         0         0         0         0
## 38   -8        0           0         0         0         0         0         0
## 39   -8        0           0         0         0         0         0         0
## 40   -8        0           0         0         0         0         0         0
## 41   -8        0           0         0         0         0         0         0
## 42   -8        0           0         0         0         0         0         0
## 43   -8        0           0         0         0         0         0         0
## 44   -8        0           0         0         0         0         0         0
```

```
## 45   -8        0           0         0         0         0         0         0
## 46   -8        0           0         0         0         0         0         0
## 47   -8        0           0         0         0         0         0         0
## 48   -8        0           0         0         0         0         0         0
## 49   -8        0           0         0         0         0         0         0
## 50   -8        0           0         0         0         0         0         0
## 51   -8        0           0         0         0         0         0         0
##    RENT499_R RENT549_R RENT599_R RENT649_R RENT699_R RENT749_R RENT799_R
## 1          0         0         0         0         0         0         0
## 2          0         0         0         0         0         0         0
## 3          0         0         0         0         0         0         0
## 4          0         0         0         0         0         0         0
## 5          0         0         0         0         0         0         0
## 6          0         0         0         0         0         0         1
## 7          0         0         0         0         0         0         0
## 8          0         0         0         0         0         0         0
## 9          0         0         0         0         0         0         0
## 10         0         0         0         0         0         0         0
## 11         0         0         0         0         0         0         0
## 12         0         0         0         0         0         0         0
## 13         0         0         0         0         0         0         0
## 14         0         0         0         0         0         0         0
## 15         0         0         0         0         0         0         0
## 16         0         0         0         0         0         0         0
## 17         0         0         0         0         0         0         0
## 18         0         0         0         0         0         0         0
## 19         0         0         0         0         0         0         0
## 20         0         0         0         0         0         0         0
## 21         0         0         0         0         0         0         0
## 22         0         0         0         0         0         0         0
## 23         0         0         0         0         0         0         0
## 24         0         0         0         0         0         0         0
## 25         0         0         0         0         0         0         0
## 26         0         0         0         0         0         0         0
## 27         0         0         0         0         0         0         0
## 28         0         0         0         0         0         0         0
## 29         0         0         0         0         0         0         0
## 30         0         0         0         0         0         0         0
## 31         0         0         0         0         0         0         0
## 32         0         0         0         0         0         0         0
## 33         0         0         0         0         0         0         0
## 34         0         0         0         0         0         0         0
## 35         0         0         0         0         0         0         0
## 36         0         0         0         0         0         0         0
## 37         0         0         0         0         0         0         0
## 38         0         0         0         0         0         0         0
## 39         0         0         0         0         0         0         0
## 40         0         0         0         0         0         0         0
## 41         0         0         0         0         0         0         0
## 42         0         0         0         0         0         0         0
## 43         0         0         0         0         0         0         0
## 44         0         0         0         0         0         0         0
## 45         0         0         0         0         0         0         0
## 46         0         0         0         0         0         0         0
```

```
## 47         0         0         0         0         0         0         0
## 48         0         0         0         0         0         0         0
## 49         0         0         0         0         0         0         0
## 50         0         0         0         0         0         0         0
## 51         0         0         0         0         0         0         0
##    RENT899_R RENT999_R RENT1099_R RENT1249_R RENT1499_R RENT1999_R RENT2499_R
## 1          0         0          0          0          0          0          0
## 2          0         0          0          0          0          0          0
## 3          0         0          0          0          0          0          0
## 4          0         0          0          0          0          0          3
## 5          0         0          0          0          0          0          0
## 6          0         0          0          0          0          0          0
## 7          0         0          0          0          0          0          0
## 8          0         0          0          0          0          0          0
## 9          0         0          0          0          0          0         12
## 10         1         0          0          0          0          0          0
## 11         0         1          1          0          0          1          0
## 12         0         0          0          0          0          0          0
## 13         0         0          0          0          0          0          0
## 14         0         0          0          0          0         12         16
## 15         0         0          0          0          0        100        105
## 16         0         0          0          0          0          0          0
## 17         0         0          0        116        111          0        398
## 18         0         0          0          0          0         38         38
## 19         0         0          0          0          0          0          2
## 20         0         0          0          0          0          0          1
## 21         0         0          0          0          0          0          0
## 22         0         0          0          0          0          0          0
## 23         0         0          0          0          0          0          0
## 24         0         0          0          0          0          0          0
## 25         0         0          0          0          0          0         12
## 26         0         0          0          0          0          0          0
## 27         0         0          0          0          0          0          0
## 28         0         0          0          0          0          0          0
## 29         0         0          0          0          5          3          0
## 30         0         0          0          0          0          0          0
## 31         0         0          0          0          0          0          0
## 32         0         0         40         25         25         60         30
## 33         0         0          0          0          0          0         12
## 34         0         0          0          0          1        121         56
## 35         0         0          0          0         11         22         27
## 36         0         0          0          0          0         39          0
## 37         0         0          0          0          0        203        398
## 38         0         1          0          1         18         17          1
## 39         0         0          0          0         43         86        172
## 40         0         0          0          0         59        167        243
## 41         0         0          0          0          0          0        149
## 42         0         0          0          0          0          0          0
## 43         0         0          0          0          0          0          0
## 44         0         0          0          0          0          0          0
## 45         0         0          0          0          0          0          0
## 46         0         0          0         93         27        102         17
## 47         0         0          4          3          7          7          3
## 48         0         0          0          0          0          0          0
```

```
## 49         0         0          0          0          0        123         61
## 50         0         0          0          0          0          0          0
## 51         0         0          0          0          0          0          0
##    RENT2999_R RENT3499_R RENT3999_R RENT4499_R RENT4999_R RENT5000_R
## 1           0          0          0          0          0          4
## 2           0          0          0          0          0          3
## 3           0          0          0          0          0         12
## 4           0          0          0          2          0          2
## 5           0          0          0          1          2          5
## 6           3          0          1          0          0          1
## 7           0          0          0          0          0         10
## 8           0          0          0          0          0         15
## 9           0         16          0          0          0         12
## 10          0          7         10          7          6          2
## 11          4          9          7          3          5          4
## 12          0          0          7         20          0          4
## 13          0         18         10          6          8          4
## 14          0          4          0          0          0          2
## 15         58         18         14         13          4          1
## 16          0          0          0          0          0        132
## 17        304         73        200        183         63        134
## 18         38          0         52          0          0         14
## 19         51         90         18        120         32         12
## 20         62        142          0          0         38        129
## 21          0        183          0          0          0         21
## 22         46          0         25         25          0         34
## 23          0          0         40          0          5          5
## 24          0          3         48         29         60         20
## 25        140          0          0          0         25          8
## 26          0          0          0          0         26         35
## 27         47         30          0          0          0         73
## 28          0          0          0         45         50         56
## 29          0         13          4         21         11         10
## 30          0          0          0          0          0        148
## 31         78          3         76          1         20          6
## 32         20         20         20         20          5          5
## 33         12         23         37         37         55        153
## 34          9          0          0          0          0         44
## 35         41        105         20         15         22          4
## 36        198        130          0          0          0          5
## 37        304          0         41         20          0         10
## 38         32         81         64         58         26         73
## 39         77        169        228        183        116         45
## 40         66          5          0          0          0          7
## 41         45        141          3         23          6          1
## 42          0         87        133        111         34          3
## 43          0        253          0         93          0         37
## 44          0         24         48         98         80        119
## 45          0          0          0          0          0         90
## 46          0          0          0          0          0          1
## 47          8          4          5         10          4         11
## 48          0          0          0          0         46         76
## 49         64          0          0         30         60         35
## 50          0          0          0          0         29        153
```

```
## 51         45          2         137          0         116          6
```

```
# RENT's 27 rent-band variables are only applicable to properties with
# NUMUNITS >= 2. For single-unit properties, RENT provides the monthly rent,
# which can be used to determine the corresponding rent bands. Because RENT is
# rounded, some observations' possible original rents span multiple rent bands,
# preventing a unique assignment. Cells that must be 0 or 1 are imputed;
# ambiguous cells and properties with unreported RENT remain unchanged. RENT
# values topcoded as 5644 are assigned to RENT5000_R.

# Lower endpoints of the 27 rent bands, in ascending order.
rent_starts <- c(
  RENTNO_R = 0,
  RENTLT249_R = 1,
  RENT274_R = 250,
  RENT299_R = 275,
  RENT349_R = 300,
  RENT399_R = 350,
  RENT449_R = 400,
  RENT499_R = 450,
  RENT549_R = 500,
  RENT599_R = 550,
  RENT649_R = 600,
  RENT699_R = 650,
  RENT749_R = 700,
  RENT799_R = 750,
  RENT899_R = 800,
  RENT999_R = 900,
  RENT1099_R = 1000,
  RENT1249_R = 1100,
  RENT1499_R = 1250,
  RENT1999_R = 1500,
  RENT2499_R = 2000,
  RENT2999_R = 2500,
  RENT3499_R = 3000,
  RENT3999_R = 3500,
  RENT4499_R = 4000,
  RENT4999_R = 4500,
  RENT5000_R = 5000
)

# The corresponding upper endpoints.
rent_ends <- set_names(
  c(unname(rent_starts[-1]) - 1, Inf),
  names(rent_starts)
)

rent_cols <- names(rent_starts)

# Snapshot for comparison after the imputations.
# Useful for checks that are not explicitly included in this .Rmd.
rent_before <- rhfs |>
  select(NUMUNITS_R, RENT, all_of(rent_cols))
```

```r
rhfs <- rhfs |>
  mutate(
    .rent_low = case_when(
      NUMUNITS_R != 1 | RENT < 0 ~ NA_real_,
      RENT == 5644 ~ 5000,
      RENT == 1000 ~ 975,
      RENT < 1000 ~ pmax(RENT - 25, 0),
      TRUE ~ RENT - 50
    ),
    .rent_high = case_when(
      is.na(.rent_low) ~ NA_real_,
      RENT == 5644 ~ Inf,
      RENT < 1000 ~ RENT + 24,
      TRUE ~ RENT + 49
    ),
    across(
      all_of(rent_cols),
      \(x) case_when(
        .rent_low >= rent_starts[[cur_column()]] &
          .rent_high <= rent_ends[[cur_column()]] ~ 1L,
        .rent_high < rent_starts[[cur_column()]] |
          .rent_low > rent_ends[[cur_column()]] ~ 0L,
        TRUE ~ x
      )
    )
  ) |>
  select(-.rent_low, -.rent_high)


# If DEBT is 2 (no mortgage or similar debt), then DEBTNUM (number of mortgages
# or similar debts) is 0. This expands DEBTNUM's universe.
rhfs$DEBTNUM <- ifelse(rhfs$DEBT == 2, 0, rhfs$DEBTNUM)


# RCONTROL's 377 -8s are due to these values being outside of its universe of
# NUMUNITS >= 2. 311 of them can be derived from RCONTROLAHS.
rhfs <- rhfs |>
  mutate(
    RCONTROL = case_when(
      NUMUNITS_R == 1 & RCONTROL == -8 & RCONTROLAHS == "1" ~ 1,
      NUMUNITS_R == 1 & RCONTROL == -8 & RCONTROLAHS == "2" ~ 0,
      TRUE ~ RCONTROL
    )
  )


# LOWRENT and HIRENT are -8 for NUMUNITS >= 2. For 14 vacant-for-rent properties
# with reported, non-topcoded RENT (the topcoded value is RENT = 5644), both
# LOWRENT and HIRENT equal RENT.
rhfs |>
  select(UNITSTAT, LOWRENT, HIRENT, RENT) |>
  filter(UNITSTAT == 2)
```

```
##    UNITSTAT LOWRENT HIRENT RENT
## 1         2      -8     -8 1200
## 2         2      -8     -8 2000
## 3         2      -8     -8 2000
## 4         2      -8     -8  750
## 5         2      -8     -8 5644
## 6         2      -8     -8 1700
## 7         2      -8     -8 3000
## 8         2      -8     -8  750
## 9         2      -8     -8 1200
## 10        2      -8     -8 1500
## 11        2      -8     -8 1700
## 12        2      -8     -8 2000
## 13        2      -8     -8 3000
## 14        2      -8     -8 2000
## 15        2      -8     -8 1600
## 16        2      -8     -8   -9
## 17        2      -8     -8   -9
```

```
rhfs <- rhfs |>
  mutate(
    across(
      c(LOWRENT, HIRENT),
      \(x) if_else(
        NUMUNITS_R == 1 & UNITSTAT == "2" &
          RENT >= 0 & RENT != 5644 & x == -8,
        RENT, x
      )
    )
  )


# Viewing the targets for imputation
rhfs |>
  select(OPEX_R, OPPAY, OPINSUR, OPELGAS, OPWTRSWR, OPREP, OPMNG,
         OPPROTH, OPSECUR, OPLAND, OPOTHEX)
```

```
##        OPEX_R   OPPAY OPINSUR OPELGAS OPWTRSWR   OPREP   OPMNG OPPROTH OPSECUR
## 1          -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2          -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3          -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4          -9      -9      -9      -9       -9      -9      -9      -9      -9
## 5          -9      -9      -9      -9       -9      -9      -9      -9      -9
## 6          -9      -9      -9      -9       -9      -9      -9      -9      -9
## 7          -9      -9      -9      -9       -9      -9      -9      -9      -9
## 8          -9      -9      -9      -9       -9      -9      -9      -9      -9
## 9          -9      -9      -9      -9       -9      -9      -9      -9      -9
## 10         -9      -9      -9      -9       -9      -9      -9      -9      -9
## 11         -9      -9      -9      -9       -9      -9      -9      -9      -9
## 12         -9      -8      -8      -8       -9      -8      -8      -8      -8
## 13       3000      -8      -9      -9       -9    3000      -8      -8      -8
## 14         -9      -9      -9      -9       -9      -9      -9      -9      -9
## 15         -9      -9      -9      -9       -9      -9      -9      -9      -9
## 16         -9      -8      -8      -8       -8      -8      -8      -8      -8
```

```
## 17     -9    -9    -9    -9    -9    -9    -9    -9    -9
## 18     -9    -8    -8    -9    -9    -8    -8    -8    -8
## 19     -9    -8    -8    -8    -8    -8    -8    -8    -8
## 20   4500    -8    -9    -9    -9  3100  1400    -8    -8
## 21     -9    -8    -8    -8    -8    -8    -8    -8    -8
## 22   4500    -9  2000    -9    -9  2000    -9   500    -9
## 23    350    -8    -8     0     0   350    -8    -8    -8
## 24     -9    -9    -9    -9    -9    -9    -9    -9    -9
## 25     -9    -9    -9    -9    -9    -9    -9    -9    -9
## 26   2000  1000    -9     0     0     0  1000     0     0
## 27     -9    -9    -9    -9    -9    -9    -9    -9    -9
## 28     -9    -8    -8    -8    -8    -8    -8    -8    -8
## 29  12400    -8  2400    -8    -8 10000    -8    -8    -8
## 30  15108    -9    -9    -9    -9  7000    -9  3533    -9
## 31     -9    -9    -9    -9    -9    -9    -9    -9    -9
## 32   8733    -8    -8    -8  1200  4000    -8  3533    -8
## 33     -9    -9    -9    -9    -9    -9    -9    -9    -9
## 34     -9    -8    -8    -8    -8    -8    -8    -8    -8
## 35     -9    -9    -9    -9    -9    -9    -9    -9    -9
## 36  14500    -8    -8    -8  5200  1800  6000  1500    -8
## 37     -9    -9    -9    -9    -9    -9    -9    -9    -9
## 38     -9    -8    -9    -9    -9    -9    -9    -9    -9
## 39     -9    -9    -9    -9    -9    -9    -9    -9    -9
## 40    680     0   480    -8    -8   200    -8    -8    -8
## 41     -9    -8    -8    -8    -8    -8    -8    -8    -8
## 42   3200    -8    -9    -8    -9  3000    -8    -8    -8
## 43   5000    -8    -8    -8    -9  5000    -8    -8    -8
## 44     -9    -9    -9    -9    -9    -9    -9    -9    -9
## 45     -9    -9    -9    -9    -9    -9    -9    -9    -9
## 46   4300    -8   800    -8    -8  3500    -8    -8    -8
## 47   2950     0   950     0     0  1800     0   200     0
## 48   5000    -8    -8    -8    -9  5000    -8    -8    -8
## 49   9633  3600  1500    -8    -8   500    -8  3533    -8
## 50   5060    -8    -8    -9    -9    -8   960    -8    -8
## 51     -9    -8    -8    -8    -8    -8    -8    -8    -8
## 52     -9    -9    -9    -9    -9    -9    -9    -9    -9
## 53  10000  2000  1500   600  1400  2500    -8  1000    -8
## 54    990    -8    -9    -8    -9     0   960    30     0
## 55     -9    -9    -9    -9    -9    -9    -9    -9    -9
## 56     -9    -9    -9    -9    -9    -9    -9    -9    -9
## 57   4300    -9  1500    -9    -9  1000  1600   200    -9
## 58    850     0   450    -8    -8   200    -8   200    -8
## 59     -9    -8    -9    -9    -9    -9    -9    -9    -9
## 60   3400    -8    -8    -8    -9  2500    -8    -8    -8
## 61     -9    -8    -8    -8    -8    -8    -8    -8    -8
## 62   3000    -8  1000    -8    -8  2000    -8    -8    -8
## 63   2150    -8   760    50    80   490    -8   180    -8
## 64     -9    -9    -9    -9    -9    -9    -9    -9    -9
## 65     -9    -8    -8    -8    -8    -8    -8    -8    -8
## 66  19100    -8   800    -8  2000 11000    -8   800    -8
## 67   3200    -9   950   400    -8  1000    -8    -8    -8
## 68     -9    -9    -9    -9    -9    -9    -9    -9    -9
## 69     -9    -9    -9    -9    -9    -9    -9    -9    -9
## 70   8380    -8     0     0     0  6000  1500    -8   180
```

```
## 71       6520    1200     930     200      670    2400     660     160       0
## 72       3570      -8     730      -8       -8    1600    1000     240      -8
## 73         -9      -8      -9      -9       -9      -9      -9      -9      -8
## 74       5810      -9     210      -9       -9    3000    2100      -9      -9
## 75       9705      -8     700       0      190     250      -8     200      -8
## 76       1150      -8     150      -8       -8    1000      -8      -8      -8
## 77       6000      -8    1500      -8       -8    4500      -8      -8      -8
## 78       1350      -8     250     600       -9      -9      -8     500      -8
## 79         -9      -8      -8      -8       -8      -8      -8      -8      -8
## 80       4010      -8     650      -8       -8    2600      -8      -8      -8
## 81      15624      -8    5524      -8       -9    8900      -8      -8      -8
## 82       4700      -8    3700      -8       -8      -8      -8      -8      -8
## 83       5400       0    2300       0        0     600     600       0       0
## 84         -9      -9      -9      -9       -9      -9      -9      -9      -9
## 85       4400      -8    2400      -9       -9    1000      -8      -8      -8
## 86        500      -9      -9      -9       -9     500      -9      -9      -9
## 87       2520       0     790     200        0     760       0       0      -9
## 88         -9      -9      -9      -9       -9      -9      -9      -9      -9
## 89         -9      -9      -9      -9       -9      -9      -9      -9      -9
## 90      18567    5667    1500     200     1200   10000       0       0       0
## 91         -9      -8      -8      -8       -8      -8      -8      -8      -8
## 92         -9      -8      -8      -8       -8      -8      -8      -8      -8
## 93       8200      -8    3000      -8     5200      -8      -8      -8      -8
## 94         -9      -9      -9      -9       -9      -9      -9      -9      -9
## 95       5000      -8    2500      -8     1000    1000      -8      -8      -8
## 96         -9      -8      -8      -8       -8      -8      -8      -8      -8
## 97      24828      -9    2000      -9      250    8200      -9     450    2953
## 98       8000      -8     500      -9       -9    5000    2500      -8      -8
## 99      12700    1000    1300      -8       -8    8000      -8      -8      -8
## 100      3400       0    1400       0        0       0       0    2000       0
## 101     21599       0    5524       0     2200    4100    5200       0       0
## 102     14073       0    3800       0        0    2400    4200      -8    2953
## 103        -9       0      -9       0        0       0      -9       0       0
## 104      1500      -8    1000      -8       -8     500      -8      -8      -8
## 105        -9      -8      -8      -8       -8      -8      -8      -8      -8
## 106        -9      -8      -8      -8       -8      -8      -8      -8      -8
## 107        -9      -8      -8      -8       -8      -8      -8      -8      -8
## 108        -9      -8      -8      -8       -8      -8      -8      -8      -8
## 109       600      -8      -9      -8       -8      -8      -8      -8      -8
## 110      1490       0     890       0        0     600       0       0       0
## 111      3600       0    3000       0        0     600       0       0       0
## 112        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 113        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 114      3500       0     800       0        0       0       0       0       0
## 115      4980       0    2500      -8       -8    2000      -8     100      -8
## 116        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 117      5460       0     510     480        0    2400       0    1400      -9
## 118        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 119      5160      -8     660      -8       -8    4500      -8      -8      -9
## 120     19174      -8    1100    1400      600   15574      -8      -8      -8
## 121      6700       0    1800    1800       -9     500    2600      -8      -8
## 122      5610       0     410      -8       -8    4000    1200       0       0
## 123        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 124     10300      -8    1900      -8       -8      -8    1800      -8      -8
```

```
## 125      1600       0    1000      -8      600      -8      -8      -8      -8
## 126        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 127        -9      -8      -8      -8       -8      -8      -8      -8      -8
## 128        -9      -8      -8      -8       -8      -8      -8      -8      -8
## 129        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 130        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 131      4800       0       0       0        0      -9      -9       0       0
## 132        -9      -8      -8      -8       -8      -8      -8      -8      -8
## 133      4230       0    1900       0      780     350       0       0       0
## 134      1700       0    1000       0      700       0       0       0       0
## 135      2000       0     850       0        0    1000       0     150       0
## 136      7480      -8     360      -8       -9     230      -8     290      -8
## 137        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 138        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 139      2700      -8      -8      -8       -8    1500      -8      -8      -8
## 140      2450      -8     980       0      820     250       0       0       0
## 141        -9      -8      -8      -8       -8      -8      -8      -8      -8
## 142        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 143      2700      -8      -8      -8       -8      -8      -8      -8      -8
## 144      1500      -8    1300      -8       -9     200      -8      -8      -8
## 145      1240       0      -8      -8      840       0       0       0       0
## 146     16454      -9     640      -9       -9   15574     240      -9      -9
## 147      1800      -8     880      -8       -8     920      -8      -8      -8
## 148       800      -8     800      -8       -8      -8      -8      -8      -8
## 149        -9      -8      -8      -8       -8      -8      -8      -8      -8
## 150      3300      -8     700      -8       -8     900    1700      -8      -8
## 151      3650      -9     950      -8       -8     500    2100     100      -8
## 152      9620    5667      -8      -8       -8    1000      -8      -8    2953
## 153      3500    2000    1100      -9       -9      -9      -8     400      -8
## 154      6200      -9     200      -9       -9      -9      -9      -9      -9
## 155        -9      -9      -8      -8       -8      -8      -9      -8      -8
## 156      4840      -9     540      -9       -9    1000      -9      -9      -9
## 157        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 158      7500      -8    1500       0        0    5000       0    1000       0
## 159      3000      -8     800      -8       -9    1000      -8      -8      -8
## 160        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 161        -9      -8      -8      -8       -8      -8      -8      -8      -8
## 162        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 163        -9      -8      -8      -8       -8      -8      -8      -8      -8
## 164        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 165      1070       0     320      -9       -9      -9     750      -9      -9
## 166      8849      -8     390      -8       -8      -8    7959     500      -8
## 167      3700      -8    1000      -8       -8    1500      -8     850      -8
## 168     11400      -8     400      -8       -8    8000    3000      -8      -8
## 169        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 170        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 171       710      -8      -8      -8      710      -8      -8      -8      -8
## 172        -9      -8      -8      -8       -8      -8      -8      -8      -8
## 173      2120       0    1400      -8      720      -8      -8      -8      -8
## 174      1700      -8    1200      -8       -8     500      -8      -8      -8
## 175     17304      -8      -9     130      250   15574      -8     700      -8
## 176     21274      -8    3000      -8       -8   15574    2500      -8      -8
## 177        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 178      3190      -8      -8      -8      490      -8    2700      -8      -8
```

```
## 179   1670   -8   770    0    0   300    0    0    0
## 180     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 181     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 182     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 183     -9   -8    -9   -8   -9    -9   -9   -9   -8
## 184     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 185     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 186   3780 3500   280   -8   -8     0   -8    0   -8
## 187     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 188   1900   -8    -8   -8 1900    -8   -8   -8   -8
## 189   1400   -8    -8   -8 1400    -8   -8   -8   -8
## 190     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 191   7400    0     0    0    0     0    0    0    0
## 192     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 193     -9    0    -9   -9   -9    -9   -9   -9   -9
## 194  11400   -8  1400   -8   -8  9000 1000   -8   -8
## 195     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 196   8900   -8    -8   -8   -9  3800   -8   -8   -8
## 197   1600   -8    -8   -8   -8    -8 1600   -8   -8
## 198     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 199      0    0     0    0    0     0    0    0    0
## 200     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 201     -9   -8    -8   -8   -9    -8   -8   -8   -8
## 202     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 203     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 204     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 205     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 206    500   -8   500   -9   -9    -8   -8   -8   -8
## 207   3730   -8   500   -8   -8   530   -8   -8   -8
## 208     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 209    190   -8   190   -8   -8    -8   -8   -8   -8
## 210     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 211   7600   -8  4000   -8   -8  3200   -8   -8   -8
## 212   6900    0  1400    0    0  3000    0    0    0
## 213     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 214     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 215  12417 5667  2200    0  100   850 2900  150    0
## 216     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 217     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 218    350   -9   350   -9   -9    -9   -9   -9   -9
## 219   7200   -8  1000   -9   -9  1000 5200   -8   -8
## 220   3500   -8  1500   -8  500  1000   -8   -8   -8
## 221   4470   -8   170   -8   -8    -8 4200   -8   -8
## 222     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 223   4200   -8   500   -9   -9  1500 2000  200   -8
## 224     -9   -8    -8   -8   -9    -8   -8   -8   -8
## 225   6420   -9   320   -8   -9    -9 2300   -8   -8
## 226     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 227   6724    0  5524   -8   -8    -8   -8   -8   -8
## 228   3200   -9  3200   -9   -9    -9   -9   -9   -9
## 229   1980   -8   980   -8   -8  1000   -8   -8   -8
## 230   1120   -8     0   -8  720   400   -8    0   -8
## 231     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 232   3600   -8   400   -8   -9   200   -8   -9   -8
```

```
## 233      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 234    1100     -8    500     -9     -9    600     -8     -8     -8
## 235    2250     -8    350     -8     -9      0   1600    300     -8
## 236    8910     -8   3100   1800   1100    830   2000     -8     -8
## 237      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 238    6120      0   1700   3200    720    500      0      0      0
## 239    4550     -9    250     -9    300   1000     -9     -9     -9
## 240   11100   1800   3000     -9     -9   1500     -9     -9     -9
## 241    3070     -8    570     -8     -8   2000     -8     -8     -8
## 242    2490     -8    570     -8   1200     -8    720     -8     -8
## 243      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 244      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 245    6120     -8    820    100     -8   1500   2100     -8     -8
## 246      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 247      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 248      -9     -8     -8     -8     -9     -8     -8     -8     -8
## 249    5470     -8   1000     -9     -9    620   3700    150     -9
## 250    3030     -8    130     -8     -8    500     -8     -8     -8
## 251      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 252    2300      0   1100     -8     -8     -8     -8     -8     -8
## 253   27439      0   1000      0   2000  15574      0    500      0
## 254      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 255    2200     -8   1200     -8     -8   1000     -8     -8     -8
## 256    7000     -8   3900     -8     -8     -8   3100     -8     -8
## 257    4900     -8   1700     -8     -8   1000   2200     -8     -8
## 258      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 259    6500     -8   1500     -8     -9   5000     -8     -8     -8
## 260      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 261    2550     -8    150     -8     -9     -8     -9     -9     -9
## 262      -9     -8     -8     -8     -8     -8     -9     -8     -8
## 263      -9     -8     -8     -9     -9     -8     -8     -8     -8
## 264    1200     -8     -8     -8   1200     -8     -8     -8     -8
## 265    1700     -9    700     -9     -9     -9   1000     -9     -9
## 266      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 267     500    500     -9      0      0     -9     -8     -9     -8
## 268      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 269      -9     -8     -8     -9     -9     -8     -8     -8     -9
## 270      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 271      -9      0     -9     -9     -9     -9     -9     -9     -9
## 272      -9      0     -9     -8     -8     -8     -8     -8     -8
## 273    3800     -8     -8     -8     -8   2000   1800     -8     -8
## 274      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 275      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 276      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 277      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 278   13559      0    600      0      0   5000   7959      0      0
## 279      -9      0     -9     -9     -9     -9     -9     -9     -9
## 280    2850     -8    850     -8     -8   2000     -8     -8     -8
## 281      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 282      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 283    5000     -8    900     -8     -9   1500   2600     -8     -8
## 284      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 285      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 286    3500     -8   1500     -8     -8   2000     -8     -8     -8
```

```
## 287   7000   -8  1000   -9   -9   -9   -9   -9   -8
## 288     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 289   1400    0  1400   -8   -8   -8   -8   -8   -8
## 290   7550    0   450   -9   -9 1200   -9  900   -9
## 291     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 292   3000   -8  2500   -8   -8  500   -8   -8   -8
## 293     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 294    620   -8   170   -9   -9  450   -9   -9   -9
## 295   2550    0   870   -9   -9   50 1300   -9   -9
## 296   3200   -8   200 1500   -9  500  900  100   -8
## 297  17700 1500  1000 3200 2000 10000   -8   -8   -8
## 298   7300   -8  2300   -8 1800   -8   -8   -8   -8
## 299     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 300   4030   -8   170   -9   -9  990 2800   -8   -8
## 301     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 302     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 303     -9   -8    -9   -9   -9   -9   -9   -9   -9
## 304     -9   -8    -8   -8   -8   -8   -8   -8   -8
## 305     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 306     -9   -8    -9   -9   -9   -9   -9   -9   -9
## 307     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 308   6370 2400  1400   -8   -9 2000  320  250   -8
## 309     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 310     -9   -8    -8   -8   -8   -8   -8   -8   -8
## 311     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 312   4900   -8  1200   -8   -8 3700   -8   -8   -8
## 313     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 314     -9   -8    -8   -8   -8   -8   -8   -8   -8
## 315     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 316     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 317   6230   -8   530   -8 1400 3200   -8   -8   -8
## 318  11000   -8  3500   -9   -8 5000 2500   -8   -8
## 319     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 320   2600   -8  1100    0 1200  300   -8   -8   -8
## 321     -9   -8    -9   -9   -9   -8   -8   -9   -8
## 322   1820   -8   560   -8   -8  130 1000  130   -8
## 323   2450   -8   450   -8   -8 1000   -8 1000   -8
## 324   1600   -8    -8   -8 1600   -8   -8   -8   -8
## 325   2300    0  2000    0  300    0    0    0    0
## 326   3620    0  3200    0    0  420    0    0    0
## 327  13000   -8    -9   -9   -9 10000 3000   -9   -9
## 328     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 329   4700   -9    -8   -8   -8 1600 2000   -8   -8
## 330     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 331   3100    0   600    0 1000    0 1500    0    0
## 332  11879   -8   920   -8   -8 2000 7959 1000   -8
## 333   8750   -8  3000   -8 5200   -8   -8   -8   -8
## 334     -9   -8    -9   -8   -8   -8   -8   -9   -8
## 335  12300   -8  1500   -9 1800 1500   -8   -8   -8
## 336     -9   -8    -8   -9   -9   -8   -8   -8   -8
## 337     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 338     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 339     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 340   9100   -9    -9 1300   -9 7800   -9   -9   -9
```

```
## 341   9260   -8   680   -8   -9   380  1300   -8   -8
## 342   2000   -9  1200   -8   -8    -8    -8  200   -8
## 343  23024   -8  1600   -8   -8 15574  2500   -8   -8
## 344     -9   -9    -9   -9   -9    -9    -9   -9   -9
## 345   2530 2000   250   -8   -9   280    -8   -8   -8
## 346     -9   -9    -9   -9   -9    -9    -9   -9   -9
## 347  20165   -8   500   -8   -9  7500  3800   -8   -8
## 348     -9   -9    -9   -9   -9    -9    -9   -9   -9
## 349   6000   -9    -9   -8   -8  5000    -8   -8   -8
## 350   6960   -8  2500   -8  960  3500    -8   -8   -8
## 351   2580   -9  1200   -9  660    -9    -9   -9   -9
## 352     -9   -9    -9   -9   -9    -9    -9   -9   -9
## 353   5524    0  5524   -8   -8    -8    -8   -8   -8
## 354     -9   -9    -9   -9   -9    -9    -9   -9   -9
## 355   7800   -9  1400   -9   -9  4000  2400   -9   -9
## 356   2500    0  2000    0    0     0    -8   -8   -8
## 357   8720   -8   510   -8   -9   860  5200  350   -8
## 358     -9   -8    -8   -8   -8    -8    -8   -8   -8
## 359     -9   -9    -9   -8   -8    -8    -9   -9   -8
## 360   3800   -9    -9 1800   -9    -9    -9   -9   -9
## 361   6760   -8   940   40   -9    80  2100   -8   -8
## 362  26074   -8  1500 3200 1800 15574  4000   -8   -8
## 363   7575   -8  3000   -8   -9    -8    -8   -8   -8
## 364   7150   -8  1200   -8   -9    50  5900   -8   -8
## 365     -9   -8    -8   -8   -8    -8    -8   -8   -8
## 366     -9   -8    -8   -8   -8    -8    -8   -8   -8
## 367  10700   -8  2000 1500 5200  2000    -8   -8   -8
## 368   1000   -8    -8   -8   -9    -8    -8   -8   -8
## 369   1500   -8  1000   -8   -8    -8    -8  500   -8
## 370     -9   -8    -9   -9   -9    -8    -9   -8   -8
## 371   8140    0   790    0    0   500   400  750   -8
## 372   3200   -8   600   -8   -9  2000    -8   -8   -8
## 373   8100   -9  2400 1400 1200  2500    -8   -8   -8
## 374   6800 4000   800 1000   -9  1000    -9   -8   -8
## 375   4800   -8  2000  800   -9  1500    -8  500   -8
## 376  10580   -8  2200   -8   -8  2200  5400   -8   -8
## 377     -9   -9    -9   -9   -9    -9    -9   -9   -9
## 378     -9   -9    -9   -9   -9    -9    -9   -9   -9
## 379     -9   -9    -9   -9   -9    -9    -9   -9   -9
## 380     -9   -9    -9   -9   -9    -9    -9   -9   -9
## 381     -9   -9    -9   -9   -9    -9    -9   -9   -9
## 382     -9   -9    -9   -9   -9    -9    -9   -9   -9
## 383     -9   -9    -9   -9   -9    -9    -9   -9   -9
## 384     -9   -8    -8   -8   -8    -8    -8   -8   -8
## 385     -9   -9    -9   -9   -9    -9    -9   -9   -9
## 386     -9   -8    -8   -8   -8    -8    -8   -8   -8
## 387     -9   -9    -9   -9   -9    -9    -9   -9   -9
## 388   5000    0    -9    0 3000     0  1500    0    0
## 389     -9   -8    -8   -8   -8    -8    -8   -8   -8
## 390     -9   -9    -9   -9   -9    -9    -9   -9   -9
## 391     -9   -9    -9   -9   -9    -9    -9   -9   -9
## 392     -9   -9    -9   -9   -9    -9    -9   -9   -9
## 393     -9    0    -9   -9   -9    -9    -9   -9   -9
## 394  23400   -8  5000 4000 1200 12000    -8   -8   -8
```

```
## 395      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 396   14700      0   3000   1700   2400   5000      0   1800      0
## 397      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 398      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 399    6500     -8   1400     -8    600   2000     -8     -8     -8
## 400    6600     -8   2600   2000   2000     -8     -8     -8     -8
## 401      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 402      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 403    4600      0     -9      0      0   3200   1400      0      0
## 404      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 405    5350     -9   3000     -8    250     -9     -9    300     -9
## 406      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 407      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 408      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 409    5400     -8   2000     -8    900   2000     -8    500     -8
## 410   10700     -8   2200   1700   2300     -9   3100     -9     -8
## 411    7990   1400   2100     -8    540   2400    740      0      0
## 412   15700     -8   1300     -8   1400   8000     -8   1000     -8
## 413       0      0      0      0      0      0      0      0      0
## 414    3600     -8   3000     -8     -8     -8     -8     -8     -8
## 415      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 416   17314      0     -9     -9     -9     -9     -8     -9     -9
## 417    4800     -8   1300     -8   2500     -8     -8   1000     -8
## 418    3000     -9   3000     -9     -9     -9     -9     -9     -9
## 419      -9     -8     -9     -9     -9     -9     -9     -9     -9
## 420    1800     -8   1800     -9     -9     -9     -8     -8     -8
## 421      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 422   12500     -8   5000     -9   2500   5000     -8     -9     -8
## 423      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 424      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 425      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 426      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 427   30500  23550    650     -8   3300     -8     -8   2000     -8
## 428      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 429    2000     -8     -9     -9   2000     -9     -9     -9     -9
## 430      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 431    3300     -8   2000     -8   1000    300     -8     -8     -8
## 432    8100     -9   3600     -9   4500     -9     -9     -9     -9
## 433   10900     -8   2000   3700   5200     -9     -8     -8     -8
## 434      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 435   11200     -8   2000      0   1200   6000     -8    800     -8
## 436      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 437    5900     -8    400     -8     -8   3500   1600    100     -8
## 438      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 439    3500     -8   1500     -9   2000     -9     -8     -9     -9
## 440      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 441   13400     -8   1300     -8   6100   6000     -8     -8     -8
## 442    3300     -8    900     -8   1200     -9     -8   1200     -8
## 443      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 444    6000     -8   1000     -8     -9   5000     -8     -8     -8
## 445    5320    450    800    100   1700   1800      0     50      0
## 446      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 447    2200     -8   1200     -8    400     -8     -8     -8     -8
## 448    2400     -9    900     -9     -9   1500     -9     -9     -9
```

```
## 449   13600     0  5000     0  5400  2000     0     0     0
## 450      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 451    4000    -8   500    -8   500  2500    -8    -8    -8
## 452      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 453      -9    -9    -9    -9    -9    -9    -9    -9    -9
## 454      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 455      -9    -9    -9    -9    -9    -9    -9    -9    -9
## 456   10150    -8  1300    -8  1100  6500    -8    -8    -8
## 457      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 458      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 459   20040    -9  2200   240  1600 15000    -9  1000     0
## 460      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 461   36867    -8  3700  1500   840 28697    -8   130    -8
## 462      -9    -8    -9    -9    -9    -9    -9    -9    -9
## 463    2800    -8  1800    -8    -8  1000    -8    -8    -8
## 464    8300    -8  2000    -8    -9  5000    -8    -8    -8
## 465      -9    -8    -8    -8    -9    -8    -8    -8    -8
## 466    2600   500   750    -8   600   500    -8   250    -8
## 467    6960    -8  1000    -8   960  5000    -8    -8    -8
## 468      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 469      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 470      -9    -8    -8    -8    -9    -8    -8    -8    -9
## 471      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 472   44263 23550  8713    -8  2000 10000    -8    -8    -8
## 473    4480    -8  1300     0  1200  1100    -8   880    -8
## 474   10500    -8  5500    -8  4000    -8    -8    -8    -8
## 475      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 476    3400    -8    -8    -8  2400    -8    -8    -8    -8
## 477   27000    -8  4000  2000  3000 12000    -8  1200    -8
## 478      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 479   10700     0  2400    -8  1800  6200    -9    -9    -9
## 480   14483    -8  1600    -8  9683  2000    -9    -8    -8
## 481    2200    -8  1200    -8    -9  1000    -9    -8    -8
## 482      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 483   10400    -8  1500  4600  1200  2500    -8   600    -8
## 484      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 485      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 486    4120    -8   920   100    -8  2000    -8   650    -8
## 487    2200    -8  2200    -9    -9    -9    -9    -9    -9
## 488    8000     0  3300  1100  1400  1000     0  1200     0
## 489    7500    -8  2500    -8  5000    -8    -8    -8    -8
## 490   15500    -8  3500  4500  7500    -8    -8    -8    -8
## 491      -9    -8    -9    -9    -9    -9    -9    -9    -9
## 492      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 493      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 494      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 495    8000    -8  5000    -8  3000    -8    -8    -8    -8
## 496      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 497   11900    -8    -9     0  2700  3300  2700    -9    -9
## 498      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 499    4500    -8    -9    -8  2000    -9    -8  1300    -8
## 500    9940    -8   570  4200   260  3400    -8   920    -8
## 501      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 502   12200    -8    -8    -8  4000  3000  4000    -9    -9
```

```
## 503   7000   -8  2000  1500  3500   -8   -8   -8   -8
## 504   3100   -8   800   -8  2000   -8   -8   -8   -8
## 505     -9   -8    -8   -8    -8   -8   -8   -8   -8
## 506   5500   -8  3000   -8  2500   -8   -8   -8   -8
## 507   9000   -9  4200   -8  4800   -8   -8   -8   -8
## 508     -9   -8    -8   -8    -8   -8   -8   -8   -8
## 509  13000    0  4000    0  2700  4300    0 2000    0
## 510  23350   -8  4500    0  1700 12000   -8   -8   -8
## 511  18400   -8  2600 5000  1400  7200   -8   -8   -8
## 512     -9   -9    -9   -9    -9   -9   -9   -9   -9
## 513  23900   -8  1500  300  1900  6500 8000 2200   -8
## 514  10500   -8  3000 1500  2500  2500   -8   -8   -8
## 515   7600   -8  2000   -8  1300   -9   -8  800   -8
## 516  11700   -8  3000   -8  1200  6000   -8   -8   -8
## 517     -9   -8    -8   -8    -8   -8   -8   -8   -8
## 518     -9   -9    -9   -9    -9   -9   -9   -9   -9
## 519     -9   -9    -9   -9    -9   -9   -9   -9   -9
## 520  11350 5000  2000   -8   960  2200   -8   90   -8
## 521  17800 1700  2300  590  4300   470 4200 1000 1440
## 522     -9   -8    -8   -8    -8   -8   -8   -8   -8
## 523     -9   -8    -8   -8    -8   -8   -8   -8   -8
## 524     -9   -9    -9   -9    -9   -9   -9   -9   -9
## 525     -9   -8    -8   -8    -8   -8   -8   -8   -8
## 526   3500    0   500    0  2000  1000    0    0    0
## 527     -9   -8    -8   -8    -8   -8   -8   -8   -8
## 528   4710   -8  1100   -8  2400   760   -8   -8   -8
## 529   7710   -8  1400   -8  2100   350   -8   60   -8
## 530  29100   -8  4000   -8  2600 15000 6300   -8   -8
## 531  17313    0  8713 3600   600  2400    0 1500    0
## 532  14900    0  3500    0     0  3000   -8 3250   -8
## 533   8200    0  1200 4000  2000  1000    0    0    0
## 534     -9   -9    -9   -9    -9   -9   -9   -9   -9
## 535  16000   -8  5000 7000  4000   -8   -8   -8   -8
## 536  10500   -8  5000   -8    -9  5500   -8   -9   -8
## 537  65211 5000  4800 5000  2000 28697   -9    0    0
## 538  36627    0  3300 1200  9683  7500 14054  140   70
## 539  11400 5000  2200   -9  1200  3000   -8   -8   -8
## 540     -9   -8    -8   -8    -8   -8   -8   -8   -8
## 541     -9   -8    -8   -8    -8   -8   -8   -8   -8
## 542   5400   -8  4200   -8    -8   600   -8   -8   -8
## 543  41597 5000  4000 1500  2400 28697   -8   -8   -8
## 544  26000 1500   700 2000   800  1000 8000 2000   -8
## 545  38504   -8  4000  250    -8 18500 14054 1000   -8
## 546     -9   -9    -9   -9    -9   -9   -9   -9   -9
## 547   1280   -8   760   -8    -8   520   -8   -8   -8
## 548  16820   -8  1000   -8   420 14800   -8  400   -8
## 549     -9   -8    -8   -8    -8   -8   -8   -8   -8
## 550  14400   -8  2800   -8  1000 10000   -8   -8   -8
## 551  12700   -8  3600 3500  1400  3000   -8   -8   -8
## 552   3720  580  1100    0   240  1300    0    0    0
## 553   1500   -8  1200   -8   300   -8   -8   -8   -8
## 554  12000   -8  5500 1500  5000   -8   -8   -8   -8
## 555  17400    0  4600    0  1800  6500 1200    0    0
## 556     -9   -9    -9   -9    -9   -9   -9   -9   -9
```

```
## 557     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 558     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 559     -9     -8     -9     -9     -9     -9     -9     -9     -9
## 560  12500   4500   1200   1500    630    970      0   1000     -9
## 561  18200     -8   1200   9000     -9   8000     -8     -8     -8
## 562   1790     -8    750     -8    450    590     -8     -8     -8
## 563  17250     -8    650   2500   4000   9500     -8    600     -8
## 564     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 565   4700     -8    500   1200     -9   3000     -8     -8     -8
## 566     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 567   2200     -8    700     -8     -8   1200     -8     -8     -8
## 568   1400     -8     -9     -8     -9    700     -8    500     -8
## 569   3150     -8   1800     -8     -9    350     -8     -8     -8
## 570   4700     -8   1700     -8     -9   3000     -8     -8     -8
## 571   4200     -8    640     -8    260   2300     -8   1000     -8
## 572     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 573   3900     -8   1900     -8     -9   2000     -8     -8     -8
## 574  10510      0    510   3000   2000   2000      0    600     -9
## 575   2100     -9   1000      0    600    500     -9     -9     -9
## 576     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 577   5500     -8   1000     -8   2500   2000     -8     -8     -8
## 578  10000   2000   1000   1000     -9   6000     -8     -8     -8
## 579     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 580     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 581     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 582  14113     -8   8713     -8   2400   3000     -8     -8     -8
## 583    800     -9    800     -9     -9     -9     -9     -9     -9
## 584    800     -9     -9     -9     -9     -9     -9     -9     -9
## 585   7500     -8   2900     -8   1600   3000     -8     -8     -8
## 586   5000     -8     -9     -8     -8   5000     -8     -8     -8
## 587     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 588   2960     -8   1200    660     -9   1000     -8     -8     -8
## 589   3000     -8   3000     -8     -8     -8     -8     -8     -8
## 590     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 591     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 592     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 593     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 594     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 595     -9     -8     -8     -9     -9     -8     -8     -8     -8
## 596   6450     -8     -8     -8   1200   2500   2500     -8     -8
## 597  13700     -8    900   3500   3000   4000     -8    300     -8
## 598   6450     -8   1200     -8   3000   1600     -8     -8    400
## 599     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 600     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 601   2000      0      0      0    800      0      0      0      0
## 602   2990     -9   1700    250      0    590      0    450      0
## 603   2950     -8    450     -8    500   2000     -8     -8     -8
## 604     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 605   1680    580   1100     -8     -8     -8     -8     -8     -8
## 606   4450      0   4000      0      0    450      0      0      0
## 607   4230    600   1200     -9     -8   1500     -8     80     -8
## 608     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 609  15230     -8   4400    290   4300   5000     -8    180     -8
## 610     -9     -9     -9     -9     -9     -9     -9     -9     -9
```

```
## 611   7170    0   690    0   830  5000   -8  350   -8
## 612   6480   -8     0 1200   380  2300   -8  800    0
## 613     -9   -8    -9   -9    -9    -9   -9   -9   -9
## 614   8200   -8  2900 1100  2700  1500   -9   -9   -9
## 615     -9   -9    -9   -9    -9    -9   -9   -9   -9
## 616  15200   -8  1500   -8    -8  7900   -8 1900   -8
## 617   4400    0  1200 2200     0   500    0    0    0
## 618     -9   -9    -9   -9    -9    -9   -9   -9   -9
## 619  20100   -8  5400 6800  1500  2600   -8 2800   -8
## 620   3740   -8  1800  170   570     0   -8    0    0
## 621   4200   -9    -9   -9  1200  1000   -8   -8   -8
## 622     -9   -8    -8   -8    -8    -8   -8   -8   -8
## 623     -9   -8    -8   -8    -8    -8   -8   -8   -8
## 624   3000   -8  1500   -9    -8   500   -8   -8   -8
## 625   7570   -8   460   -8  2600  2800 1200   -8   -8
## 626   4720 1400  2000   -8   320  1000   -8   -8   -8
## 627     -9   -8    -8   -8    -9    -8   -8   -8   -8
## 628  11220   -8   920 2400  2400  5500   -8   -8   -8
## 629     -9   -9    -9   -9    -9    -9   -9   -9   -9
## 630   3400   -8  1200   -8    -8  2200   -8   -8   -8
## 631  13000   -8    -8 8900  2000  2100   -8   -8   -8
## 632   7200    0  1200    0     0  5000    0  500    0
## 633     -9   -9    -9   -9    -9    -9   -9   -9   -9
## 634  11000   -9  1300   -8    -8  2600 3800 2300   -8
## 635     -9   -9    -9   -9    -9    -9   -9   -9   -9
## 636   1200   -8  1200    0     0     0    0    0    0
## 637     -9   -8    -8   -8    -8    -8   -8   -8   -8
## 638  23130   -8  3900 1700  2900  5100   -8  730   -8
## 639     -9   -8    -8   -8    -8    -8   -8   -8   -8
## 640   7000   -9  7000   -9    -9    -9   -9   -9   -9
## 641     -9   -9    -9   -9    -9    -9   -9   -9   -9
## 642  13500   -8  1800    0  1200 10000   -8   -8   -8
## 643     -9   -9    -9   -9    -9    -9   -9   -9   -9
## 644   2660    0  1500   -8   660   500   -8   -8   -8
## 645   4000   -8  1100  300   200  2000   -8   -8   -8
## 646     -9   -8    -8   -8    -8    -8   -8   -8   -8
## 647    800    0    -9   -9    -9    -9  800   -9   -9
## 648     -9   -9    -9   -9    -9    -9   -9   -9   -9
## 649  14000   -8  2400    0  1200  8000   -8    0 1200
## 650   6600   -8  1000   -8    -8  5500   -8  100   -8
## 651     -9   -9    -9   -9    -9    -9   -9   -9   -9
## 652  36197   -8  1400  480  2400 28697   -9  940   -9
## 653  14800   -8  1200   -8   450 12000   -8  150   -8
## 654  10100   -8  4000  630   770  2000 2300   -8   -8
## 655   2500   -8  1000   -8   500  1000   -8   -8   -8
## 656   3400   -9    -9   -9  1200    -9 2200   -9   -9
## 657   3000   -8    -8   -8    -9    -8   -8   -8   -8
## 658   2600   -9    -9   -9    -9  1200 1400   -9   -9
## 659     -9   -8    -8   -8    -8    -8   -8   -8   -8
## 660  14500   -8  2000   -8    -8 10000   -8   -8   -8
## 661     -9   -9    -9   -9    -9    -9   -9   -9   -9
## 662  13993   -8  8713   -8  3000  2000   -8   -8   -8
## 663   5800   -8  2700   -8  1900    -8   -8   -8   -8
## 664  10770   -9  6700   -9   500  3200   -9  370   -9
```

```
## 665     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 666  13500     -8   3000    500     -8  10000     -8     -8     -8
## 667   5900     -8   2200     -8     -8   2500      0      0      0
## 668  11850     -8     -9    900   3200   4000     -8   3250     -8
## 669  18350   2600   1800   4800   2400   5000     -9    250     -9
## 670  10500     -8   5000      0   1500   4000     -8     -8     -8
## 671   4900     -8   1500     -8   2900    500     -8     -8     -8
## 672   4800     -8   1800     -8     -8   2000     -8     -8     -8
## 673     -9     -8     -9     -9     -9     -9     -8     -9     -8
## 674     -9     -8     -8     -8     -9     -8     -8     -8     -9
## 675  18300     -8   2200    600   2400  12000     -8     -8     -8
## 676  27900     -8     -9     -8   3400   7500   4100     -9     -8
## 677     -9     -8     -8     -8     -9     -8     -8     -8     -8
## 678  13000     -8     -9     -9     -9     -9     -9     -9     -8
## 679   5400     -8   2000    300    600   2500     -8     -8     -8
## 680   9440     -8     -9   4300   2000   1500     -9    240     -8
## 681  18050     -8   3500   1000   1400   8000     -8    650     -8
## 682   6500     -8   2500    500   3500     -8     -8     -8     -8
## 683   7590    510   2000    240   1800   2600     -8     -8     -8
## 684   4060     -9     -9    200    900   1200     -9    360     -9
## 685   7500     -8   1800     -8   2400   3000     -8     -8     -8
## 686     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 687  15300     -8   1500   6500   2300   2500      0    500      0
## 688   6250      0   2400     -9    350   2000      0    500     -9
## 689   2500     -8   1100    400      0   1000      0      0      0
## 690   5730     -8   4100     -8     -9   1000     -8     -8     -8
## 691     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 692     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 693   6000     -9     -9     -9   3000     -9     -9     -9     -9
## 694  13113     -8   8713     -9   2400   2000     -8     -8     -8
## 695     -9      0     -8     -8     -8     -8     -8     -8     -8
## 696  15200     -8   6000   3000   1200   5000     -8     -9     -8
## 697  19610     -8   6500     -8     -8   8700   1200   1800     -8
## 698   7000     -8   3800     -8    600   1200     -8    200     -8
## 699   6940     -8   4000   1900    240    500     -8     -8     -8
## 700   6240     -8   1500     -8    940   2500     -8     -8     -8
## 701   8500     -8   3000   1500   4000     -8     -8     -8     -8
## 702     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 703   9280     -8   3000     -8   2800   3000     -8     -8     -8
## 704     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 705     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 706     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 707  27300     -8   1900   3300   2000  18300     -8   1000     -8
## 708     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 709     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 710   3900     -8   2000     -8   1500    200     -8     -8     -8
## 711     -9     -8     -9     -9     -9     -9     -9     -9     -9
## 712     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 713   4240     -8   2800     -8    890     -8     -8     -8     -8
## 714     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 715  14100     -9   1500      0      0   8800   3800      0      0
## 716  14200     -8   4000   2400   1600   5000     -8     -8   1200
## 717   7150     -8   3000     -8   2500    150     -8     -8     -8
## 718  11800      0   1200   5800   2400   1800      0      0      0
```

```
## 719    21100    5500    2300    2300    2400    1800      -8    1600      -8
## 720     2100      -8      -8      -8      -9      -8      -8      -8      -8
## 721    12000       0    3500    2000    1500    2500       0       0       0
## 722     4000      -8      -9      -9    2000    2000      -8      -8      -8
## 723     9500      -8    4500     500    2500    2000      -8      -8      -8
## 724    35316      -8    5000   11933    9683    8000      -8      -8      -8
## 725    24750      -8    3500     500    4500    8000      -9    3250      -9
## 726       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 727       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 728    31864       0    1800       0    4900    3100    2300     750       0
## 729       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 730     6000      -8      -8      -8    2400      -8    2400      -8      -8
## 731       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 732     5000      -8    2500      -8    2500      -8      -8      -8      -8
## 733       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 734    12020       0    1700     320    2200    2400    4600      -9      -9
## 735       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 736       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 737     3000      -8      -8      -8      -9      -8      -8      -8      -8
## 738    18783       0    8713    2000    1900    5200       0       0     100
## 739     3400      -8    3000      -9     200     200      -8      -8      -8
## 740     8160      -8    1500     950    1700    3000      -8     110      -8
## 741       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 742    12420      -8    3900     770    2200    4400      -8     150      -8
## 743    11209      -8    1400     330    3700    2000      -8     330    1089
## 744    10500      -8    3000    2500    5000      -8      -8      -8      -8
## 745       -9      -8      -9      -9      -9      -9      -9      -9      -9
## 746       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 747       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 748       -9      -8      -8      -8      -9      -8      -8      -8      -8
## 749       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 750    13400      -8    5000      -8    1600    5000      -8    1200      -8
## 751    10200       0    4000    1000     600     600       0    1500       0
## 752    14300      -8    1900    1100    2700    3700    4700      -8      -8
## 753    25300      -8    4000    1100    4000   15000      -8    1200      -8
## 754    32353    2000    3200    1000    1000   15000      -8     550    5003
## 755    17200      -8    1200    3400    3400    5000    3200    1000      -8
## 756     3200      -9    1100      -8    1600     500      -8      -8      -8
## 757     6000      -8    2000    1000    2000      -8      -8      -8      -8
## 758     8000      -8      -9      -8    3000    5000      -8      -8      -8
## 759    15850     650    2500    7600    2500    2000      -8     300      -8
## 760    23100      -8    5800    7200    2900    3600      -8      -8      -8
## 761       -9      -8      -9      -9      -9      -9      -9      -9      -9
## 762       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 763    16500      -8    2600    1200     700   12000      -8      -8      -8
## 764       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 765     8800      -8    1500    4000    3300      -8      -8      -8      -8
## 766       -9      -8      -9      -9      -9      -9      -9      -9      -9
## 767    34897      -8    2600      -8    2200   28697      -8      -8      -8
## 768     3500      -8    1500      -8    2000      -8      -8      -8      -8
## 769     6500      -8    2700    2000    1800      -8      -8      -8      -8
## 770       -9      -8      -9      -9      -9      -9      -8      -9      -8
## 771     3800      -8      -9      -9      -9      -9    3800      -9      -9
## 772     8800      -8    2500      -8    2000    3300      -8      -8      -8
```

```
## 773      8000      -8    5000      -8     2500     500      -8      -8      -8
## 774        -9      -8      -8      -8       -9      -8      -8      -8      -9
## 775        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 776     10000      -9    4000    3500       -8    2500      -8      -8      -8
## 777     10600      -8     900    5400     1700    2600      -8      -8      -9
## 778      2750      -8    1500      -8     1000     250      -8      -8      -8
## 779     10700      -8    3000    1500     5000      -8      -8      -8      -8
## 780      6800      -8    3000    1000     1800      -8      -8      -8      -8
## 781     11950    3800    2200       0     1800    2500      -8     150      -8
## 782      8480      -8    4800     480       -9    2000      -8      -8      -8
## 783     21683      -8    3500    8500     9683      -8      -8      -8      -8
## 784      2500      -8      -8      -8     2500      -8      -8      -8      -8
## 785     10150      -8    3000     550     3000    2000      -8      -8      -8
## 786        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 787        -9      -8      -8      -8       -8      -8      -8      -8      -8
## 788     13600      -8    1500      -9     3200    5100      -8    2700      -8
## 789      1000      -8      -9      -8       -8    1000      -8      -8      -8
## 790     30154      -8    4100    8500       -8    2000   14054      -8      -8
## 791     42297      -8    1100      -8       -8   28697      -8    1500      -8
## 792     22383      -8    5200      -8     9683    5500      -8      -8      -8
## 793     12850      -8     650      -8     1500    8000      -8     300      -8
## 794        -9      -8      -8      -8       -8      -8      -8      -8      -8
## 795      5300      -8    3500      -8     1000      -8      -8      -8      -8
## 796        -9      -8      -8      -8       -8      -8      -8      -8      -8
## 797     11000      -8      -9    4000       -9    7000      -8      -8      -8
## 798        -9      -8      -8      -8       -8      -8      -8      -8      -8
## 799      6500      -8    3000      -8     1500    2000      -8      -8      -8
## 800      5800      -9    3300      -9       -9    2500      -9      -9      -9
## 801        -9      -8      -8      -8       -8      -8      -8      -8      -8
## 802      8600      -8    4000    1200     2400    1000      -8      -8      -8
## 803     20300      -8    3000    5200     3000    7100      -8      -8      -8
## 804        -9      -8      -9      -8       -8      -9      -9      -9      -8
## 805        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 806      6000      -8    2300     200      200    1500      -8      -8      -8
## 807     21483      -8    2000   11933     1600      -9      -8      -8      -8
## 808      5600      -8    1000      -8     3600    1000      -8      -8      -8
## 809      1250       0      -8      -8       -9    1000       0     250       0
## 810     12820      -8    2400    3500       -9    4000      -9    2200      -8
## 811     17410      -8    3500      -8      960    6000      -8    3250      -8
## 812     23783      -8    2600   11933     1200    6000     900     500     150
## 813     13500      -8    2000     500     3000    7000      -8      -8      -8
## 814        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 815     11300      -8    3500    2000     4200    1600      -8      -8      -8
## 816      9700       0    2400      -8     2800    2500      -8      -8      -8
## 817        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 818     67850   23550    5300    8000     6900   13200    6200      -8      -8
## 819     16000       0       0    2000     2000    2000    8000       0       0
## 820     23713    2800    6700      -8       -9    2600    5500     680    5003
## 821      9200      -8    2200    1800     1200    3000      -8    1000      -8
## 822     12900      -8    4000     300     3600    3000      -8      -8      -8
## 823     22200      -8    2000    2000     2000    8200    6800      -9      -9
## 824     17900      -8    4100    2600     4100    5000      -8      -8      -8
## 825     38336    6000    7000    5500     9683      -8      -8      -8    5003
## 826        -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 827      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 828      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 829      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 830      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 831      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 832      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 833      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 834      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 835      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 836      -9     -9     -9     -9     -9     -9     -9     -9      0
## 837      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 838      -9     -8     -8     -8     -9     -8     -8     -8     -8
## 839      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 840      -9     -8     -9     -9     -9     -9     -9     -9     -9
## 841      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 842      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 843      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 844      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 845   22600     -9   2300   9700   2000   6800   1800     -9     -9
## 846      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 847      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 848      -9     -8     -8     -8     -9     -8     -8     -8     -8
## 849      -9      0      0      0      0      0      0      0     -8
## 850   19730   3600    850    280   4500  10500     -8     -9     -8
## 851      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 852   55550     -9     -9  15000   6000  14100  13000    450     -8
## 853      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 854   74000     -8   8000   6400   6100  18400     -8   1000     -8
## 855      -9     -8     -8     -8     -9     -8     -8     -8     -8
## 856      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 857   76900     -8  13300   2300   9400   1000  10700    200     -8
## 858   68300     -9  17800  17300  26500     -9     -9     -9     -9
## 859      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 860      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 861      -9      0      0      0      0      0     -9     -9     -9
## 862      -9      0      0      0      0      0      0     -8     -8
## 863      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 864      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 865      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 866      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 867      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 868      -9     -8     -8     -8     -9     -8     -8     -8     -8
## 869      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 870      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 871      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 872      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 873      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 874      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 875      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 876      -9     -9     -9     -9     -9     -9     -9     -9      0
## 877      -9     -9     -9     -9     -9     -8     -8     -9     -9
## 878      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 879      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 880      -9     -9     -9     -9     -9     -9     -9     -9     -9
```

```
## 881       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 882       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 883       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 884   114880    9200    4100     190    15500   60000   14100      -8      -8
## 885       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 886       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 887       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 888   101000       0   16000      -9    10000   50000   20000    5000      -8
## 889       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 890       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 891       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 892       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 893       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 894       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 895       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 896       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 897    55600   31000    7500    2200     9400      -9      -9      -9       0
## 898       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 899       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 900       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 901       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 902       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 903       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 904       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 905       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 906    89000   30000    7200    2400    10600   14700   13600    6300      -8
## 907       -9      -8      -9      -9       -9      -9      -9      -9      -9
## 908    15620      -8    2400      -8     3200    8200      -8     320      -8
## 909       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 910       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 911       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 912       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 913     7240      -8    5800     720       -9      -8      -8      -8      -8
## 914       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 915    22300      -8    6500      -8       -8   10000    5800      -8      -8
## 916    19000   11000    8000      -8       -8      -8      -8      -8      -8
## 917       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 918    65500   30000    9000    5000     4000   15000      -8      -8      -8
## 919       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 920       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 921       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 922    26270       0    4000    8800     1200    7800       0     220       0
## 923       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 924       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 925       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 926       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 927   120000      -9      -9      -9       -9      -9      -9      -9      -9
## 928       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 929    69746      -8   18200    5500    38546    7500      -8      -8      -8
## 930       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 931       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 932   106300    6600    7800   24500    23900   33000      -8    1700      -8
## 933       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 934   193271   44000      -9   23000    27000   27000      -8    3800   11880
```

```
## 935    51880     -8    3000    1100    6500   18800   18000      -8      -8
## 936       -9     -8      -8      -9      -8      -8      -8      -8      -8
## 937   134900      0   11000    2300    6200  100000   15000       0       0
## 938       -9     -9      -9      -9      -9      -9      -9      -9      -9
## 939    85500     -9   22000      -9   24000   12000   24000      -9      -9
## 940       -9     -9      -9      -9      -9      -9      -9      -9      -9
## 941   151098 139098      -9   12000      -9      -9      -9      -9      -9
## 942   110600  46000   12000       0   12000   30000   10000       0       0
## 943       -9     -9      -9      -9      -9      -9      -9      -9      -9
## 944       -9     -9      -9      -9      -9      -9      -9      -9      -9
## 945     2700     -9      -9      -9      -9      -9      -9      -9      -9
## 946       -9     -9      -9      -9      -9      -9      -9      -9      -9
## 947       -9     -9      -9      -9      -9      -9      -9      -9      -9
## 948       -9     -9      -9      -9      -9      -9      -9      -9      -9
## 949       -9     -9      -9      -9      -9      -9      -9      -9      -9
## 950       -9     -8      -8      -8      -8      -8      -8      -8      -8
## 951       -9     -8      -8      -8      -8      -8      -8      -8      -8
## 952       -9     -9      -9      -9      -9      -9      -9      -9      -9
## 953       -9     -8      -8      -8      -8      -8      -8      -8      -8
## 954     3000      0      -9    3000      -9      -9       0       0       0
## 955       -9     -8      -8      -8      -8      -8      -8      -8      -8
## 956       -9     -8      -8      -8      -8      -8      -8      -8      -8
## 957       -9     -8      -8      -8      -8      -8      -8      -8      -8
## 958    12000     -9      -9      -9      -9      -9      -9      -9      -8
## 959     1200     -8    1200      -9      -9       0      -8      -8      -8
## 960       -9     -8      -8      -8      -8      -8      -8      -8      -8
## 961       -9     -9      -9      -9      -9      -9      -9      -9      -9
## 962   105095  32000   33295      -8   12900    6000   12500      -8      -8
## 963       -9     -9      -9      -9      -9      -9      -9      -9      -9
## 964       -9     -8      -8      -8      -8      -8      -8      -8      -8
## 965       -9     -8      -8      -8      -8      -8      -8      -8      -8
## 966       -9     -9      -9      -9      -9      -9      -9      -9      -9
## 967       -9     -9      -9      -9      -9      -9      -9      -9      -9
## 968       -9     -8      -8      -8      -8      -8      -8      -8      -8
## 969       -9     -8      -8      -8      -9      -8      -8      -8      -8
## 970   208600  45000   18000   15600   25000   44000   36000    6000      -8
## 971       -9     -8      -8      -8      -8      -8      -8      -8      -8
## 972       -9     -9      -9      -9      -9      -9      -9      -9      -9
## 973    60460    9100    4200   12400    6200    9800    3800    3200      -9
## 974       -9     -8      -8      -8      -8      -8      -8      -8      -8
## 975       -9     -8      -8      -8      -8      -8      -8      -8      -8
## 976       -9     -8      -8      -8      -8      -8      -8      -8      -8
## 977       -9     -8      -8      -8      -8      -8      -8      -8      -8
## 978       -9     -9      -9      -9      -9      -9      -9      -9      -9
## 979       -9     -8      -8      -8      -8      -8      -8      -8      -8
## 980       -9     -8      -8      -8      -8      -8      -8      -8      -8
## 981       -9     -8      -8      -8      -8      -8      -8      -8      -8
## 982       -9     -8      -8      -8      -8      -8      -8      -8      -8
## 983       -9     -8      -8      -8      -8      -8      -8      -8      -8
## 984       -9     -9      -9      -9      -9      -9      -9      -9      -9
## 985       -9     -8      -8      -8      -8      -8      -8      -8      -8
## 986       -9     -9      -9      -9      -9      -9      -9      -9      -9
## 987    57250  35000    5000     250    5000   12000      -8      -8      -8
## 988    17000    5000    4800      -8    1000    5000      -8      -8      -9
```

```
## 989      7900      -8      -8      -9      -9      -8      -8      -8      -8
## 990     21700      -9    2900    2300    5000    4500      -9      -9      -9
## 991        -9      -9      -9      -9      -9      -9      -9      -9      -9
## 992    215916   18800   23100    6700    7800  147516      -8      -8      -8
## 993        -9      -9      -9      -9      -9      -9      -9      -9      -9
## 994      2540      -8    2100     240       0      -9      -8     200      -8
## 995    102400   12800    6400   15000   10300    9000   20000    6900    2000
## 996    103100   21600   14900    4900   14900   18700      -8    2100      -8
## 997        -9      -8      -8      -8      -8      -8      -8      -8      -8
## 998     10640      -8    2000     240    3000    5000      -8     400      -8
## 999     89600    9000    6900    9700    6700   13000   20600    5100      -8
## 1000       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1001   274546   75000    4000   25000   38546  100000      -8   12000    8000
## 1002       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1003       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1004       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1005    99400   19700   14000   19800   15700    2000   10700   10900      -8
## 1006    59000      -8      -8      -8      -8   41800   13000      -8      -8
## 1007       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1008   125940      -8   10700    8000   10300   35400   13100    4400       0
## 1009    46420   22900    4400     300    3900    4500    8700       0       0
## 1010       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1011       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1012       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1013     5540    2500      -8      -8      -8    2500      -8      -8      -8
## 1014   101000   41000    8400   10300    8900   10900    9100    2700       0
## 1015       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1016    66500   10500    8100    1800      -9    4900   16400    6600      -8
## 1017       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1018       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1019       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1020    10500       0    1200    2000      -9    6500       0      -8      -8
## 1021    77800      -8    5900   12400    5000   24200   14800    5600      -8
## 1022       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1023     4600      -8    4600      -9      -9      -8      -8      -8      -8
## 1024    45700      -9    3100    8000    5000   21600    5000    3000      -8
## 1025       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1026       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1027       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1028       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1029       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1030       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1031    20550      -9    2500      -9    6000    5000    4300     750      -9
## 1032     2400      -9      -9      -9      -8      -9      -8      -9      -8
## 1033    79461      -8   18000   43461   18000      -9      -9      -9      -8
## 1034    41100    9700    7800    2700    4300    2400       0     500       0
## 1035       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1036    92800   45300    6600    6900   10100   14900       0    2700       0
## 1037       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1038    34566       0      -9    6000   11500    3700    6100      -9     966
## 1039    86869      -8   14900   12100   11300    5100      -8   31369       0
## 1040    24570       0    4200     400    2100   13100      -8     770      -8
## 1041   191949      -8      -8      -9      -9      -8      -8      -8      -8
## 1042       -9      -8      -8      -8      -8      -8      -8      -8      -8
```

```
## 1043    32400       0    5400    1100    2700    4900    4200    6600      -9
## 1044    28600      -8    3500    5700      -9   14000      -8    1000     300
## 1045    15000      -8      -9      -8    3000    5000      -8    1000      -8
## 1046       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1047       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1048   153332      -8   23300   32100    5600   44800   16600   13600     132
## 1049    73200      -8    9600   20400   13500   21500    6700    1500      -8
## 1050    40800      -8    9300    5200    5000   12700      -8    1200      -8
## 1051    28680      -8    7400     680    1500    4200   10600     400      -8
## 1052       -9      -8      -9      -9      -9      -9      -9      -9      -9
## 1053       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1054   179591   59000   15100    6800    3800   48200   14000   18500      -8
## 1055   196516      -8    7100    7300    9600  147516   22000    3000       0
## 1056       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1057       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1058    63790       0   25000   15900    9500    6500    5000     490       0
## 1059       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1060       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1061   102330   12400   11200   11900   12000   20500   11800     530      -8
## 1062    27620   12000    6200      -9    1800     700      -8     620       0
## 1063    58000   15900   10900    2500    4800    4400    5200    2700       0
## 1064    32600      -8    4000    7500    4500   11200      -8    1800      -8
## 1065   309240   23400   12100   11600   15100   19600   18800    2500       0
## 1066    35400    5000    8000    9600    8000    3000      -8      -8      -8
## 1067   123500   31700   11200   21500    6200   18500   11800    7100       0
## 1068    65500      -8    3200    4700    4600   26000   15000    8400      -8
## 1069    89900      -8    8300   29700    7800   33000    4100    5000      -8
## 1070       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1071       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1072    51800      -8    7000    1500   15000   25000      -8      -8    1500
## 1073    92661    9200   20000   43461   10000    8000      -8    2000      -8
## 1074    17990      -8      -8     600    6300    3300    5800     650       0
## 1075    18300      -8    6100    1500    2600    7500      -8      -8      -8
## 1076       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1077       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1078    30200      -9    9300    3300    3900    7700      -8    1000      -9
## 1079    32200   12000    2500      -8    6000   10000      -8     600      -8
## 1080       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1081    55000      -8    3000    5000    5000   20000   18000    1000      -8
## 1082       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1083       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1084       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1085       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1086       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1087       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1088       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1089   183180   13100   16000    8700   12800   41200   17600    8800    2880
## 1090       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1091       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1092       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1093   148012   16600   33295    2000   17500   10900   25400    3800    7317
## 1094    26000       0    5000    3000    6000    3000    8000    1000       0
## 1095    34000       0    8000    1400    2600   10000      -8      -8      -8
## 1096   210295   73000   33295    3800    9500    9800   22400    7800      -8
```

```
## 1097      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1098  118800    9600   13000    3000    7600   63000   18700      -8      -8
## 1099   89800   17000   12800   12100   13000   13500      -8    1800    3600
## 1100  231100  107000   23800   31800    6700   51000      -8   10800      -8
## 1101  300999      -8   13900    3200   11900   10600   50750   11200      -8
## 1102   81400      -8   10900    3100   12300   51000      -9    1300      -9
## 1103      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1104  136800      -8   12300   28800      -9   67000   27800     900      -8
## 1105   58730   20000    9400     480    2900   13200      -8     150      -8
## 1106      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1107  130000   12000   12000   23000   23000   30000   25000    5000      -8
## 1108   73490       0    9100    3700   20700   10500   20200     890      -8
## 1109      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1110      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1111      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1112    2600       0    1800       0       0       0       0       0      -8
## 1113      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1114   72000      -9   10000   34000    8000   20000      -9      -9      -9
## 1115      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1116  216065    5400   16400   13600    9300   87000    1200    3800    3065
## 1117   23300      -8      -8    4000      -9    5000   10500    1800      -8
## 1118      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1119      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1120      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1121  196769   34100    6300   27200    9100   65000   21900   31369       0
## 1122   41100      -8    7900    4900    9700   14500      -8      -9      -8
## 1123   48200      -8   13200    2400    7600   18000    6000    1000       0
## 1124      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1125      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1126   26000    5000    3500    5000   10000    2500      -8      -8      -8
## 1127      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1128  256890   53000   33295   43461   11700   57000   34300   11000    4334
## 1129  257016   75000   12000   10000    2500  147516      -8   10000      -8
## 1130      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1131   62000      -8    5000   12000   24000   10000      -8      -8      -8
## 1132      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1133      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1134      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1135      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1136  189369   15000   15000   20000   10000   30000   30000   31369    8000
## 1137      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1138      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1139      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1140  120546      -8      -8   34800   38546   30000    7200   10000      -8
## 1141   33000    3600    5000       0   15000    7000       0       0       0
## 1142      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1143  106300   18900    7400    1400   24300   17700   27600      -9      -9
## 1144  232500   67000   16600   10000   24000   80000   24000    7000      -8
## 1145   86840   30000    9800    2600   12600   14100   12200     540      -9
## 1146  119195   34000   33295    5600    1500    9100   31000      -9      -9
## 1147      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1148      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1149      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1150      -9      -9      -9      -9      -9      -9      -9      -9      -9
```

```
## 1151     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1152     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1153  49140     -8   2000    140   7900  33100     -8     -8     -8
## 1154     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1155  40000     -8  10800  18000   8200   3000     -8     -8     -8
## 1156 230616  23000  13100   2000  13400 147516  20100     -8     -8
## 1157   8500     -8   3500   2000   3000     -8     -8     -8     -8
## 1158  75700     -8   3300   1100   7400  47000   9600    800     -8
## 1159   2800     -8   1200     -9     -9    400     -8     -8     -8
## 1160     -9     -8     -8     -8     -9     -8     -8     -8     -8
## 1161  43300     -8   5000   7000   1300  30000     -8     -8     -8
## 1162  23000     -8   5500   2400   4100   2000     -8     -8     -8
## 1163     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1164  50030     -8   3900    230   7900   3100  15900     -9     -8
## 1165  37600   6000   5000   1600  11000  10000     -8   1500     -8
## 1166     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1167     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1168  47500     -8   6500     -8   4000  20000  12000   2000   3000
## 1169  98290  56000  10700  13600   9800   5200     -8     -8     -8
## 1170  12000     -9  12000     -8     -8     -8     -8     -8     -8
## 1171   7200     -8     -8     -8   5400     -8     -8     -8     -8
## 1172     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1173  25000     -8   4000   5000   3000   8000   5000     -8     -8
## 1174     -9     -8     -8     -8     -9     -8     -8     -8     -8
## 1175  20980     -8     -9  11000   5600   4300     -8     80     -8
## 1176  32350   5300    950   1900   8500  10500     -8     -8     -8
## 1177  11400     -9   3400     -9     -9     -9     -9     -9     -9
## 1178  61100  29000   3300     -9   2500  25000     -9   1300     -9
## 1179  63620  25000   7000    300   2400  24000     -8     -8    720
## 1180     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1181  31800     -8   7000   4000   4300  15000     -8     -8     -8
## 1182  84700  65000     -8   3900  15800     -8     -8     -8     -8
## 1183  18000     -8   2500   7500   5500   2500     -8     -8     -8
## 1184   6070     -8     -9    670     -8   5400     -8     -8     -8
## 1185  76000  19200   6300  10000   2000  34000      0    500     -9
## 1186     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1187     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1188  19000     -9   2900   1400   2200   2400   7700   1200     -9
## 1189   1300     -8     -9    300     -9   1000     -8     -8     -8
## 1190      0      0      0      0      0      0      0      0      0
## 1191 177469     -8     -9  10600   8000  84000  38000  31369     -9
## 1192     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1193   1740     -9     -9     -9     -9    540   1200     -9     -9
## 1194     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1195  34940     -8     -8    640     -8  31000     -8     -8     -8
## 1196  19600   6000   7000      0   3600   1500      0      0      0
## 1197  69600      0   8100  10500   4700  35000      0   6600      0
## 1198  96928  27300   6000  22100   4300   9900  15900   2800    128
## 1199     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1200     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1201  12500     -8  10000     -8     -8     -9     -8   2000     -8
## 1202  32310     -8   2300    610   4500  21900     -9    600      0
## 1203  37200     -8   4000   5600   1400  15500     -8     -9     -8
## 1204     -9     -9     -9     -9     -9     -9     -9     -9     -9
```

```
## 1205     -9     -8     -9     -9     -9     -9     -8     -9     -8
## 1206  15000     -8   6900   1000   2800   1400     -8    910     -8
## 1207  46600      0   5000    600   7000  30000      0      0      0
## 1208 126000      0  19200  14000   8800  74000   7600   2400      0
## 1209     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1210     -9     -8     -9     -9     -9     -9     -9     -9     -9
## 1211  37000  10000   3500  10000   2900   5000     -8   2000     -8
## 1212     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1213  51600     -8   8300   4600   6000  19800   5300   1600     -8
## 1214  63100     -8   2900   3100   3100  47700     -8    100     -8
## 1215     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1216     -9     -8     -8     -8     -9     -8     -8     -8     -8
## 1217  24130     -8   1500    630   6200  11500     -8     -8     -8
## 1218  41600      0   5200  11400      0  12000   3300   3800      0
## 1219  29000     -8   3800     -8   8200   3500   8500     -8     -8
## 1220     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1221     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1222  38500     -9   3100  11300     -9  16000   8100     -9     -9
## 1223  55400     -8  13000  11100   1900  10000  11700   1700     -8
## 1224  45500     -8   7000   3000   7700  25000     -8     -8     -8
## 1225  24100     -8   5600   2000   1500  10000     -8   5000     -8
## 1226  19150     -8   1500    450   4500   7000   5700     -8     -8
## 1227  10300   7200     -8     -8   3100     -8     -8     -8     -8
## 1228  28500     -8   7200   3800   1000   7700   2000   1800     -8
## 1229     -9     -8     -8     -8     -9     -8     -8     -8     -8
## 1230     -9     -9     -9     -9     -9     -9     -9     -9      0
## 1231  52250     -8   7700   3500   7300  16000  17000     -8     -8
## 1232  25900      0   4100   5300   2900  12000      0      0      0
## 1233     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1234 142200  50000  19000   5000   9000  25000     -8  14000     -8
## 1235  10000     -8   2400   1100   3300     -9     -8     -9     -9
## 1236 125550  38500   5000  20700  11600  30600   6400   1800   8550
## 1237  15880     -8    500    480   2300  11300     -9     -9     -9
## 1238     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1239  27900     -8   4800     -8   9400   6500     -8   3100     -8
## 1240   3750     -8     -8   3000     -9    750     -8     -9     -9
## 1241     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1242  37800  15000   3000   1400   7200  10000     -9     -9     -9
## 1243  71890     -8   3500   3700   3500  56000   2200    590     -8
## 1244 181800  43700   5000  34200   8100  43000  22400   8000     -8
## 1245  51000      0  10000   6000   5000  10000      0   5000      0
## 1246     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1247     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1248  15900     -8   6500     -9   3500   5000     -8     -8     -8
## 1249     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1250   6600      0   1800   2000   1800    500     -8     -8     -8
## 1251     -9     -8     -9     -9     -9     -9     -8     -9     -8
## 1252  29300     -8     -9  18500     -9     -9   9000     -9     -8
## 1253     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1254     -9     -8     -8     -8     -9     -8     -8     -8     -8
## 1255  53400     -8  14200   2100   3600  31700     -8     -8     -8
## 1256     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1257  51300      0   4500   4000   1200  40000      0      0    400
## 1258  84300     -8   5900  17100  19800  37000     -8     -8     -8
```

```
## 1259    14800       0    1800    7000    2000      -8    4000       0       0
## 1260    25700      -8   11000    1200    7500    2000    3500     500      -8
## 1261    46000      -8   15000      -8   12000    6000      -8    4000      -8
## 1262       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1263    46070      -8   19900    1500   17200    7300      -8       0       0
## 1264    13300   10800    2500      -9      -9      -9      -9      -9      -9
## 1265    70000      -9      -9      -9      -9      -9      -9      -9      -9
## 1266    24300    1200    7000     600    1400   12000      -8     900      -8
## 1267   158600      -8    7100   11500   24700   95000   13600    2100      -8
## 1268       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1269    12000       0    2000    1000    2000    3000      -8       0      -8
## 1270       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1271    14500    5000    2700    2100    2700    1000      -8      -8      -8
## 1272       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1273       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1274       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1275    36200   10000    4000    1200   10000   10000      -9    1000      -9
## 1276    46500   15000    5000    3500   10000    3000   10000      -8      -8
## 1277    21000      -8    3000    5000    3000    6000      -8    1000      -8
## 1278       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1279   113200   80000    8000    7200    6000   12000      -8      -8      -8
## 1280    19600      -8    7000     600   12000      -8      -8      -8      -8
## 1281       -9      -8      -8      -9      -9      -8      -8      -8      -8
## 1282    43000      -8    6000      -8    6000   25000      -8    1000      -8
## 1283       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1284     7500      -9    7500      -9      -9      -9      -9      -9      -9
## 1285    22760       0    3000    2100    2600    2800    5700     260       0
## 1286       -9      -8      -8      -8      -9      -8      -8      -8      -8
## 1287    27600      -8    7600      -8    2900   10500    4400     900      -8
## 1288    13820      -8    2500     600    4000    6000      -9      -9     360
## 1289       -9      -8      -9      -9      -9      -9      -9      -9      -9
## 1290    49000      -8    5500    1500    5000   30000    4000    3000      -8
## 1291   185746       0   24500   36900   38546   48200   27600    4200       0
## 1292   117200   98000      -8      -8   19200      -8      -8      -8      -8
## 1293       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1294    22800       0    9500    3000    4200    4000       0       0       0
## 1295       -9      -8      -8      -9      -9      -9      -9      -8      -8
## 1296    48500   14600    3500    5200    7200   13000       0    4000      -9
## 1297       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1298   116700      -8   21500   25800    8500   22300   27700    2500      -8
## 1299    18700      -8    8500    6000    4200      -9      -8      -9      -8
## 1300    25000   25000      -9      -9      -9      -9      -9       0       0
## 1301       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1302    82400       0    7500    9600   24600   19300    6000    3000      -8
## 1303    85300   11000    3300   10000    9000   15000   10000   12000       0
## 1304    17450      -8    6000      -8    2800    5000    2400     750      -8
## 1305    46300      -8    5600    8000    4600    5700   14800    1800      -8
## 1306       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1307       -9      -8       0       0       0       0      -8       0       0
## 1308       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1309       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1310    15480       0     580    3800      -9    8200    2900      -9      -9
## 1311    32100       0    6600    7000    4000    2000    7800    1200    2500
## 1312    32500   15000   10000    2500    5000      -8      -8      -8      -8
```

```
## 1313   53400      0   5000  21200   9000   6000     -8   3000     -8
## 1314   24000     -8   4000     -8     -9  20000     -8      0      0
## 1315      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1316      -9     -8     -9     -9     -9     -9     -9     -9     -9
## 1317   37750   5000   2300     -9  10500   8000     -8    750     -8
## 1318   23380     -8   5200    580   1300   8500     -8   1300     -8
## 1319   15800     -8   7700      0    500     -9      0      0     -9
## 1320   27300     -8   4300   3900   4900  10800     -8   1000     -8
## 1321   95500     -8  23000  32000     -9  30000     -8   4000     -8
## 1322      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1323   66800     -8   5500   8800   3500  16800  15000   1500     -9
## 1324   50900  45000     -8     -8   3800     -8     -8     -8     -8
## 1325      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1326   21000   8000   1500   4000   3500   4000     -9     -9     -9
## 1327   28100      0   9000   7500   5000   3000      0      0      0
## 1328    6000   6000     -8     -8     -9     -8     -8     -8     -8
## 1329   15300     -8     -8   3800  11500     -8     -8     -8     -8
## 1330  105247     -8   7000  43461  38546  15000     -8   1000     -8
## 1331   51250  20000   5500   2000   1500  15000     -8   3000     -8
## 1332   12900      0   3400      0   3000   1600      0      0      0
## 1333   22710      0   7600    700    610   8800     -9     -9     -9
## 1334   24400     -8  12000   2400     -8  10000     -8     -8     -8
## 1335  185546   9000  15000  24000  38546  50000     -8     -8     -8
## 1336      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1337      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1338   42300      0   2100   1500  10400  11500  12800   2000      0
## 1339      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1340      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1341      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1342   63701     -8   4000  43461     -9  12200     -8    540     -8
## 1343  115540     -8  14700   1700  16100  49000  25500   2500     -8
## 1344      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1345   56800     -8  12900  21600   8300  14000     -8     -8     -8
## 1346  237861 100000   6400  43461   3600  14400  26600   2800      0
## 1347      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1348   25550     -9   3500     -8     -8  13800   4300    350     -8
## 1349   94495     -9  33295   4000  24000  30000     -9    700     -9
## 1350   18300     -8     -8   3700   2700   2300     -8     -8     -8
## 1351  139200  21900  21300     -9  11200  16200   6100     -9     -9
## 1352   56500     -8   4400   2500   8000  18000  15100   3500     -8
## 1353  244616  32500   7500  21500  18600 147516  15000   2000      0
## 1354   54500   3500  13000   3000  12000   5000  13000   2000     -8
## 1355      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1356  172200  80000   5200   5000  10000  30000  20000   1000     -9
## 1357   45710     -8   3500      0   8300  13600  16600    410     -9
## 1358  175441      0  25000   3500  22700  56000  50750   3300      0
## 1359      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1360      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1361   19300   3000   4500   1000    800   9000      0      0      0
## 1362      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1363      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1364    8500     -8   3500   1500   3500     -8     -8     -8     -8
## 1365  128000  17900  23700  14500  16200  38600     -8     -8   7600
## 1366      -9     -9     -9     -9     -9     -9     -9     -9     -9
```

```
## 1367    45800      -8    7200      -9     4100    3500   25000      -8      -8
## 1368       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1369    75900   20000    5000    5400    12000    5000   25000      -8      -8
## 1370       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1371       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1372    20560      -8    1500     960    13200    3500      -8      -8      -8
## 1373    29400      -8    3400    6000     6000   11000      -8     600      -8
## 1374   107600    7700   13200    1700    13800   50000   16200    1800      -9
## 1375    71500   11900    6900    3400     4200    8500    9100   15900      -8
## 1376    39500      -8    9800      -8    11000    2500   15000      -8      -8
## 1377   191949      -9      -9      -9       -9      -9      -9      -9      -9
## 1378       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1379       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1380    17170      -8    1600     420    10300    1700      -8    1000      -8
## 1381    34600      -8   18000    5800     4800    6000       0       0       0
## 1382    37840      -8    4700     610     4600   16200      -8    7100      -8
## 1383    63200    3000      -9    4600     7200   20800   21600    6000      -9
## 1384    22160      -8    3100    3700      560   12200      -8      -9      -8
## 1385    24700      -8   10500     200     8000    5000      -8     500       0
## 1386       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1387    37300    7000   11500     500     5400    8800      -8    1900      -9
## 1388    84500   12000   16000   28000     8000   15000      -8    2500      -8
## 1389    86500   22000    9000    6000     5500    7000   10000   12000       0
## 1390   158746       0   13900   18200    38546   70000   11800     900      -9
## 1391    37600      -9    6500    1400      800   28900      -9      -9      -9
## 1392    38200      -8    4200    7400     5700   10800    7900      -8      -8
## 1393       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1394    84000   10000    8000   13000     7000   40000      -8    2000       0
## 1395       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1396       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1397       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1398    27000      -9    5000    4000     5000   10000      -9      -9      -9
## 1399    44500    9300    2400   12800     1900   15000      -8      -8      -8
## 1400    22700    8000    6400      -8     2400    3000      -8    1500      -8
## 1401    64110      -8    4900     810    13700   18500    7800    3400       0
## 1402   125691      -8   20200    2400    12200   55000      -9    9200      -9
## 1403       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1404       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1405       -9      -8      -9      -8       -9      -9      -8      -8      -8
## 1406    35350      -9    6200    3500    13800    8600      -9     750      -9
## 1407    75800   25000    3500     300     3000    6000   15000   10000       0
## 1408    36000      -9      -9      -9       -9   36000      -9      -9      -9
## 1409       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1410       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1411    34100      -8    3100    1200     3200    8200   11000    3000      -8
## 1412    34900   17000    4900    3000     3000    3000    2000       0       0
## 1413       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1414       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1415   125500   20000   17500   10000    10000   20000   20000   15000       0
## 1416    29220    2500    3600    4400     7500   11000       0     220       0
## 1417       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1418    83746      -8   12000   17000    38546   15000      -8    1200      -8
## 1419    78800      -8   23200      -8     7500   29000   19100      -8      -8
## 1420    42000       0    4000   11000       -9   15000      -9    6000      -9
```

```
## 1421     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1422     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1423     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1424  50200     -8   9100   8600   3500  25000     -8   1200     -8
## 1425     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1426     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1427  30000     -8   6100   3300   5600   3500   4500   4000     -8
## 1428     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1429 141568  16100  14500  16400   8400  38600  20300   2100   1868
## 1430  38500     -9  12000     -9   5000   7400  10000    500     -9
## 1431     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1432  95000  50000  12000   5000   5000   4000   4000   5000      0
## 1433  25000     -8   5000     -8   2200   6000   9800     -9     -8
## 1434  30390      0   3000   1100    720  10900  10900    770     -8
## 1435 257593 139098  33295  26900  15600  12400     -8   5100     -8
## 1436  47700     -8   1800    800  15400   5100  15600     -8     -8
## 1437  48800     -9  13000     -8     -9   7200  25000     -9     -9
## 1438 244107     -8  14500   5700  21500 147516  35400   4100     -8
## 1439 253578 139098   1400  30400  10400  16000  35100   4500  11880
## 1440  45900     -8     -8   1000   7900  19400  15700     -8     -8
## 1441  18710  14400     -9    530     -9   2100     -9    480     -9
## 1442  45500     -8   7000  14200   9800   6000   6000     -8     -8
## 1443  89000  18000  25000   6500  12000  15000     -9   5000     -9
## 1444     -9     -8     -8     -8     -9     -8     -8     -8     -8
## 1445   8000     -9     -9     -9     -9     -9     -9     -9     -9
## 1446  62800   9800   5000  12700   4800  16400  10500   1200      0
## 1447 112300  15000   7500   5600  13000  48000  20500   2700      0
## 1448  32700     -9   6500     -9   8200  10000     -8     -8     -9
## 1449 148900     -8   8500   4300   9900  69000  23600   1600     -8
## 1450 122600     -8   3700     -8  20000  91000     -8   3000     -8
## 1451 104800  64000   4000   2600   4000  10200   8700   7800      0
## 1452  81000  70000   4000      0      0   1000   5000      0      0
## 1453   6400     -8     -8     -8   6400     -8     -8     -8     -8
## 1454     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1455     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1456     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1457     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1458  29690     -8  11900     -8  13000   3600     -8     -8     -8
## 1459     -9      0     -9     -9     -9     -9     -9     -9     -9
## 1460 107800  35700   6600   8200  11000  25600  13000   1300     -8
## 1461     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1462 107700  37500   7500  10400   5600  19500   7500     -9     -9
## 1463  67400      0   6000   7000  26400  10000  12000   6000      0
## 1464     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1465 151400  65000  10800   5100  21500  19000     -9   5000     -9
## 1466  51500      0  14900   4000   4800   6000  17000   4800     -9
## 1467     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1468     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1469  97400  19900   9900   2700  16200  19800  11300   1600     -8
## 1470  17000  17000     -9     -9     -9     -9     -9     -9     -9
## 1471 129750     -8  15000   7000   7000  50000  50750     -8     -8
## 1472 197061  47000  23000  43461  21000  40000  14000   6600   2000
## 1473  30400     -9     -9     -9     -9     -9     -9     -9     -9
## 1474 227800  12000   5000   5000  17400  25000  31000   4000     -8
```

```
## 1475      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 1476      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1477  264330   26000   19500   43461   23000   68000   28000   31369      -8
## 1478  167480      -8   21000    7200   23000   75000   12600   15000   11880
## 1479      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1480   94100   44200    9400    2400   13900   15900      -8      -8      -8
## 1481      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1482      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1483      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1484  138700      -8   16700   25400    5600   36900   19900   11100      -8
## 1485  160750       0      -8   10000      -9  100000   50750      -8      -8
## 1486      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1487      -9      -8      -9      -8      -8      -9      -9      -8      -8
## 1488      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1489      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1490      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1491      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1492      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1493      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1494      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1495      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1496      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1497      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1498      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1499      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1500      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1501      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1502      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1503      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1504      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1505      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1506      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1507      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1508      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1509      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1510      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1511      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1512      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1513      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1514      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1515      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1516      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1517      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1518      -9      -8      -9      -9      -9      -9      -9      -9      -9
## 1519      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1520      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1521      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1522      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 1523      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1524      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1525      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1526      -9      -9      -9      -9      -9      -9      -9      -9      -8
## 1527      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1528      -9      -9      -9      -9      -9      -9      -9      -9      -9
```

```
## 1529      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1530      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1531      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1532      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1533      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1534      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1535      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1536      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1537      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1538      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1539      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1540      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1541      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1542      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1543      -9      -8      -8      -9      -9      -8      -8      -8      -8
## 1544      -9      -9       0       0       0       0       0       0       0
## 1545  212000   29200   30100   47000   10700   41600   25100    6300       0
## 1546      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1547  192500   30200   20900   21900   10500   74000   17000    7400      -8
## 1548      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1549      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1550  379648  164000   21400   12900   15800   67000   31700    8700   22848
## 1551  202200      -9   45100   20100    8700   12000   35700    5400       0
## 1552  281096   30000   10000   18000   12000  206096       0       0      -9
## 1553      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1554  595000  162000   47000   92000   65000  153000   65000   10000      -8
## 1555  209800   20000   22700    6300   24400   86000   14200    3200      -8
## 1556      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1557  197300      -8   26000    8900   18300   98000   24500    4000      -8
## 1558      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1559      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1560  189400   50000   33000    5400   16000   30000      -8    3000    4000
## 1561      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1562      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1563      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1564      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1565      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1566      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1567      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1568      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1569      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1570      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1571      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1572      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1573      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1574      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1575      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1576  402571      -8      -8      -9      -9      -8      -8      -8      -8
## 1577      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1578      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1579      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1580      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1581      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1582  212900   44600   40400    9200    6100   68000   30000    4800       0
```

```
## 1583      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1584  216206      -9   23800    8700   77516   10200   71000    8200    1990
## 1585      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1586  289470   23700   24000    7100   48400  181000      -9     470      -8
## 1587      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1588      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1589      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1590      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1591      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1592      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1593      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1594      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1595  116500   48000      -9   26900   25600   16000      -9      -9       0
## 1596      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1597      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1598      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1599      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1600      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1601  377653  239253   23800   18900   28800   37200      -8   10400      -8
## 1602      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1603      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1604  206400   28600   32000   49100   25300   23600   21800    6600      -9
## 1605      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1606      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1607  688653  112000   61432   91000   77516  206096   53000   52761   22848
## 1608      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1609      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1610      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1611  192000      -9   44000   55000   20000   65000      -8    8000      -8
## 1612  156800   44800   14100   16300   16700   46900   11500    6500      -8
## 1613      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1614      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1615      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1616      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1617      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1618      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1619   91100       0   21500    7300   23500      -9   33800     500      -9
## 1620      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1621      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1622  282696      -8   20800    5000   12000  206096   25000    5000       0
## 1623      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1624  167111   45900    5000   20300    4000   27700   14400   14100    1711
## 1625      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1626  488294      -8   41800   65000   22100   70000   47000   23000    4057
## 1627      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1628      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1629      -9       0       0      -8      -9      -8      -8      -8      -8
## 1630      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1631      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1632  152761      -8   24400    3000    5900   40700   14600   52761       0
## 1633      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1634      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 1635      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1636      -9      -9      -9      -9      -9      -9      -9      -9      -9
```

```
## 1637     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1638 204300  63000  11100   5200  14700  48900  36100   4600     -8
## 1639     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1640     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1641 145400  31900  13200   5300   6000  41100  17200   7200     -8
## 1642     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1643 179500  35800  12500  13700  11500  69000  25900   2200      0
## 1644     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1645     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1646     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1647 101500      0  24000   2500   9200  27500  14800   5800      0
## 1648     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1649     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1650 158700  25600  12600   6700  21400  17500  24600   8500     -8
## 1651 201634  62000  17000   3900  12000  32600  20300  20000   2034
## 1652 203100  16200  31500   2200   2600  38400  28700   1100     -8
## 1653 242356  72000  23600   8300  28900  23800  22700   8700    656
## 1654     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1655     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1656 250100  60000  19700  14800   8100  83000  11000  20000     -8
## 1657 208200  20000  27700  42000  33000  32500  34000  19000     -8
## 1658     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1659 165200     -8  13100  28500  14900  48900  30300   9500     -8
## 1660     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1661     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1662 120700     -8   6200  34400   5100  50000  18000   7000     -8
## 1663     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1664 145200  25800  31900  10000   7600  51000     -8   5000     -8
## 1665     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1666 282000 150000     -9     -9  32000  75000     -9   5000     -9
## 1667     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1668 200261  37700     -8  50000   6700  30600  22500  52761     -9
## 1669     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1670     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1671     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1672  77600     -8  15300     -8  27500  15000  11400   1800     -8
## 1673     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1674 380878 107000  61432  15400   9100  73000  38500   9800   3346
## 1675     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1676 308632      0  22400  24000  10600 104000  18000  11300    932
## 1677 128700   8000  47000  18500  17500   7300   5000   5000      0
## 1678 125000   5100  21600   8100   6300  36800  20000   5900      0
## 1679     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1680 387707  73000  27900  69000  14800 106000  29700   8200   1407
## 1681 198241  34300   5900 139741   3800   4300     -8   1800   1200
## 1682 270300     -8  10300  71000     -9     -9     -9     -9     -9
## 1683 277300  61000  10200  46400  25900  45200  20200   7200     -8
## 1684 209100  47500  21800  14500  25100  25400  40100   5600     -8
## 1685     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1686 331200  82000   7600  11800  14800  78000     -9     -9     -9
## 1687 314700 102000  27900  30800  20100  40900     -8     -8      0
## 1688 224300  14500   9800     -9     -9     -9  22000     -9     -9
## 1689 284656     -8   9100  33300  15000  58000  35200  14800   2256
## 1690 217900  53000  14200   4300  41800  36200  17700  21800     -9
```

```
## 1691   200600  109000   28100   31900      -9      -8      -8      -8      -8
## 1692       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1693   108148      -8      -8   22600      -9   45100   14000      -8   13548
## 1694   205895   58000   35800    7300    9800   15700   18500   16700    1495
## 1695      300      -8      -9      -9      -9      -9      -9      -8      -8
## 1696       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1697       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1698   211118   47300   21200   13100    4500   52000   59000    8700    5318
## 1699       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1700       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1701       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1702       -9      -8      -8      -9      -9      -8      -8      -8      -8
## 1703   165600   18600    9100   28800    8200   45900   16100    7000       0
## 1704       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1705       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1706   425582      -8   16500   89000    6400  109000   97742     940      -8
## 1707   235601   22500   16600  139741   10200   19700   13100     560      -9
## 1708   219000  125000   15000    5000   40000   25000      -8    2000      -8
## 1709   196900  112000   10700   19100   15800   18900      -8    7500      -8
## 1710   158000      -8   17700   20600   17700   47100   14100    3900       0
## 1711   143900   27900   14800   21200    9400   30600   14900    3400       0
## 1712   124830   26000   20100    6300   24200   10800   14300    2600      -8
## 1713       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1714   277500   42200   12700   16000   12000  148000   34700    4500    3400
## 1715   386896   57000   26700    2900   21800  206096   18000   18300     200
## 1716   261541      -8   34000  139741    4000   23500   15600    2300       0
## 1717   272960      -8   27100   66000   38000   39500   16400   12900     660
## 1718       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1719       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1720   260500  108000   31600      -8   25200   63000      -8    7200      -8
## 1721   245800   37600   18100    8200   24600   88000   18100   11300       0
## 1722   217080   75000   28900   21200    4400   66000    5800     680      -8
## 1723       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1724   126000   47000   18000   30000   10000   15000       0    2500       0
## 1725   448571       0       0    8700   12200    8600       0      -8      -8
## 1726       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1727   439610   51000   44900  139741   63000   30100       0   12600    3269
## 1728   235600   69000    8600    4700   18000   81000    8000    2500      -8
## 1729   363800  135000   40200   25500      -9   35500   19100   22500      -8
## 1730       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1731    17500      -8   12000    3500    2000      -8      -8      -9      -8
## 1732       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1733   281000  175000   50000   10000    5000    5000    5000   10000    3000
## 1734    18100   18100      -9      -9      -9      -9      -9      -9      -9
## 1735       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1736   141500      -8   26800    3500    3000  104000      -8      -8      -8
## 1737    33500      -8   13500      -9      -9   10000      -8      -8      -8
## 1738   334703   61000   16000   10100   44400  103000   56000    3900    2203
## 1739       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1740   480193   99000    6500   16900   20600  151000   23800   16100    8493
## 1741       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1742   115300   19000   17000    4000   15000   13900   16000    9400      -8
## 1743       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1744       -9      -9      -9      -9      -9      -9      -9      -9      -9
```

```
## 1745   252742   25000   50000   25000    55000     -8   97742     -8     -8
## 1746   162000      -8    8700    5100    15000 104000   26000   1200     -8
## 1747   297000   94000   27500    3600    17900  40600   76000     -8     -8
## 1748       -9      -8      -8      -8       -8     -8      -8     -8     -8
## 1749       -9      -8      -8      -8       -8     -8      -8     -8     -8
## 1750   345900  133000   34100   51000    25800  51000   27200   2800     -9
## 1751   176912   30000   19600    7700    17800  35100   23700  11600   3612
## 1752       -9      -9      -9      -9       -9     -9      -9     -9     -9
## 1753       -9      -8      -8      -8       -8     -8      -8     -8     -8
## 1754   178342      -8   10000   11200    13300  37300   97742   4200     -9
## 1755       -9      -8      -8      -8       -8     -8      -8     -8     -8
## 1756    75000   50000   10000    5000        0   5000    5000      0      0
## 1757       -9      -8      -8      -8       -8     -8      -8     -8     -8
## 1758   187420   20100   19300    7200    13900  53000   37800    820     -8
## 1759       -9      -8      -8      -8       -8     -8      -8     -8     -8
## 1760   276100       0   15800    9800    18200 116000   25300   8200     -9
## 1761       -9      -8      -8      -8       -8     -8      -8     -8     -8
## 1762   426900  130000   36700   91000    34000  68000   38000  27000     -8
## 1763       -9      -9      -9      -9       -9     -9      -9     -9     -9
## 1764   133000      -8   15000      -8    18000  60000   33000     -8     -8
## 1765   228000   31600   38000   25000    20000  66000   43800     -8     -8
## 1766   337900   92000   26400    9900    47900  79000   52000   3200     -8
## 1767   299000  185000   21000   63000       -9  10000      -8   8000     -8
## 1768       -9      -9      -9      -9       -9     -9      -9     -9     -9
## 1769       -9      -8      -8      -8       -8     -8      -8     -8     -8
## 1770       -9      -8      -8      -8       -8     -8      -8     -8     -8
## 1771   408600  114000   14800   42500    21100 114000   27300  31200      0
## 1772   430533  239253   61432   48000    27000  12000      -8     -8  22848
## 1773   153500   35500   23100   16300    43800  32600      -9   2200     -9
## 1774   258639   23000   28000    9300    27700 106000   44300    790   5349
## 1775       -9      -8      -8      -8       -8     -8      -8     -8     -8
## 1776   135000      -8   10000   25000    50000      -8   50000     -8     -8
## 1777       -9      -8       0       0        0      0       0      0     -8
## 1778   561257  140000    9900  139741    77516 170000      -9  20200     -8
## 1779    38600      -9   20000    5000     3000  10000      -8    600     -8
## 1780       -9      -8      -8      -8       -8     -8      -8     -8     -8
## 1781       -9      -8      -8      -8       -8     -8      -8     -8     -8
## 1782   205500   50000    7500   52000    48000  25000   18000   5000      0
## 1783   237500   13000   50000   25800    17600  38800   58000  24700     -8
## 1784       -9      -8      -8      -8       -8     -8      -8     -8     -8
## 1785   675958   49700   37000  100000    77516  79000   97742  23000     -8
## 1786     3000      -9      -9      -9       -9   3000      -9     -9     -9
## 1787   219500   36000    8400   50000    11100  45000   30500   2500      0
## 1788   186900      -8      -8   13700    23000 116000   34200     -8     -8
## 1789       -9      -9      -9      -9       -9     -9      -9     -9     -9
## 1790       -9      -9      -9      -9       -9     -9      -9     -9     -9
## 1791   138000      -8      -9    5000    50000  50000   33000     -8     -8
## 1792       -9      -9      -9      -9       -9     -9      -9     -9     -9
## 1793       -9      -9      -9      -9       -9     -9      -9     -9     -9
## 1794       -9      -9      -9      -9       -9     -9      -9     -9     -9
## 1795       -9      -8      -8      -8       -9     -8      -8     -8     -8
## 1796       -9      -9      -9      -9       -9     -9      -9     -9     -9
## 1797    32000   10000   10000    4500     7500     -8      -8     -8     -8
## 1798   239100   92000   15700    9400    17800  51000   21900   3800     -8
```

```
## 1799   277380  133000   28200   25400   18100   13400   19900   12800       0
## 1800       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1801   162500      -8   15700    6200   66000   52000   15000    1500      -8
## 1802   142200      -8      -8   44300   15700   13300   38800    7300      -8
## 1803   112600   47000    7600   17500    9500   22200      -8       0       0
## 1804   467561  159000   43400   38200   32200  138000      -8   52761      -8
## 1805    39000       0      -9   12000    8000   12000      -8    2000      -8
## 1806       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1807   350500      -8   18300   38500   35700  123000   69000      -9      -8
## 1808       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1809   170686   35900   13400    4200       0   68000   31500    7400     486
## 1810   172500   39400   16200   12700   20000   56000   22400    2600      -9
## 1811       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1812       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1813    53300      -8   11000    2900    6900    2800   25300    1200      -9
## 1814       -9      -8      -8      -8      -9      -8      -8      -8      -9
## 1815   109866   23900   41900      -8      -9   12000   25400    4400    2266
## 1816       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1817       -9      -8      -8      -8      -9      -8      -9      -8      -8
## 1818   244000      -8   15000    5000   45000  150000       0   25000     500
## 1819   374037   92000   24500    6000   51000   49100   24600    2500       0
## 1820       -9      -9      -9      -9      -9      -9      -9      -9      -8
## 1821    13500      -8   12000      -9      -9    1500      -9      -8      -8
## 1822     7000      -8      -8      -8    7000      -8      -8      -8      -8
## 1823       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1824   228700   46400   16900    9800   20500   33700   33000   17700       0
## 1825       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1826   106000      -9      -9      -9      -9      -9      -9      -9      -9
## 1827   191860   45800   17900     860   16000   67000   14400   11000      -8
## 1828   100500   50000      -9   10500      -9   30000      -8      -8      -8
## 1829       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1830       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1831       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1832       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1833   274400   15000   22000   48000   36000  120000   26400    3000      -8
## 1834       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1835       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1836   469171      -8   20000      -8      -9   10000   36000     600      -8
## 1837       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1838   117400      -8   20000    5100   36000   37000      -8    3000    6300
## 1839   179400   24000   43800   13200    8400   41600   42400    1600      -8
## 1840   156200   10800   10600   11300   37000   59000   21400    3500      -8
## 1841       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1842       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1843       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1844       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1845   207340   44300    9500    3800   28100   61000    6000   17200    3540
## 1846   134500   14000   12100    3200   30000   31000   27000    2500      -9
## 1847       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1848   178500  112000      -8    3500   63000      -8      -8      -8      -8
## 1849       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1850       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1851   214000   48000   12000   18000   25000   62000   24000    5000      -9
## 1852    92970      -8      -9   21400   12300   37500   14800     870      -9
```

```
## 1853     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1854  24000  24000     -9     -9     -9     -9     -9     -9     -9
## 1855  60000     -8     -8     -8     -9     -8     -8     -8     -8
## 1856     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1857     -9     -8     -8     -8     -9     -8     -8     -8     -8
## 1858     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1859     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1860 203900     -8  15500  37500  21500  35000  69000  13900     -8
## 1861 114100  12100   3900   4000  19100  44300     -8   4000     -8
## 1862     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1863 196716  31200  18900   7200  77516  14000     -9     -9     -9
## 1864 281500     -8  16000   6100  43600 112000  63000   4800     -8
## 1865     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1866  93000     -8  10000   6800  16100  27800  16800   1500     -8
## 1867 122685  11600  21400  18800  21700  13400     -9  10400    285
## 1868 213600     -9  12400  36800  28300  54000  37900   8500     -9
## 1869 107700     -9  10800  33500  21400  31000     -9     -9     -9
## 1870 237500 110000  13600  20100  21000  53000      0  17800      0
## 1871 343300  79000  32100  51000  58000 107000      0   6700      0
## 1872     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1873  94246      0   8400   3900  29000  27600     -9      0  10546
## 1874 136700      0  13500  14500  22400  61000  19400   1800      0
## 1875 169000  85000  43000   5000   5000   8000  10000   5000      0
## 1876     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1877 184900     -8   8700  34000  26500  77000     -8   2800     -8
## 1878 160200  81000  30000  20000  16000   4200     -8   9000     -8
## 1879 332742 100000     -8  75000  35000  15000  97742     -8     -8
## 1880     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1881     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1882     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1883     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1884 183700   9000  23000  11300  22600  52000  52000   3800     -8
## 1885   2400     -8     -8     -8     -9     -8     -8     -8     -8
## 1886 230300  45900  29800  25900  14100  58000     -8  19800     -8
## 1887  80000   5000  13000  51000     -9  11000     -8     -8     -8
## 1888  89900     -8  15000   2600   7300  35000  30000     -9     -8
## 1889 207400  16400  21400  20300  28800  36500  19600   1600     -8
## 1890     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1891 399789  58000  13700  60000   6600 111000  27900   6600   1689
## 1892     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1893     -9     -8     -8     -9     -9     -8     -8     -8     -8
## 1894 171700  38600  18500  11700  17000  15700  40000   3200  10000
## 1895 266500  38500  30200  23600  22000  45400  35200  16600     -8
## 1896 152000     -8  13000  20000  14000  50000  40000   5000     -8
## 1897     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1898  42300  34000     -8     -9   4500     -8     -8     -8     -8
## 1899     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1900     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1901 161400     -9  13000   4000  35000 100000     -9     -9     -9
## 1902     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1903 331397     -8  23900  30100  29800  82000  97742  10500   4355
## 1904     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1905     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1906     -9     -9     -9     -9     -9     -9     -9     -9     -9
```

```
## 1907   417253  239253   40000   15000    10000   15000   25000   35000   10000
## 1908       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1909   123200   10200   14400   39600    17000   36000      -8      -8      -8
## 1910   294932   19800   61432   39500    30700   65000   36000   12500      -9
## 1911   303200   15800   10600    5500    30300  180000   35100    2400      -8
## 1912   281900   10500   45000   47900    52000   67000   40000   19500       0
## 1913       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1914       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1915    27700      -8      -8    4900    22800      -8      -8      -8      -8
## 1916    68500      -8    7300    6600     4300   28300   15300      -8      -8
## 1917    53300      -9   14000      -8    16800    2500   20000      -9      -8
## 1918   163800   18500   46000    6000    47000    4800   33000    6000      -8
## 1919   164500      -9   20000   26000    15500   61000   39000    3000      -9
## 1920   166600      -8   13000   35600    17500   41100   18300    4100       0
## 1921   218700   42000   19700   12600    25100   64000   38800    9300      -8
## 1922       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1923   169800   13000   32900   42400    35400   25800      -8    8400      -8
## 1924   331569   37200   61432   23900    63000   17800   42900    8000    2000
## 1925       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1926   104600   14600   26800   25200    12300   19100      -8      -8      -8
## 1927       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1928   224700   12900   20700    5000    19900  105000   34000    3400      -8
## 1929   337533      -8   11100   30000    57000  206096      -9      -9      -8
## 1930   211680   19300   11700   16900    11600   84000   56000     300     380
## 1931       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1932    81800   12000    4800   25000    10500   26500      -8      -8      -8
## 1933       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1934       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1935       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1936       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1937       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1938       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1939       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1940       -9      -9      -9      -9       -9      -9      -9      -9       0
## 1941       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1942       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1943       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1944       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1945       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1946       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1947       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1948       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1949       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1950       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1951       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1952       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1953       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1954       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1955       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1956       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1957       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1958       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1959       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1960       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 1961      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1962      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1963      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1964      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1965      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1966      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1967      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1968      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1969      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1970      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1971      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1972      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1973      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1974      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1975      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1976      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1977      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1978      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1979      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1980      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1981      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1982      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1983      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1984      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1985      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1986      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1987      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1988      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1989      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1990      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1991      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1992      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1993  655000      -9      -9  436000  219000      -8      -8      -9      -9
## 1994      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1995      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1996      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1997      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1998      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1999      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2000      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2001      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2002      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2003      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2004      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2005      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2006      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2007      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2008      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2009      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2010      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2011      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2012      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2013      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2014      -9      -8      -8      -8      -8      -8      -8      -8      -8
```

```
## 2015      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2016      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2017      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2018      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2019      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2020      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2021      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 2022      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2023      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2024      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2025  684800  160000   37600   71000   71000  225000   93000       0       0
## 2026      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2027      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2028  548300      -8   35000  185000   47800  200000   45000   15500      -8
## 2029      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2030      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2031      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2032      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2033      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2034      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2035      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2036      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2037  255500   79000   38400   62000   49000      -9   24000      -9      -9
## 2038      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2039  384300   78000  120000   14500  123000   32000      -8   12300      -8
## 2040      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2041      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2042      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2043      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2044      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2045      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2046      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2047      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2048      -9      -9      -9      -9      -9      -9      -8      -9      -8
## 2049   88200   23900      -9    2400   22200   12200   21000      -9      -8
## 2050      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2051      -9      -8      -8      -9      -9      -8      -8      -8      -8
## 2052      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2053  269000  181000      -9      -9   88000      -9      -9      -9       0
## 2054      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2055      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2056      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2057      -9      -9      -9      -9      -9      -9      -9      -9      -8
## 2058      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2059      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2060      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2061      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2062      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 2063      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2064      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2065  285000      -9  225000      -9      -9      -9      -9      -9      -8
## 2066  440500      -8   47000   23000   78000  110000      -8  149000    7500
## 2067      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2068      -9      -9      -9      -9      -9      -9      -9      -9      -9
```

```
## 2069       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2070       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2071  1120300      -8  299000   90000   352000   52000  225000   21700      -8
## 2072       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2073       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2074     2100    2100      -9      -9       -9      -9      -9      -9      -9
## 2075       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2076       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2077       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2078       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2079       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2080       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2081       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2082  3554740      -8  182000   35400   759371 1180000 1148153  166000   48216
## 2083       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2084       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2085       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2086       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2087       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2088       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2089       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2090       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2091       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2092       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2093       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2094       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2095       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2096       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2097       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2098       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2099       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2100       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2101       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2102       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2103       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2104       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2105       -9      -8      -8      -8       -9      -9      -9      -9      -9
## 2106       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2107       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2108  1337247  358000  104000  302000   189000  161000   87000   49600   53747
## 2109   672000      -8      -9  240000       -9  120000      -9      -9      -9
## 2110       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2111       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2112       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2113       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2114       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2115       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2116       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2117       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2118  1472100      -9  297000  225000   240000  574000      -9   42100      -9
## 2119   401000  168000   45000   19000    59000   36000   42000      -8   20000
## 2120       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2121       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2122       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 2123      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2124      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2125      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2126      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2127      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 2128      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2129      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2130      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2131      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2132 2820000  542000  405000   45000  306000  528000  329000  148000       0
## 2133      -9      -8       0       0       0       0       0       0       0
## 2134  355600   92000   27400   16100   32800  126000   39600    1300      -8
## 2135      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2136      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2137      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2138      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2139  218800   50000   43700   19700   26300   18900   39100    6000       0
## 2140      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2141      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2142      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2143      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2144      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2145      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2146      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2147      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2148      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2149      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2150      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2151      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2152      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2153      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2154      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2155      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2156      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2157      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2158      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2159      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2160      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2161  184500      -9      -9   19200  133000      -9      -9      -9      -9
## 2162 1666900  195000  110000   52000  103000 1100000  102000    4900      -8
## 2163 2611371 1020000  277000  118000  759371  287000      -8   52000      -8
## 2164      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2165 4066017      -8  183000  157000  395000 1602217   67000  101000   47000
## 2166      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2167      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2168      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2169      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2170      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2171      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2172  317900   77000   24800   38300   36400   84000      -8   35200      -8
## 2173      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2174      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2175      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2176      -9      -9      -9      -9      -9      -9      -9      -9      -9
```

```
## 2177       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2178   231046      -9   81000   24600    96000      -9      -9      -9   11146
## 2179       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2180       -9       0       0       0        0       0       0       0      -8
## 2181       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2182       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2183  1515000  467000  285000   25000   121000  213000      -8  153000    6000
## 2184       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2185       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2186       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2187       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2188       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2189    40000      -9      -9      -9       -9   40000      -9      -9      -9
## 2190  1376400  457000  174000   40300   153000  292000  159000      -8   38100
## 2191       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2192       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2193       -9      -8      -8      -9       -9      -8      -8      -8      -8
## 2194       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 2195       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2196       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2197  1583214  474000  153000   59000   208000  265000  195000  114000   16214
## 2198       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2199       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2200       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2201       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2202       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2203       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2204       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2205       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2206  2447000 1020000      -8  247000   393000  560000      -8   65000   60000
## 2207  1976350  468000  130000  486000   205000  266000  175000   17200   12150
## 2208       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2209       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2210       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2211   876000      -9      -9  426000   231000  107000      -9    4300      -9
## 2212  2053000  753000  369000   92000   131000  312000  240000   48000      -9
## 2213       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2214   417000      -8   73000      -9       -9  118000  149000    4000      -8
## 2215   198500      -8      -9    9100    53000   79000      -9      -9      -8
## 2216       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2217       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2218       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2219       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2220       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2221       -9      -9      -9      -9       -9      -9       0      -9      -9
## 2222       -9      -8      -8      -9       -8      -8      -8      -8      -8
## 2223       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2224       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2225       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2226       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2227       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2228       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2229       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2230       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 2231      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2232      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2233 1514000  451000  318000   46000  305000  141000  191000   14000      -9
## 2234      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2235      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2236      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2237      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2238      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2239      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2240      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2241      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2242      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2243       0       0       0       0       0       0       0       0       0
## 2244      -9       0       0      -8      -8       0       0       0       0
## 2245      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2246      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2247      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2248      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2249      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2250      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 2251      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2252      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2253      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2254      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2255      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2256      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2257      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2258      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2259      -9      -9      -9      -9      -9      -9      -9      -9       0
## 2260      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2261      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2262      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2263      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2264      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2265  424800      -9      -9   26100  148000      -9      -9    2700      -9
## 2266      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2267  395000  100000   91000   18100   68000   39700   50000    3500      -8
## 2268      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2269      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2270      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2271      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2272      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2273      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2274      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2275      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2276      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2277   40000      -9      -9      -9      -9      -9      -9      -9      -9
## 2278      -9      -8      -9      -8      -9      -8      -8      -8      -8
## 2279      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2280      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2281      -9      -8      -8      -9      -9      -8      -8      -8      -8
## 2282      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2283      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2284      -9      -8      -8      -8      -8      -8      -8      -8      -8
```

```
## 2285       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2286       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2287  1979153      -9  240000   70000   120000  256000 1148153      -8   42000
## 2288       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2289    36500      -8      -8      -8       -9      -8      -8      -8      -8
## 2290       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2291       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2292  2180500  920000  151000   73000   328000  329000  331000    8300    9000
## 2293       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2294       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2295       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2296       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2297       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2298       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2299       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2300   383800      -8      -8   38000    73000  218000      -8    1800      -8
## 2301       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2302       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2303       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2304       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2305       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2306       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2307       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2308       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2309       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2310       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2311       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2312       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2313       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2314       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2315       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2316       -9      -8      -9      -9       -9      -9      -9      -9      -9
## 2317       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2318       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2319       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2320       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2321  2687217      -8  266000   75000   380000 1602217  264000      -8      -8
## 2322       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2323       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2324       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2325       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2326  1808950  837000  127000  102000   257000  290000      -8   10000   40950
## 2327       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2328       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2329       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2330       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2331       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2332       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2333       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2334       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2335       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2336   205100      -9      -9   23800   140000      -9      -9   24500      -9
## 2337  1067000  271000  299000   86000       -9  411000      -8      -8      -8
## 2338       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 2339      -9      -8      -8      -9      -8      -8      -8      -8      -8
## 2340 2669046  800000  149000   87000  438000  320000      -8      -8   16000
## 2341      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2342       0       0       0       0       0       0       0       0       0
## 2343   32000      -8      -8      -9      -9      -8      -8      -8      -8
## 2344 1282201  516000  132000   46300   60000  123000  141000    7100    7801
## 2345      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2346      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2347      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2348  547900  300000      -8   15900   33000   52000   90000      -8    5000
## 2349      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2350      -9      -9      -9      -9      -9      -9      -9      -9      -8
## 2351      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2352      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 2353      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2354 8292116 4105867  655222  847124  759371  723000      -8  102000  293486
## 2355 3539054  369000      -9   61000   73000  237000  138000   56000      -9
## 2356      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2357      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2358      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2359      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 2360      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2361 1313200      -9  133000  134000  402000   24400  400000   87000      -9
## 2362      -9      -9      -9      -9      -9      -9      -9      -9       0
## 2363      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2364      -9      -8      -8      -8      -8      -8      -9      -8      -8
## 2365      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2366 1344000  593000  128000   85000  180000   90000  193000      -9    6000
## 2367      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2368      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2369      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2370      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2371      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2372  684605      -8   42200   19500  110000   63000  149000    9000    1605
## 2373      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2374 1110000  296000  188000   60000   85000   88000  227000   27000      -8
## 2375      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2376      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2377      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2378      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2379   77000   40600      -8      -8      -8      -8      -8   25000      -8
## 2380      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2381      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2382 1570012  526000  156000   63000  209000  210000      -8   35100   39212
## 2383      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2384      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2385  105920   59000    8100   16300   11400    8500      -8     120      -8
## 2386      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2387      -9      -8      -9      -8      -8      -8      -8      -8      -8
## 2388      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2389      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2390 1590313  633000  141000   24400  150000  277000  216000   36300   45613
## 2391      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2392      -9      -8      -8      -9      -9      -8      -8      -8      -8
```

```
## 2393      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2394      -9      -9      -9      -9      -9      -9      -9      -9      -8
## 2395      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2396       0       0       0       0       0       0       0       0       0
## 2397  805600   64000   69000   55000  276000  150000      -8  150000      -8
## 2398      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2399      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2400  626100   63000   44500   61000  126000  198000  103000    3000      -8
## 2401      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2402      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2403      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2404      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2405      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2406      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2407      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2408      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2409      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2410      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2411  774900  100000   28000   75000   38000  289000  140000   42600      -8
## 2412      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2413      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2414      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 2415      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2416      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2417 6740630 4105867   44500  212000  112000 1602217      -8   79000   57000
## 2418 2153622  400000  655222  129000  215000  384000      -9    4400      -9
## 2419      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2420      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2421      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2422      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2423 1380400  412000  130000   37400  158000  385000      -9    2000      -8
## 2424      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2425      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2426      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2427      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2428      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2429      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2430      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2431      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 2432      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2433      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2434      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2435 1642100  132000  203000  144000  130000  625000  100000   51000   52100
## 2436      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2437  627900  307000      -8      -8      -8  133000  161000   13200      -8
## 2438      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2439 1241500  485000  170000   99000  147000   99000  146000   46000      -9
## 2440      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2441      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2442 1673600  521000  147000   45500   53000  136000  205000   48700    3400
## 2443      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2444      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2445      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 2446      -9      -9      -9      -9      -9      -9      -9      -9      -9
```

```
## 2447      -9      -9      -9      -9      -9      -9      -9      -8      -9
## 2448 2060800  779000  308000      -9      -9  100000  211000      -9   36000
## 2449      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2450      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2451      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2452      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 2453      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2454      -9      -9      -8      -8      -8      -9      -8      -9      -8
## 2455      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2456      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2457      -9      -9      -9      -8      -8      -8      -8      -8      -8
## 2458      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2459      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2460      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 2461  692500  215000   60000   67000   82000  129000  104000   10000      -8
## 2462      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2463      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2464      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2465  554050  248000      -9   11200   98000  133000      -8   14700   14650
## 2466      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2467 1766900  961000  150000  162000  379000      -8      -8      -9   72900
## 2468      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2469      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2470      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2471      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2472      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2473      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2474 6142313 4105867  432000  414000  248000  555000      -8   28400      -8
## 2475 1877883  319000   83000  130000  126000  300000  111000  771944    6039
## 2476      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2477      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2478      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2479      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2480      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2481      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2482      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2483      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2484      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2485 1374400 1140000       0   98000   59000   47400       0       0       0
## 2486  163500      -9      -9      -9      -9   50000      -9      -9   73000
## 2487      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2488      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2489  582420   52000   37900   39800   43100  271000  100000      -8    6020
## 2490      -9      -8      -9      -9      -9      -9      -9      -9      -9
## 2491      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2492      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2493  655000      -9  100000   85000  100000  340000      -9      -9      -9
## 2494 1287000  416000   52000   97000  218000  180000  132000   22000      -8
## 2495  336934  116000   66000   40900   53000   27000      -8    6600    3134
## 2496      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2497      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2498 1245780  458000   30800   66000  152000   83000  249000   83000   14880
## 2499      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2500      -9      -9      -9      -9      -9      -9      -9      -9      -9
```

```
## 2501      -9      -9      -9      -8      -8      -9      -8      -9      -9
## 2502      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2503   18200      -9      -9      -9      -9      -9      -9      -9      -9
## 2504      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2505      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2506   64000      -9    7000      -9      -9      -8   57000      -8      -8
## 2507  403400   33000      -9      -8  360000      -9      -9      -9      -8
## 2508   22000      -9      -9      -9      -9   10000      -9      -9      -8
## 2509 2887796  755000  371000   62000  192000  862000  248000   70000  214796
## 2510      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2511  517500  181000   20200   32900  126000   52000   42800    5300   13600
## 2512  982814  295000  152000   84000  121000  147000  130000   13800    3514
## 2513      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2514      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2515      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2516      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2517      -9      -9      -9      -9      -9      -9      -8      -9      -9
## 2518      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2519      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2520      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2521      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2522      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2523      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2524  465600  158000   33900   14300   72000   47700   66000   13500   26400
## 2525      -9      -9      -9      -9      -9      -9      -9      -9      -8
## 2526      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2527      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2528      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2529  831773      -8   71000   97000   45700  298000  164000   46700   34373
## 2530      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2531      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2532      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2533      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2534      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2535      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2536      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2537      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2538   12000      -9      -9      -9      -9      -9      -9      -9      -9
## 2539  286000      -9  100000  104000      -9   20000   41000   12000      -9
## 2540      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2541      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2542 1357682  375000  129000  211000   43100  194000  168000   40800   22782
## 2543 1097812  227000  167000  235000      -9  202000      -8   65000   91812
## 2544      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2545      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2546      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2547  855088  283000  109000   17100  273000   77000      -8   12100   13888
## 2548      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2549      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2550  374000  182000      -9      -9      -9      -9      -9      -9  192000
## 2551      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2552  553436  188000   57000   14200   66000   99000   31700   90000    7536
## 2553      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2554      -9      -9      -9      -9      -9      -9      -9      -9      -9
```

```
## 2555      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2556      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2557      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2558      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2559      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2560      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2561      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2562      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2563      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2564      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2565      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2566      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2567      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2568      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2569  254212      -8   15600   12000   56000   68000   72000    6700    1912
## 2570      -9      -9      -9      -9      -9      -9      -9      -9      -8
## 2571      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2572      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2573      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2574  496067      -8   34300   57000   41500  134000   91000   93000    8267
## 2575      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2576      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2577      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2578      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2579      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2580      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2581      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2582      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2583      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2584      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2585      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2586      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2587      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2588      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2589      -9      -8      -9      -9      -9      -9      -9      -9      -9
## 2590 1206000  363000  179000   21000  100000  323000  127000      -8   13000
## 2591 1629100  402000  298000  370000  128000  200000  170000    5100   16000
## 2592      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2593  915000  391000  262000   75000  187000      -9      -9      -9      -9
## 2594      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2595 1350629  309000  302000  237000  115000  144000  105000      -9  123529
## 2596      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2597      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2598      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2599      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2600      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2601      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2602  505000      -9   99000   80000  150000   87000      -9      -9   49000
## 2603      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2604      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2605 1315640  446000  298000   74000      -8  333000  113000      -8   14640
## 2606 1678210  425000  157000   64000  214000  510000  199000   24700   17510
## 2607      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2608 1136965  232000   28700  283000  169000  370000      -9   17400    4065
```

```
## 2609       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2610       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2611       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2612       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2613       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2614       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2615    40000      -9      -9      -9       -9      -9      -9      -9      -9
## 2616   687300  377000  260000      -9       -9      -9      -8    6300      -8
## 2617  1066000  200000  242000  330000       -9   83000  112000      -8   20000
## 2618  2213000  358000  250000   75000       -9  511000  150000    5000      -8
## 2619   811160  380000  177000   20700    47300   18600  122000     460      -8
## 2620       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2621       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2622       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2623  1267900  476000  160000  135000       -8  305000  106000      -8      -8
## 2624       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2625       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2626       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2627       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2628       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2629       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2630       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2631       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2632       -9      -8      -8      -9       -9      -8      -8      -8      -8
## 2633       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2634       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2635       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2636  1205500  526000   54000   54000    41600  196000  138000    5200      -8
## 2637       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2638       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2639       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2640       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2641       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2642  1805000  373000  249000   27000   218000  652000  266000   20000      -8
## 2643       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2644       -9      -9      -8      -8       -8      -8      -9      -8      -8
## 2645       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2646       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2647       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2648   934963      -9      -9  328000   451000      -9      -8   44700   86363
## 2649  7060020 4105867  178000  412000   117000  899000 1148153   58000      -8
## 2650  1170000      -9   86000  317000    79000  559000      -8  129000      -8
## 2651  1056200   77000  205000  100000    26700  480000   80000   39700      -9
## 2652       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2653       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2654       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2655   804900      -8  332000   73000   122000  231000      -8      -8      -8
## 2656       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2657  1106900  490000  140000   47900       -9  187000  175000      -9      -9
## 2658  1902587  506000  655222   42200   187000  123000  123000   15600    9565
## 2659   546000      -9  125000      -9       -9  342000      -9      -9      -9
## 2660   413000      -9  413000      -8       -9      -9      -9      -9      -9
## 2661  4099240 1290000  397000  288000   759371  192000  290000  771944   57925
## 2662       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 2663  7453365 4105867  655222  287000   759371  969000  513000    5400   49505
## 2664       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2665       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2666       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2667       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2668       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2669       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2670       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2671       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2672       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2673       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2674       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2675       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2676  1234000  438000  313000   18000   180000   88000  137000   15000      -8
## 2677       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2678       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2679       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2680   588000  160000      -8   18000   320000      -8      -8      -8      -8
## 2681       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2682       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2683       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2684       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2685       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2686       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2687       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2688       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2689       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2690       -9      -8      -8      -9       -8      -8      -8      -8      -8
## 2691       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2692       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2693       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2694       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2695       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2696       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2697       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2698       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2699       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2700       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2701       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2702       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2703       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2704       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2705       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2706       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2707       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2708       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2709       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2710       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2711       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2712       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2713       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2714       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2715       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2716       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 2717      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2718      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2719      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2720      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2721      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2722      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2723      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2724  224000      -9      -9   84000      -9  110000      -9      -9    6000
## 2725      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2726      -9      -8      -9      -8      -8      -8      -8      -8      -8
## 2727 2665000  600000  400000  325000  200000  315000      -8  160000  140000
## 2728      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2729      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2730 2548054      -9      -9      -9      -9      -9      -9      -9      -9
## 2731      -9      -8      -9      -9      -9      -9      -9      -9      -9
## 2732      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2733      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2734      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2735  321466   37700   28300   17600   40900  121000   49500    8000   18466
## 2736  337728      -8      -9   31500      -9   66000  176000      -9   13228
## 2737      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2738      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2739      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2740      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2741      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2742      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2743      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2744   15000      -9      -9    9000    6000      -9      -9      -9      -9
## 2745      -9       0       0      -9      -9      -9      -9      -9      -9
## 2746      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2747      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2748      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2749      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2750      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2751      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2752      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2753      -9      -8      -8      -9      -9      -8      -8      -8      -8
## 2754      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2755      -9      -9      -9      -9      -9      -9      -8      -9      -9
## 2756      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2757      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2758      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2759      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2760      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2761      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2762      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2763      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2764  219000      -9      -9    7000   75000  110000      -9      -9      -9
## 2765      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2766      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2767      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2768      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2769      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2770 3503684  433000  160000   43300  246000 1602217  179000   80000  596367
```

```
## 2771      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2772      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2773  251796      -8      -9      -9  148000   55000      -8      -9   14796
## 2774      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2775      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2776      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2777      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2778      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2779      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2780      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2781      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2782      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2783  548300      -9   61000   63000   62000  109000      -9   13000      -9
## 2784      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2785      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2786      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2787      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2788      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2789      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2790      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2791      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2792      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2793      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2794      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2795      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2796      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2797  638400  210000   68000   24400   38000   71000   97000   73000    2500
## 2798      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2799      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 2800      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2801  628000  146000   79000  147000  112000   42500   79000   17100      -8
## 2802      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2803      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2804  469243  143000   52000      -8   50000   69000   48500   17300    2743
## 2805      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2806      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2807      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2808      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2809      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2810      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2811      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2812      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2813      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2814      -9      -8      -8      -9      -9      -8      -9      -9      -9
## 2815      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2816      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2817      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2818      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2819      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2820      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2821      -9       0       0      -8       0       0       0       0      -8
## 2822 2880040  335000  655222  324000  255000  414000  277000  113000  211818
## 2823      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2824 1634000  500000  128000   70000  214000  372000  160000      -9  109000
```

```
## 2825      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2826      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2827  908000  250000   44000  330000   70000   92000   70000   12000      -8
## 2828   60000      -9      -9      -9   60000      -9      -9      -9      -9
## 2829      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2830      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2831      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2832      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2833      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2834      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2835      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2836      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2837      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2838      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2839      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2840      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2841      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2842  304500      -8      -9   15500  128000  161000      -8      -9      -8
## 2843      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2844      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2845  558700  181000  143000   16500   44000   96000   28800   28400      -8
## 2846      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2847      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2848      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2849      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2850  935900  168000   59000  115000  162000  299000   76000   24000   32900
## 2851      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2852      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2853      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2854      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2855      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2856      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2857      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2858      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2859      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2860      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2861      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2862      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2863      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2864      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2865      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2866      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2867      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2868  787799      -9      -9   29600   18000  733000      -9      -9    7199
## 2869      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2870      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2871      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2872      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2873      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2874      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2875      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2876      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2877    6000      -9      -9      -9      -9      -9      -9      -9      -9
## 2878      -9      -8      -8      -8      -8      -8      -8      -8      -8
```

```
## 2879      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 2880  831224 354000  16300 100000 101000  61000     -8 105000   1924
## 2881      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2882      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2883      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 2884      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 2885      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 2886      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 2887      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2888      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2889      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2890      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2891      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2892      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2893      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2894      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2895      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2896      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 2897      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2898      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2899      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2900      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2901  130000 120000     -9     -9     -9  10000     -9     -9     -9
## 2902  886832 403000 104000  31700 128000 193000     -8     -8  15132
## 2903      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2904      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2905      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2906      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2907      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2908      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 2909      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2910      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2911      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 2912      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 2913      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2914      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 2915      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 2916      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2917      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 2918      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 2919      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 2920      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2921      -9     -8     -8     -9     -9     -8     -8     -8     -8
## 2922      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 2923      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2924      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2925      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 2926      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2927      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 2928      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2929      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 2930      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 2931      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 2932      -9     -8     -8     -8     -8     -8     -8     -8     -8
```

```
## 2933      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2934      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2935      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2936      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2937      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2938      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2939      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2940      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2941      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2942      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2943      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2944      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2945      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2946      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2947      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2948      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2949      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2950      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2951      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2952      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2953      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2954      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2955      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2956      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2957      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2958      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2959      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2960      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2961      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2962      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2963      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2964      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2965      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2966      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2967      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2968      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2969      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2970      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2971      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2972      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2973      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2974      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2975      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2976      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2977      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2978      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2979      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2980      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2981      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2982      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2983      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2984      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2985      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2986      -9      -8      -8      -8      -8      -8      -8      -8      -8
```

```
## 2987       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2988       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2989       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2990       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2991       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2992       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2993       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2994       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2995       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2996       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2997       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2998       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2999  5883349 1280000  259000  847124   463000 1602217 1148153   20200  131655
## 3000       -9      -8      -9      -9       -9      -8      -8      -8      -8
## 3001       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3002       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3003       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3004       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3005       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3006       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3007  1680653       0   62000  363000    45800   13600 1148153    2100       0
## 3008       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3009       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3010       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3011       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3012       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3013       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3014  2087097      -8      -8   60000    40000   67000 1148153  771944      -8
## 3015    18900       0      -9      -9       -9      -9   18900      -9      -9
## 3016       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3017       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3018       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3019       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3020       -9      -8      -8      -8       -8      -8      -9      -8      -8
## 3021       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3022       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3023   425000      -9   23000   19000    55000  170000   35000   96000      -8
## 3024       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3025       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3026       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3027       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3028       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3029       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3030       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3031       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3032       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3033       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3034       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3035       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3036       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3037  1950000  822000  144000  272000   100000  518000       0   15000       0
## 3038       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3039       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3040       -9      -8      -8      -8       -8      -8      -8      -8      -8
```

```
## 3041       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3042       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3043    98000      -9      -9      -9       -9   98000      -9      -9      -9
## 3044   179200   52000   21500      -8    38400   43000      -8   10000    6600
## 3045       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3046       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3047       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3048       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3049       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3050       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3051       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3052  2076768  686000  215000  108000       -9  624000  235000   28300   52568
## 3053       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3054       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3055       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3056        0       0       0       0        0       0       0       0       0
## 3057       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3058       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3059   915700  369000      -8   38000   249000  208000      -9   15500      -9
## 3060       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3061       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3062       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3063       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3064       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3065       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3066       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3067       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3068       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3069       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3070       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3071       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3072   846000      -9      -9   96000       -9      -9      -9      -9      -9
## 3073       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3074       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3075       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3076       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3077       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3078       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3079       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3080       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3081       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3082  2094900  161000  108000   54000   426000 1150000  148000   47900      -8
## 3083       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3084       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3085  2548054      -9      -9      -9       -9      -9      -9      -9      -9
## 3086       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3087       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3088       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3089       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3090       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3091       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3092   178000      -8  120000      -9       -9      -9      -9      -9      -8
## 3093       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3094       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 3095      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3096      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3097      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3098      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3099      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3100      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3101      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3102      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3103      -9      -8      -8      -9      -9      -8      -8      -8      -8
## 3104      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3105   95430      -9      -9      -8      -9      -9      -9      -9   13430
## 3106      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3107      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3108      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3109 3253000  838000  181000  248000  207000 1140000  522000   64000      -8
## 3110      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3111      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3112      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3113      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3114      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3115 1918853  238000  163000  202000  127000  778000  224000    8200   59153
## 3116      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3117      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3118      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3119      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3120      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3121      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3122      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3123      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3124      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3125      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3126      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3127      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3128      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3129      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3130      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3131      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3132      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3133      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3134      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3135      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3136      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3137      -9      -9      -9      -8      -9      -9      -9      -9      -9
## 3138      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3139      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3140      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3141  509900   70000  172000   25000   26300  166000   32400    8400      -8
## 3142      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3143      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3144      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3145  292500  100000   10000   10000   75000      -8   75000      -8    7500
## 3146      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 3147      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3148      -9      -9      -9      -9      -9      -9      -9      -9      -9
```

```
## 3149   666164  211000   89000   42000    42000   67000   52000    1500    5064
## 3150       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3151   174371   41400   21500   11100    10600   22400   17800   21800    2271
## 3152   452600   61000   22600    6000     6000   65000  258000   10000      -9
## 3153       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3154       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3155       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3156       -9      -9      -9      -9       -9      -9      -9      -9      -8
## 3157       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3158   405000  185000   20000   18500    14000   75000   45000   10000   17500
## 3159   714997  164000   52000    9100    38000  144000   47600   15600    6197
## 3160       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3161       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3162   177600   27300   25400    3600    25300   11000   36100    6600       0
## 3163       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3164       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3165       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3166       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3167       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3168       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3169   125000   50000   10000   35000    20000      -8      -8      -8   10000
## 3170       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3171       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3172       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3173       -9      -8      -8      -9       -9      -8      -8      -8      -8
## 3174       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3175    68760   19000    8900    8300       -9   16200   11900     960      -8
## 3176   115000   25000   15000   15000    25000      -8   15000   10000      -8
## 3177       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3178       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3179       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3180       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3181       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3182       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3183   142500   30000      -8    2500    50000   40000      -8   10000      -8
## 3184       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3185   492823  105000   73000    7400    72000   99000   89000    4100    1023
## 3186        0       0       0       0        0       0       0       0       0
## 3187       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3188   270700   96000      -8    8200    14500      -9   53000      -9      -8
## 3189       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3190       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3191   360900  199000    5900      -8     9700  120000   10100      -8      -8
## 3192   491117  166000   27800   13100    58000  119000   43800      -8   25217
## 3193       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3194   507665  138000   79000   13500    35500   67000   66000   16300    2365
## 3195   569000  175000   57000   19000    32000  142000   53000   18000   44000
## 3196   613929  180000   38500   10000    35800  150000   46300   16700  136629
## 3197       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3198       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3199       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3200       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3201       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3202       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 3203      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3204  892190  437000   12700   18100   93000  288000      -9     490      -9
## 3205      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3206  808400   85000   90000  148000   49500  164000  181000   30900      -8
## 3207  313800      -8   22600   10700   55000   84000  121000      -8      -8
## 3208  320700  135000   53000   11400   20500   23800   21500   30000      -8
## 3209      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3210      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3211      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3212      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3213      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3214      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3215      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3216  609000  107000  100000  143000      -9  154000   29000   26000      -8
## 3217      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3218      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3219      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3220  477100       0   42600   16800   29800  128000  145000   72000      -8
## 3221      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3222  356000   80000    7000   74000   52000   31400   46500    4200      -8
## 3223      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3224  497200  148000  138000   14200      -8  149000   24000      -8      -8
## 3225 1040000  250000  190000   80000   90000   70000   70000   80000   50000
## 3226  545116  129000   56000   23500   45100  123000   56000   11200    1816
## 3227      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3228  189250   72000   20000    6300   35600   26500    9700     550      -8
## 3229      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3230  312458   56000   18200   15900    8300   35000   29000   33000     458
## 3231      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3232  336655  105000   29200   46800   22200   75000   42200    7100    5655
## 3233      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3234  137500   30800   11300    5400    9800   17800   10400      -9       0
## 3235      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3236  518000   70000   26300   52000   39000  145000   82000    1600       0
## 3237      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3238      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3239  399577      -8   39500   28300   10700   43000   30100   12700   12077
## 3240  407400      -9   10000   14400   15000  290000   24000   18000      -9
## 3241      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3242      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3243      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3244      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3245  591300   88000   56000   17300  150000  150000  110000      -8      -8
## 3246      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3247      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3248 3143624  302000  193000  847124  218000  633000  155000   18500      -8
## 3249      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3250      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3251 1735095  347000  101000  168000  235000  435000  148000   18600   31695
## 3252      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3253      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3254      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3255      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3256  193200   62000   12200   21500   20600   37800   22500    7400      -8
```

```
## 3257      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3258      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3259      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3260      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3261  382900   76000   30400   11000   57000  121000   40000    7500    9000
## 3262      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3263      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3264  782200      -9  187000   27500  167000  304000   43000    1700   16000
## 3265  464119  148000   81000   25000   47600   62000   40000   11100     419
## 3266  296100  115000   23800   51000    9300   97000      -8      -8      -8
## 3267  403953  121000   39800   29500   51000   47500       0   28100   24353
## 3268  853100  291000   61000   77000   68000  190000   80000   30100      -9
## 3269  464700  143000    5900   83000   49600  137000   14600   16600       0
## 3270      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3271  777100  137000   71000   81000   77000  241000  130000   25100      -8
## 3272      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3273 1489000  923000  333000      -8      -8  136000      -8      -8      -8
## 3274  109000      -8      -8   50000   49000      -8      -8      -8      -8
## 3275   10000      -8      -8      -8      -8      -8      -8      -8      -8
## 3276      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3277      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3278      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3279      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3280  340100  114000   37000  145000   44100      -8      -8      -8      -8
## 3281  655500  235000   37500  104000   42700  133000   56000    2600       0
## 3282      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3283      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3284  596500  142000   54000   58000   59000  108000   98000    5500      -8
## 3285      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3286 4605867 4105867  130000      -8      -8  200000  170000      -9      -8
## 3287      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3288      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3289      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3290      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3291  281100  140000      -9   32400   78000      -9       0    3000    2500
## 3292      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3293 1261222  180000  655222   84000  130000  140000      -8   20000      -8
## 3294      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3295      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3296      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3297      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3298      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3299      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3300      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3301      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3302  628300  127000   75000   22100   77000  151000   49500   10000       0
## 3303      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3304  760000  137000   55000   81000   77000  240000  130000   25000       0
## 3305 1635300  278000  153000  241000  126000  308000  208000   31200       0
## 3306  331000  210000      -8      -8      -9   25000      -8      -8      -8
## 3307      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3308  468000  300000   55000   20000   50000   10000   10000   10000    5000
## 3309      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3310      -9      -8      -8      -8      -8      -8      -8      -8      -8
```

```
## 3311   611000  200000   53000   51000    58000  200000   10000   19000       0
## 3312       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3313       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3314       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3315       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3316  7053921 4105867   46000  139000    22000   25000      -8   90000      -8
## 3317  1287300  328000   61000   74000    62000  381000      -8   97000  149000
## 3318       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3319   202600      -8      -8   27600    96000      -8      -8      -8   62400
## 3320   239200   57000   62000    2800     2200   33500   46500    7900      -8
## 3321   387000   22400   41100   60000    50000  133000   79000    1500      -8
## 3322   414404       0   51000   36700    33900   52000   54000   35000    3804
## 3323   407500  111000   14100   22700    36400  105000   59000   18600      -8
## 3324   174000   35000   50000   10000    10000   50000   10000      -8    5500
## 3325       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3326    92000      -8   20000   25000    15000   10000   15000    4000      -8
## 3327       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3328  1555900  446000  143000  314000    72000  247000  126000   32400      -8
## 3329       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3330       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3331   495267  177000   58000   94000     8100   34000   44900   33200    4567
## 3332       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3333       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3334       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3335   918200  235000  110000   84000   136000  178000  133000   40000      -8
## 3336   737730   96000       0   39000    33800   89000   43400   15300    3830
## 3337       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3338   474500      -8       0   97000        0   89000  112000   30000  115200
## 3339       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3340       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3341   720950  268000   93000   14300    61000  140000   70000    3500   10150
## 3342       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3343   558400      -8   29700   75000    36400   57000   69000    5900    1000
## 3344       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3345       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3346       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3347       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3348       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3349   735000  110000   62000  167000       -9  182000  182000   32000      -8
## 3350       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3351       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3352       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3353  1138375  350000   76000  128000    45900  251000   79000   20100   35275
## 3354  2262200  369000  174000  193000   122000 1000000      -9   72000  288000
## 3355   326300   49600   23700   35800    22900  112000   39200   33100      -8
## 3356   332500   88000   14000   38800    16700   97000   29400   16800      -8
## 3357       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3358       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3359   556142   99000   47600   25100    12300  174000   71000   15800    3742
## 3360   568136  146000   31100   71000     4700   43100   36400   19300  123136
## 3361   819064   97000   50000  113000    87000  132000   69000   99000   37164
## 3362   574520   93000  111000   28100    44600  147000   86000   40100    1920
## 3363       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3364  1011000  228000   77000  102000    74000  322000  188000   10000       0
```

```
## 3365       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3366       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3367       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3368   200000      -9      -9      -9       -9      -9      -9      -9      -9
## 3369       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3370       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3371       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3372       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3373       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3374       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3375       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3376       -9      -8      -8      -9       -9      -8      -8      -8      -8
## 3377       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3378  1806000  388000  305000   79000   164000  404000  151000       0       0
## 3379       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3380   276000   65000   36000   25000    18000   78000   28000   13000       0
## 3381   566500  103000   28000   67000    33800  121000   73000   19700      -8
## 3382  1843100  278000  122000  337000   170000  717000  158000   27800       0
## 3383       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3384    67000   25000   10000   15000     7000    5000      -8      -8      -8
## 3385       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3386       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3387   252000  163000   48000      -8       -9   20000      -8      -8    6000
## 3388       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3389       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3390  1316100  227000   75000  112000    63000  496000  114000   29400      -8
## 3391       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3392       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3393       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3394       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3395       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3396       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3397   437000   98000   35000   32000    28000  126000   81000   17000      -8
## 3398       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3399       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3400  1104300      -8  310000   49300   155000  152000  186000      -9      -8
## 3401       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3402  1080070   71000   76000   70000   224000  422000  116000   31100   36670
## 3403       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3404  1255600  272000  106000  135000    49600  344000  148000   41000      -8
## 3405  2415046  514000      -8  504000   193000  727000  172000   46000      -8
## 3406  1069000  255000  177000  224000       -9  220000   92000   37000      -8
## 3407   323500  173000    7700   19000    39000   13200   59000    3200       0
## 3408       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3409       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3410       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3411       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3412       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3413       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3414       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3415       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3416       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3417  1077000  295000   85000  230000    50000  175000  125000   25000      -8
## 3418    52200      -9      -9   33300       -9   18900      -9      -9      -9
```

```
## 3419      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3420      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3421      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3422 1643900  539000  113000   79000  138000  343000  210000  118000   25900
## 3423      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3424      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3425      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3426      -9      -8      -8      -9      -9      -8      -8      -9      -8
## 3427      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3428      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3429  500000  135000   40000   60000   58000   53000   53000   18900      -8
## 3430      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3431      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3432  589683  141000   43800   17200   48100  120000   44400   38500   37983
## 3433      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3434  606088  156000  103000   18600   61000   80000   69000    3900    9788
## 3435      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3436  639293  181000  139000   20600   30100   26400   70000   10000  118093
## 3437  585000  100000   72000   80000   20000  190000   50000   20000   12000
## 3438 2818000      -8  240000  530000   55000 1010000  332000   91000      -8
## 3439  541000  143000   47700   15000    8400  284000      -8    2900      -8
## 3440      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3441      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3442 1182400  226000  298000  104000   46000  164000  121000   96000      -8
## 3443   32200      -9      -9      -9      -9      -9      -9      -9      -9
## 3444      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3445 1218000  330000  161000   19200  136000  203000  148000      -8      -8
## 3446      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3447 2548054      -8      -8      -8      -8      -8      -8      -8      -8
## 3448      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3449      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3450  571700      -8   61000   28700   74000  324000   80000      -8      -8
## 3451  445600   84000      -9    3600   30000  288000      -9   10000      -9
## 3452      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3453      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3454      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3455      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3456 1144352  245000   62000   20000  115000  368000       0  136000   76752
## 3457  516100  157000   17800   48600   66000  100000   46500   21700      -8
## 3458  504500      -8   42600   39100   20200   47500   41800   22300      -8
## 3459  520000  205000   36000  249000      -9   17000      -8      -9      -8
## 3460      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3461      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3462  842100  230000   36000   13200   79000   43000   57000   26700      -8
## 3463  457700  156000   28000    2900   64000   34800      -8   28500       0
## 3464      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3465      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3466      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3467  387224      -9  120000   72000   75000   35700   62000      -9   12124
## 3468      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3469      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3470      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3471      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3472  945388  185000  119000   28100  138000  197000   93000   20600    1688
```

```
## 3473  1280225  487000  139000  105000    61000  395000      -8   38400   48225
## 3474       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3475       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3476       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3477   714200  187000   87000   44400   124000  116000       0    8900       0
## 3478   107000  107000      -8      -9       -9      -8      -8      -8      -8
## 3479   522684  161000   80000   16800    60000   78000   42900   15800     584
## 3480       -9      -9      -9      -8       -9      -9      -9      -9      -9
## 3481       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3482       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3483       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3484       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3485  1091263  344000  132000   44700    35200  108000   93000   16200   12963
## 3486       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3487   891800       0   82000   19900    45400  454000  162000   44000       0
## 3488   369335  143000   38900   21700    57000   22600      -9    6900    4535
## 3489       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3490       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3491       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3492   319500   95000   30000   14700    24000  113000   38700    4100      -9
## 3493       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3494       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 3495   266600   80000   33000   30000    28000   31000      -8   48000    1600
## 3496       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3497       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3498  1108900  525000  108000   54000    30700  288000      -8   14400   52800
## 3499       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3500  1664000  500000  350000   35000   220000  250000  150000   74000      -8
## 3501   859600  236000  193000   22000   116000  108000   91000   20800      -8
## 3502   762819  209000  106000   36000    58000   74000   79000   16200   90719
## 3503       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3504   783608   88000   54000  132000    36400  330000       0  114000   21608
## 3505  1191000  475000  117000   36000   125000  400000      -9      -9      -9
## 3506       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3507       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3508       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3509  4293380  605000   79000  123000   183000  369000       0   96000   31280
## 3510       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3511       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3512       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3513   891500  157000   39900  100000    93000  114000  106000   47300      -8
## 3514       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3515       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3516       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3517       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3518   461000  118000   45000   20000    28000   87000  104000   19000      -8
## 3519  2716217  499000  250000   41000   148000 1602217  158000   18000      -8
## 3520       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3521  1677395  624000   68000  166000    65000  222000   90000   21500   16195
## 3522       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3523   698300  170000   74000   12300    81000  222000      -9      -9      -9
## 3524       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3525       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3526       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 3527       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3528  2517700  674000  159000  243000  170000  593000  390000  104000      -8
## 3529       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3530       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3531       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3532       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3533       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3534  1673857  403000  279000   37300  266000  247000  142000    7800  114757
## 3535       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3536       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3537       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3538   881599      -8   99000  255000   98000  288000   86000   12500   10899
## 3539   687907  144000   65000   20100   91000   82000   77000   81000   63907
## 3540  1413200  348000  142000   25200  105000  639000   62000   92000      -8
## 3541  3850305  414000  112000  349000  183000  167000       0   23200   50551
## 3542       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3543  1634000  500000   20000  300000  200000  200000  350000   35000    9000
## 3544   159000       0   40000   11000   34000   10000   40000    4000       0
## 3545  1670000      -8   90000  127000  138000  686000  249000  109000      -8
## 3546  1227000  379000   62000      -8   91000  337000  153000    1800      -8
## 3547       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3548   907000  360000   95000  136000   36000  165000   75000      -9      -8
## 3549   298900   85000   46000   52000   25000   68000      -8    2800       0
## 3550       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3551       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3552  1083523       0  250000  269000   86000  271000  156000   16600     923
## 3553   668000  460000      -8      -8      -8   80000   45000      -8   23000
## 3554       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3555  1989800  364000  161000  179000  301000  583000  158000   33200  169100
## 3556   817458  139000   56000   17900   94000  253000  152000   10200    1458
## 3557       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3558       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3559       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3560       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3561   607300  251000   75000   14500   74000  110000   63000    3600      -8
## 3562  2831268 1130000  147000  170000  186000  524000  223000      -9   38268
## 3563       -9      -8      -8      -9      -9      -8      -8      -8      -8
## 3564       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3565  2731687  809000  297000  466000  161000  394000  305000   31400   81287
## 3566   976993  193000  141000  262000  134000   34800      -9   20600    1093
## 3567   984800  192000   97000   10300  107000  286000  195000    7900      -8
## 3568  1745000  150000  200000  350000  165000  350000  240000  100000      -8
## 3569  3647730  312000   76000   78000   95000 1602217  244000   12300   35213
## 3570       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3571       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3572       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3573       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3574   255500   30000  100000   10500  100000   10000      -8      -8      -8
## 3575       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3576   370000  110000   38100   23700   37800   67000   75000    8500       0
## 3577       -9       0      -9      -9      -9      -9      -9      -9      -9
## 3578   763217  115000   18000  137000   68000  274000      -9      -9   76717
## 3579       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3580   955413      -9      -9      -9      -9      -9      -9      -9  596367
```

```
## 3581   488700  145000   23700   32600    39500   24000   98000    5200      -9
## 3582   506010  153000   46000   36400    12200   65000   88000   18300   12410
## 3583   733000  733000      -9      -9       -9      -9      -9      -9      -9
## 3584  1397100  432000  356000   42700   124000   63000  105000   75000       0
## 3585   108000       0   10000    6000    40000   10000       0    3000       0
## 3586       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3587   527322   92000   23500   10200    61000  202000   67000   21600    1622
## 3588       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3589       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3590       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3591       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3592       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3593  1663200  216000   87000  118000   404000  613000  161000   25000      -8
## 3594   801700   66000   83000  137000    71000  203000   99000   84000      -9
## 3595  4566391  871000  161000  847124   381000 1080000  243000   20900  596367
## 3596  1923305  463000  222000   11900   257000  567000  180000    6700    6705
## 3597       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3598       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3599   759364      -8   25000   72000    30600  306000  180000    6900    2264
## 3600       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3601       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3602  4459457  683000  655222  847124   125000  424000  249000  771944  596367
## 3603       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3604  3048124  674000  339000  847124   215000  660000  128000   30000   60000
## 3605  1707000  486000  158000   72000   168000   76000  149000   10000      -9
## 3606       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3607       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3608       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3609       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3610  1306100  401000  171000  263000   100000  160000   93000    7100      -9
## 3611       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3612       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3613       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3614       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3615       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 3616       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3617  1766000  159000  350000  272000   106000  420000  249000   92000      -8
## 3618       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3619       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3620       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3621  1769671  284000  252000  221000   270000  235000   95000   23700   68971
## 3622       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3623  2834667  431000  241000  274000   176000  833000  106000   39300  596367
## 3624   809900  167000   23400  126000    56000  142000   70000    5500      -8
## 3625  1166888  406000  139000   36200   107000   96000   94000    7400   86688
## 3626   913600  370000   46000   45800    48100   60000  136000   17700      -9
## 3627   953600  144000   79000   93000    65000  380000   93000   19500      -9
## 3628       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3629       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3630       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3631       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3632       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3633  2099300  641000  254000      -9       -9  543000  143000    5400    2900
## 3634  3445317  603000  183000   80000   339000 1602217  317000   46600      -8
```

```
## 3635  1416559  180000   77000   11500   405000  383000  241000   15200   13959
## 3636       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3637  1551300  517000   92000  307000    78000  110000  176000   47200      -8
## 3638  2325096  547000  230000   37700   353000  431000  347000   13500    6850
## 3639       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3640  2548054      -9      -9      -9       -9      -9      -9      -9      -9
## 3641  3848848 1230000  116000  847124   351000  885000      -8   45700   87924
## 3642  2205353  309000  126000   31000   181000  347000 1148153    6200      -9
## 3643   900546      -8  147000   40000   109000   97000  152000   35600    4846
## 3644   639828  116000  108000   53000    84000  143000   85000   16500    5128
## 3645  4569710 1210000  263000  403000   326000  169000 1148153  771944   19567
## 3646       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3647  1267426  465000   67000   49300   189000  140000  152000   20100    9026
## 3648       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3649       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3650       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3651       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3652       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3653       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3654       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3655       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3656   656065  185000   39600   21400   141000   92000   61000   16100   10265
## 3657  1275051  402000  153000   33000   141000  214000  229000   20400   28651
## 3658  4363668  952000  276000  847124   489000  243000  559000  771944      -9
## 3659   937856      -9  192000  478000   194000      -9      -9      -9   17856
## 3660       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3661       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3662       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3663       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3664       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3665   915800      -8  116000   69000    38800  270000  250000   82000   40000
## 3666  2097046 1000000  220000   50000   100000  240000      -8      -8  228000
## 3667  1015688  262000   48700  149000    94000  272000  136000   11800   37388
## 3668       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3669       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3670  2826222  432000  655222   50000   169000 1000000  285000   30000   15000
## 3671       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3672       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3673   544861  115000      -8  152000       -9   51000   95000   22600   79761
## 3674       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3675  1835700  482000  115000  242000   174000   72000  156000   46700      -8
## 3676       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3677   290000      -8   38000  176000       -8   42100      -8    7000      -8
## 3678   910000      -9  140000   60000    45000  110000  470000      -9      -9
## 3679       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3680       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3681       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3682       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3683   410000  360000   50000      -8       -9      -8      -8      -8      -8
## 3684       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3685  1107508  312000   82000   36000   203000  182000      -8   84000    2508
## 3686       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3687   521300      -8  108000   40000    66000   86000  174000   12500      -8
## 3688   832215  254000   57000   36200    82000   58000  119000   16800   26215
```

```
## 3689       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3690       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3691       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3692  3682124 1330000  302000  847124   218000  655000  278000       0       0
## 3693       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3694       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3695       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3696  1164600  379000   90000   56000    55000   61000  130000   22000    9600
## 3697       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3698   682900       0   97000    8400    90000  148000  124000    8500       0
## 3699       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3700  3499217  532000  440000   58000   227000 1602217  371000   30000   19000
## 3701       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3702       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3703  1233200  421000   77000  115000        0  188000  201000   12200       0
## 3704       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3705       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3706       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3707       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3708       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3709       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3710       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3711       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3712  2659371  500000  250000  500000   759371  500000      -8   50000      -8
## 3713       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3714   996215  141000   48600   33400   213000  283000  130000   37600     915
## 3715  3327944  800000  400000  150000   116000  300000  200000  771944  100000
## 3716       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3717       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3718  1016800  334000   96000   34800    78000  211000  157000    5300      -8
## 3719       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3720  1031000  520000  342000   86000    83000      -8      -8      -8      -8
## 3721  2088933  422000  186000   28700   194000  370000  569000    1600    2133
## 3722       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3723  2616083  950000  162000   98000   105000  196000  405000   27800   87283
## 3724       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3725       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3726       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3727       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3728       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3729       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3730       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3731       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3732       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3733       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3734       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3735   889160  318000  123000   64000    64000   81000  161000   16200   11960
## 3736       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3737  1421320  572000  171000   55000   198000   63000  194000  133000   10820
## 3738  8232387 4105867  274000  545000   167000  129000 1148153  100000  596367
## 3739  1645000  306000  198000  117000   195000  232000  242000   68000      -8
## 3740       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3741       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3742       -9      -8      -8      -8       -8      -8      -8      -8      -8
```

```
## 3743       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3744       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3745       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3746       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3747       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3748  1609000  685000  110000  100000   256000  165000  205000   40000      -8
## 3749       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3750       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3751  1004799      -8  141000      -8   233000  221000  351000      -8   18799
## 3752       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3753       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3754  1087200  386000   44900  319000       -9  321000      -9   16300      -9
## 3755       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3756  1667500  439000   80000   26500   283000   89000  211000    4100   24900
## 3757  7371867 4105867   73000  308000   120000  295000  385000      -8      -8
## 3758       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3759       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3760  3556341      -8  410000  847124   360000 1602217   33000      -8  150000
## 3761       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3762  1527352  562000  145000   11800   193000  398000  171000    4000    3752
## 3763  3026059      -8  118000  205000   159000  738000  339000   13800   20259
## 3764  1466600  444000  436000      -8   283000   42000  216000      -8      -8
## 3765  7521032 4105867  332000  349000   759371  427000  505000  771944  102850
## 3766  1947746  454000  134000   76000    82000  586000  186000  125000   10700
## 3767       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3768       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3769       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3770  4771183 1130000  369000  255000   394000 1150000  508000  771944   50239
## 3771       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3772       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3773       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3774  2256423  535000  164000  148000   367000  373000  601000   47900   20523
## 3775       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3776  1638000  634000  194000   43000   205000      -9  291000      -9      -9
## 3777   781565  260000  103000   97000    27400  154000       0   55000   28265
## 3778       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3779   742000  390000   97000  108000   101000   28000      -8      -8      -8
## 3780       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3781       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3782       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3783  1737400  606000   81000   47500    38100   97000  156000    1800  108000
## 3784       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3785       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3786       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3787       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3788  2953367  200000  180000  500000   270000  750000  357000   50000  596367
## 3789  2198639  744000   96000   87000   254000  219000  234000   37200   76439
## 3790       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3791  1046000   85000  101000   86000    32000  437000  118000  136000   12000
## 3792       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3793       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3794   450180  108000   23300   62000   102000   57000       0   44500    9480
## 3795        0       0       0       0        0       0       0       0       0
## 3796       -9      -8      -8      -8       -8      -8      -8      -8      -8
```

```
## 3797       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3798       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3799       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3800       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3801  3136804      -8  168000  241000   439000  261000   30300   20900  102604
## 3802       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3803   520300  270000      -8   14600   223000      -8      -8      -8      -8
## 3804       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3805       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3806  1209000  157000   78000   61000   441000  360000      -8      -8   12000
## 3807   948935      -8  190000   46900   345000  156000  144000   10200   15735
## 3808       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3809  6111387  573000  213000  847124   300000 1602217  637000   60000       0
## 3810       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3811  9984094 4105867  655222  847124   759371  841000 1148153  771944  596367
## 3812       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3813       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3814       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3815       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3816       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3817       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3818       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3819       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3820       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3821       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3822  5059488 1330000  290000  478000   247000 1602217  326000   12700   57571
## 3823  5858867 4105867  277000  352000   121000  456000  320000   52000      -8
## 3824       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3825       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3826       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3827       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3828   933492  446000   30000  106000    28400   83000       0   18300  168492
## 3829       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3830       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3831   312875   43100   51000   20000    23900   40100   37800    1900   48375
## 3832       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3833   200000      -9      -9      -9   200000      -9      -9      -9      -9
## 3834       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3835   337100      -9   37700   10000   122000  152000      -9      -9      -9
## 3836       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3837  7802237 4105867  248000  346000   225000 1602217 1148153      -9      -9
## 3838       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3839       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3840       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3841       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3842  5400887 1130000   92000  229000   759371  865000  379000   30800   96670
## 3843   363800  183000      -9   19600    87000   72000       0       0       0
## 3844   372500   82000   75000   20000    24000   54000   28000    6000      -8
## 3845       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3846   352100  116000   49700    8400    21200   67000   26200    5400       0
## 3847       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3848       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3849   400000   80000  100000  100000    50000   50000       0    5000       0
## 3850    68000      -8      -8      -8    68000      -8      -8      -8      -8
```

```
## 3851   605000   50000  200000  150000   50000      -8  100000      -8   55000
## 3852       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3853       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3854       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3855       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3856       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3857   562200       0   30700   41200  129000  219000  106000    3500       0
## 3858       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3859       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3860       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3861       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3862       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3863       -9      -8      -8      -9      -9      -8      -8      -8      -8
## 3864       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3865       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3866       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3867       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3868       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3869   187160      -8   43600      -9   29800   59000   23100     760      -8
## 3870       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3871   352000   55000   47000   40000   84000  120000      -8    2000      -8
## 3872   315500   60000      -9   20500   92000   78000   49000      -9      -9
## 3873       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3874       -9      -9      -9      -9      -9      -9      -9      -9       0
## 3875     1800      -8      -8      -8      -9      -8      -8      -8      -8
## 3876   365000  100000   15000  100000   50000      -8  100000      -8      -8
## 3877   303200  150000  108000      -8      -8   18000      -8      -8      -8
## 3878  2232600  264000   70000   54000   31600   86000  119000      -9      -9
## 3879   117900   45000   35000    9600    7200   15600      -8    3000      -8
## 3880       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3881   231000  100000   15000    6000   50000   10000   20000   10000    5000
## 3882  2284921  546000  655222   85000  323000  530000       0    8700   41999
## 3883   476000   80000   35000  105000  109000   82000      -8    8000      -8
## 3884   209400       0      -8   44000   18000   40000  100000    1000    6400
## 3885   750000  300000   40000   30000   30000   60000   70000   80000   50000
## 3886  1318000  900000  100000  100000  100000   30000   20000   20000   15000
## 3887   494400      -9   20200   11000   78000   32000   64000    6100       0
## 3888       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3889       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3890       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3891   942200  330000  167000   40700  112000   89000   99000   79000      -8
## 3892       -9      -8      -8      -8      -9      -8      -8      -8      -8
## 3893  1173086   36400   49100   90000   54000  722000  131000   43900    1786
## 3894       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3895       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3896       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3897   405000  117000   61000   58000   19000   64000   42000   34000       0
## 3898   134300      -8   23000   16700   25500   24600    7200    3400      -8
## 3899       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3900   423700  150000   20000    9000   36700   75000   40000   21000       0
## 3901       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3902       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3903   280000   37000   30000  102000      -9  102000      -8      -8      -8
## 3904   443000  176000   60000   60000   65000   18000   52000   12000      -8
```

```
## 3905   318200   81000   23400    9100    32100   99000   29000   10400       0
## 3906       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3907       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3908    19000      -8      -8    3800    12000      -8      -8      -8      -8
## 3909       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3910       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3911   419000  300000   20000   15000    15000   10000   20000   15000    8000
## 3912  1008520  523000      -8  123000   246000   40000      -8      -8   11520
## 3913   775060  236000   45700   77000    30200  161000   94000   52000    8160
## 3914   282000  201000   56000       0        0   10000    5000       0       0
## 3915   265200   80000   15000    4200   156000   10000      -8      -8      -8
## 3916       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3917   731600  182000   53000   17000    57000  215000  178000    3900       0
## 3918       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3919  1365000  286000   38000   89000    70000  116000   99000   13000    5000
## 3920   486900   92000   73000   36000    96000   94000   76000    1600      -8
## 3921       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3922       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3923       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3924       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3925  3078534  551000  655222  179000   255000  688000  551000   21200  178112
## 3926   602000  106000   38000   97000    29000  242000   46000    6000      -8
## 3927       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3928    32400      -8      -8      -8    32400      -8      -8      -8      -8
## 3929   125000   70000   12000    6000     6000   12000    7000    3000    2000
## 3930       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3931   298700  143000   16500   17100    26600   34400   42100    4900       0
## 3932   677000   58000   70000   87000    99000  288000   67000    8000      -8
## 3933   600000  500000   12000    5000    50000   10000   10000    5000       0
## 3934       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3935   718000  250000   40000   60000    60000   65000  200000       0    6000
## 3936   566000  240000      -8   16000   150000   20000  110000      -9   30000
## 3937   270000   60000   50000   35000    50000      -8   75000      -8      -8
## 3938       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3939       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3940       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3941   649650   76000   26500      -9       -9   77000  443000     750   26400
## 3942   307600   84000   56000    9500    40800   63000   29200    4100      -8
## 3943   335000      -9   48000   52000    41000  145000   49000      -9      -9
## 3944   350380   44200   14800   48500    43200  102000      -8    3600   29080
## 3945   207300      -8    8500    8100    33700  140000      -8      -8      -8
## 3946   819000  141000  113000  109000    66000  136000   93000    9300       0
## 3947       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3948       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3949   337500      -8   28700   13400    37800  113000  105000   36100      -8
## 3950   808434  292000  190000   24700    11700   61000   67000    9300    2134
## 3951   177100   48000      -8   60000    24000      -8   33600      -8      -8
## 3952   532500  100000  200000  100000   100000      -8   25000      -8      -8
## 3953  1549899      -8   88000   60000   229000  201000  204000  147000   12399
## 3954   393300   82000   62000   75000    86000   46200   30000   12100      -9
## 3955       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3956  1333500  144000   58000  255000   146000  483000  138000   14600      -9
## 3957  1217000  598000   39000  130000    50000  184000   89000      -9       0
## 3958   374500  143000   55000   49600    48000   65000       0    8500       0
```

```
## 3959   596977  136000   43000   62000    40900  105000  109000   11800   13377
## 3960   704834  247000   45800   26200    75000   89000   69000    2600     534
## 3961   241100   88000      -9   18600    52000   48400      -8      -8      -8
## 3962       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3963   652400  292000   10800   36500   118000  138000      -8   25800      -8
## 3964   377000  160000      -8  180000    37000      -8      -8      -8      -8
## 3965       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3966       -9      -8      -8      -9       -9      -8      -8      -8      -8
## 3967       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3968  1067696  334000   39500   58000   162000  222000   89000   94000    8196
## 3969   416600  105000   25900    8900    74000   27700   44500   35800       0
## 3970       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3971   491500  156000  106000   37900    35900  119000      -8   29700      -8
## 3972       -9       0       0       0        0       0      -9       0       0
## 3973       -9      -8      -8      -9       -9      -8      -8      -8      -8
## 3974   198200   50000   17000   24500    22500   49000      -8    2200      -8
## 3975       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3976       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3977       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3978       -9      -8      -8      -9       -9      -8      -8      -8      -8
## 3979   415000  250000   50000   20000    10000   50000       0    5000      -9
## 3980   787500  208000   41400  123000    58000  106000   67000   33000      -8
## 3981  1814000  580000   81000  517000       -9  112000  187000      -8      -8
## 3982  1326003  353000  101000   87000   105000  400000  220000    9600    9303
## 3983       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 3984       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3985       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3986       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3987       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3988  1069521  353000   98000   97000   108000  138000   23300   25500    2721
## 3989       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 3990  1924496  460000   91000  130000   144000  204000      -9  771944     552
## 3991       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3992       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3993   966698  194000  193000   91000   107000  227000   63000   60000    3198
## 3994       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3995   825500       0   48800   45700    36300   56000   80000    3500       0
## 3996   254000   84000      -8      -9    40200   20000   55000      -8   31200
## 3997       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3998  1624000  415000  302000  296000   270000  200000      -9      -9      -9
## 3999       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4000       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 4001       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4002   351100   39500   59000   52000    38600   74000   24900   14800   12000
## 4003   154500      -8   90000    6000    54000      -8      -8    2500      -8
## 4004   518900  176000   54000   19700    41800   51000   65000      -9      -9
## 4005     7000      -9      -9      -9       -9      -9      -9      -9    7000
## 4006   351200   55000   14400   18000    26800  136000   41700    6200       0
## 4007   852000      -9   88000  248000       -9   56000  425000      -9   15000
## 4008       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4009   493800  148000  111000   32900    28300  139000      -8    1500      -8
## 4010   585000  120000   50000   10000    80000  100000  150000   50000    5000
## 4011       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4012  1833589  420000  149000   19500   233000  658000  174000   54000    2089
```

```
## 4013   776371     -8     -8   17000   759371     -8     -8     -8     -8
## 4014   365000 266000     -8   22000    77000     -8     -8     -8     -8
## 4015   407550 149000  27700    8900    58000  90000  45000    750      0
## 4016  1148216 283000  46300  206000    64000  28800 283000  20400 142016
## 4017       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4018   769737  51000 128000   11800    54000 212000 146000  10500   1337
## 4019       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4020       -9     -8     -8      -8       -8     -8     -8     -8     -8
## 4021       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4022   237100  37500     -8    8000    70000 100000     -8     -8     -8
## 4023   164800  95000     -9    3600    48000  11000     -8     -8     -8
## 4024   617724 144000  83000    2800    72000 136000     -8  64000  25924
## 4025       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4026  3085137 675000 416000  118000   464000 208000 225000 771944  39193
## 4027  2382000 581000 123000  246000    74000 760000 423000 158000   5000
## 4028   167400  57000  86000    5800    18600     -9     -9     -9     -9
## 4029   789000 183000  47600   64000   266000  26500  59000   3400     -8
## 4030   745940     -8  62000  155000    88000 225000 187000     -9   1040
## 4031  2362879     -8 114000   79000   285000 425000 1148153 44300  19426
## 4032       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4033  2418315     -8 171000  291000   132000 547000 202000 177000  25315
## 4034       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4035       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4036   602000 360000  30000   12000    60000  20000 100000     -8     -8
## 4037       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4038   672500 125000 104000   18000   102000 119000  48800   8600     -9
## 4039   446200 154000  46200   35900    33200 169000     -8   2200     -8
## 4040       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4041       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4042       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4043       -9     -8     -8      -8       -9     -8     -8     -8     -8
## 4044       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4045       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4046   621600 227000 116000      -8   182000  69000     -8     -8     -8
## 4047       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4048       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4049       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4050       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4051   813000 242000  51000   14000    54000 190000  52000  13000   8000
## 4052       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4053   148000  15000  10000    6000    70000  12000  15000  10000      0
## 4054       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4055       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4056       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4057   615900 208000  25900   17200    59000  92000  79000  14900      0
## 4058   410000  71000  28700    9200    68000  23800  69000   2500     -8
## 4059       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4060  2536761     -8  88000   80000   286000 594000 1148153 41400  41208
## 4061       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4062       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4063  2485500 522000 175000  376000   226000 543000 233000  20500      0
## 4064       -9     -9     -9      -9       -9     -9     -9     -9     -9
## 4065  1145050 282000 135000   45500   125000 143000     -8   6500   3050
## 4066       -9     -9     -9      -9       -9     -9     -9     -9     -9
```

```
## 4067       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4068   399000  225000   20000   14000  110000   15000       0      -8      -8
## 4069       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4070       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4071       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4072       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 4073  1208500      -9      -9   60000  413000  284000  141000   16500   22000
## 4074       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4075       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4076   392000  189000      -8   10000   80000   20000   60000      -8   15000
## 4077       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4078       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 4079  3130124  810000  153000  847124   85000  862000  130000   61000  170000
## 4080   332000  170000   23000   12000  100000   15000       0       0       0
## 4081       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4082       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4083   878033  193000   48000   40000   89000  137000   78000   26500   27533
## 4084       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4085  1477100  373000  185000  166000  130000  180000  180000   10800      -9
## 4086       -9       0      -9      -9      -9      -9      -9      -9      -9
## 4087     3300      -9      -8      -8      -8      -8      -8      -8      -8
## 4088       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4089  1265700  384000   66000  300000  144000  250000   52000   50000      -8
## 4090   986700  240000  113000   24000   89000  341000  112000    2700       0
## 4091    51000   33000      -9      -9      -9      -9      -9      -9      -9
## 4092   853000  400000  130000   30000  110000  110000      -8   50000      -8
## 4093       -9      -8      -8      -8      -9      -8      -8      -8      -8
## 4094       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4095       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4096       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4097   970000  160000  124000  162000      -9  288000  121000   35000       0
## 4098   242000       0      -9    7000   97000  120000      -9       0       0
## 4099       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4100       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4101       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 4102   324300      -8      -8  214000   50000      -8      -8   31300      -8
## 4103   989300  274000  187000   11800  240000   70000  114000   20000      -8
## 4104   781200   84000   96000      -9  201000  218000  125000   15200      -9
## 4105       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4106  1601244  420000  218000   32000  131000  296000  151000   71000   17444
## 4107    80500   15000   10000    5500   25000      -8   15000      -8   10000
## 4108   614400      -8  164000   59000  144000   70000   92000   50000     400
## 4109   169000      -8   17000   13000  100000   20800      -9   11000      -8
## 4110       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4111       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 4112       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 4113       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4114   438600  125000   60000  150000   15000   10000   60000      -8     600
## 4115       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4116   756900  184000   67000   20700  161000  295000      -8      -8      -8
## 4117       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4118   161000      -8   14000      -8   36000   70000   15000   20000      -8
## 4119  1476027  452000   41900   51000   83000  351000  161000   16000   20127
## 4120       -9      -9      -9      -9      -9      -9      -9      -9      -9
```

```
## 4121   838600      -8  459000   49700    26900  113000  190000      -8      -8
## 4122   790200  144000   54000    5600    26100  112000   86000    5500      -8
## 4123   360200      -8   95000   17400    99000      -8   67000    6000      -9
## 4124       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4125       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4126  1272193  371000   77000  108000   125000  210000   96000  153000   78193
## 4127       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4128   154700      -9      -9   15900    15800   55000      -9      -9      -9
## 4129   347200   66000   47200   42900    79000   55000      -8   16200      -8
## 4130  1863331  361000  227000   18000   134000  127000  137000   32300    1031
## 4131   602500  207000   82000  179000    43100   58000   20400      -8      -9
## 4132       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4133       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4134       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4135   398200   22800   60000  129000    86000   90000      -8      -8      -8
## 4136       -9      -9      -9      -8       -8      -9      -9      -9      -8
## 4137       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4138       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4139   795500  177000   67000   85000    88000  192000  131000   12300   11400
## 4140       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4141       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4142  1611756  519000   91000   75000   205000  129000  181000   98000   46056
## 4143   597840  214000   92000   26000    72000  155000   16000     640      -8
## 4144   428600  156000   79000    3900    17100   37600   54000    1200      -8
## 4145       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 4146       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4147       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4148  1299000  250000   79000  100000   120000  100000  300000   50000  100000
## 4149       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4150       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4151       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4152   549500      -8   43000  130000    31500   35000  285000      -8   10000
## 4153       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4154       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4155       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4156       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4157  3621800  294000  419000  222000    35800  734000  406000   51000      -8
## 4158  1263119  216000  114000  114000    41200  383000   94000  116000   13319
## 4159       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4160       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4161   369000      -9   30000   84000    31000  220000      -9    4000      -8
## 4162       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4163       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4164       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4165       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4166  1362000      -8  124000  326000   101000  151000  155000   54000      -8
## 4167  1038000       0  107000   75000    95000  166000   82000   13100      -8
## 4168       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4169  1002043      -8  134000  268000    80000  208000  130000      -9   16043
## 4170       -9      -8      -9      -9       -9      -9      -9      -9      -9
## 4171  1074600       0   91000  161000    57000  194000   32400   27200      -9
## 4172       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4173  4290352  787000  655222  337000   319000  886000  424000   43900  193630
## 4174  3247190  949000  232000  330000   152000  816000  243000    5700  105490
```

```
## 4175  2958091  753000  147000   19300  759371  564000  250000   26200   20220
## 4176       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4177       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4178       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4179       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4180       -9      -8      -8      -8      -9      -8      -8      -8      -8
## 4181   850358  234000   99000   55000   61000  162000  102000     550   20808
## 4182  1230000      -9  300000   29000  477000  190000  150000      -8      -8
## 4183   358650   40000   67000   60000   51000   22800   63000   37000     950
## 4184   609400  260000   46000   64000   68000   15000  130000      -8      -8
## 4185   790700  107000   45200  159000  142000  167000  148000      -8      -8
## 4186   200000      -9      -9      -9      -9      -9      -9      -9      -9
## 4187  2600600  765000      -9  186000  304000  653000  318000  179000       0
## 4188       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4189       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4190       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4191  1526909  494000  139000   63000  176000  501000   81000   15100    5809
## 4192       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4193       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4194   124000      -9      -9   20000   22000   30000      -9   30000      -9
## 4195       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4196       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4197       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4198       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4199       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4200  1601700  558000  178000   45800  230000  240000  143000      -9      -9
## 4201  3629254  375000  655222  491000  381000  349000  243000   72000   24032
## 4202       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 4203  1561000  289000  180000  115000  128000  500000  100000    5000  144000
## 4204       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 4205       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 4206   569600  159000   59000   15100   68000   74000   89000    6400      -8
## 4207       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4208       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4209       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4210  1480000  270000  210000   50000  115000  150000  235000   95000   10000
## 4211  4733906  969000      -9  283000  111000  331000  339000   40900   54952
## 4212       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 4213       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4214  1965000 1200000  100000  100000   75000  200000  250000      -8      -8
## 4215  1054993  285000   97000   69000  109000  291000  140000    8100   11493
## 4216       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 4217   830000  700000   43000      -8      -8   87000      -8      -8      -8
## 4218   325000      -9   25000   25000  150000   20000   60000      -8      -8
## 4219  1388900      -8  173000  374000  251000  148000  255000       0  162600
## 4220       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4221  1078000  301000  251000   87000  168000  106000   95000      -8   10000
## 4222       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4223       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4224       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4225       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4226       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4227       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4228       -9      -9      -9      -9      -9      -9      -9      -9      -9
```

```
## 4229       -9      -8      -9      -9       -9      -9      -9      -9      -9
## 4230       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4231       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4232   560000  160000   30000   65000   180000   10000   85000      -8      -9
## 4233       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4234       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4235       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4236       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4237  3256524  732000  146000  171000   186000  582000  395000  771944   13534
## 4238       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4239  1404753  409000   43400   52000   234000  328000  225000   15600   17753
## 4240       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4241  2893875  497000  415000   47600   348000  245000  263000   41600   56875
## 4242       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4243       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4244    66000      -9      -9      -9       -9      -9      -9      -9      -9
## 4245  1900400 1110000      -9  290000   160000  287000      -8    5000      -8
## 4246       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4247       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4248       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4249  1150400  336000  113000   60000   175000  180000  203000   30400       0
## 4250       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4251   823000  166000  101000   52000    95000  299000      -8      -8      -8
## 4252       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4253       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4254       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4255       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4256   749000  214000  330000   15000   100000   80000      -8      -8      -8
## 4257       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4258  1618603  255000  180000   48700   348000  435000  165000   22700   10203
## 4259   867895       0  167000   62000   239000   95000  188000    4700    6195
## 4260       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4261       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4262  1463286  430000   54000   54000   144000  219000  246000   27300   27986
## 4263       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4264       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4265   959623  118000  103000  113000   142000  328000   55000   64000    2923
## 4266  1472000  341000  162000   90000   125000  379000  169000   60000       0
## 4267       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4268       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4269       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4270   871000  270000   59000  113000    48000  142000   74000  165000      -9
## 4271       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4272  1863000  441000  210000  118000   460000  462000      -8      -8      -8
## 4273       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4274       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4275  1105000  169000  147000  249000    84000  347000      -8      -8      -8
## 4276       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4277  1252000  458000  131000  165000   255000  100000      -8      -8      -9
## 4278       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4279       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4280   791000  345000  133000      -8   180000   50000      -8   24000      -8
## 4281       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4282       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 4283       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4284       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4285       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4286       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4287  1561000  500000   95000   50000   130000  410000  200000   15000    1000
## 4288       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4289       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4290       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4291  3479600  608000  157000  286000   115000  545000   86000   22600      -8
## 4292       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4293  1346000  173000      -8  351000       -8  748000      -8      -8      -8
## 4294       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4295  2724074  593000  172000  173000   475000  337000  327000   70000   65274
## 4296  4841590  796000  655222   42400   759371   48800  390000   94000   40797
## 4297       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4298  1379523  528000  116000  163000   241000   75000  189000   18200    1023
## 4299   649740  331000   96000   15900    98000   36400      -8    4000    5940
## 4300       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4301   826000  300000   30000   58000   274000   15000   96000      -8      -8
## 4302  1090750  350000  136000   60000   110000  220000      -9     750   14000
## 4303       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4304       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4305       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4306       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4307  1475500  449000  300000  149000   144000  186000  152000       0       0
## 4308       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4309  1343000  600000   40000   55000   338000  200000  100000      -8      -8
## 4310       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4311       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4312  2303000  580000  170000  105000   330000  560000  300000   16000   13000
## 4313  1040800  455000   67000   39600    88000  176000  151000    3200       0
## 4314       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4315       -9      -8      -8      -9       -9      -8      -8      -8      -8
## 4316  2098838  549000  288000  290000   210000  445000  222000    5200   22638
## 4317       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4318       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4319       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4320       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4321       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4322       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4323       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4324  1322850  482000   62000   76000   167000  202000  173000    8600   19250
## 4325  1611100  504000  151000   75000   187000   98000  197000    3400      -8
## 4326       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4327       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 4328       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4329       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4330  1017263  312000  162000   17300    52000  128000  148000   70000    5363
## 4331       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4332       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4333       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4334  1801000  500000  111000    5000   250000  500000  120000   15000   50000
## 4335       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4336       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 4337       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4338       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4339  1034022  280000  161000   40300   126000  165000  149000   19900    8222
## 4340  2394231  602000  655222  127000   457000   89000  104000    1400   69009
## 4341  1923000  420000  365000   88000   123000  810000      -8      -8   72000
## 4342   963828      -8  187000      -8   249000   45400  159000      -8   16428
## 4343   673200      -8  315000   51000   125000   79000   78000      -8      -8
## 4344       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4345  1019040  148000      -9   78000   349000  126000  163000      -8   13140
## 4346    64500   12600    5700    5500     4800   21000    5800    1600      -8
## 4347       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4348       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4349  8744554 4105867  655222  847124   759371 1602217  388000   31200   29553
## 4350  3564487 1400000  220000  122000   759371  277000  320000   39200  188916
## 4351  1606000  484000  120000   75000   190000  250000  168000  109000      -8
## 4352       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4353       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4354       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4355       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4356  2342000  600000  288000   47000   270000  337000  260000   27000   36000
## 4357       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4358       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4359       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4360  1534232  393000  172000   56000   240000  139000  129000  187000   10232
## 4361   182300   80000   44500      -8       -8   37500      -8    5300      -8
## 4362       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4363       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4364   450400  110000  127000   75000    33100   47700   36500    2700    7000
## 4365       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4366  1818900  578000   90000  140000   223000  468000  151000   73000      -8
## 4367  6575867 4105867  200000  100000   500000 1000000  500000   50000       0
## 4368  2045103  621000  243000   13700    96000  503000  209000   25100    9303
## 4369       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4370       -9      -8      -9      -9       -9      -9      -9      -9      -9
## 4371       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4372       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4373       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4374       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4375       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4376       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4377   482000  200000      -8   12000   260000   10000      -8      -8       0
## 4378       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4379       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4380       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4381  3267124  698000  197000  847124   420000  800000  203000   58000   17100
## 4382  5543017      -8  655222  847124   759371 1602217 1148153  125000  146884
## 4383       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4384       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4385       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4386       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4387       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4388   800971      -8      -8   21000   759371      -8      -8      -8      -8
## 4389  2548054      -9      -9      -9       -9      -9      -9      -9      -9
## 4390       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 4391       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4392       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4393       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4394       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4395       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4396  2111700  507000  295000   43000   166000  590000  169000   24700      -8
## 4397       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4398       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4399  3349341  500000      -8  847124   100000 1602217      -8      -8  200000
## 4400       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4401  3773571  846000  219000  472000   759371 1140000  276000   26700      -8
## 4402   147201      -9      -9      -9    76000      -9      -9      -9   28501
## 4403  2263944  753000  261000  113000   143000  222000      -8  771944      -8
## 4404       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4405       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4406       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4407       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4408  2620160  741000  209000  128000   306000  518000      -8  134000   20160
## 4409       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4410       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4411       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4412  2689939  695000  145000  847124   759371  167000      -8   56000    8244
## 4413       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4414  1927300      -8  327000  486000   344000   96000  346000   36300   74000
## 4415       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4416  1770666  555000  133000  102000   132000  436000  281000   34300   43366
## 4417       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4418       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4419  5032307  958000  351000  847124   284000  463000  392000   30000  207983
## 4420 10858717 4105867  655222  220000   100000 1602217      -8  771944  596367
## 4421       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4422       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4423  2367000  844000  355000  421000   391000  281000      -8      -8      -8
## 4424  2836217  377000  266000  123000   261000 1602217      -8      -8      -8
## 4425       -9      -8      -8      -8       -8      -8      -8      -8      -8
##       OPLAND OPOTHEX
## 1         -9      -9
## 2         -9      -9
## 3         -9      -9
## 4         -9      -9
## 5         -9      -9
## 6         -9      -9
## 7         -9      -9
## 8         -9      -9
## 9         -9      -9
## 10        -9      -9
## 11        -9      -9
## 12        -8      -8
## 13        -8      -9
## 14        -9      -9
## 15        -9      -9
## 16        -8      -8
## 17        -9      -9
## 18        -8      -8
```

```
## 19       -8      -8
## 20       -8      -9
## 21       -8      -8
## 22       -9      -9
## 23       -8      -8
## 24       -9      -9
## 25       -9      -9
## 26        0      -8
## 27       -9      -9
## 28       -8      -8
## 29       -8      -8
## 30     4575      -9
## 31       -9      -9
## 32       -8      -8
## 33       -9      -9
## 34       -8      -8
## 35       -9      -9
## 36       -8      -8
## 37       -9      -9
## 38       -9      -9
## 39       -9      -9
## 40       -8      -8
## 41       -8      -8
## 42      200      -9
## 43       -8      -8
## 44       -9      -9
## 45       -9      -9
## 46       -8      -8
## 47        0       0
## 48       -8      -8
## 49       -8     500
## 50       -8    4100
## 51       -8      -8
## 52       -9      -9
## 53      500     500
## 54       -8      -8
## 55       -9      -9
## 56       -9      -9
## 57       -9      -9
## 58       -8      -8
## 59       -9      -9
## 60      900      -8
## 61       -8      -8
## 62       -8      -8
## 63      590      -8
## 64       -9      -9
## 65       -8      -8
## 66       -8    4500
## 67      600     250
## 68       -9      -9
## 69       -9      -9
## 70      700       0
## 71      110     190
## 72       -8      -8
```

```
## 73       -8      -8
## 74       -9     500
## 75       -9    8365
## 76       -8      -8
## 77       -8      -8
## 78       -8      -8
## 79       -8      -8
## 80      400     360
## 81     1200      -8
## 82     1000      -8
## 83        0    1900
## 84       -9      -9
## 85     1000      -8
## 86       -9      -9
## 87      770      -9
## 88       -9      -9
## 89       -9      -9
## 90        0       0
## 91       -8      -8
## 92       -8      -8
## 93       -8      -8
## 94       -9      -9
## 95      500      -8
## 96       -8      -8
## 97     4575    6400
## 98       -8      -9
## 99     2400      -8
## 100       0       0
## 101    4575      -9
## 102     720       0
## 103       0      -9
## 104      -8      -8
## 105      -8      -8
## 106      -8      -8
## 107      -8      -8
## 108      -8      -8
## 109     600      -9
## 110       0       0
## 111       0       0
## 112      -9      -9
## 113      -9      -9
## 114    2700       0
## 115     380      -8
## 116      -9      -9
## 117     670       0
## 118      -9      -9
## 119      -8      -8
## 120     500      -8
## 121      -8      -8
## 122       0       0
## 123      -9      -9
## 124      -8    6600
## 125      -8      -8
## 126      -9      -9
```

```
## 127      -8      -8
## 128      -8      -8
## 129      -9      -9
## 130      -9      -9
## 131       0    4800
## 132      -8      -8
## 133    1200       0
## 134       0       0
## 135       0       0
## 136      -8    6600
## 137      -9      -9
## 138      -9      -9
## 139    1200      -9
## 140     400      -8
## 141      -8      -8
## 142      -9      -9
## 143      -8    2700
## 144      -8      -8
## 145     400      -8
## 146      -9      -9
## 147      -8      -8
## 148      -8      -8
## 149      -8      -8
## 150      -8      -8
## 151      -8      -9
## 152      -8      -8
## 153      -8      -8
## 154      -9    6000
## 155      -8      -8
## 156      -9    3300
## 157      -9      -9
## 158       0       0
## 159      -8    1200
## 160      -9      -9
## 161      -8      -8
## 162      -9      -9
## 163      -8      -8
## 164      -9      -9
## 165      -9      -9
## 166      -8      -8
## 167     350      -8
## 168      -8      -8
## 169      -9      -9
## 170      -9      -9
## 171      -8      -8
## 172      -8      -8
## 173      -8      -8
## 174      -8      -8
## 175      -8     650
## 176      -8     200
## 177      -9      -9
## 178      -8      -8
## 179     600      -8
## 180      -9      -9
```

```
## 181      -9      -9
## 182      -9      -9
## 183      -8      -8
## 184      -9      -9
## 185      -9      -9
## 186      -8      -8
## 187      -9      -9
## 188      -8      -8
## 189      -8      -9
## 190      -9      -9
## 191       0    7400
## 192      -9      -9
## 193      -9      -9
## 194      -8      -8
## 195      -9      -9
## 196      -8    5100
## 197      -8      -8
## 198      -9      -9
## 199       0       0
## 200      -9      -9
## 201      -8      -8
## 202      -8      -8
## 203      -9      -9
## 204      -8      -8
## 205      -9      -9
## 206      -8      -8
## 207      -8    2700
## 208      -9      -9
## 209      -8      -8
## 210      -9      -9
## 211     400      -8
## 212    2500       0
## 213      -9      -9
## 214      -9      -9
## 215     250     300
## 216      -9      -9
## 217      -9      -9
## 218      -9      -9
## 219      -8      -8
## 220     500      -8
## 221      -8     100
## 222      -9      -9
## 223      -8      -9
## 224      -8      -9
## 225      -9    3800
## 226      -9      -9
## 227    1200      -8
## 228      -9      -9
## 229      -8      -8
## 230      -8       0
## 231      -8      -8
## 232      -8    3000
## 233      -9      -9
## 234      -8      -8
```

```
## 235      -8      -8
## 236      80      -9
## 237      -8      -8
## 238       0       0
## 239      -9    3000
## 240    1200    3600
## 241     500      -9
## 242      -8      -8
## 243      -8      -8
## 244      -8      -8
## 245    1600      -9
## 246      -9      -9
## 247      -9      -9
## 248      -8      -8
## 249      -8      -8
## 250    2400      -8
## 251      -9      -9
## 252    1200      -8
## 253       0    8365
## 254      -8      -8
## 255      -8      -8
## 256      -8      -8
## 257      -8      -8
## 258      -9      -9
## 259      -8      -8
## 260      -9      -9
## 261      -8    2400
## 262      -8      -8
## 263      -8      -8
## 264      -8      -8
## 265      -9      -9
## 266      -9      -9
## 267      -8      -9
## 268      -9      -9
## 269      -8      -8
## 270      -8      -8
## 271      -9      -9
## 272      -8      -8
## 273      -8      -8
## 274      -9      -9
## 275      -9      -9
## 276      -9      -9
## 277      -9      -9
## 278       0       0
## 279      -9      -9
## 280      -8      -8
## 281      -9      -9
## 282      -8      -8
## 283      -8      -8
## 284      -8      -8
## 285      -8      -8
## 286      -8      -8
## 287      -9    6000
## 288      -9      -9
```

```
## 289      -8      -8
## 290      -9    5000
## 291      -9      -9
## 292      -8      -8
## 293      -9      -9
## 294      -9      -9
## 295      -9     330
## 296      -8      -8
## 297      -8      -8
## 298    3200      -8
## 299      -9      -9
## 300      -8      70
## 301      -9      -9
## 302      -9      -9
## 303      -9      -9
## 304      -8      -8
## 305      -9      -9
## 306      -9      -9
## 307      -9      -9
## 308      -8      -9
## 309      -9      -9
## 310      -8      -8
## 311      -9      -9
## 312      -8      -8
## 313      -9      -9
## 314      -8      -8
## 315      -9      -9
## 316      -9      -9
## 317    1100      -8
## 318      -8      -8
## 319      -9      -9
## 320      -8      -8
## 321      -9      -8
## 322      -8      -8
## 323      -8      -8
## 324      -8      -8
## 325       0       0
## 326       0       0
## 327      -9      -9
## 328      -9      -9
## 329    1100      -9
## 330      -9      -9
## 331       0       0
## 332      -8      -8
## 333     550      -8
## 334      -8      -8
## 335    2500    5000
## 336      -8      -8
## 337      -9      -9
## 338      -9      -9
## 339      -9      -9
## 340      -9      -9
## 341      -8    6900
## 342      -8     600
```

```
## 343    2500     850
## 344      -9      -9
## 345      -8      -8
## 346      -9      -9
## 347      -8    8365
## 348      -9      -9
## 349    1000      -9
## 350      -8      -8
## 351     720      -9
## 352      -9      -9
## 353      -8      -8
## 354      -9      -9
## 355      -9      -9
## 356     500       0
## 357      -8    1800
## 358      -8      -8
## 359      -8      -9
## 360      -9    2000
## 361      -8    3600
## 362      -8      -8
## 363    4575      -8
## 364      -8      -9
## 365      -8      -8
## 366      -8      -8
## 367      -8      -9
## 368    1000      -8
## 369      -8      -8
## 370      -8      -8
## 371      -8    5700
## 372     600      -8
## 373     600      -9
## 374      -8      -9
## 375      -8      -8
## 376     780      -9
## 377      -9      -9
## 378      -9      -9
## 379      -9      -9
## 380      -9      -9
## 381      -9      -9
## 382      -9      -9
## 383      -9      -9
## 384      -8      -8
## 385      -9      -9
## 386      -8      -8
## 387      -9      -9
## 388     500      -9
## 389      -8      -8
## 390      -9      -9
## 391      -9      -9
## 392      -9      -9
## 393      -9      -9
## 394    1200      -8
## 395      -9      -9
## 396     800      -8
```

```
## 397      -8      -8
## 398      -9      -9
## 399    2500      -9
## 400      -8      -8
## 401      -9      -9
## 402      -9      -9
## 403       0       0
## 404      -9      -9
## 405    1800      -9
## 406      -9      -9
## 407      -8      -8
## 408      -8      -8
## 409      -8      -8
## 410    1400      -9
## 411     470     340
## 412    4000      -8
## 413       0       0
## 414     600      -8
## 415      -9      -9
## 416      -9   17314
## 417      -9      -8
## 418      -9      -9
## 419      -9      -9
## 420      -8      -8
## 421      -9      -9
## 422      -8      -9
## 423      -9      -9
## 424      -9      -9
## 425      -8      -8
## 426      -8      -8
## 427    1000      -9
## 428      -8      -8
## 429      -9      -9
## 430      -9      -9
## 431      -8      -8
## 432      -9      -9
## 433      -8      -8
## 434      -9      -9
## 435    1200      -9
## 436      -8      -8
## 437     300      -8
## 438      -8      -8
## 439      -8      -9
## 440      -8      -8
## 441      -8      -8
## 442      -9      -9
## 443      -8      -8
## 444      -8      -8
## 445     420      -8
## 446      -8      -8
## 447     600      -9
## 448      -9      -9
## 449    1200       0
## 450      -8      -8
```

```
## 451     500      -8
## 452      -8      -8
## 453      -9      -9
## 454      -8      -8
## 455      -9      -9
## 456     900     350
## 457      -8      -8
## 458      -8      -8
## 459       0       0
## 460      -8      -8
## 461    2000      -8
## 462      -9      -9
## 463      -8      -8
## 464    1300      -8
## 465      -8      -8
## 466      -8      -8
## 467      -8      -8
## 468      -8      -8
## 469      -8      -8
## 470      -8      -8
## 471      -8      -8
## 472      -8      -8
## 473      -8      -8
## 474    1000      -8
## 475      -8      -8
## 476      -8    1000
## 477    1800    3000
## 478      -8      -8
## 479     300      -9
## 480    1200      -8
## 481      -8      -8
## 482      -8      -8
## 483      -8      -8
## 484      -8      -8
## 485      -8      -8
## 486     450      -8
## 487      -9      -9
## 488       0       0
## 489      -8      -8
## 490      -8      -8
## 491      -9      -9
## 492      -8      -8
## 493      -8      -8
## 494      -8      -8
## 495      -8      -8
## 496      -8      -8
## 497    1800    1400
## 498      -8      -8
## 499    1200      -8
## 500     590      -8
## 501      -8      -8
## 502    1200      -9
## 503      -8      -8
## 504     300      -8
```

```
## 505      -8      -8
## 506      -8      -8
## 507      -8      -8
## 508      -8      -8
## 509      -9      -9
## 510    5150      -9
## 511      -8    2200
## 512      -9      -9
## 513    3500      -9
## 514    1000      -8
## 515    3500      -8
## 516    1500      -8
## 517      -8      -8
## 518      -9      -9
## 519      -9      -9
## 520    1100      -8
## 521    1800      -9
## 522      -8      -8
## 523      -8      -8
## 524      -9      -9
## 525      -8      -8
## 526       0       0
## 527      -8      -8
## 528     300     150
## 529    2000    1800
## 530    1200      -9
## 531     500       0
## 532    5150      -8
## 533       0       0
## 534      -9      -9
## 535      -8      -8
## 536      -8      -8
## 537    2400   17314
## 538     520     160
## 539      -8      -8
## 540      -8      -8
## 541      -8      -8
## 542     600      -8
## 543      -8      -8
## 544      -8   10000
## 545     700      -9
## 546      -9      -9
## 547      -8      -9
## 548      -8     200
## 549      -8      -8
## 550      -8     600
## 551    1200      -8
## 552     500       0
## 553      -9      -8
## 554      -8      -8
## 555    3000     300
## 556      -9      -9
## 557      -9      -9
## 558      -9      -9
```

```
## 559      -9      -9
## 560      -9    2700
## 561      -8      -8
## 562      -8      -9
## 563      -8      -8
## 564      -8      -8
## 565      -8      -8
## 566      -9      -9
## 567     300      -8
## 568     200      -9
## 569    1000      -8
## 570      -8      -8
## 571      -8      -8
## 572      -9      -9
## 573      -8      -8
## 574    2400       0
## 575      -9      -9
## 576      -9      -9
## 577      -8      -8
## 578      -8      -8
## 579      -9      -9
## 580      -8      -8
## 581      -9      -9
## 582      -8      -8
## 583      -9      -9
## 584     800      -9
## 585      -8      -9
## 586      -8      -9
## 587      -9      -9
## 588     100      -8
## 589      -8      -8
## 590      -9      -9
## 591      -9      -9
## 592      -9      -9
## 593      -9      -9
## 594      -9      -9
## 595      -8      -8
## 596     250      -8
## 597    1000    1000
## 598     250      -8
## 599      -9      -9
## 600      -9      -9
## 601    1200       0
## 602       0       0
## 603      -8      -8
## 604      -9      -9
## 605      -8      -8
## 606       0      -8
## 607     600     250
## 608      -9      -9
## 609     400     660
## 610      -9      -9
## 611     300      -9
## 612    1800       0
```

```
## 613      -9      -9
## 614      -9      -9
## 615      -9      -9
## 616    2900    1000
## 617     500       0
## 618      -9      -9
## 619    1000      -8
## 620    1200       0
## 621    2000      -8
## 622      -8      -8
## 623      -9      -8
## 624    1000      -8
## 625     510      -8
## 626      -9      -8
## 627      -8      -8
## 628      -8      -8
## 629      -9      -9
## 630      -8      -8
## 631      -8      -8
## 632     500      -9
## 633      -9      -9
## 634    1000      -8
## 635      -9      -9
## 636       0       0
## 637      -8      -8
## 638    1200    7600
## 639      -8      -8
## 640      -9      -9
## 641      -9      -9
## 642     500      -8
## 643      -9      -9
## 644      -8      -8
## 645     400      -9
## 646      -8      -8
## 647      -9      -9
## 648      -9      -9
## 649    1200       0
## 650      -8      -8
## 651      -9      -9
## 652     480    1800
## 653    1000      -8
## 654     400      -8
## 655      -8      -8
## 656      -9      -9
## 657      -8    3000
## 658      -9      -9
## 659      -8      -8
## 660     500    2000
## 661      -9      -9
## 662      -8     280
## 663    1200      -8
## 664      -9      -9
## 665      -9      -9
## 666      -8      -8
```

```
## 667    1200       0
## 668     500      -8
## 669       0    1500
## 670      -8      -8
## 671      -8      -8
## 672    1000      -8
## 673      -8      -9
## 674      -8      -8
## 675    1100      -8
## 676    2400   10500
## 677      -8      -8
## 678      -9   13000
## 679      -8      -8
## 680    1400      -8
## 681    3500      -8
## 682      -8      -8
## 683     440      -8
## 684    1400      -9
## 685     300      -8
## 686      -9      -9
## 687    2000       0
## 688    1000      -9
## 689       0      -8
## 690     630      -8
## 691      -9      -9
## 692      -9      -9
## 693    3000      -9
## 694      -8      -9
## 695      -8      -8
## 696      -8      -8
## 697    1000     410
## 698      -9    1200
## 699     300      -8
## 700      -8    1300
## 701      -8      -8
## 702      -9      -9
## 703     480      -8
## 704      -9      -9
## 705      -9      -9
## 706      -9      -9
## 707     400     400
## 708      -8      -8
## 709      -9      -9
## 710     200      -8
## 711      -9      -9
## 712      -9      -9
## 713     550      -8
## 714      -9      -9
## 715       0       0
## 716      -8      -8
## 717    1500      -8
## 718     600      -9
## 719    1200    4000
## 720      -8    2100
```

```
## 721    2500       0
## 722      -8      -8
## 723      -8      -8
## 724     700      -8
## 725    3000    2000
## 726      -9      -9
## 727      -8      -8
## 728    1700   17314
## 729      -9      -9
## 730    1200      -8
## 731      -9      -9
## 732      -8      -8
## 733      -8      -8
## 734     800      -9
## 735      -9      -9
## 736      -9      -9
## 737    3000      -8
## 738     600     270
## 739      -8      -8
## 740     900      -8
## 741      -9      -9
## 742      -8    1000
## 743    1400     960
## 744      -8      -8
## 745      -9      -9
## 746      -8      -8
## 747      -9      -9
## 748      -8      -8
## 749      -9      -9
## 750     600      -8
## 751    1000    1500
## 752     200      -8
## 753      -8      -8
## 754    4000     600
## 755      -8      -8
## 756      -8      -8
## 757    1000      -8
## 758      -9      -8
## 759     300      -8
## 760    3600      -8
## 761      -9      -9
## 762      -9      -9
## 763      -8      -9
## 764      -9      -9
## 765      -8      -8
## 766      -9      -9
## 767    1400      -8
## 768      -8      -8
## 769      -8      -8
## 770      -8      -9
## 771      -8      -8
## 772    1000      -8
## 773      -8      -8
## 774      -8      -8
```

```
## 775      -9      -9
## 776      -8      -8
## 777      -9      -9
## 778      -8      -8
## 779    1200      -8
## 780    1000      -8
## 781     500    1000
## 782    1200      -8
## 783      -8      -8
## 784      -8      -8
## 785     600    1000
## 786      -9      -9
## 787      -8      -8
## 788      -8    1100
## 789      -8      -9
## 790    1500      -8
## 791    4000    7000
## 792      -8    2000
## 793    2400      -8
## 794      -8      -8
## 795     800      -8
## 796      -8      -8
## 797      -9      -8
## 798      -8      -8
## 799       0      -8
## 800      -9      -9
## 801      -8      -8
## 802      -8      -8
## 803    2000      -8
## 804      -9      -9
## 805      -9      -9
## 806    1800      -8
## 807    5150     800
## 808      -8      -8
## 809       0      -9
## 810     720      -8
## 811    2500    1200
## 812     500      -8
## 813    1000      -8
## 814      -9      -9
## 815      -8      -8
## 816    2000      -9
## 817      -9      -9
## 818      -8    4700
## 819       0    2000
## 820     430      -8
## 821      -8      -8
## 822      -8    2000
## 823    1200      -9
## 824    1100    1000
## 825    5150      -8
## 826      -9      -9
## 827      -9      -9
## 828      -9      -9
```

```
## 829      -9      -9
## 830      -9      -9
## 831      -9      -9
## 832      -9      -9
## 833      -9      -9
## 834      -9      -9
## 835      -9      -9
## 836      -9      -9
## 837      -9      -9
## 838      -8      -8
## 839      -9      -9
## 840      -9      -9
## 841      -9      -9
## 842      -9      -9
## 843      -9      -9
## 844      -9      -9
## 845      -9      -9
## 846      -9      -9
## 847      -9      -9
## 848      -8      -8
## 849      -8      -8
## 850      -8      -8
## 851      -9      -9
## 852    7000      -8
## 853      -8      -8
## 854    2400   31700
## 855      -8      -8
## 856      -9      -9
## 857      -8   40000
## 858    6700      -9
## 859      -9      -9
## 860      -9      -9
## 861      -9      -9
## 862       0      -8
## 863      -9      -9
## 864      -9      -9
## 865      -9      -9
## 866      -8      -8
## 867      -9      -9
## 868      -8      -8
## 869      -9      -9
## 870      -9      -9
## 871      -9      -9
## 872      -9      -9
## 873      -9      -9
## 874      -9      -9
## 875      -9      -9
## 876      -9      -9
## 877      -8      -9
## 878      -9      -9
## 879      -9      -9
## 880      -9      -9
## 881      -9      -9
## 882      -9      -9
```

```
## 883      -9      -9
## 884   11300     490
## 885      -9      -9
## 886      -8      -8
## 887      -9      -9
## 888      -8      -8
## 889      -9      -9
## 890      -9      -9
## 891      -9      -9
## 892      -9      -9
## 893      -9      -9
## 894      -9      -9
## 895      -9      -9
## 896      -9      -9
## 897    1500    4000
## 898      -9      -9
## 899      -9      -9
## 900      -9      -9
## 901      -9      -9
## 902      -9      -9
## 903      -9      -9
## 904      -9      -9
## 905      -8      -8
## 906    4200      -9
## 907      -9      -9
## 908    1500      -8
## 909      -8      -8
## 910      -9      -9
## 911      -9      -9
## 912      -9      -9
## 913     720      -9
## 914      -9      -9
## 915      -8      -8
## 916      -8      -9
## 917      -9      -9
## 918    2500      -8
## 919      -9      -9
## 920      -8      -9
## 921      -9      -9
## 922     750    3500
## 923      -8      -8
## 924      -8      -8
## 925      -8      -8
## 926      -8      -8
## 927      -9  120000
## 928      -9      -9
## 929      -8      -8
## 930      -9      -9
## 931      -9      -9
## 932    1800    7000
## 933      -9      -9
## 934   14191   42400
## 935    4300     180
## 936      -8      -8
```

```
## 937     400       0
## 938      -9      -9
## 939    3500      -9
## 940      -9      -9
## 941      -9      -9
## 942     600       0
## 943      -9      -9
## 944      -8      -9
## 945      -9    2700
## 946      -9      -9
## 947      -9      -9
## 948      -9      -9
## 949      -9      -9
## 950      -8      -8
## 951      -8      -8
## 952      -9      -9
## 953      -8      -8
## 954       0      -9
## 955      -8      -8
## 956      -8      -8
## 957      -8      -8
## 958   12000      -9
## 959      -8      -9
## 960      -8      -8
## 961      -9      -9
## 962    5400    3000
## 963      -9      -9
## 964      -8      -8
## 965      -8      -8
## 966      -9      -9
## 967      -9      -9
## 968      -8      -8
## 969      -8      -8
## 970      -8   19000
## 971      -8      -8
## 972      -9      -9
## 973     460   11300
## 974      -8      -8
## 975      -8      -8
## 976      -8      -8
## 977      -8      -8
## 978      -9      -9
## 979      -8      -8
## 980      -8      -8
## 981      -8      -8
## 982      -8      -8
## 983      -8      -9
## 984      -9      -9
## 985      -8      -8
## 986      -9      -9
## 987      -8      -8
## 988    1200      -8
## 989      -8    7900
## 990    4200    2800
```

```
## 991      -9      -9
## 992   12000      -8
## 993      -9      -9
## 994      -8      -8
## 995    5000   15000
## 996      -8   26000
## 997      -8      -8
## 998      -8      -8
## 999    2400   16200
## 1000     -8      -8
## 1001   2000   10000
## 1002     -8      -8
## 1003     -8      -8
## 1004     -8      -8
## 1005   1600    5000
## 1006   4200      -8
## 1007     -9      -9
## 1008    340   43700
## 1009    720    1000
## 1010     -8      -8
## 1011     -8      -8
## 1012     -8      -8
## 1013    540      -8
## 1014   2500    7200
## 1015     -8      -8
## 1016   8500    9700
## 1017     -8      -8
## 1018     -8      -8
## 1019     -8      -8
## 1020    800      -8
## 1021   6500    3400
## 1022     -8      -8
## 1023     -8      -8
## 1024     -8      -8
## 1025     -9      -9
## 1026     -8      -8
## 1027     -8      -8
## 1028     -8      -8
## 1029     -9      -9
## 1030     -9      -9
## 1031   2000      -9
## 1032   2400      -8
## 1033     -8      -9
## 1034     -8   13700
## 1035     -9      -9
## 1036      0    6300
## 1037     -8      -8
## 1038     -9    6300
## 1039      0   12100
## 1040   2100    1900
## 1041     -8  191949
## 1042     -8      -8
## 1043     -9    7500
## 1044   1100    3000
```

```
## 1045   4000    2000
## 1046     -9      -9
## 1047     -8      -8
## 1048   9800    7400
## 1049     -8      -8
## 1050   2200    5200
## 1051   3300     600
## 1052     -9      -9
## 1053     -8      -8
## 1054  14191      -9
## 1055     -8      -8
## 1056     -9      -9
## 1057     -9      -9
## 1058      0    1400
## 1059     -8      -8
## 1060     -8      -8
## 1061   5200   16800
## 1062   5000    1300
## 1063   2900    8700
## 1064   2200    1400
## 1065  14191  191949
## 1066   1800      -9
## 1067   4400   11100
## 1068   3600       0
## 1069   2000      -8
## 1070     -8      -8
## 1071     -8      -8
## 1072   1000     800
## 1073     -8      -9
## 1074    810     530
## 1075    600      -8
## 1076     -9      -9
## 1077     -8      -8
## 1078   2100    2900
## 1079   1100      -8
## 1080     -8      -8
## 1081   3000      -8
## 1082     -8      -8
## 1083     -9      -9
## 1084     -9      -9
## 1085     -8      -8
## 1086     -9      -9
## 1087     -9      -9
## 1088     -8      -8
## 1089   7100   55000
## 1090     -9      -9
## 1091     -8      -8
## 1092     -9      -9
## 1093      0   31200
## 1094      0       0
## 1095     -9   12000
## 1096  11100   39600
## 1097     -8      -8
## 1098   3900      -9
```

```
## 1099   7000    9000
## 1100     -8      -8
## 1101   7500  191949
## 1102   2800      -9
## 1103     -8      -8
## 1104     -8      -8
## 1105   3300    9300
## 1106     -8      -8
## 1107     -8      -8
## 1108   8400      -8
## 1109     -9      -9
## 1110     -9      -9
## 1111     -8      -8
## 1112    800       0
## 1113     -8      -8
## 1114     -9      -9
## 1115     -8      -8
## 1116   1300   75000
## 1117   2000      -8
## 1118     -9      -9
## 1119     -8      -8
## 1120     -8      -8
## 1121   1800      -9
## 1122   4100      -8
## 1123      0       0
## 1124     -9      -9
## 1125     -8      -8
## 1126     -8      -8
## 1127     -9      -9
## 1128   6300    2500
## 1129     -8      -8
## 1130     -9      -9
## 1131   6000    5000
## 1132     -9      -9
## 1133     -8      -8
## 1134     -8      -8
## 1135     -8      -8
## 1136  10000   20000
## 1137     -8      -8
## 1138     -8      -8
## 1139     -8      -8
## 1140     -8      -8
## 1141   2400       0
## 1142     -9      -9
## 1143   2900    6100
## 1144   3900      -8
## 1145   5000      -9
## 1146   4700      -9
## 1147     -8      -8
## 1148     -9      -9
## 1149     -9      -9
## 1150     -9      -9
## 1151     -9      -9
## 1152     -8      -8
```

```
## 1153   2400    3600
## 1154     -9      -9
## 1155     -8      -8
## 1156   4000    7500
## 1157     -8      -8
## 1158   6500      -8
## 1159   1200      -8
## 1160     -8      -8
## 1161     -8      -8
## 1162     -8    9000
## 1163     -9      -9
## 1164   5400   13600
## 1165   2500      -8
## 1166     -8      -8
## 1167     -9      -9
## 1168     -8      -8
## 1169    490    2500
## 1170     -8      -8
## 1171   1800      -8
## 1172     -8      -8
## 1173     -8      -8
## 1174     -8      -8
## 1175     -9      -8
## 1176   2500    2700
## 1177     -9    8000
## 1178     -9      -9
## 1179   4200      -8
## 1180     -9      -9
## 1181   1500      -8
## 1182     -8      -9
## 1183     -8      -8
## 1184     -8      -8
## 1185   2000    2000
## 1186     -9      -9
## 1187     -9      -9
## 1188     -9    1200
## 1189     -8      -8
## 1190      0       0
## 1191   5500      -9
## 1192     -8      -8
## 1193     -9      -9
## 1194     -9      -9
## 1195   3300      -8
## 1196   1500       0
## 1197   1400    3300
## 1198   8500      -8
## 1199     -9      -9
## 1200     -9      -9
## 1201    500      -8
## 1202   1300    1100
## 1203   6900    3800
## 1204     -9      -9
## 1205     -8      -9
## 1206    890    1100
```

```
## 1207   4000       0
## 1208      0      -9
## 1209     -8      -8
## 1210     -9      -9
## 1211   3600      -8
## 1212     -9      -9
## 1213   6000      -8
## 1214     -8    6200
## 1215     -9      -9
## 1216     -8      -9
## 1217   2300    2000
## 1218   1200    4700
## 1219   5000      -8
## 1220     -9      -9
## 1221     -9      -9
## 1222     -9      -9
## 1223   6000      -8
## 1224   2800      -8
## 1225     -8      -8
## 1226     -8       0
## 1227     -8      -9
## 1228   5000      -8
## 1229     -8      -8
## 1230     -9      -9
## 1231    750      -8
## 1232   1600       0
## 1233     -8      -8
## 1234   2200   18000
## 1235    600    2600
## 1236   2400      -9
## 1237   1300      -9
## 1238     -8      -8
## 1239     -8    4100
## 1240     -9      -9
## 1241     -9      -9
## 1242     -9    1200
## 1243   2400      -9
## 1244   6200   11200
## 1245   5000   10000
## 1246     -8      -8
## 1247     -9      -9
## 1248    900      -8
## 1249     -9      -9
## 1250    500      -9
## 1251     -8      -9
## 1252   1800      -9
## 1253     -9      -9
## 1254     -8      -8
## 1255   1800      -8
## 1256     -9      -9
## 1257   1200       0
## 1258   4500      -8
## 1259      0       0
## 1260     -8      -8
```

```
## 1261   9000      -8
## 1262     -9      -9
## 1263      0     170
## 1264     -9      -9
## 1265     -9   70000
## 1266   1200      -8
## 1267     -8    4600
## 1268     -8      -8
## 1269   1000    3000
## 1270     -9      -9
## 1271   1000      -8
## 1272     -9      -9
## 1273     -8      -8
## 1274     -9      -9
## 1275     -9      -9
## 1276     -8      -8
## 1277   3000      -8
## 1278     -8      -8
## 1279     -8      -8
## 1280     -9      -8
## 1281     -8      -8
## 1282   5000      -8
## 1283     -9      -9
## 1284     -9      -9
## 1285   1400    4900
## 1286     -8      -8
## 1287   1300      -8
## 1288    360      -9
## 1289     -9      -9
## 1290     -9      -9
## 1291      0    5800
## 1292     -8      -9
## 1293     -9      -9
## 1294    700    1400
## 1295     -9      -9
## 1296   1000      -8
## 1297     -9      -9
## 1298   4300    4100
## 1299     -9      -9
## 1300     -9      -9
## 1301     -9      -9
## 1302   4900    7500
## 1303   5000   10000
## 1304    500      -8
## 1305   2600    3200
## 1306     -9      -9
## 1307      0       0
## 1308     -9      -9
## 1309     -9      -9
## 1310     -9      -9
## 1311   1000      -8
## 1312     -8      -9
## 1313   6400    2800
## 1314      0       0
```

```
## 1315     -9      -9
## 1316     -9      -9
## 1317   2200    9000
## 1318     -8    6500
## 1319   2400    5200
## 1320     -8    2400
## 1321     -8    6500
## 1322     -9      -9
## 1323     -9   15700
## 1324   2100      -8
## 1325     -8      -8
## 1326     -9      -9
## 1327   3600       0
## 1328     -8      -8
## 1329     -8      -8
## 1330    240      -8
## 1331    750    3500
## 1332   4900       0
## 1333   5000      -9
## 1334     -8      -8
## 1335   4000   45000
## 1336     -9      -9
## 1337     -9      -9
## 1338      0    2000
## 1339     -9      -9
## 1340     -8      -8
## 1341     -8      -8
## 1342     -8    3500
## 1343    840    5200
## 1344     -9      -9
## 1345     -8      -8
## 1346   6300   34300
## 1347     -9      -9
## 1348   3600      -8
## 1349   2500      -9
## 1350   3600    6000
## 1351   5500   57000
## 1352   1100    3900
## 1353      0      -9
## 1354   3000      -8
## 1355     -8      -8
## 1356   1000   20000
## 1357     -9    3300
## 1358  14191       0
## 1359     -9      -9
## 1360     -9      -9
## 1361   1000       0
## 1362     -9      -9
## 1363     -9      -9
## 1364     -8      -8
## 1365   9500      -8
## 1366     -9      -9
## 1367   6000      -8
## 1368     -9      -9
```

```
## 1369   3500      -8
## 1370     -8      -8
## 1371     -9      -9
## 1372   1400      -8
## 1373   2400      -8
## 1374   3200      -9
## 1375   1200   10400
## 1376   1200      -8
## 1377     -9  191949
## 1378     -9      -9
## 1379     -9      -9
## 1380    850    1300
## 1381      0       0
## 1382    930    3700
## 1383     -9      -9
## 1384   2600      -8
## 1385    500      -8
## 1386     -9      -9
## 1387   2200      -9
## 1388     -8    3000
## 1389   5000   10000
## 1390   5400      -9
## 1391     -9      -9
## 1392   2200      -8
## 1393     -9      -9
## 1394   4000      -8
## 1395     -9      -9
## 1396     -9      -9
## 1397     -9      -9
## 1398   3000      -9
## 1399   3100      -8
## 1400   1400      -8
## 1401   6200    8800
## 1402  14191   12500
## 1403     -9      -9
## 1404     -9      -9
## 1405     -9      -9
## 1406   1100    1400
## 1407   5000    8000
## 1408     -9      -9
## 1409     -9      -9
## 1410     -9      -9
## 1411   2100    2300
## 1412   2000       0
## 1413     -8      -8
## 1414     -9      -9
## 1415   5000    8000
## 1416     -8      -9
## 1417     -9      -9
## 1418     -8      -8
## 1419     -8      -8
## 1420   3500    2500
## 1421     -9      -9
## 1422     -9      -9
```

```
## 1423     -9      -9
## 1424   1000    1800
## 1425     -8      -8
## 1426     -8      -8
## 1427   3000      -8
## 1428     -9      -9
## 1429   2000   21300
## 1430   3600      -9
## 1431     -9      -9
## 1432   4000    6000
## 1433   2000      -8
## 1434   3000      -8
## 1435   1200   24000
## 1436     -8    9000
## 1437   3600      -8
## 1438  14191    1200
## 1439   4800      -8
## 1440   1900      -9
## 1441   1200      -9
## 1442   2500      -8
## 1443   7500      -9
## 1444     -8      -8
## 1445   8000      -9
## 1446   1200    1200
## 1447      0      -9
## 1448   3000    5000
## 1449   8000   24000
## 1450   1800    3100
## 1451   1200    2300
## 1452      0    1000
## 1453     -8      -8
## 1454     -8      -8
## 1455     -8      -8
## 1456     -9      -9
## 1457     -9      -9
## 1458    960     230
## 1459     -9      -9
## 1460   6400      -9
## 1461     -9      -9
## 1462   8800   10900
## 1463      0       0
## 1464     -9      -9
## 1465     -9   25000
## 1466     -9      -9
## 1467     -9      -9
## 1468     -9      -9
## 1469   9300    6700
## 1470     -9      -9
## 1471     -8      -9
## 1472     -8      -8
## 1473     -9   30400
## 1474   4400  124000
## 1475     -8      -8
## 1476     -9      -9
```

```
## 1477     -8   25000
## 1478   1800      -8
## 1479     -9      -9
## 1480     -8    8300
## 1481     -9      -9
## 1482     -9      -9
## 1483     -9      -9
## 1484     -8   23100
## 1485     -8       0
## 1486     -9      -9
## 1487     -9      -9
## 1488     -9      -9
## 1489     -9      -9
## 1490     -9      -9
## 1491     -9      -9
## 1492     -9      -9
## 1493     -9      -9
## 1494     -9      -9
## 1495     -9      -9
## 1496     -9      -9
## 1497     -9      -9
## 1498     -9      -9
## 1499     -9      -9
## 1500     -9      -9
## 1501     -9      -9
## 1502     -9      -9
## 1503     -9      -9
## 1504     -9      -9
## 1505     -9      -9
## 1506     -9      -9
## 1507     -8      -8
## 1508     -9      -9
## 1509     -9      -9
## 1510     -9      -9
## 1511     -9      -9
## 1512     -9      -9
## 1513     -9      -9
## 1514     -9      -9
## 1515     -9      -9
## 1516     -9      -9
## 1517     -8      -8
## 1518     -9      -9
## 1519     -9      -9
## 1520     -8      -8
## 1521     -9      -9
## 1522     -8      -8
## 1523     -9      -9
## 1524     -9      -9
## 1525     -9      -9
## 1526     -9      -9
## 1527     -9      -9
## 1528     -9      -9
## 1529     -9      -9
## 1530     -9      -9
```

```
## 1531     -9      -9
## 1532     -9      -9
## 1533     -9      -9
## 1534     -9      -9
## 1535     -9      -9
## 1536     -9      -9
## 1537     -9      -9
## 1538     -9      -9
## 1539     -9      -9
## 1540     -9      -9
## 1541     -9      -9
## 1542     -9      -9
## 1543     -8      -8
## 1544      0       0
## 1545   8000   14000
## 1546     -9      -9
## 1547   1200    9400
## 1548     -9      -9
## 1549     -9      -9
## 1550   9700   25600
## 1551   5200   70000
## 1552   5000      -9
## 1553     -9      -9
## 1554   1000      -8
## 1555  18000   15000
## 1556     -9      -9
## 1557   7700    9900
## 1558     -9      -9
## 1559     -9      -9
## 1560  18000   30000
## 1561     -9      -9
## 1562     -9      -9
## 1563     -9      -9
## 1564     -9      -9
## 1565     -9      -9
## 1566     -9      -9
## 1567     -9      -9
## 1568     -9      -9
## 1569     -9      -9
## 1570     -8      -8
## 1571     -8      -8
## 1572     -8      -8
## 1573     -8      -8
## 1574     -8      -8
## 1575     -8      -8
## 1576     -8  402571
## 1577     -8      -8
## 1578     -8      -8
## 1579     -9      -9
## 1580     -9      -9
## 1581     -9      -9
## 1582   9800       0
## 1583     -8      -8
## 1584  11800    3000
```

```
## 1585     -8      -8
## 1586   4800      -9
## 1587     -8      -8
## 1588     -8      -8
## 1589     -9      -9
## 1590     -9      -9
## 1591     -8      -8
## 1592     -8      -8
## 1593     -8      -8
## 1594     -8      -8
## 1595     -9      -9
## 1596     -9      -9
## 1597     -9      -9
## 1598     -9      -9
## 1599     -9      -9
## 1600     -9      -9
## 1601     -8   19300
## 1602     -8      -8
## 1603     -9      -9
## 1604  12600    6800
## 1605     -8      -8
## 1606     -8      -8
## 1607  12000      -9
## 1608     -9      -9
## 1609     -9      -9
## 1610     -8      -8
## 1611     -8      -9
## 1612     -8      -8
## 1613     -8      -8
## 1614     -8      -8
## 1615     -8      -8
## 1616     -8      -8
## 1617     -8      -8
## 1618     -8      -8
## 1619   4500      -9
## 1620     -9      -9
## 1621     -8      -8
## 1622   8800      -9
## 1623     -8      -8
## 1624  10700   23300
## 1625     -8      -8
## 1626  33337  182000
## 1627     -8      -8
## 1628     -8      -8
## 1629     -8      -9
## 1630     -8      -8
## 1631     -9      -9
## 1632   7100    4300
## 1633     -8      -8
## 1634     -8      -8
## 1635     -8      -8
## 1636     -9      -9
## 1637     -8      -8
## 1638   9100   11600
```

```
## 1639     -8      -8
## 1640     -9      -9
## 1641   4300   19200
## 1642     -8      -8
## 1643   8900      -8
## 1644     -8      -8
## 1645     -9      -9
## 1646     -8      -8
## 1647   4300   13400
## 1648     -8      -8
## 1649     -8      -8
## 1650   6400   35400
## 1651   8800   23000
## 1652   9400   73000
## 1653   5000   48700
## 1654     -8      -8
## 1655     -9      -9
## 1656   7200   26300
## 1657     -8      -8
## 1658     -8      -8
## 1659   7800   12200
## 1660     -8      -8
## 1661     -8      -8
## 1662     -8      -8
## 1663     -8      -8
## 1664     -8   13900
## 1665     -8      -8
## 1666  20000      -9
## 1667     -8      -8
## 1668     -8      -8
## 1669     -8      -8
## 1670     -8      -8
## 1671     -8      -8
## 1672   1200    5400
## 1673     -8      -8
## 1674  17700   45600
## 1675     -8      -8
## 1676   4400  113000
## 1677  20400       0
## 1678   2400   18800
## 1679     -9      -9
## 1680  17600   40100
## 1681   1400    5800
## 1682     -9  189000
## 1683   9200   52000
## 1684   8900   20200
## 1685     -8      -8
## 1686     -9  137000
## 1687      0   93000
## 1688     -9  178000
## 1689  13000  104000
## 1690  15700   13200
## 1691  13700   17900
## 1692     -8      -8
```

```
## 1693     -8   12900
## 1694   8100   34500
## 1695    300      -8
## 1696     -8      -8
## 1697     -9      -9
## 1698     -9      -8
## 1699     -8      -8
## 1700     -9      -9
## 1701     -9      -9
## 1702     -8      -8
## 1703   7800   24100
## 1704     -8      -8
## 1705     -8      -8
## 1706   6000  100000
## 1707   1300   11900
## 1708   7000      -8
## 1709   1300   11600
## 1710  14200   22700
## 1711  13800    7900
## 1712    830   19700
## 1713     -8      -8
## 1714   4000      -8
## 1715   2400   33500
## 1716  16300   26100
## 1717   8400   64000
## 1718     -8      -8
## 1719     -9      -9
## 1720   6600   18900
## 1721      0   39900
## 1722  15100      -9
## 1723     -9      -9
## 1724   2000    1500
## 1725  16500  402571
## 1726     -8      -8
## 1727      0   95000
## 1728     -8   43800
## 1729  17000   69000
## 1730     -8      -8
## 1731     -8      -8
## 1732     -8      -8
## 1733   3000   15000
## 1734     -9      -9
## 1735     -8      -8
## 1736   4200      -8
## 1737  10000      -8
## 1738     -8   38100
## 1739     -8      -8
## 1740  10800  127000
## 1741     -8      -8
## 1742   5000   16000
## 1743     -8      -8
## 1744     -9      -9
## 1745     -8      -8
## 1746   2000      -9
```

```
## 1747     -8   37400
## 1748     -8      -8
## 1749     -8      -8
## 1750  12400    8600
## 1751   6500   21300
## 1752     -9      -9
## 1753     -8      -8
## 1754   2900    1700
## 1755     -8      -8
## 1756      0       0
## 1757     -8      -8
## 1758   5500   29800
## 1759     -8      -8
## 1760  20800   62000
## 1761     -8      -8
## 1762   2200      -8
## 1763     -9      -9
## 1764   7000      -8
## 1765   3600      -9
## 1766  10800   16700
## 1767  12000      -8
## 1768     -9      -9
## 1769     -8      -8
## 1770     -8      -8
## 1771      0   43700
## 1772  20000      -8
## 1773     -9      -9
## 1774     -8   14200
## 1775     -8      -8
## 1776     -8      -8
## 1777     -8      -8
## 1778   3900      -9
## 1779     -8      -9
## 1780     -8      -8
## 1781     -8      -8
## 1782     -8      -8
## 1783   3800    5800
## 1784     -8      -8
## 1785  12000  200000
## 1786     -9      -9
## 1787  16000   20000
## 1788     -9      -8
## 1789     -9      -9
## 1790     -9      -9
## 1791     -8      -8
## 1792     -9      -9
## 1793     -9      -9
## 1794     -9      -9
## 1795     -8      -8
## 1796     -9      -9
## 1797     -8      -8
## 1798  10300   17200
## 1799    980   25600
## 1800     -9      -9
```

```
## 1801   6100      -8
## 1802   4600   18200
## 1803     -8    8800
## 1804     -8    4000
## 1805   5000      -8
## 1806     -9      -9
## 1807     -8   66000
## 1808     -8      -8
## 1809   9800      -8
## 1810   3200      -9
## 1811     -9      -9
## 1812     -9      -9
## 1813     -9    3200
## 1814     -8      -9
## 1815     -8      -8
## 1816     -9      -9
## 1817     -8      -8
## 1818   1500    2000
## 1819  33337   91000
## 1820     -8      -9
## 1821     -8      -9
## 1822     -8      -9
## 1823     -9      -9
## 1824   8000   42700
## 1825     -9      -9
## 1826     -9  106000
## 1827   5600   13300
## 1828  10000      -8
## 1829     -9      -9
## 1830     -9      -9
## 1831     -9      -9
## 1832     -9      -9
## 1833   4000      -8
## 1834     -9      -9
## 1835     -9      -9
## 1836     -8  402571
## 1837     -9      -9
## 1838  10000      -9
## 1839   4400      -8
## 1840   2600      -8
## 1841     -9      -9
## 1842     -9      -9
## 1843     -9      -9
## 1844     -9      -9
## 1845   8900   25000
## 1846   2700   12000
## 1847     -9      -9
## 1848     -8      -8
## 1849     -8      -8
## 1850     -9      -9
## 1851   6000   14000
## 1852   2000    4100
## 1853     -9      -9
## 1854     -9      -9
```

```
## 1855     -8   60000
## 1856     -9      -9
## 1857     -8      -8
## 1858     -9      -9
## 1859     -9      -9
## 1860  11500      -9
## 1861     -8   26700
## 1862     -9      -9
## 1863     -9   47900
## 1864   9800   26200
## 1865     -9      -9
## 1866   4000   10000
## 1867   1000   24100
## 1868  10000   25700
## 1869      0   11000
## 1870   2000      -9
## 1871   6600    2900
## 1872     -9      -9
## 1873   1300   13500
## 1874   1900    2200
## 1875   3000    5000
## 1876     -9      -9
## 1877   8000   27900
## 1878     -8      -8
## 1879  10000      -8
## 1880     -9      -9
## 1881     -9      -9
## 1882     -9      -9
## 1883     -8      -8
## 1884     -8   10000
## 1885   2400      -8
## 1886    800   36000
## 1887     -8      -8
## 1888     -8      -9
## 1889   1800   61000
## 1890     -9      -9
## 1891   1300  113000
## 1892     -9      -9
## 1893     -8      -8
## 1894  17000      -8
## 1895     -8   55000
## 1896   5000    5000
## 1897     -9      -9
## 1898   3800      -8
## 1899     -8      -8
## 1900     -9      -9
## 1901   7000    2400
## 1902     -8      -8
## 1903      0   53000
## 1904     -9      -9
## 1905     -9      -8
## 1906     -9      -9
## 1907   8000   20000
## 1908     -8      -8
```

```
## 1909   6000      -8
## 1910     -9   30000
## 1911   8500   15000
## 1912      0      -8
## 1913     -9      -9
## 1914     -9      -9
## 1915     -8      -8
## 1916   4500    2200
## 1917     -8      -9
## 1918   2500      -9
## 1919     -9      -9
## 1920   2000   35000
## 1921   7200      -8
## 1922     -9      -9
## 1923   1500   10400
## 1924  33337   42000
## 1925     -9      -9
## 1926   2600    4000
## 1927     -9      -9
## 1928   5700   18100
## 1929  33337      -9
## 1930  11500      -9
## 1931     -8      -8
## 1932   3000      -8
## 1933     -9      -9
## 1934     -9      -9
## 1935     -9      -9
## 1936     -9      -9
## 1937     -9      -9
## 1938     -9      -9
## 1939     -9      -9
## 1940     -9      -9
## 1941     -9      -9
## 1942     -9      -9
## 1943     -9      -9
## 1944     -9      -9
## 1945     -9      -9
## 1946     -9      -9
## 1947     -9      -9
## 1948     -9      -9
## 1949     -9      -9
## 1950     -9      -9
## 1951     -9      -9
## 1952     -9      -9
## 1953     -9      -9
## 1954     -9      -9
## 1955     -9      -9
## 1956     -9      -9
## 1957     -9      -9
## 1958     -9      -9
## 1959     -9      -9
## 1960     -9      -9
## 1961     -9      -9
## 1962     -9      -9
```

```
## 1963     -9      -9
## 1964     -9      -9
## 1965     -9      -9
## 1966     -9      -9
## 1967     -9      -9
## 1968     -9      -9
## 1969     -9      -9
## 1970     -9      -9
## 1971     -9      -9
## 1972     -9      -9
## 1973     -9      -9
## 1974     -9      -9
## 1975     -9      -9
## 1976     -9      -9
## 1977     -9      -9
## 1978     -9      -9
## 1979     -9      -9
## 1980     -9      -9
## 1981     -9      -9
## 1982     -9      -9
## 1983     -9      -9
## 1984     -9      -9
## 1985     -9      -9
## 1986     -9      -9
## 1987     -9      -9
## 1988     -9      -9
## 1989     -9      -9
## 1990     -9      -9
## 1991     -9      -9
## 1992     -9      -9
## 1993     -9      -9
## 1994     -9      -9
## 1995     -9      -9
## 1996     -9      -9
## 1997     -9      -9
## 1998     -9      -9
## 1999     -9      -9
## 2000     -9      -9
## 2001     -9      -9
## 2002     -9      -9
## 2003     -9      -9
## 2004     -9      -9
## 2005     -9      -9
## 2006     -9      -9
## 2007     -9      -9
## 2008     -9      -9
## 2009     -9      -9
## 2010     -9      -9
## 2011     -9      -9
## 2012     -8      -8
## 2013     -8      -8
## 2014     -8      -8
## 2015     -9      -9
## 2016     -9      -9
```

```
## 2017     -8      -8
## 2018     -9      -9
## 2019     -9      -9
## 2020     -9      -9
## 2021     -8      -8
## 2022     -9      -9
## 2023     -9      -9
## 2024     -9      -9
## 2025     -9   27200
## 2026     -9      -9
## 2027     -9      -9
## 2028  20000      -8
## 2029     -9      -9
## 2030     -9      -9
## 2031     -9      -9
## 2032     -9      -9
## 2033     -9      -9
## 2034     -9      -9
## 2035     -9      -9
## 2036     -9      -9
## 2037   3100      -9
## 2038     -9      -9
## 2039     -8    4500
## 2040     -9      -9
## 2041     -9      -9
## 2042     -9      -9
## 2043     -9      -9
## 2044     -8      -8
## 2045     -9      -9
## 2046     -9      -9
## 2047     -9      -9
## 2048     -9      -9
## 2049   6500      -9
## 2050     -9      -9
## 2051     -8      -8
## 2052     -9      -9
## 2053     -9      -9
## 2054     -9      -9
## 2055     -9      -9
## 2056     -8      -8
## 2057     -9      -9
## 2058     -9      -9
## 2059     -9      -9
## 2060     -9      -9
## 2061     -9      -9
## 2062     -8      -8
## 2063     -9      -9
## 2064     -9      -9
## 2065  60000      -8
## 2066  26000      -9
## 2067     -9      -9
## 2068     -9      -9
## 2069     -9      -9
## 2070     -9      -9
```

```
## 2071  58000   22600
## 2072     -9      -9
## 2073     -9      -9
## 2074     -9      -9
## 2075     -9      -9
## 2076     -8      -8
## 2077     -9      -9
## 2078     -9      -9
## 2079     -9      -9
## 2080     -9      -9
## 2081     -8      -8
## 2082  35600      -8
## 2083     -9      -9
## 2084     -9      -9
## 2085     -9      -9
## 2086     -9      -9
## 2087     -9      -9
## 2088     -9      -9
## 2089     -9      -9
## 2090     -9      -9
## 2091     -9      -9
## 2092     -9      -9
## 2093     -9      -9
## 2094     -9      -9
## 2095     -8      -8
## 2096     -9      -9
## 2097     -9      -9
## 2098     -9      -9
## 2099     -9      -9
## 2100     -9      -9
## 2101     -9      -9
## 2102     -9      -9
## 2103     -8      -8
## 2104     -9      -9
## 2105     -9      -9
## 2106     -8      -8
## 2107     -9      -9
## 2108  32900      -9
## 2109     -9  312000
## 2110     -9      -9
## 2111     -9      -9
## 2112     -9      -9
## 2113     -8      -8
## 2114     -9      -9
## 2115     -9      -9
## 2116     -9      -9
## 2117     -9      -9
## 2118  94000      -9
## 2119  12000      -9
## 2120     -9      -9
## 2121     -9      -9
## 2122     -9      -9
## 2123     -9      -9
## 2124     -9      -9
```

```
## 2125     -9      -9
## 2126     -9      -9
## 2127     -8      -8
## 2128     -9      -9
## 2129     -9      -9
## 2130     -9      -9
## 2131     -9      -9
## 2132  66000  451000
## 2133      0       0
## 2134  20400      -8
## 2135     -9      -9
## 2136     -8      -8
## 2137     -9      -9
## 2138     -9      -9
## 2139  15100       0
## 2140     -9      -9
## 2141     -9      -9
## 2142     -9      -9
## 2143     -9      -9
## 2144     -9      -9
## 2145     -9      -9
## 2146     -9      -9
## 2147     -9      -9
## 2148     -9      -9
## 2149     -9      -9
## 2150     -9      -9
## 2151     -9      -9
## 2152     -9      -9
## 2153     -9      -9
## 2154     -9      -9
## 2155     -9      -9
## 2156     -8      -8
## 2157     -9      -9
## 2158     -9      -9
## 2159     -9      -9
## 2160     -9      -9
## 2161  32300      -9
## 2162     -8      -8
## 2163  98000      -8
## 2164     -9      -9
## 2165  13800 1500000
## 2166     -9      -9
## 2167     -9      -9
## 2168     -9      -9
## 2169     -9      -9
## 2170     -9      -9
## 2171     -9      -9
## 2172  22200      -9
## 2173     -9      -9
## 2174     -9      -9
## 2175     -9      -9
## 2176     -9      -9
## 2177     -9      -9
## 2178     -9   18300
```

```
## 2179     -9      -9
## 2180      0      -8
## 2181     -9      -9
## 2182     -9      -9
## 2183  65000  180000
## 2184     -9      -9
## 2185     -9      -9
## 2186     -8      -8
## 2187     -9      -9
## 2188     -9      -9
## 2189     -9      -9
## 2190  63000      -8
## 2191     -9      -9
## 2192     -9      -9
## 2193     -8      -9
## 2194     -8      -8
## 2195     -9      -9
## 2196     -9      -9
## 2197  99000      -8
## 2198     -8      -8
## 2199     -9      -9
## 2200     -8      -8
## 2201     -9      -9
## 2202     -9      -9
## 2203     -9      -9
## 2204     -9      -9
## 2205     -9      -9
## 2206 102000      -8
## 2207  67000  150000
## 2208     -9      -9
## 2209     -9      -9
## 2210     -9      -9
## 2211  63000   44700
## 2212 108000      -8
## 2213     -9      -9
## 2214  73000      -9
## 2215  42600   14800
## 2216     -9      -9
## 2217     -9      -9
## 2218     -9      -9
## 2219     -9      -9
## 2220     -9      -9
## 2221     -9      -9
## 2222     -8      -8
## 2223     -9      -9
## 2224     -9      -9
## 2225     -9      -9
## 2226     -9      -9
## 2227     -9      -9
## 2228     -8      -8
## 2229     -9      -9
## 2230     -9      -9
## 2231     -9      -9
## 2232     -9      -9
```

```
## 2233  48000      -9
## 2234     -9      -9
## 2235     -9      -9
## 2236     -8      -8
## 2237     -9      -9
## 2238     -9      -9
## 2239     -9      -9
## 2240     -8      -8
## 2241     -9      -9
## 2242     -9      -9
## 2243      0       0
## 2244      0       0
## 2245     -8      -8
## 2246     -9      -9
## 2247     -9      -9
## 2248     -8      -8
## 2249     -9      -9
## 2250     -8      -8
## 2251     -9      -9
## 2252     -9      -9
## 2253     -9      -9
## 2254     -9      -9
## 2255     -9      -9
## 2256     -9      -9
## 2257     -9      -9
## 2258     -9      -9
## 2259     -9      -9
## 2260     -9      -9
## 2261     -9      -9
## 2262     -9      -9
## 2263     -9      -9
## 2264     -8      -8
## 2265  67000  181000
## 2266     -8      -8
## 2267  24700      -9
## 2268     -9      -9
## 2269     -9      -9
## 2270     -9      -9
## 2271     -9      -9
## 2272     -9      -9
## 2273     -9      -9
## 2274     -9      -9
## 2275     -9      -9
## 2276     -8      -8
## 2277  40000      -9
## 2278     -8      -8
## 2279     -9      -9
## 2280     -9      -9
## 2281     -8      -8
## 2282     -9      -9
## 2283     -9      -9
## 2284     -8      -8
## 2285     -9      -9
## 2286     -8      -8
```

```
## 2287  81000   22000
## 2288     -9      -9
## 2289     -8   36500
## 2290     -9      -9
## 2291     -9      -9
## 2292  31200       0
## 2293     -9      -9
## 2294     -9      -9
## 2295     -9      -9
## 2296     -9      -9
## 2297     -9      -9
## 2298     -8      -8
## 2299     -9      -9
## 2300  53000      -8
## 2301     -8      -8
## 2302     -9      -9
## 2303     -9      -9
## 2304     -9      -9
## 2305     -8      -8
## 2306     -9      -9
## 2307     -9      -9
## 2308     -9      -9
## 2309     -9      -9
## 2310     -9      -9
## 2311     -9      -9
## 2312     -9      -9
## 2313     -9      -9
## 2314     -9      -9
## 2315     -9      -9
## 2316     -9      -9
## 2317     -8      -8
## 2318     -8      -8
## 2319     -9      -9
## 2320     -9      -9
## 2321 100000      -8
## 2322     -9      -9
## 2323     -9      -9
## 2324     -9      -9
## 2325     -9      -9
## 2326 110000   35000
## 2327     -9      -9
## 2328     -9      -9
## 2329     -9      -9
## 2330     -9      -9
## 2331     -9      -9
## 2332     -9      -9
## 2333     -9      -9
## 2334     -9      -9
## 2335     -9      -9
## 2336  16800      -9
## 2337     -8      -8
## 2338     -9      -9
## 2339     -8      -8
## 2340 259046  600000
```

```
## 2341     -9      -9
## 2342      0       0
## 2343  32000      -8
## 2344  79000  170000
## 2345     -9      -9
## 2346     -9      -9
## 2347     -9      -9
## 2348  52000      -9
## 2349     -9      -9
## 2350     -9      -9
## 2351     -9      -9
## 2352     -8      -8
## 2353     -9      -9
## 2354 259046  547000
## 2355  57000 2548054
## 2356     -9      -9
## 2357     -9      -9
## 2358     -9      -9
## 2359     -8      -8
## 2360     -9      -9
## 2361  49800   83000
## 2362     -9      -9
## 2363     -8      -8
## 2364     -8      -8
## 2365     -9      -9
## 2366  69000      -9
## 2367     -9      -9
## 2368     -9      -9
## 2369     -9      -9
## 2370     -9      -9
## 2371     -9      -9
## 2372  45300  245000
## 2373     -9      -9
## 2374  95000   44000
## 2375     -9      -9
## 2376     -9      -9
## 2377     -8      -8
## 2378     -9      -9
## 2379  11400      -8
## 2380     -9      -9
## 2381     -9      -9
## 2382  39700  292000
## 2383     -9      -9
## 2384     -9      -9
## 2385   2500      -9
## 2386     -9      -9
## 2387     -8      -8
## 2388     -9      -9
## 2389     -9      -9
## 2390  67000      -8
## 2391     -9      -9
## 2392     -8      -8
## 2393     -9      -9
## 2394     -9      -9
```

```
## 2395     -9      -9
## 2396      0       0
## 2397  21600   20000
## 2398     -9      -9
## 2399     -9      -9
## 2400  27600      -8
## 2401     -9      -9
## 2402     -8      -8
## 2403     -8      -8
## 2404     -9      -9
## 2405     -9      -9
## 2406     -8      -8
## 2407     -9      -9
## 2408     -9      -9
## 2409     -9      -9
## 2410     -9      -9
## 2411  12300   50000
## 2412     -9      -9
## 2413     -9      -9
## 2414     -8      -8
## 2415     -9      -9
## 2416     -9      -9
## 2417 259046  269000
## 2418 111000  255000
## 2419     -9      -9
## 2420     -8      -8
## 2421     -9      -9
## 2422     -9      -9
## 2423  65000  191000
## 2424     -9      -9
## 2425     -8      -8
## 2426     -9      -9
## 2427     -9      -9
## 2428     -9      -9
## 2429     -9      -9
## 2430     -9      -9
## 2431     -8      -8
## 2432     -9      -9
## 2433     -9      -9
## 2434     -8      -8
## 2435 108000   97000
## 2436     -9      -9
## 2437  13700      -9
## 2438     -8      -8
## 2439  49500      -9
## 2440     -9      -9
## 2441     -9      -9
## 2442  69000  445000
## 2443     -9      -9
## 2444     -9      -9
## 2445     -8      -8
## 2446     -9      -9
## 2447     -9      -9
## 2448  49800  577000
```

```
## 2449     -9      -9
## 2450     -8      -8
## 2451     -8      -8
## 2452     -8      -8
## 2453     -9      -9
## 2454     -9      -9
## 2455     -9      -9
## 2456     -9      -9
## 2457     -8      -8
## 2458     -9      -9
## 2459     -8      -8
## 2460     -8      -8
## 2461  25500      -9
## 2462     -9      -9
## 2463     -9      -9
## 2464     -9      -9
## 2465  34500      -9
## 2466     -9      -9
## 2467  42000      -9
## 2468     -9      -9
## 2469     -9      -9
## 2470     -9      -9
## 2471     -9      -9
## 2472     -9      -9
## 2473     -9      -9
## 2474 259046  100000
## 2475  30900      -9
## 2476     -9      -9
## 2477     -9      -9
## 2478     -9      -9
## 2479     -9      -9
## 2480     -9      -9
## 2481     -9      -9
## 2482     -9      -9
## 2483     -9      -9
## 2484     -9      -9
## 2485  30000       0
## 2486  40500      -9
## 2487     -9      -9
## 2488     -9      -9
## 2489  32600      -8
## 2490     -9      -9
## 2491     -9      -9
## 2492     -9      -9
## 2493  30000      -9
## 2494 112000   58000
## 2495     -8   24300
## 2496     -9      -9
## 2497     -9      -9
## 2498  83000   26100
## 2499     -9      -9
## 2500     -9      -9
## 2501     -9      -8
## 2502     -9      -9
```

```
## 2503  18200      -9
## 2504     -9      -9
## 2505     -9      -9
## 2506     -8      -8
## 2507  10400      -9
## 2508  12000      -9
## 2509 113000      -8
## 2510     -9      -9
## 2511  15800   27900
## 2512  36500      -9
## 2513     -9      -9
## 2514     -9      -9
## 2515     -9      -9
## 2516     -9      -9
## 2517     -9      -9
## 2518     -8      -8
## 2519     -9      -9
## 2520     -9      -9
## 2521     -9      -9
## 2522     -9      -9
## 2523     -9      -9
## 2524  14400   19400
## 2525     -9      -9
## 2526     -9      -9
## 2527     -9      -9
## 2528     -9      -9
## 2529  75000      -8
## 2530     -9      -9
## 2531     -9      -9
## 2532     -9      -9
## 2533     -9      -9
## 2534     -9      -9
## 2535     -9      -9
## 2536     -9      -9
## 2537     -9      -9
## 2538  12000      -9
## 2539   9000      -9
## 2540     -9      -9
## 2541     -9      -9
## 2542  65000  109000
## 2543     -8  110000
## 2544     -9      -9
## 2545     -9      -9
## 2546     -9      -9
## 2547  70000      -8
## 2548     -9      -9
## 2549     -9      -9
## 2550     -9      -9
## 2551     -9      -9
## 2552     -8      -8
## 2553     -9      -9
## 2554     -9      -9
## 2555     -9      -9
## 2556     -9      -9
```

```
## 2557     -9      -9
## 2558     -9      -9
## 2559     -9      -9
## 2560     -9      -9
## 2561     -9      -9
## 2562     -9      -9
## 2563     -9       0
## 2564     -9      -9
## 2565     -9      -9
## 2566     -9      -9
## 2567     -9      -9
## 2568     -9      -9
## 2569  22000      -8
## 2570     -9      -9
## 2571     -9      -9
## 2572     -9      -9
## 2573     -9      -9
## 2574  24100   12900
## 2575     -8      -8
## 2576     -8      -8
## 2577     -9      -9
## 2578     -9      -9
## 2579     -9      -9
## 2580     -9      -9
## 2581     -9      -9
## 2582     -9      -9
## 2583     -9      -9
## 2584     -9      -9
## 2585     -9      -9
## 2586     -9      -9
## 2587     -9      -9
## 2588     -9      -9
## 2589     -9      -9
## 2590  80000      -8
## 2591  40000      -8
## 2592     -9      -9
## 2593     -9      -9
## 2594     -9      -8
## 2595  15100      -9
## 2596     -9      -9
## 2597     -9      -9
## 2598     -9      -9
## 2599     -9      -9
## 2600     -9      -9
## 2601     -9      -9
## 2602  30000   10000
## 2603     -9      -9
## 2604     -9      -9
## 2605  37000      -8
## 2606  67000      -8
## 2607     -9      -9
## 2608  32800      -9
## 2609     -8      -8
## 2610     -9      -9
```

```
## 2611     -9      -9
## 2612     -8      -8
## 2613     -9      -9
## 2614     -9      -9
## 2615  40000      -9
## 2616  44000      -9
## 2617  59000   20000
## 2618  20000  844000
## 2619  30600   14500
## 2620     -8      -8
## 2621     -9      -9
## 2622     -9      -9
## 2623  23900   62000
## 2624     -9      -9
## 2625     -9      -9
## 2626     -9      -9
## 2627     -9      -9
## 2628     -9      -9
## 2629     -9      -9
## 2630     -9      -9
## 2631     -9      -9
## 2632     -8      -8
## 2633     -9      -9
## 2634     -9      -9
## 2635     -8      -8
## 2636  27700  163000
## 2637     -9      -9
## 2638     -9      -9
## 2639     -9      -9
## 2640     -9      -9
## 2641     -8      -8
## 2642     -8      -8
## 2643     -8      -8
## 2644     -8      -8
## 2645     -9      -9
## 2646     -9      -9
## 2647     -8      -8
## 2648     -8   24900
## 2649     -8  142000
## 2650     -8      -9
## 2651  47800      -9
## 2652     -9      -9
## 2653     -9      -9
## 2654     -9      -9
## 2655  46900      -9
## 2656     -9      -9
## 2657  67000      -9
## 2658 110000  131000
## 2659  79000      -9
## 2660     -9      -9
## 2661  53000      -9
## 2662     -9      -9
## 2663 109000      -9
## 2664     -9      -9
```

```
## 2665     -9      -9
## 2666     -9      -9
## 2667     -9      -9
## 2668     -9      -9
## 2669     -9      -9
## 2670     -9      -9
## 2671     -9      -9
## 2672     -9      -9
## 2673     -8      -8
## 2674     -9      -9
## 2675     -9      -9
## 2676  45000      -8
## 2677     -9      -9
## 2678     -9      -9
## 2679     -9      -9
## 2680  90000      -8
## 2681     -9      -9
## 2682     -9      -9
## 2683     -9      -9
## 2684     -9      -9
## 2685     -9      -9
## 2686     -9      -9
## 2687     -9      -9
## 2688     -9      -9
## 2689     -9      -9
## 2690     -8      -8
## 2691     -9      -9
## 2692     -9      -9
## 2693     -9      -9
## 2694     -9      -9
## 2695     -9      -9
## 2696     -9      -9
## 2697     -9      -9
## 2698     -9      -9
## 2699     -9      -9
## 2700     -9      -9
## 2701     -9      -9
## 2702     -9      -9
## 2703     -9      -9
## 2704     -9      -9
## 2705     -9      -9
## 2706     -9      -9
## 2707     -9      -9
## 2708     -9      -9
## 2709     -9      -9
## 2710     -9      -9
## 2711     -9      -9
## 2712     -9      -9
## 2713     -9      -9
## 2714     -9      -9
## 2715     -9      -9
## 2716     -9      -9
## 2717     -9      -9
## 2718     -9      -9
```

```
## 2719     -9      -9
## 2720     -9      -9
## 2721     -9      -9
## 2722     -9      -9
## 2723     -9      -9
## 2724  24000      -9
## 2725     -9      -9
## 2726     -8      -9
## 2727  25000  500000
## 2728     -9      -9
## 2729     -9      -9
## 2730     -9 2548054
## 2731     -9      -9
## 2732     -8      -8
## 2733     -8      -8
## 2734     -8      -8
## 2735     -9      -9
## 2736  51000      -8
## 2737     -8      -8
## 2738     -8      -8
## 2739     -8      -8
## 2740     -8      -8
## 2741     -9      -9
## 2742     -9      -9
## 2743     -8      -8
## 2744     -9      -9
## 2745     -9      -9
## 2746     -9      -9
## 2747     -8      -8
## 2748     -9      -9
## 2749     -9      -9
## 2750     -8      -8
## 2751     -8      -8
## 2752     -9      -9
## 2753     -8      -8
## 2754     -9      -9
## 2755     -9      -9
## 2756     -9      -9
## 2757     -8      -8
## 2758     -8      -8
## 2759     -9      -9
## 2760     -8      -8
## 2761     -8      -8
## 2762     -8      -8
## 2763     -9      -9
## 2764  27000      -9
## 2765     -9      -9
## 2766     -8      -8
## 2767     -8      -8
## 2768     -8      -8
## 2769     -9      -9
## 2770 122000   41800
## 2771     -8      -8
## 2772     -9      -9
```

```
## 2773  34000      -9
## 2774     -9      -9
## 2775     -8      -8
## 2776     -8      -8
## 2777     -9      -9
## 2778     -9      -9
## 2779     -9      -9
## 2780     -9      -9
## 2781     -8      -8
## 2782     -9      -9
## 2783  12300  228000
## 2784     -8      -8
## 2785     -8      -8
## 2786     -9      -9
## 2787     -9      -9
## 2788     -8      -8
## 2789     -9      -9
## 2790     -9      -9
## 2791     -8      -8
## 2792     -8      -8
## 2793     -8      -8
## 2794     -8      -8
## 2795     -9      -9
## 2796     -8      -8
## 2797  14400   40100
## 2798     -8      -8
## 2799     -8      -9
## 2800     -8      -8
## 2801   5400      -9
## 2802     -9      -9
## 2803     -8      -8
## 2804  16700   70000
## 2805     -9      -9
## 2806     -8      -8
## 2807     -8      -8
## 2808     -8      -8
## 2809     -9      -9
## 2810     -8      -8
## 2811     -9      -9
## 2812     -9      -9
## 2813     -8      -8
## 2814     -9      -9
## 2815     -8      -8
## 2816     -9      -9
## 2817     -8      -8
## 2818     -8      -8
## 2819     -9      -9
## 2820     -9      -9
## 2821      0      -8
## 2822  57000  238000
## 2823     -8      -8
## 2824  81000      -9
## 2825     -8      -8
## 2826     -9      -9
```

```
## 2827  40000      -8
## 2828     -9      -9
## 2829     -8      -8
## 2830     -8      -8
## 2831     -8      -8
## 2832     -8      -8
## 2833     -8      -8
## 2834     -9      -9
## 2835     -9      -9
## 2836     -8      -8
## 2837     -9      -9
## 2838     -9      -9
## 2839     -9      -9
## 2840     -8      -8
## 2841     -8      -8
## 2842     -8      -9
## 2843     -8      -8
## 2844     -8      -8
## 2845   9700   11300
## 2846     -8      -8
## 2847     -9      -9
## 2848     -8      -8
## 2849     -8      -8
## 2850     -8      -8
## 2851     -8      -8
## 2852     -8      -8
## 2853     -8      -8
## 2854     -9      -9
## 2855     -9      -9
## 2856     -8      -8
## 2857     -8      -8
## 2858     -8      -8
## 2859     -9      -9
## 2860     -9      -9
## 2861     -8      -8
## 2862     -8      -8
## 2863     -8      -8
## 2864     -8      -8
## 2865     -9      -9
## 2866     -9      -9
## 2867     -8      -8
## 2868     -9      -9
## 2869     -8      -8
## 2870     -8      -8
## 2871     -8      -8
## 2872     -8      -8
## 2873     -9      -9
## 2874     -8      -8
## 2875     -9      -9
## 2876     -8      -8
## 2877     -9    6000
## 2878     -8      -8
## 2879     -9      -9
## 2880  92000      -9
```

```
## 2881     -8      -8
## 2882     -8      -8
## 2883     -9      -9
## 2884     -9      -9
## 2885     -9      -9
## 2886     -9      -9
## 2887     -8      -8
## 2888     -8      -8
## 2889     -8      -8
## 2890     -8      -8
## 2891     -8      -8
## 2892     -8      -8
## 2893     -8      -8
## 2894     -8      -8
## 2895     -8      -8
## 2896     -9      -9
## 2897     -8      -8
## 2898     -8      -8
## 2899     -8      -8
## 2900     -8      -8
## 2901     -9      -9
## 2902  12000      -8
## 2903     -8      -8
## 2904     -8      -8
## 2905     -8      -8
## 2906     -8      -8
## 2907     -8      -8
## 2908     -9      -9
## 2909     -8      -8
## 2910     -8      -8
## 2911     -9      -9
## 2912     -9      -9
## 2913     -8      -8
## 2914     -9      -9
## 2915     -9      -9
## 2916     -8      -8
## 2917     -9      -9
## 2918     -9      -9
## 2919     -9      -9
## 2920     -8      -8
## 2921     -8      -8
## 2922     -9      -9
## 2923     -8      -8
## 2924     -8      -8
## 2925     -9      -9
## 2926     -8      -8
## 2927     -9      -9
## 2928     -8      -8
## 2929     -9      -9
## 2930     -8      -8
## 2931     -9      -9
## 2932     -8      -8
## 2933     -8      -8
## 2934     -9      -9
```

```
## 2935     -8      -8
## 2936     -9      -9
## 2937     -8      -8
## 2938     -8      -8
## 2939     -8      -8
## 2940     -8      -8
## 2941     -8      -8
## 2942     -9      -9
## 2943     -9      -9
## 2944     -8      -8
## 2945     -9      -9
## 2946     -8      -8
## 2947     -8      -8
## 2948     -8      -8
## 2949     -8      -8
## 2950     -9      -9
## 2951     -8      -8
## 2952     -8      -8
## 2953     -8      -8
## 2954     -8      -8
## 2955     -8      -8
## 2956     -9      -9
## 2957     -9      -9
## 2958     -9      -9
## 2959     -9      -9
## 2960     -8      -8
## 2961     -9      -9
## 2962     -8      -8
## 2963     -9      -9
## 2964     -8      -8
## 2965     -8      -8
## 2966     -9      -9
## 2967     -9      -9
## 2968     -8      -8
## 2969     -9      -9
## 2970     -9      -9
## 2971     -8      -8
## 2972     -8      -8
## 2973     -8      -8
## 2974     -9      -9
## 2975     -8      -8
## 2976     -8      -8
## 2977     -8      -8
## 2978     -9      -9
## 2979     -9      -9
## 2980     -9      -9
## 2981     -9      -9
## 2982     -8      -8
## 2983     -8      -8
## 2984     -9      -9
## 2985     -8      -8
## 2986     -8      -8
## 2987     -8      -8
## 2988     -9      -9
```

```
## 2989     -9      -9
## 2990     -8      -8
## 2991     -8      -8
## 2992     -8      -8
## 2993     -8      -8
## 2994     -8      -8
## 2995     -8      -8
## 2996     -9      -9
## 2997     -9      -9
## 2998     -9      -9
## 2999  69000   63000
## 3000     -8      -9
## 3001     -9      -9
## 3002     -8      -8
## 3003     -8      -8
## 3004     -9      -9
## 3005     -8      -8
## 3006     -8      -8
## 3007  28600   17400
## 3008     -8      -8
## 3009     -9      -9
## 3010     -8      -8
## 3011     -8      -8
## 3012     -9      -9
## 3013     -9      -9
## 3014     -8      -8
## 3015     -9      -9
## 3016     -8      -8
## 3017     -9      -9
## 3018     -8      -8
## 3019     -8      -8
## 3020     -8      -8
## 3021     -9      -9
## 3022     -8      -8
## 3023  27000      -9
## 3024     -9      -9
## 3025     -8      -8
## 3026     -8      -8
## 3027     -8      -8
## 3028     -9      -9
## 3029     -9      -9
## 3030     -9      -9
## 3031     -9      -9
## 3032     -8      -8
## 3033     -9      -9
## 3034     -9      -9
## 3035     -8      -8
## 3036     -8      -8
## 3037  22000   57000
## 3038     -8      -8
## 3039     -8      -8
## 3040     -8      -8
## 3041     -8      -8
## 3042     -9      -9
```

```
## 3043     -9      -9
## 3044   7700      -8
## 3045     -8      -8
## 3046     -8      -8
## 3047     -9      -9
## 3048     -8      -8
## 3049     -9      -9
## 3050     -8      -8
## 3051     -8      -8
## 3052  46900   81000
## 3053     -8      -8
## 3054     -8      -8
## 3055     -8      -8
## 3056      0       0
## 3057     -9      -9
## 3058     -8      -8
## 3059  36200      -9
## 3060     -9      -9
## 3061     -9      -9
## 3062     -8      -8
## 3063     -8      -8
## 3064     -8      -8
## 3065     -8      -8
## 3066     -8      -8
## 3067     -8      -8
## 3068     -8      -8
## 3069     -9      -9
## 3070     -8      -8
## 3071     -9      -9
## 3072     -9  750000
## 3073     -8      -8
## 3074     -8      -8
## 3075     -8      -8
## 3076     -9      -9
## 3077     -8      -8
## 3078     -9      -9
## 3079     -8      -8
## 3080     -8      -8
## 3081     -8      -8
## 3082     -8      -8
## 3083     -9      -9
## 3084     -8      -8
## 3085     -9 2548054
## 3086     -9      -9
## 3087     -8      -8
## 3088     -9      -9
## 3089     -8      -8
## 3090     -9      -9
## 3091     -8      -8
## 3092  58000      -9
## 3093     -9      -9
## 3094     -9      -9
## 3095     -8      -8
## 3096     -8      -8
```

```
## 3097     -8      -8
## 3098     -8      -8
## 3099     -9      -9
## 3100     -9      -9
## 3101     -8      -8
## 3102     -8      -8
## 3103     -8      -8
## 3104     -9      -9
## 3105  82000      -9
## 3106     -9      -9
## 3107     -8      -8
## 3108     -8      -8
## 3109  53000      -8
## 3110     -8      -8
## 3111     -8      -8
## 3112     -8      -8
## 3113     -8      -8
## 3114     -9      -9
## 3115   3500  116000
## 3116     -9      -9
## 3117     -9      -9
## 3118     -9      -9
## 3119     -8      -8
## 3120     -9      -9
## 3121     -8      -8
## 3122     -8      -8
## 3123     -9      -9
## 3124     -8      -8
## 3125     -9      -9
## 3126     -8      -8
## 3127     -8      -8
## 3128     -8      -8
## 3129     -8      -8
## 3130     -9      -9
## 3131     -8      -8
## 3132     -8      -8
## 3133     -8      -8
## 3134     -8      -8
## 3135     -8      -8
## 3136     -8      -8
## 3137     -9      -9
## 3138     -8      -8
## 3139     -8      -8
## 3140     -9      -9
## 3141   9800      -8
## 3142     -9      -9
## 3143     -8      -8
## 3144     -8      -8
## 3145  15000      -8
## 3146     -8      -8
## 3147     -8      -8
## 3148     -9      -9
## 3149  16600  140000
## 3150     -8      -8
```

```
## 3151  12300   13200
## 3152  24000      -9
## 3153     -8      -8
## 3154     -8      -8
## 3155     -8      -8
## 3156     -9      -9
## 3157     -8      -8
## 3158  20000      -8
## 3159  13500  225000
## 3160     -9      -9
## 3161     -8      -8
## 3162   3700   38600
## 3163     -8      -8
## 3164     -8      -8
## 3165     -8      -8
## 3166     -8      -8
## 3167     -9      -9
## 3168     -8      -8
## 3169     -8      -8
## 3170     -9      -9
## 3171     -9      -9
## 3172     -8      -8
## 3173     -8      -8
## 3174     -9      -9
## 3175     -9    3500
## 3176  10000      -8
## 3177     -9      -9
## 3178     -8      -8
## 3179     -8      -8
## 3180     -9      -9
## 3181     -8      -8
## 3182     -8      -8
## 3183  10000      -8
## 3184     -8      -8
## 3185  16400   25900
## 3186      0       0
## 3187     -8      -8
## 3188     -9   99000
## 3189     -8      -8
## 3190     -9      -9
## 3191  16200      -8
## 3192  20900   17300
## 3193     -9      -9
## 3194  18000   72000
## 3195  29000      -8
## 3196     -9      -9
## 3197     -8      -8
## 3198     -8      -8
## 3199     -9      -9
## 3200     -9      -9
## 3201     -8      -8
## 3202     -9      -9
## 3203     -9      -9
## 3204  42900      -9
```

```
## 3205     -9      -9
## 3206     -8   60000
## 3207  20500      -8
## 3208  25500      -8
## 3209     -9      -9
## 3210     -9      -9
## 3211     -9      -9
## 3212     -9      -9
## 3213     -9      -9
## 3214     -8      -8
## 3215     -9      -9
## 3216     -8   50000
## 3217     -9      -9
## 3218     -8      -8
## 3219     -9      -9
## 3220   5700   37200
## 3221     -8      -8
## 3222  23200   37700
## 3223     -8      -8
## 3224  24000      -8
## 3225  60000  100000
## 3226  25500   74000
## 3227     -8      -8
## 3228  15900    2700
## 3229     -8      -8
## 3230  20600   96000
## 3231     -8      -8
## 3232   3500      -8
## 3233     -8      -8
## 3234     -9   52000
## 3235     -8      -8
## 3236  40100   62000
## 3237     -9      -9
## 3238     -9      -9
## 3239  12200  211000
## 3240  36000      -9
## 3241     -8      -8
## 3242     -8      -8
## 3243     -8      -8
## 3244     -8      -8
## 3245  20000      -8
## 3246     -8      -8
## 3247     -8      -8
## 3248  63000  714000
## 3249     -8      -8
## 3250     -9      -9
## 3251  33800  217000
## 3252     -9      -9
## 3253     -8      -8
## 3254     -8      -8
## 3255     -8      -8
## 3256   9200      -8
## 3257     -9      -9
## 3258     -9      -9
```

```
## 3259     -8      -8
## 3260     -9      -9
## 3261  31000       0
## 3262     -9      -9
## 3263     -9      -9
## 3264  36000      -8
## 3265  10700   38300
## 3266     -8      -8
## 3267   1700   61000
## 3268  56000      -9
## 3269  10000    5000
## 3270     -8      -8
## 3271  15000      -8
## 3272     -8      -8
## 3273  97000      -8
## 3274  10000      -8
## 3275  10000      -8
## 3276     -8      -8
## 3277     -9      -9
## 3278     -9      -9
## 3279     -8      -8
## 3280     -8      -8
## 3281   7700   37000
## 3282     -9      -9
## 3283     -8      -8
## 3284     -8   72000
## 3285     -8      -8
## 3286     -8      -8
## 3287     -8      -8
## 3288     -9      -9
## 3289     -9      -9
## 3290     -9      -9
## 3291  22200    3000
## 3292     -8      -8
## 3293  12000   40000
## 3294     -8      -8
## 3295     -8      -8
## 3296     -8      -8
## 3297     -9      -9
## 3298     -8      -8
## 3299     -8      -8
## 3300     -8      -8
## 3301     -8      -8
## 3302  10700  106000
## 3303     -9      -9
## 3304  15000      -9
## 3305  11100  279000
## 3306  48000   48000
## 3307     -9      -9
## 3308   6000    2000
## 3309     -8      -8
## 3310     -8      -8
## 3311  20000      -8
## 3312     -8      -8
```

```
## 3313     -8      -8
## 3314     -9      -9
## 3315     -9      -9
## 3316  78000 2548054
## 3317  18300  117000
## 3318     -8      -8
## 3319   7200    9400
## 3320  12100   15200
## 3321     -8      -8
## 3322  28000  120000
## 3323     -8   40700
## 3324   3500      -8
## 3325     -9      -9
## 3326   3000      -9
## 3327     -8      -8
## 3328  14500  161000
## 3329     -9      -9
## 3330     -8      -8
## 3331  24900   16600
## 3332     -8      -8
## 3333     -8      -8
## 3334     -8      -8
## 3335     -8    2200
## 3336  11400  406000
## 3337     -8      -8
## 3338  31300      -8
## 3339     -8      -8
## 3340     -9      -9
## 3341  16500   44500
## 3342     -8      -8
## 3343  30400  254000
## 3344     -9      -9
## 3345     -9      -9
## 3346     -8      -8
## 3347     -8      -8
## 3348     -8      -8
## 3349     -8      -8
## 3350     -8      -8
## 3351     -8      -8
## 3352     -9      -9
## 3353   3100  150000
## 3354  44200      -9
## 3355  10000      -8
## 3356     -8   31800
## 3357     -8      -8
## 3358     -9      -9
## 3359  30600   77000
## 3360  12400   81000
## 3361  17900  117000
## 3362   8000   14800
## 3363     -9      -9
## 3364  10000       0
## 3365     -8      -8
## 3366     -9      -9
```

```
## 3367     -8      -8
## 3368     -9  200000
## 3369     -9      -9
## 3370     -8      -8
## 3371     -9      -9
## 3372     -8      -8
## 3373     -8      -8
## 3374     -8      -8
## 3375     -9      -9
## 3376     -8      -8
## 3377     -9      -9
## 3378 146000  169000
## 3379     -8      -8
## 3380  13000       0
## 3381   5000  116000
## 3382  23300   10000
## 3383     -9      -9
## 3384   5000      -8
## 3385     -8      -8
## 3386     -9      -9
## 3387  15000      -8
## 3388     -8      -8
## 3389     -9      -9
## 3390  45700  154000
## 3391     -8      -8
## 3392     -8      -8
## 3393     -8      -8
## 3394     -8      -8
## 3395     -9      -9
## 3396     -8      -8
## 3397     -8   20000
## 3398     -8      -8
## 3399     -8      -8
## 3400  12000  240000
## 3401     -8      -8
## 3402  33300      -9
## 3403     -8      -8
## 3404  23000  137000
## 3405 259046      -9
## 3406     -8   64000
## 3407   2700    6700
## 3408     -9      -9
## 3409     -9      -9
## 3410     -8      -8
## 3411     -8      -8
## 3412     -8      -8
## 3413     -8      -8
## 3414     -9      -9
## 3415     -8      -8
## 3416     -8      -8
## 3417  12000   80000
## 3418     -9      -9
## 3419     -9      -9
## 3420     -9      -9
```

```
## 3421     -8      -8
## 3422  13000   65000
## 3423     -8      -8
## 3424     -8      -8
## 3425     -9      -9
## 3426     -8      -8
## 3427     -8      -8
## 3428     -9      -9
## 3429  13100   69000
## 3430     -9      -9
## 3431     -9      -9
## 3432  10700   88000
## 3433     -9      -9
## 3434  26800   78000
## 3435     -9      -9
## 3436  10700   33400
## 3437  30000   11000
## 3438     -8  560000
## 3439     -8   40000
## 3440     -8      -8
## 3441     -9      -9
## 3442  18400  109000
## 3443     -9   32200
## 3444     -8      -8
## 3445  28800  192000
## 3446     -8      -8
## 3447     -8 2548054
## 3448     -8      -8
## 3449     -8      -8
## 3450     -8    4000
## 3451  30000      -9
## 3452     -8      -8
## 3453     -8      -8
## 3454     -8      -8
## 3455     -8      -8
## 3456  47600   74000
## 3457   9700   48800
## 3458   6000  285000
## 3459  13000      -8
## 3460     -9      -9
## 3461     -8      -8
## 3462  22200  335000
## 3463  13500  130000
## 3464     -8      -8
## 3465     -9      -9
## 3466     -9      -9
## 3467  10400      -9
## 3468     -8      -8
## 3469     -8      -8
## 3470     -9      -9
## 3471     -8      -8
## 3472  58000  105000
## 3473   6600      -8
## 3474     -9      -9
```

```
## 3475     -8      -8
## 3476     -9      -9
## 3477  10900  136000
## 3478     -8      -8
## 3479  15600   52000
## 3480     -9      -9
## 3481     -9      -9
## 3482     -9      -9
## 3483     -9      -9
## 3484     -9      -9
## 3485  16200  289000
## 3486     -8      -8
## 3487  23500   61000
## 3488  39800   34900
## 3489     -9      -9
## 3490     -8      -8
## 3491     -8      -8
## 3492     -8      -8
## 3493     -9      -9
## 3494     -8      -8
## 3495   6000    9000
## 3496     -8      -8
## 3497     -8      -8
## 3498  36000      -9
## 3499     -8      -8
## 3500  60000   25000
## 3501  67000    5800
## 3502  41900   52000
## 3503     -8      -8
## 3504   7600      -9
## 3505  38000      -9
## 3506     -9      -9
## 3507     -8      -8
## 3508     -8      -8
## 3509 259046 2548054
## 3510     -8      -8
## 3511     -9      -9
## 3512     -8      -8
## 3513  30300  204000
## 3514     -8      -8
## 3515     -9      -9
## 3516     -8      -8
## 3517     -8      -8
## 3518     -8   40000
## 3519     -8      -8
## 3520     -9      -9
## 3521  38700  366000
## 3522     -9      -9
## 3523     -9  139000
## 3524     -8      -8
## 3525     -8      -8
## 3526     -9      -9
## 3527     -8      -8
## 3528 143000   41700
```

```
## 3529     -8      -8
## 3530     -8      -8
## 3531     -9      -9
## 3532     -8      -8
## 3533     -8      -8
## 3534  51000  126000
## 3535     -9      -9
## 3536     -8      -8
## 3537     -9      -9
## 3538  32200      -8
## 3539  12900   51000
## 3540     -8       0
## 3541   3500 2548054
## 3542     -8      -8
## 3543  20000      -8
## 3544   5000   15000
## 3545     -8  271000
## 3546  18200  185000
## 3547     -8      -8
## 3548     -9   40000
## 3549   6000   14100
## 3550     -9      -9
## 3551     -9      -9
## 3552  34000      -8
## 3553  60000      -8
## 3554     -8      -8
## 3555   6900   34600
## 3556  44500   49400
## 3557     -8      -8
## 3558     -9      -9
## 3559     -9      -9
## 3560     -8      -8
## 3561  16200      -8
## 3562  94000  319000
## 3563     -8      -8
## 3564     -9      -9
## 3565  81000  106000
## 3566  16500  174000
## 3567  38600   51000
## 3568  40000  150000
## 3569  53000 1140000
## 3570     -8      -8
## 3571     -8      -8
## 3572     -8      -8
## 3573     -9      -9
## 3574   5000      -8
## 3575     -8      -8
## 3576   9900      -9
## 3577     -9      -9
## 3578  12500   62000
## 3579     -8      -8
## 3580 259046  100000
## 3581  18700  102000
## 3582  13700   61000
```

```
## 3583     -9      -9
## 3584  32400  167000
## 3585  24000   15000
## 3586     -8      -8
## 3587  44100    4300
## 3588     -8      -8
## 3589     -8      -8
## 3590     -9      -9
## 3591     -8      -8
## 3592     -9      -9
## 3593  39200      -8
## 3594  15500   43200
## 3595  44000  322000
## 3596  52000  157000
## 3597     -8      -8
## 3598     -8      -8
## 3599  40600   96000
## 3600     -8      -8
## 3601     -9      -9
## 3602  32800   75000
## 3603     -8      -8
## 3604  95000      -8
## 3605  56000  532000
## 3606     -8      -8
## 3607     -8      -8
## 3608     -8      -8
## 3609     -9      -9
## 3610     -9  111000
## 3611     -9      -9
## 3612     -9      -9
## 3613     -9      -9
## 3614     -8      -8
## 3615     -8      -8
## 3616     -8      -8
## 3617  24000   94000
## 3618     -8      -8
## 3619     -9      -9
## 3620     -9      -9
## 3621 102000  218000
## 3622     -8      -8
## 3623     -8  138000
## 3624   2000  218000
## 3625  34600  160000
## 3626  60000  130000
## 3627   7100   73000
## 3628     -9      -9
## 3629     -9      -9
## 3630     -8      -8
## 3631     -8      -8
## 3632     -8      -8
## 3633 116000  394000
## 3634  39500  235000
## 3635  29900   60000
## 3636     -8      -8
```

```
## 3637  47100  177000
## 3638 259046  100000
## 3639     -8      -8
## 3640     -9 2548054
## 3641  20100  266000
## 3642  57000      -9
## 3643  11100  304000
## 3644  29200      -8
## 3645 259046      -9
## 3646     -8      -8
## 3647  57000  119000
## 3648     -9      -9
## 3649     -8      -8
## 3650     -9      -9
## 3651     -8      -8
## 3652     -8      -8
## 3653     -9      -9
## 3654     -8      -8
## 3655     -8      -8
## 3656  30700   59000
## 3657  54000      -8
## 3658  41600  184000
## 3659  56000      -9
## 3660     -9      -9
## 3661     -8      -8
## 3662     -9      -9
## 3663     -8      -8
## 3664     -9      -9
## 3665  50000      -9
## 3666 259046      -8
## 3667   4800      -8
## 3668     -9      -9
## 3669     -9      -9
## 3670  80000  110000
## 3671     -9      -9
## 3672     -9      -9
## 3673     -8   29500
## 3674     -9      -9
## 3675  99000  449000
## 3676     -9      -9
## 3677  26900      -8
## 3678  85000      -9
## 3679     -8      -8
## 3680     -8      -8
## 3681     -9      -9
## 3682     -8      -8
## 3683     -8      -8
## 3684     -8      -8
## 3685  64000  142000
## 3686     -9      -9
## 3687  34800      -8
## 3688  78000  105000
## 3689     -8      -8
## 3690     -8      -8
```

```
## 3691     -8      -8
## 3692  52000      -9
## 3693     -9      -9
## 3694     -8      -8
## 3695     -8      -8
## 3696  78000  284000
## 3697     -9      -9
## 3698  51000  156000
## 3699     -9      -9
## 3700 100000  120000
## 3701     -9      -9
## 3702     -8      -8
## 3703  63000  156000
## 3704     -8      -8
## 3705     -9      -9
## 3706     -8      -8
## 3707     -8      -8
## 3708     -8      -8
## 3709     -9      -9
## 3710     -8      -8
## 3711     -8      -8
## 3712 100000      -8
## 3713     -9      -9
## 3714  42700   66000
## 3715  90000  400000
## 3716     -8      -8
## 3717     -8      -8
## 3718  67000   33700
## 3719     -9      -9
## 3720     -8      -8
## 3721  43500  272000
## 3722     -9      -9
## 3723  81000  504000
## 3724     -8      -8
## 3725     -8      -8
## 3726     -8      -8
## 3727     -8      -8
## 3728     -9      -9
## 3729     -8      -8
## 3730     -9      -9
## 3731     -9      -9
## 3732     -8      -8
## 3733     -8      -8
## 3734     -8      -8
## 3735  50000      -9
## 3736     -8      -8
## 3737  24500      -8
## 3738  77000 1090000
## 3739  59000  228000
## 3740     -9      -9
## 3741     -9      -9
## 3742     -8      -8
## 3743     -8      -8
## 3744     -8      -8
```

```
## 3745     -8      -8
## 3746     -8      -8
## 3747     -9      -9
## 3748  48000      -8
## 3749     -8      -8
## 3750     -8      -8
## 3751  30000   10000
## 3752     -9      -9
## 3753     -8      -8
## 3754     -9      -9
## 3755     -9      -9
## 3756  97000  413000
## 3757  85000 2000000
## 3758     -8      -8
## 3759     -8      -8
## 3760 154000      -8
## 3761     -9      -9
## 3762  38800      -8
## 3763 143000 1290000
## 3764  45600      -8
## 3765  85000   83000
## 3766 259046   35000
## 3767     -9      -9
## 3768     -8      -8
## 3769     -8      -8
## 3770  75000   68000
## 3771     -9      -9
## 3772     -8      -8
## 3773     -8      -8
## 3774     -9      -9
## 3775     -8      -8
## 3776  82000  189000
## 3777   3900   53000
## 3778     -8      -8
## 3779  18000      -8
## 3780     -9      -9
## 3781     -8      -8
## 3782     -8      -8
## 3783  88000  514000
## 3784     -8      -8
## 3785     -9      -9
## 3786     -9      -9
## 3787     -8      -8
## 3788  50000      -9
## 3789  73000  378000
## 3790     -8      -8
## 3791  24000   15000
## 3792     -8      -8
## 3793     -9      -9
## 3794  43900       0
## 3795      0       0
## 3796     -8      -8
## 3797     -8      -8
## 3798     -9      -9
```

```
## 3799     -9      -9
## 3800     -8      -8
## 3801  74000 1800000
## 3802     -9      -9
## 3803  12700      -8
## 3804     -9      -9
## 3805     -8      -8
## 3806 100000      -8
## 3807  41100      -8
## 3808     -8      -8
## 3809 259046 1620000
## 3810     -9      -9
## 3811 259046      -8
## 3812     -8      -8
## 3813     -8      -8
## 3814     -9      -9
## 3815     -8      -8
## 3816     -9      -9
## 3817     -8      -8
## 3818     -8      -8
## 3819     -8      -8
## 3820     -9      -9
## 3821     -8      -8
## 3822 145000  571000
## 3823 100000   75000
## 3824     -8      -8
## 3825     -9      -9
## 3826     -9      -9
## 3827     -9      -9
## 3828   5300   48000
## 3829     -9      -9
## 3830     -9      -9
## 3831  46700      -8
## 3832     -8      -8
## 3833     -9      -9
## 3834     -9      -9
## 3835     -9   15400
## 3836     -8      -8
## 3837     -9  127000
## 3838     -9      -9
## 3839     -9      -9
## 3840     -9      -9
## 3841     -9      -9
## 3842 259046 1560000
## 3843   2200       0
## 3844   9500   74000
## 3845     -9      -9
## 3846   9900   48300
## 3847     -9      -9
## 3848     -9      -9
## 3849  15000      -9
## 3850     -8      -8
## 3851     -8      -8
## 3852     -9      -9
```

```
## 3853     -9      -9
## 3854     -9      -9
## 3855     -9      -9
## 3856     -9      -9
## 3857  20000   12800
## 3858     -9      -9
## 3859     -9      -9
## 3860     -9      -9
## 3861     -9      -9
## 3862     -9      -9
## 3863     -8      -8
## 3864     -9      -9
## 3865     -9      -9
## 3866     -9      -9
## 3867     -9      -9
## 3868     -9      -9
## 3869   2900   28000
## 3870     -9      -9
## 3871   4000      -8
## 3872  16000      -9
## 3873     -9      -9
## 3874     -9      -9
## 3875   1800      -8
## 3876     -8      -8
## 3877  27200      -8
## 3878  88000 1520000
## 3879   2500      -8
## 3880     -9      -9
## 3881   5000   10000
## 3882  95000       0
## 3883  22000   35000
## 3884      0       0
## 3885  40000   50000
## 3886  13000   20000
## 3887  25100  258000
## 3888     -9      -9
## 3889     -9      -9
## 3890     -9      -9
## 3891  25500      -8
## 3892     -8      -8
## 3893   6500   38400
## 3894     -9      -9
## 3895     -9      -9
## 3896     -9      -9
## 3897  10000      -9
## 3898  33900      -8
## 3899     -9      -9
## 3900  22000   50000
## 3901     -8      -8
## 3902     -9      -9
## 3903   4000    5000
## 3904     -8      -9
## 3905  29200    5000
## 3906     -9      -9
```

```
## 3907     -9      -9
## 3908   3200      -8
## 3909     -9      -9
## 3910     -9      -9
## 3911   6000   10000
## 3912  65000      -8
## 3913  71000      -9
## 3914      0   10000
## 3915     -8      -8
## 3916     -9      -9
## 3917   6000   19700
## 3918     -9      -9
## 3919  21000  628000
## 3920  16300    2000
## 3921     -9      -9
## 3922     -9      -9
## 3923     -8      -8
## 3924     -9      -9
## 3925      0       0
## 3926  38000      -8
## 3927     -9      -9
## 3928     -8      -8
## 3929   2000    5000
## 3930     -9      -9
## 3931  14100      -9
## 3932     -8      -8
## 3933   3000    5000
## 3934     -9      -9
## 3935  12000   25000
## 3936     -8      -8
## 3937     -8      -8
## 3938     -9      -9
## 3939     -9      -9
## 3940     -9      -9
## 3941     -8      -9
## 3942  12500    8500
## 3943     -9      -9
## 3944  17400   47600
## 3945  17000      -9
## 3946  32700  119000
## 3947     -9      -9
## 3948     -9      -9
## 3949     -8    3500
## 3950   9600  141000
## 3951  11500      -8
## 3952   7500      -8
## 3953  30500  578000
## 3954     -9      -9
## 3955     -9      -9
## 3956  25900   69000
## 3957   5000  122000
## 3958   5400      -9
## 3959  41800   34100
## 3960  45700  104000
```

```
## 3961     -9   34100
## 3962     -8      -8
## 3963  16200   15100
## 3964     -8      -8
## 3965     -9      -9
## 3966     -8      -8
## 3967     -8      -8
## 3968  61000      -9
## 3969  40800   54000
## 3970     -9      -9
## 3971   7000       0
## 3972      0       0
## 3973     -8      -8
## 3974  18000   15000
## 3975     -8      -8
## 3976     -9      -9
## 3977     -9      -9
## 3978     -8      -8
## 3979  30000      -9
## 3980  17100  134000
## 3981  67000  270000
## 3982     -9   41100
## 3983     -8      -8
## 3984     -9      -9
## 3985     -9      -9
## 3986     -9      -9
## 3987     -9      -9
## 3988  60000  164000
## 3989     -8      -8
## 3990 123000      -8
## 3991     -9      -9
## 3992     -9      -9
## 3993     -8   28500
## 3994     -8      -8
## 3995  38200  517000
## 3996  21500    2100
## 3997     -9      -9
## 3998  59000   82000
## 3999     -9      -9
## 4000     -8      -8
## 4001     -9      -9
## 4002  36300      -8
## 4003   2000      -8
## 4004  28400   83000
## 4005     -9      -9
## 4006  19000   34100
## 4007  20000      -8
## 4008     -9      -9
## 4009   5700   27400
## 4010   5000   15000
## 4011     -9      -9
## 4012  71000   53000
## 4013     -8      -8
## 4014     -8      -8
```

```
## 4015  16700   11500
## 4016  21700   53000
## 4017     -9      -9
## 4018  26100  129000
## 4019     -9      -9
## 4020     -8      -8
## 4021     -9      -9
## 4022  12000    9600
## 4023   7200      -8
## 4024  32000   58000
## 4025     -9      -9
## 4026  52000  116000
## 4027   5000    7000
## 4028     -9      -9
## 4029  17500  122000
## 4030  27900      -9
## 4031  38000  210000
## 4032     -9      -9
## 4033  91000  782000
## 4034     -9      -9
## 4035     -9      -9
## 4036  20000      -8
## 4037     -9      -9
## 4038  28100  119000
## 4039   5700      -8
## 4040     -9      -9
## 4041     -9      -9
## 4042     -9      -9
## 4043     -8      -8
## 4044     -9      -9
## 4045     -9      -9
## 4046  27600      -8
## 4047     -9      -9
## 4048     -9      -9
## 4049     -9      -9
## 4050     -9      -9
## 4051  25000  164000
## 4052     -9      -9
## 4053   5000    5000
## 4054     -9      -9
## 4055     -9      -9
## 4056     -9      -9
## 4057  25900   94000
## 4058  30800  107000
## 4059     -9      -9
## 4060  50000  208000
## 4061     -9      -9
## 4062     -9      -9
## 4063  42000  348000
## 4064     -9      -9
## 4065  81000  324000
## 4066     -9      -9
## 4067     -9      -9
## 4068  15000      -8
```

```
## 4069     -9      -9
## 4070     -9      -9
## 4071     -9      -9
## 4072     -8      -8
## 4073  15000  257000
## 4074     -9      -9
## 4075     -9      -9
## 4076  18000      -9
## 4077     -9      -9
## 4078     -8      -8
## 4079  12000      -9
## 4080  12000       0
## 4081     -9      -9
## 4082     -9      -9
## 4083     -8  239000
## 4084     -9      -9
## 4085  19300  233000
## 4086     -9      -9
## 4087   3300      -8
## 4088     -9      -9
## 4089  19700      -8
## 4090  65000      -9
## 4091  18000      -9
## 4092  23000      -9
## 4093     -8      -8
## 4094     -9      -8
## 4095     -9      -9
## 4096     -9      -9
## 4097  46000   34000
## 4098  18000       0
## 4099     -9      -9
## 4100     -9      -9
## 4101     -8      -8
## 4102  29000      -8
## 4103  22500   50000
## 4104  17000   25000
## 4105     -9      -9
## 4106  41800  223000
## 4107     -8      -8
## 4108  35000      -8
## 4109   7200      -8
## 4110     -9      -9
## 4111     -8      -8
## 4112     -8      -8
## 4113     -9      -9
## 4114  18000      -8
## 4115     -9      -9
## 4116     -8   29200
## 4117     -9      -9
## 4118   6000      -8
## 4119  89000  211000
## 4120     -9      -9
## 4121     -8      -8
## 4122     -8  357000
```

```
## 4123  39100   36700
## 4124     -9      -9
## 4125     -9      -9
## 4126  54000      -8
## 4127     -9      -9
## 4128  68000      -9
## 4129   6900   34000
## 4130  44000  782000
## 4131  13000      -8
## 4132     -9      -9
## 4133     -8      -8
## 4134     -9      -9
## 4135     -8   10400
## 4136     -9      -9
## 4137     -9      -9
## 4138     -9      -9
## 4139  31800       0
## 4140     -9      -9
## 4141     -9      -9
## 4142  14700  253000
## 4143  22200      -8
## 4144  12800   67000
## 4145     -8      -8
## 4146     -9      -9
## 4147     -9      -9
## 4148 100000  100000
## 4149     -9      -9
## 4150     -8      -8
## 4151     -9      -9
## 4152  15000      -8
## 4153     -9      -9
## 4154     -9      -9
## 4155     -9      -9
## 4156     -9      -8
## 4157  60000 1400000
## 4158  16600  155000
## 4159     -9      -9
## 4160     -9      -9
## 4161     -8      -8
## 4162     -9      -9
## 4163     -9      -9
## 4164     -9      -9
## 4165     -8      -8
## 4166  56000  395000
## 4167  16900  483000
## 4168     -8      -8
## 4169     -9  166000
## 4170     -9      -9
## 4171  27000  485000
## 4172     -9      -9
## 4173  19600  625000
## 4174  64000  350000
## 4175  69000  350000
## 4176     -9      -9
```

```
## 4177     -9      -9
## 4178     -9      -9
## 4179     -9      -9
## 4180     -8      -8
## 4181  52000   64000
## 4182  84000      -8
## 4183   3100   13800
## 4184  26400      -8
## 4185  22500      -8
## 4186     -9  200000
## 4187  48600  147000
## 4188     -9      -9
## 4189     -9      -9
## 4190     -9      -9
## 4191  52000      -9
## 4192     -9      -9
## 4193     -9      -9
## 4194  22000      -9
## 4195     -9      -9
## 4196     -9      -9
## 4197     -9      -9
## 4198     -9      -9
## 4199     -9      -9
## 4200  41900  165000
## 4201  99000  940000
## 4202     -8      -8
## 4203  20000   80000
## 4204     -8      -8
## 4205     -8      -8
## 4206  17100   82000
## 4207     -9      -9
## 4208     -9      -9
## 4209     -9      -9
## 4210  45000  300000
## 4211  57000 2548054
## 4212     -8      -8
## 4213     -9      -9
## 4214  40000      -8
## 4215  44400       0
## 4216     -8      -8
## 4217     -8      -8
## 4218  45000      -8
## 4219  25300       0
## 4220     -9      -9
## 4221  60000      -9
## 4222     -9      -9
## 4223     -9      -9
## 4224     -9      -9
## 4225     -9      -9
## 4226     -9      -9
## 4227     -9      -9
## 4228     -9      -9
## 4229     -9      -9
## 4230     -9      -9
```

```
## 4231     -9      -9
## 4232  30000      -8
## 4233     -8      -8
## 4234     -9      -9
## 4235     -8      -8
## 4236     -9      -9
## 4237 259046      -9
## 4238     -9      -9
## 4239  80000      -9
## 4240     -9      -9
## 4241  49800  930000
## 4242     -8      -8
## 4243     -8      -8
## 4244  66000      -9
## 4245  48400      -8
## 4246     -9      -9
## 4247     -9      -9
## 4248     -9      -9
## 4249  53000       0
## 4250     -9      -9
## 4251     -8  110000
## 4252     -9      -9
## 4253     -9      -9
## 4254     -9      -9
## 4255     -9      -9
## 4256  10000      -8
## 4257     -8      -8
## 4258 154000      -9
## 4259 106000      -9
## 4260     -9      -9
## 4261     -8      -8
## 4262  71000  190000
## 4263     -9      -9
## 4264     -9      -9
## 4265  33700      -9
## 4266  52000   94000
## 4267     -9      -9
## 4268     -9      -9
## 4269     -8      -8
## 4270     -9      -9
## 4271     -9      -9
## 4272     -8  172000
## 4273     -9      -9
## 4274     -9      -9
## 4275     -8  109000
## 4276     -9      -9
## 4277  36000  107000
## 4278     -8      -8
## 4279     -9      -9
## 4280  59000      -8
## 4281     -9      -9
## 4282     -9      -9
## 4283     -9      -9
## 4284     -9      -9
```

```
## 4285     -9      -9
## 4286     -9      -9
## 4287  60000  100000
## 4288     -9      -9
## 4289     -8      -8
## 4290     -9      -9
## 4291     -8 1660000
## 4292     -9      -9
## 4293  74000      -8
## 4294     -9      -9
## 4295  29800  482000
## 4296  75000 1940000
## 4297     -9      -9
## 4298  48300      -8
## 4299  43300   19200
## 4300     -8      -8
## 4301  53000      -8
## 4302  70000  130000
## 4303     -8      -8
## 4304     -9      -9
## 4305     -9      -9
## 4306     -9      -9
## 4307  46200   49300
## 4308     -9      -9
## 4309  10000      -8
## 4310     -9      -9
## 4311     -9      -9
## 4312  67000  162000
## 4313  61000      -9
## 4314     -9      -9
## 4315     -8      -8
## 4316  67000      -8
## 4317     -9      -9
## 4318     -9      -9
## 4319     -9      -9
## 4320     -8      -8
## 4321     -8      -8
## 4322     -9      -9
## 4323     -9      -9
## 4324  58000   75000
## 4325  46700  349000
## 4326     -9      -9
## 4327     -8      -8
## 4328     -9      -9
## 4329     -9      -9
## 4330  41600   81000
## 4331     -9      -9
## 4332     -9      -9
## 4333     -9      -9
## 4334  50000  200000
## 4335     -9      -9
## 4336     -9      -9
## 4337     -9      -9
## 4338     -9      -9
```

```
## 4339  34600   50000
## 4340  39600  250000
## 4341  45000      -8
## 4342  54000  253000
## 4343  25200      -8
## 4344     -9      -9
## 4345 103000   38900
## 4346   4800    2700
## 4347     -9      -9
## 4348     -9      -9
## 4349 115000  211000
## 4350  98000  140000
## 4351  54000  156000
## 4352     -9      -9
## 4353     -9      -9
## 4354     -9      -9
## 4355     -9      -9
## 4356  71000  406000
## 4357     -9      -9
## 4358     -9      -9
## 4359     -9      -9
## 4360  66000  142000
## 4361  15000      -9
## 4362     -9      -9
## 4363     -9      -8
## 4364  11400       0
## 4365     -9      -9
## 4366  52000   43900
## 4367 120000      -8
## 4368  78000  247000
## 4369     -9      -9
## 4370     -9      -9
## 4371     -9      -9
## 4372     -9      -9
## 4373     -9      -9
## 4374     -9      -9
## 4375     -9      -9
## 4376     -9      -9
## 4377      0       0
## 4378     -9      -9
## 4379     -8      -8
## 4380     -9      -9
## 4381  26900      -8
## 4382 259046      -8
## 4383     -9      -9
## 4384     -9      -9
## 4385     -8      -8
## 4386     -9      -9
## 4387     -9      -9
## 4388  20600      -8
## 4389     -9 2548054
## 4390     -9      -9
## 4391     -9      -9
## 4392     -9      -9
```

```
## 4393     -9      -9
## 4394     -9      -9
## 4395     -9      -9
## 4396 121000  196000
## 4397     -9      -9
## 4398     -8      -8
## 4399 100000      -8
## 4400     -9      -9
## 4401  34500      -8
## 4402  42700      -9
## 4403     -8      -8
## 4404     -9      -9
## 4405     -9      -9
## 4406     -9      -9
## 4407     -8      -8
## 4408 163000  401000
## 4409     -8      -8
## 4410     -9      -9
## 4411     -9      -9
## 4412  12200      -8
## 4413     -9      -9
## 4414  36000  182000
## 4415     -9      -9
## 4416  54000      -9
## 4417     -9      -9
## 4418     -9      -9
## 4419  49200 1450000
## 4420 259046 2548054
## 4421     -9      -9
## 4422     -9      -9
## 4423  75000      -8
## 4424     -8  207000
## 4425     -8      -8
```

```r
# A component of OPEX_R coded -8 contributes zero when OPEX_R is observed.
rhfs <- rhfs |>
  mutate(
    across(
      all_of(c("OPPAY", "OPINSUR", "OPELGAS", "OPWTRSWR", "OPREP", "OPMNG",
               "OPPROTH", "OPSECUR", "OPLAND", "OPOTHEX")),
      \(x) if_else(OPEX_R >= 0 & x == -8, 0L, x)
    )
  )



write.csv(rhfs, "RHFS-2024-Cleaned.csv", row.names = FALSE)
```

```r
# Brings back all of the columns that were dropped for convenience at the
# beginning of Cleaning the Dataset
rhfs <- rhfs |>
  left_join(
    rhfs.original |> select(-any_of(names(rhfs)), CONTROLPUF),
    join_by(CONTROLPUF)
```

```
  )

# Loads the data frame with srvyr
df <- rhfs |>
  select(-REPWGT0) |> # REPWGT0 is equivalent to WEIGHT
  as_survey_rep(
    repweights = starts_with("REPWGT"),
    weights = WEIGHT,
    type = "successive-difference",
    combined_weights = TRUE,
    # Decided to set to T to follow SDR as outlined by Ash (to match what was
    # done in Census's Methodology)
    mse = TRUE
  )

# Sanity check for if the data was successfully loaded with srvyr
# using Table 5 of Census's RHFS Methodology
df |>
  summarize(
    properties = survey_total(vartype = "se") # Applies weights and calculates se of properties
  ) |>
  mutate( # Dividing by 1000 to match Table 5's scale
    properties = properties / 1000,
    properties_se = properties_se / 1000,
    moe = properties_se * 1.645 # 90% margin of error
  )
```

```
## # A tibble: 1 x 3
##   properties properties_se   moe
##        <dbl>         <dbl> <dbl>
## 1     18965.          165.  271.
```

```
missing_by_var <- data.frame(
  variable = names(rhfs),
  percent_missing = sapply(
    rhfs,
    \(x) mean(x %in% c(-9))
  )
) |>
  arrange(desc(percent_missing))


# Evidence for MAR (missing at random) in MRKTVAL_R
df |>
  mutate(
    mrktNA = MRKTVAL_R == -9
  ) |>
  summarize(
    survey_mean(mrktNA),
    .by = NUMCAT_R
  )
```

```
## # A tibble: 5 x 3
```

```
##   NUMCAT_R      coef  `_se`
##   <fct>        <dbl>  <dbl>
## 1 1 Unit      0.0923 0.0263
## 2 2-4 Units   0.0716 0.0224
## 3 5-24 Units  0.154  0.0224
## 4 25-49 Units 0.335  0.0547
## 5 50+ Units   0.418  0.0163
```

```
df |>
  filter(MRKTVAL_R != -9) |>
  summarize(
    mean_mrktval = survey_mean(MRKTVAL_R),
    median_mrktval = survey_median(MRKTVAL_R),
    .by = NUMCAT_R
  )
```

```
## # A tibble: 5 x 5
##   NUMCAT_R    mean_mrktval mean_mrktval_se median_mrktval median_mrktval_se
##   <fct>              <dbl>           <dbl>          <dbl>             <dbl>
## 1 1 Unit           397050.          42218.         250000            33154.
## 2 2-4 Units        569582.          33304.         425000            50491.
## 3 5-24 Units      2178309.         118540.        2000000           469164.
## 4 25-49 Units     5000928.         333704.        4000000           224403.
## 5 50+ Units      33895580.        2865340.       20000000          4987831.
```

```
# For citing R, RStudio, and the packages that were used in this analysis
cite_packages(
  pkgs = "All",
  omit = NULL,
  citation.style = "apa",
  include.RStudio = TRUE,
  cite.tidyverse = FALSE,
  out.format = "docx",
  out.dir = getwd()
)
```

```
##
## Citation report available at C:/Users/bfeler/Downloads/Theophilus College/Senior Thesis/Senior-Thesis
```

```
## [1] "C:/Users/bfeler/Downloads/Theophilus College/Senior Thesis/Senior-Thesis-WIP/grateful-report.doc
```